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A machine learning approach to sea-state prediction in the Celtic Sea using SAR satellite data

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Statement of Original Authorship

I hereby certify that the submitted work is my own work, was completed while registered as a candidate for the degree stated on the Title Page, and I have not obtained a degree elsewhere on the basis of the research presented in this submitted work.

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Abstract

Machine learning techniques have recently been employed in a number of cases to extract sea-state information from synthetic aperture radar (SAR) imagery. With increased data availability and reduced cost, it this is being explored as a potential tool for forecasting as opposed to expensive wave buoys and poorer resolution wave models. With a number of offshore wind projects in planning phases, this work could be the start of such a tool for this industry in Ireland. This work utilises Sentinel-1 C-band interferometric wide ground range detected SAR images over the Celtic Sea, collocated temporally and spatially with an Irish Marine Databuoy Observation Network wave buoy. The dataset prior to filtering contained 742 matches. The radar backscatter data extracted from the images is used as input data to a number of machine learning models. Multilayer Perceptrons (MLP), random forests and gradient boosting models are explored with the aim of predicting the significant wave height (SWH), one of the most important sea-state parameters. Architecture testing of the MLP models for $SWH < 4m$, $SWH < 3m$ and for backscatter variance < 3.5 dB was performed. For data limited to $SWH < 3m$ a RMSE of 0.458m was observed which is comparable to literature values but in all cases, the high scatter and poor correlation led to the conclusion that more information is required by the models if they are to achieve improved performance. Random forest and gradient boosting models were used to examine whether different models could exhibit improved performance. No improvement was observed which strengthened the hypothesis that features with more of a connection to the sea-state would be required to observe increased predicting power. Despite this, this work has demonstrated that with further investigation into the SAR data, there is potential for a forecasting tool to be developed.

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List of Acronyms

ANN artificial neural network.

BPNN back-propagation neural network.

CNN convolution neural network.

dB decibels.

DNN deep neural network.

EO Earth Observation.

ESA European Space Agency.

EW Extra Wide swath.

GEE Google Earth Engine.

GMES Global Monitoring for Environment and Security.

H horizontal.

IMDBON Irish Marine Databuoy Observation Network.

IW Interferometric Wide.

MLP multilayer perceptron.

MPI Max Planck Institute.

MSE mean squared error.

NRCS normalised radar cross-section.

PARSA partition rescaling and shift algorithm.

PRF pulse repetition frequency.

PRI pulse repetition interval.

RMSE root mean squared error.

SAR Synthetic Aperture Radar.

SI scatter index.

SLAR side-looking airborne radar.

SLC single-look-complex.

SM Strip Map.

SNR signal-to-noise ratio.

SWH Significant Wave Height.

TOPS Terrain Observation by Progressive Scans.

V vertical.

WV Wave.

1 Introduction

1.1 Motivation

The improvements in availability of spaceborne synthetic aperture radar (SAR) data in recent years has prompted researchers to delve deeper into its uses. One of these potential uses is in sea-state estimation and forecasting. More specifically, a number of studies have shown that machine learning techniques can be used to extract significant wave height (SWH) (see [2.2.1](#)) from the information encoded in SAR images. Sea-state data is usually captured via expensive in-situ equipment or wave models that have poor resolution when compared with SAR images ([Tapoglou et al. 2021](#)). Since satellite imagery is becoming cheaper, (such as the free datasets available on Google Earth Engine which are utilised in this work) and more satellites are being launched (ICEYE recently launched the worlds largest SAR constellation), this may offer an alternative to current methods. Algorithmic and machine learning approaches to sea-state prediction have been shown to be successful in other cases (see [2.2.2](#) and [2.2.3](#)) but no studies have attempted this from an Irish perspective. Also, the studies that explore machine learning in this vein very often do not compare multiple model types. This work aims to be the first attempt to predict significant wave height in Irish waters and will make a comparison of machine learning models to investigate performance and areas where such work can be furthered and improved. This marks a start for a potential forecasting tool to be developed that could rival current wave models.

1.2 TechWorks Marine Ltd.

Techworks Marine Ltd. are a Dublin based company offering marine data solutions to a wide range of clients, from national governments and marine research agencies, to leading engineering companies and the marine renewable sector. They provide and operate a number of services including data buoys, data acquisition systems, a data platform called CoastEye and Earth Observation related data products.

The Earth Observation (EO) team comprises expertise in satellite data products, including visible, infrared and Synthetic Aperture Radar (SAR), among others. Techworks Marine's customers can avail of CoastEye, the companies' data analysis web platform, for their product needs. A recently completed project funded the European Space Agency (ESA), called CoastEO, used a combination of Copernicus Programme Sentinel satellite data along with in-situ data from Techworks MiniBuoys to validate water quality information. The results from this project were presented by the team at the Living Planet Symposium 2022 in Bonn, Germany.

The Earth observation team are driven to explore the potential uses of EO data in all areas and to be at the cutting edge of knowledge and research. The idea for this project was borne out of that drive and interest in newly emerging studies on the application of machine learning techniques to synthetic aperture radar data.

2 Literature review

2.1 Technical Background

2.1.1 Basic Theory of Synthetic Aperture Radar

Synthetic Aperture Radar (SAR) was borne out of side-looking airborne radar (SLAR) in the 1950s, which was a modification to the previous 360° image acquisition technique used on reconnaissance aircraft during World War II. It was known that range pulse shortening could improve image resolution but azimuthal or cross-range resolution was seemingly limited by the size of the aperture (Raney 1983). Carl Wiley of Goodyear Aerospace Corporation first proposed a technique in 1954 whereby successive scans along a direction could be stored and then processed to achieve better resolution, thus mimicking the effect of a larger aperture (Wiley 1965). This incorporated coherent radar and Doppler beam sharpening techniques (Wiley 1985) which led to the current SAR principle not long after (Cutrona et al. 1961).

SAR is an active system, meaning that illumination from the Sun or other sources is not required. This means that its operation is largely independent of light conditions. Also given the microwave wavelength range (30cm - 1mm) in which SAR operates, weather phenomena such as clouds and rain do not impact on operation, permitting all-weather operation. This presents a technique perfectly suited for Earth Observation of phenomena in dynamic environments such as oceanic and coastal wave/current activity, motion of landmass or vegetation patterns (Curlander et al. 1991).

One of the major improvements of SAR over standard radar is that its resolution does not decrease with distance (Ager 2013). High resolution imagery from standard radar instruments at altitudes of Earth Observation satellites ($\sim 600 - 800\text{km}$) is not possible without comically large physical aperture sizes given how image spatial resolution decreases with distance from the target (Curlander et al. 1991). SAR achieves this high resolution through treating each transmit and receive location as if they were elements of the array of a much larger antenna.

High-power electromagnetic pulses are transmitted from a radar system mounted on a forward moving, side-looking platform. Backscattered echoes are received after the transmitted signal has interacted with the Earth's surface. SAR waveforms are typically pulsed and frequency modulated, meaning the amplitude is constant over the pulse of time τ_p (see Figure 1), while the frequency varies over a time t such that a bandwidth of

$$B_T = k_T \tau \quad (2.1)$$

is obtained, where k_T is the *chirp rate*. The receiver then awaits backscattered echoes for a period of time before emitting another pulse. The pulse repetition interval (PRI) is the temporal separation of pulses over the synthesised aperture and is defined such that pulses are repeated at at least every half length of the antenna (Ager 2013). Typical values for pulse repetition frequency (PRF) are between hundreds and thousands of Hz (Moreira et al. 2013). The received signal is reduced and when processed, information about the target object/landscape such as roughness, density, size as well as electrical properties such as conductivity can be extracted (Moreira et al. 2013). This information is encoded in the amplitude and phase of the returned signal. A simplified side-looking geometry is shown in Figure 1. The craft carrying the SAR instrument, with an antenna length L_a , is at an altitude H and moves with velocity v along an assumed straight path. A signal is transmitted

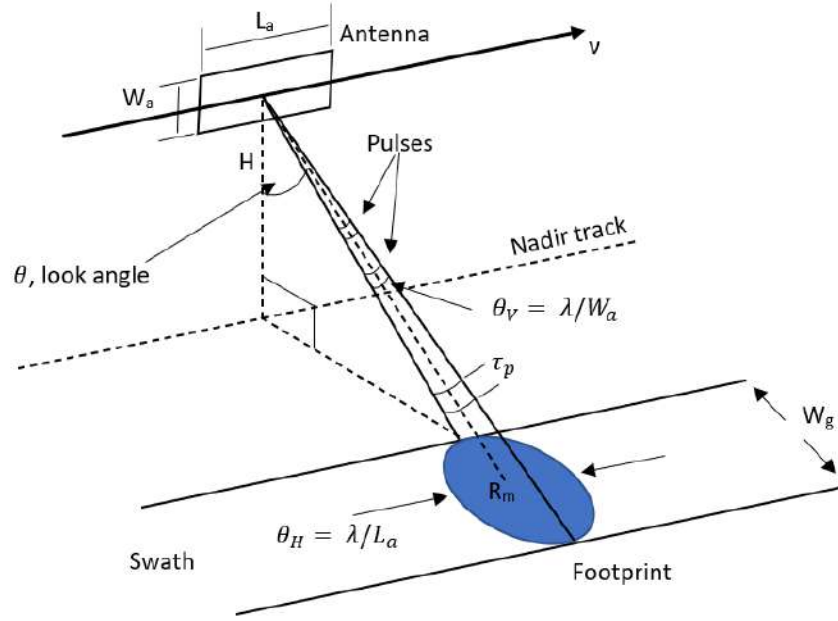


Figure 1: Simplified geometry of the side-looking aperture radar on which SAR is based.

from the antenna which is pointing at an angle θ , known as the *look angle*, from the horizontal. The *swath* width W_g is governed by the height of the antenna while the length depends on the length of observation. The vertical width of the beam is given by

$$\theta_V = \lambda/W_a. \quad (2.2)$$

The distance from the transmitter to the middle of the swath at any instant, R_m , is known as the *slant range*. Terrains are scanned both *along track* and *across track*. For the across track case (slant-range direction), the pulse is swept across the width of the swath at a velocity equal to the speed of light, c , and is converted to a digital format at the sampling frequency f_s . This gives an across track pixel size of $c/2f_s$. As shown in Figure 1, the along track (*azimuth* direction) scanning occurs at velocity v and digitization occurs at the PRI. This gives an along track pixel size of $v \cdot PRI$ (Lauknes 2004). The time in the slant-range direction is referred to as *fast time* whereas it is referred to as *slow time* in the azimuth (Moreira et al. 2013).

The slant-range resolution, ΔR , limit of the side-looking geometry is given by

$$\Delta R = \frac{c\tau_p}{2}. \quad (2.3)$$

Pulse time would need to be decreased to a unreasonable level to obtain a desirable resolution. The signal-to-noise ratio (SNR) of received echoes would be reduced too much due to the loss of energy in each pulse. Instead, pulse compression and match filtering techniques are used so that resolution and SNR are preserved (Curlander et al. 1991). The best cross track resolution then becomes

$$\Delta R = \frac{c}{2B_T}. \quad (2.4)$$

This is not unique to SAR, but where SAR is distinguished is its resolution in the azimuth. The azimuth resolution of the above geometry is

$$\Delta x = R_m \theta_H = R_m \lambda / L_a \quad (2.5)$$

where $\theta_H = \lambda / L_a$ is the angular spread of the antenna beam along the direction of movement. It is clear from this equation how large antenna apertures would have to be especially for spaceborne instruments at large R_m . The breakthrough observation by Wiley was that the reflected pulses from two targets that were at different angles with respect to the radar have their own signature Doppler shifted frequencies (Curlander et al. 1991).

At a position x along the track, the Doppler shifted frequency for a point target with relative velocity $V_r = v$ is given by

$$f_D \approx \frac{2vx}{\lambda R} \quad (\text{Factor of 2} \rightarrow \text{Two way travel}) \quad (2.6)$$

meaning that observations of points x_i that are the same distance R from the radar and within the same beam can be distinguished by analysing the Doppler spectrum of the backscatter. A ground range, R_g , coordinate is derived from this analysis so that each point has coordinates (R_g, x) with the origin located at the point on the ground directly below the moving radar (Curlander et al. 1991). By formulating an equation for R in terms of the parameters shown in Figure 2, substituting into equation 2.6 and rearranging, the expression

$$x^2 \left[\frac{2v}{\lambda f_D} - 1 \right] - R_g^2 = H^2 \quad (2.7)$$

is found to describe a hyperbola as shown in Figure 3. The target position is defined as the intersection of this hyperbola with the circle representing the pulse time delay, $\tau = \frac{2R}{c}$. The direction of the radar beam and whether or not the Doppler shift is negative or positive determines the exact position out of the four intersections. The along-track resolution Δx is related to the frequency resolution Δf by using equation 2.6

$$\Delta x = \left(\frac{\lambda R}{2v} \right) \Delta f. \quad (2.8)$$

The inverse of this is the time in which the radar beam sees the target and can be defined as follows from the relations in Figure 1

$$T = \frac{1}{\Delta f} = \frac{R \theta_H}{v} = \frac{R \lambda}{v L_a}. \quad (2.9)$$

Substituting this into equation 2.8 gives

$$\Delta x = \frac{L_a}{2}. \quad (2.10)$$

This is the theoretical limit to the azimuth resolution and represents the effect of the synthetically larger aperture. It should be noted that this rather simple derivation, for the purposes of this short summary of the theory, makes assumptions that in some cases can be invalid. Despite this, the azimuth resolution of existing SARs approaches this limit (Curlander et al. 1991).

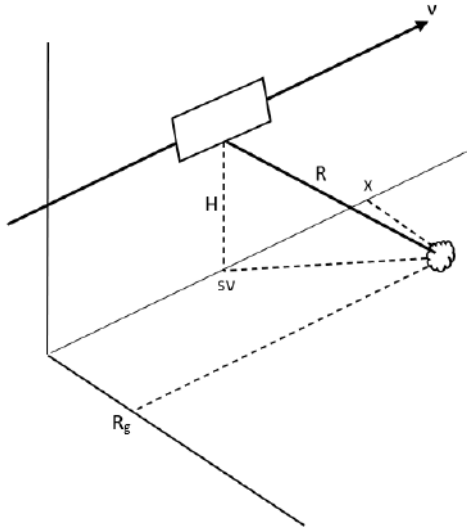


Figure 2: Coordinate system of SAR.

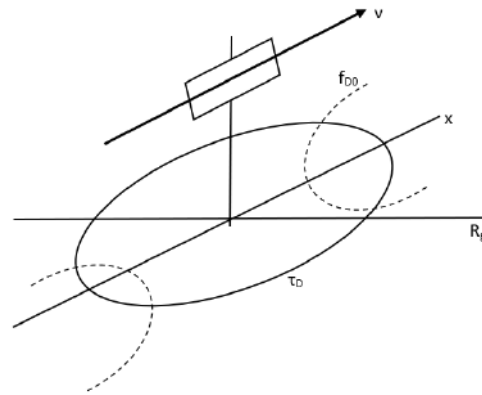


Figure 3: Target location of SAR.

2.1.2 The Sentinel-1 SAR constellation

The Sentinel-1 two-satellite constellation was developed by the ESA as part of the Global Monitoring for Environment and Security (GMES)) initiative (Torres et al. 2012) with the aim of furthering the capabilities and use cases of SAR. Each satellite has a 12-day repeat period with the constellation combining for a 6-day repeat for coverage of the entire world, however over Ireland the constellation achieves down to 2-day repeat. They have an altitude of 704 km and a ground speed of 6.8 km s^{-1} (Pleskachevsky et al. 2019). The constellation consists of Sentinel-1A which was launched in 2014 and Sentinel-1B which followed in 2016. This mission possesses the capabilities to operate without the requirement to request data. The pre-programmed modes of acquisition capture data at a rate of approximately 600 GB per day (Ali et al. 2017) and cover a vast range of global terrains including the ocean, sea-ice and landmasses. Depending on the acquisition mode, the images are optimized for resolution or swath width. The four acquisition modes provided by Sentinel-1 are Interferometric Wide (IW), Strip Map (SM), Extra Wide swath (EW) and Wave (WV). This work will make use of the IW mode for its high resolution as well as the availability of data in the region of interest.

The instrument payload of this constellation is called CSAR. This alludes to its operation in the C-band (central frequency within 5250 - 5570 MHz). The main imaging mode is the IW mode which combines a large swath width of 250 km with good ground range resolution of $5 \text{ m} \times 20 \text{ m}$. The incidence angle for IW image acquisition is in the range 32.9° - 43.1° . This is because of the mechanism used to form the images. To achieve the wide swath, the IW mode is subdivided into three Terrain Observation by Progressive Scans (TOPS) (De Zan & Monti Guarnieri 2006) modes. These sub-swaths are referred to as IW1, IW2 and IW3. Bursts are recorded using sub-echoes. This technique is illustrated in Figure 4. The first burst of the IW1 sub-swath is acquired (blue) after which the elevation of the antenna is altered such that the first burst of the IW2 sub-swath (green) comes next. IW3 then follows in a similar manner before the process is repeated as shown (Yagüe-Martínez et al. 2016).

With SAR imaging, the amount of received backscatter naturally depends on the target morphology

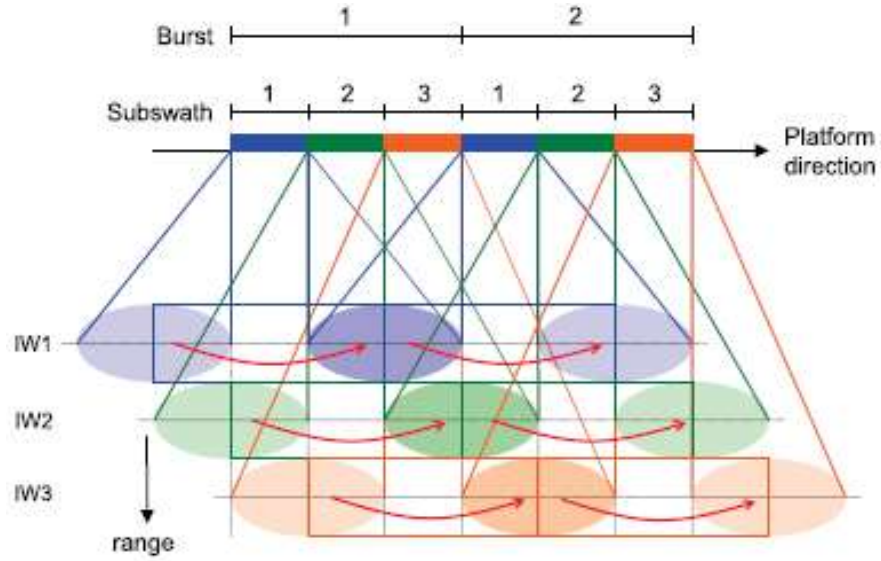


Figure 4: Illustration of the TOPS scan technique of the IW mode of Sentinel-1. *Image Credit: Yagüe-Martínez et al. (2016)*

but also on the polarization of the transmitted and received signals. The IW mode can operate in both single and dual polarizations. The polarization can be either in the horizontal (H) or vertical (V) directions. In the case of single polarization, the transmitted and received signals are in the same polarization i.e HH or VV. In dual polarization, the instrument makes use of two receive chains so that the possibilities are HH+HV and VV+VH. The next section will discuss the choice of polarization to allow the best utilization of SAR data for extracting sea-state parameters.

2.1.3 Imaging the Surface of the Ocean with SAR

For any surface imaged by radar, the received signal depends on the surface roughness, dielectric constant, conductivity and composition. Ocean waves result in radar backscattering due to variations on the order of the size of the radar wavelength. Characteristic features are produced by short waves which are wind-generated and modulated by longer gravity waves. These short waves range in wavelength up to 10s of centimetres (Holt 2004). The normalised radar cross-section (NRCS) is used to quantify the backscatter and can be described as an average of the backscattered intensity over a surface area. It is sometimes termed “sigma-naught” and, when converted to units of decibels (dB), is given by

$$\sigma_0 = 10 \log_{10}(\sigma/A) \quad (2.11)$$

where σ is the ratio of incident to received signal intensity (Meyer 2019):

$$\sigma = \frac{I_{\text{incident}}}{I_{\text{received}}} 4\pi R^2 [\text{m}^2] \quad (2.12)$$

and A is the area on the ground. SAR imaging of the ocean surface is dominated by the backscatter due to the short wave field and depends on the wavelength, orientation and tilt of the waves with

respect to the SAR instrument (Holt 2004). Backscatter increases with increasingly windy conditions and characteristic patterns in SAR images are produced by atmospheric conditions as well as the previously mentioned long wave modulation.

Ocean surface roughness accounts for a major portion of the radar return of SAR signals. Under low wind conditions, Bragg scattering is the dominant process while individual waves and their breaks can generate their own reflections in rougher conditions (Pleskachevsky et al. 2019). Bragg scattering occurs when the short wave components have wavelengths on the order of the signal wavelength

$$\lambda_B = \frac{\lambda_r}{2 \sin \theta_i} \quad (2.13)$$

where θ_i is the incidence angle. The polarization of a SAR signal influences how the signal interacts with the objects that it is directed at. This in turn has an influence on the NRCS. Since the ocean surface is primarily a rough scatterer, the polarization that has the highest scattering power from rough scatterers is of interest to this project. It turns out that VV polarization gives the highest returns for rough scatterers and therefore the most information about the interacting ocean surface (Meyer 2019). When used as input to a machine learning model, image information captured with this polarization will provide the model with the best chance of learning about the true sea-state, and hence the ability to estimate sea-state parameters from SAR images.

When dealing with SAR observations of the ocean surface, it must be considered that the target is in motion. The true wave field is altered in phase by the surface motion causing blurring. This can result in loss of information of very short wave components (Chapron & Garello 2001). The complex imaging mechanisms are divided into three modulations (Alpers & Bruening 1986, Hasselmann et al. 1985):

- Tilt modulation
- Hydrodynamic modulation
- Velocity bunching

Tilt modulation occurs when long waves, as they propagate, change the orientation (tilt) of the short waves thus varying the radar return. Hydrodynamic modulation is when the long waves change the amplitude of the short waves in a manner such that they can pile up or spread out. Finally, velocity bunching is due to the propagation of long waves parallel to the SAR instrument platform. This is related to the fact that, as previously described in 2.1.1, SAR utilises Doppler signals assuming targets that are not in motion. However, waves in motion cause an apparent change in position as interpreted by the SAR measurement. Steeper waves lead to increased velocities to a point where smearing of images results. This introduces a limit to the resolution of SAR in detecting ocean waves in a given sea-state (Holt 2004, Kerbaol et al. 1998).

2.2 State of the Art - Sea-State from SAR Imagery

2.2.1 Significant Wave Height

Significant Wave Height (SWH) is somewhat interchangeable with the phrase ‘sea-state’ in this work. It is defined as the mean wave height of the largest 1/3 of waves.

$$\text{SWH} = \frac{1}{N/3} \sum_{i=1}^{N/3} H_i. \quad (2.14)$$

Where $i = 1$ is the highest wave and N is the number of waves in a given period of time (Holthuijsen 2010). It can be equivalently described as the four times the square root of the zeroth order moment of the wave spectrum (Holthuijsen 2007).

$$\text{SWH} = 4\sqrt{m_0} \quad (2.15)$$

SWH is the most widely used sea-state parameter (Stopa & Mouche 2017) and it is used as opposed to mean wave height in many applications because the impact of larger waves is generally more significant (Tapoglou et al. 2021). This is the parameter this work aims to predict with the designed machine learning models.

2.2.2 Algorithmic Approach

Extracting sea-state information from SAR images is a relatively recent and specialised topic. Numerical models including WAM, SWAN and WAVEWATCH-III (The Wamdi Group 1988, Booij et al. 1999, Tolman et al. 2009) using the laws of physics to simulate wave fields have been the primary method for sea-state estimation and forecasting. These are widely available and provide typical spatial resolutions of 2-10km (Tapoglou et al. 2021). This is one of the limits to these models and is where SAR imagery has the opportunity to offer an alternative, given the ~ 10 m pixel sizes.

The use of SAR to make measurements of the surface of the oceans began as far back as the SEASAT mission which was launched in 1978 (Born et al. 1979). Later missions included European Remote Sensing Satellite constellation ERS-1 (Duchossois 1991) (launched in 1991) and ERS-2 (Francis et al. 1995) (launched in 1995) and ENVISAT (Louet 2001) (launched in 2002). Earlier methods of sea-state retrieval from SAR imagery include the Max Planck Institute (MPI) scheme (Hasselmann & Hasselmann 1991, Hasselmann et al. 1996). This is an iterative method which requires a first guess wave spectrum such as one of those mentioned in the above paragraph. This is one among many similar schemes for retrieving two dimensional ocean wave spectra from SAR data. The purpose of the first guess spectrum is to provide information about the direction of wave propagation and make up for losses due to the complex SAR imaging mechanisms (Wu et al. 2021). Other methods requiring first guess spectra include the semi-parametric retrieval algorithm (Mastenbroek & de Valk 2000) which employed scatterometer measurements of wind vectors to compensate for missing information in the SAR data. Schulz-Stellenfleth et al. (2005) introduced the partition rescaling and shift algorithm (PARSA) which performs a non-linear inversion on single-look-complex (SLC) SAR images but also requires a first guess spectrum such as that of a numerical model. Quasi-linear retrieval methods have also been developed which removed

the need for a first guess spectrum. The disadvantage to these methods is that they fail to resolve high-frequency waves (Chapron & Garello 2001).

Empirical algorithms have the advantage of requiring no prior information while also being less computationally complex than the non-linear approach above. These algorithms are collectively referred to as CWAVE-type algorithms. They use polynomial fitting techniques to relate SAR data to sea-state parameters (Wu et al. 2021). Many examples of these algorithms exist for SAR data from individual spaceborne SAR missions. For example CWAVE_ENV (Li et al. 2011) for ENVISAT and CWAVE_S1 (Stopa & Mouche 2017) for Sentinel-1. The input parameters to these algorithms consist of the previously described NRCS as well as a further 22 decomposed orthogonal harmonic components from the image modulation spectra (Schulz-Stellenfleth et al. 2007). In many cases, SAR image parameters are related to the sea-state in such a non-linear fashion that polynomial fitting can be insufficient (Wu et al. 2021). Hence why machine learning techniques have recently become more popular in tackling such problems.

2.2.3 Machine Learning Approach

Note: Within the literature surrounding machine learning approaches to sea-state extraction from SAR data, and possibly machine learning more broadly, there are commonly used metrics for evaluation of prediction ability of models. These are:

- bias
- root mean squared error (RMSE)
- scatter index (SI)

These parameters are defined as follows:

$$\text{Bias} = \bar{Y} - \bar{X} \quad (2.16)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum (Y_i - X_i)^2} \quad (2.17)$$

$$\text{SI} = \frac{1}{\bar{X}} \sqrt{\frac{1}{N} \sum [(Y_i - \bar{Y}) - (X_i - \bar{X})]^2} \quad (2.18)$$

where Y is the predicted SWH and X is that of the measurement used for comparison. These parameters will be used in this section to discuss the literature on this topic as well as throughout the latter sections of this document in the discussion of results.

Given the difficulties in representing the non-linear relationships between SAR data and sea-state parameters, machine learning approaches naturally arose. In terms of sea-state prediction using Sentinel-1 imagery, a number of studies have been conducted. Mostly closely related to this project are those of Tapoglou et al. (2021) and Xue et al. (2020).

Tapoglou et al. (2021) used the offshore renewable energy sector as the context for their approach. Within this sector, large costs are associated with maintenance of wind turbines since they must be switched off. This issue can be exacerbated by poor forecasting due to the lack of resolution and

accuracy of forecasting models. However if a more reliable mode of forecasting could be established to provide information about the SWH between wind turbines, the planning of maintenance could be optimised and costs minimised.

They applied 240 Sentinel-1 VV and VH polarized SLC images to an artificial neural network (ANN) in the form of a multilayer perceptron (MLP). These images were collocated and temporally matched with 3 buoys in three locations of Liverpool Bay, the Irish Sea and the South of the North Sea. These buoys were chosen for their proximity to the Burbo Bank wind farm. Using the NRCS at the buoy locations, the mean NRCS in a five pixel radius, longitude, latitude and bathymetry as inputs, a standard feed-forward network with back propagation of errors was trained. Trial and error determined that a network with one hidden layer of 10 nodes was the optimal architecture for this problem. For SWHs $< 3\text{m}$, chosen based on lack of high sea-state data, they achieved a SI of 0.21, RMSE of 0.23 and bias of -0.0041 for all locations and data.

Xue et al. (2020) applied a convolution neural network (CNN) methodology to sea-state retrieval from SAR images in locations near the US. Their goal was to tackle the issue of the lack of wave buoys at large distances from shorelines. Similar to Tapoglou et al. (2021), they utilised 1597 Sentinel-1 SLC IW mode images in both VV and VH polarizations. A total of 120 wave buoys were used to provide SWH data on which to train their model. This led to a total of 3887 collocations. Radar altimeter data and WAVEWATCH-III data were used for validation of the model after it had been trained.

In contrast to a Perceptron architecture which takes parameters extracted from images as input, CNNs are a deep learning technique typically applied used in image analysis and classification purposes (Gu et al. 2018). After selecting a region and cropping images, a homogeneity test was applied to filter out poorer quality images which may negatively impact on the model performance. Since the extraction of SWH can be affected by phenomena such as boundary layer rolls (Alpers & Brümmer 1994), biological surface slicks (Gade et al. 1998) and atmospheric fronts (Melshelmer et al. 1996), the homogeneity test is necessary to remove affected images from the dataset. An inhomogeneity parameter based on image spectral characteristics was calculated for each image and a threshold was selected to exclude those images deemed to be inhomogeneous. After image filtering and model training, Xue et al. (2020) achieved a bias of 0.03, a RMSE of 0.45 and an SI of 30.70%. The large scatter index here may suggest that the model lacks certain information. This may be due to the fact that there is no further information provided apart from the images themselves. For example, since there is variance in the incidence angle of the Sentinel-1 SAR instrument, it may be the case that similar backscatter data is observed for different SWH conditions. With no knowledge apart from image pixels, this may limit the capabilities of a CNN in this case (Wu et al. 2021).

A number of other studies applied machine learning techniques to Sentinel-1 images in other imaging modes to estimate SWH. Stopa & Mouche (2017) applied a back-propagation neural network (BPNN) to WV images in VV polarization. The nature of the image product provided in WV allowed for features to be extracted from the 2D image spectrum. This led to inclusion of features such as azimuth cutoff wavelength (see discussion in 2.1.3) and peak wavelength. Wu et al. (2021) and Quach et al. (2021) used WV images with a BPNN and EW images with a deep neural network (DNN) respectively. Both techniques involved use of NRCS and image variance as input parameters, as well as the CWAVE orthogonal features extracted from the image modulation spectra as discussed in 2.2.2. These techniques involve applying the Fast Fourier Transform and is beyond

Author	S1 SAR Images	SWH range (m)	Method	Input parameters	Bias (m)	RMSE (m)	Scatter Index (%)
Tapoglou et al. (2021)	IW SLC VV & VH pol	0 - 3	BPNN	buoy NRCS mean NRCS lat, lon bathymetry	-0.0041	0.23	21
Xue et al. (2020)	IW SLC VV pol	0 - 5	CNN	SAR images	0.03	0.45	30.70
Stopa & Mouche (2017)	WV	0 - 10	BPNN	Azimuth cutoff NRCS Image variance Image skewness peak wavelength and peak direction of 2D image spectrum	0.16	0.63	20.96
Quach et al. (2021)	WV	0 - 12	DNN	Re and Im modulation spectra CWAVE features	0.17	0.53	-
Wu et al. (2021)	EW	0 - 12	BPNN	Mean NRCS Image variance CWAVE features	0.02	0.63	21.07

Table 1: Summary of previous work using machine learning techniques to extract SWH from Sentinel-1 SAR imagery.

the scope of this work, but the results are nevertheless useful as a comparison tool. A summary of the techniques and results discussed in this section is presented in Table 1.

2.3 Extending the State of the Art

The works outlined in 2.2.3 examine a number of different machine learning approaches to extracting SWH from Sentinel-1 SAR data. This type of work has never been performed for a region near Ireland. The closest of the previous works being that of Tapoglou et al. (2021). This work also aims to compare the effectiveness of a number of different models instead of taking just a single approach. Many of the other works used engineered spectral features as input parameters. The approaches in this work do not explore this and instead use only parameters directly available from the Sentinel-1 dataset provided on Google Earth Engine (GEE) (see 3.1.1). This approach will investigate if these parameters can provide sufficient information to machine learning models to allow for their potential use as a forecasting tool for industries such as offshore wind in Ireland. Further to this, a number of models types will be examined. These include MLP, XGboost and Random Forest models. The use of multiple model types will investigate if certain models are more suited to this type of problem. Also if all models display similar results then it will allow conclusions to be made about the input data as opposed to model architecture.

3 Data and Methodology

This section outlines the collection of both SAR data from Sentinel-1 images and in-situ buoy data to use as target data for the machine learning models. The gathering of this data is outlined in 3.1.1 and 3.1.2. The spatial and temporal matching of this data is discussed in 3.1.3. Once the data had been gathered, it could be filtered and input parameters could be extracted and encoded accordingly. This is outlined in 3.2.2 and 3.2.3. Following this, the training process of the neural

networks encompassed a search for the best model architecture. This step is described in 3.2.4. The planned methodology of the project is summarised below in Figure 5.

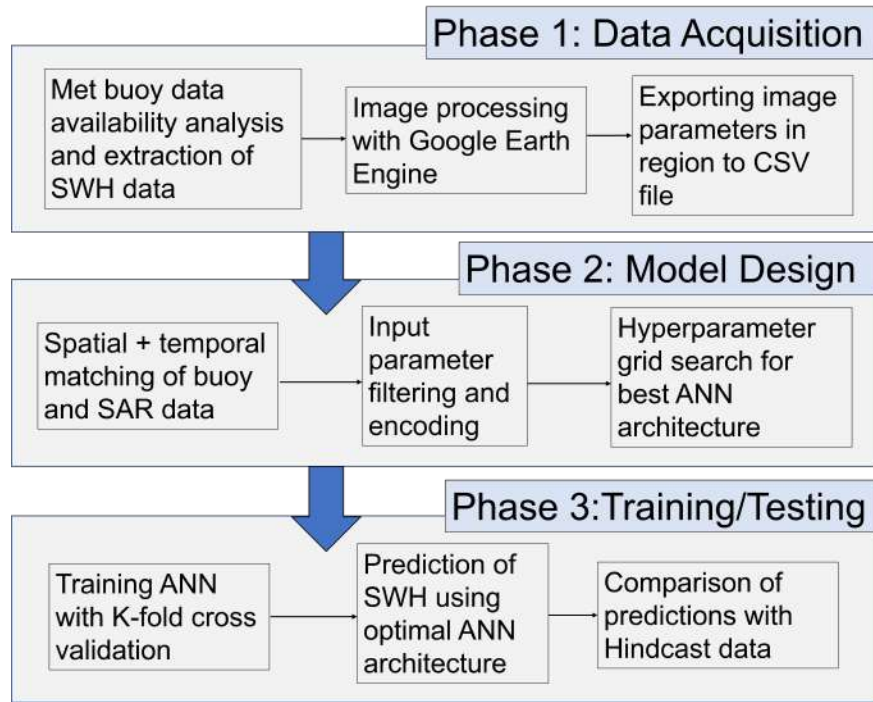


Figure 5: Outline of the planned project process from data gathering to generation of predictions of SWH.

3.1 Data

3.1.1 Sentinel-1 SAR Images

The SAR images used in this project are Sentinel-1 Ground Range Detected (GRD) IW mode images. These images were processed and their information extracted from GEE. GEE provides the ‘COPERNICUS/S1_GRD’ dataset which contains images from 02/02/2015 up to now. At the time of writing, this collection contained 818 images. The GEE code editor was used to write the JavaScript code required to perform the parameter extraction. This script can be found in appendix A. The GEE collection of Sentinel-1 imagery was filtered to include only the images that included the location of the buoy (see 3.1.2). Image information would be extracted from a 200m radius around the buoy location, but before this could be done, a smoothing function was applied to this area. This was applied to reduce speckle in the image. Speckle causes a grainy appearance and is due to interference from multiple scatterers (Meyer 2019). Speckle results in varying amplitudes in resolution cells, so smoothing will average over these.

The parameters extracted from images, are:

1. Date and time on which the image was taken.
2. Backscatter at the buoy location (NRCS).

3. Mean backscatter in the 200m radius region.
4. Backscatter variance in the 200m radius region.
5. Skewness of the backscatter in the 200m radius region.
6. Kurtosis of the backscatter in the 200m radius region.
7. Incidence angle.
8. Satellite orbit pass (ascending or descending).

The NRCS is known to be a key parameter related to wind speed and direction and thus the significant height of the waves in question (Quilfen et al. 1999). Masuko et al. (1986) finds that NRCS from SAR data is, under moderate conditions, linearly related to sea-surface wind. The image variance is another very important parameter of a SAR image for deriving SWH. In this case, an analogous parameter of the backscatter variance is used. Stopa & Mouche (2017) states that it provides information about image homogeneity as well as sea-state. The reasons for choosing a region around the buoy are similar to those described by Tapoglou et al. (2021). Backscatter readings from a single location could be affected by scattering off the buoy as well as passing ships. Data from a region may also provide additional information to the model that data from a single location may not. Skewness and kurtosis are higher-order features considered by Stopa & Mouche (2017) in a similar project. They are defined as follows:

$$skew = \frac{1}{n} \sum_{i=1}^n (\sigma_{0i} - \overline{\sigma_0})^3 / s^3 \quad (3.1)$$

$$kurt = \frac{1}{n} \sum_{i=1}^n (\sigma_{0i} - \overline{\sigma_0})^4 / s^4 \quad (3.2)$$

where s is the standard deviation of σ_0 . It was noted in the literature that including either one of these improved the model performance but including both was unnecessary. This suggests redundancy however both will be included as inputs to the model in this case.

3.1.2 Buoy Data

Met Eireann and the UK Met Office operate a number of data buoys as part of the Irish Marine Databuoy Observation Network (IMDBON) or Irish Weather Buoy Network. These 5 buoys labelled M1 though M5 are managed by the Marine Institute and funded by the Department of Agriculture, Food and Marine. They provide vital information such as for weather forecasting and shipping bulletins.

The buoys provide significant wave height measurements at hourly intervals which have been calculated as four times the root mean square wave height, which itself was calculated as an average over a period of 17.5 minutes by the sensor on the buoy. An analysis of the availability of data from each of the buoys was performed prior to selection. This is because buoys may exhibit periods of inactivity due to servicing, weather damage, vessel strikes or component failure. This included processing the CSV buoy data provided for each buoy to extract the regions of missing data and

Date-and-time	Orbit-pass	VV buoy (dB)	VV kurt (dB)	VV mean (dB)	VV skew (dB)	VV var (dB)	θ_i (°)	SWH (m)
2015-03-22 18:13:58	1	-16.922145	-0.294064	-17.509141	0.247753	0.070534	37.671	0.938
2015-03-31 06:39:04	0	-8.734265	-0.997647	-7.955076	0.143696	0.150203	39.102	4.219
2015-04-15 18:13:59	1	-19.262865	-0.601304	-19.373294	-0.522205	0.251459	37.666	1.094
2015-04-24 06:39:05	0	-18.535536	-0.862215	-18.360198	0.295748	0.227439	39.111	0.938
2015-04-27 18:13:37	1	-11.947058	0.754876	-11.991301	0.116979	0.040980	37.682	1.25

Table 2: Subset of spatially and temporally matched image parameters and M5 wave buoy significant wave height data.

compare them. Because data was only provided for times when the buoy was active, a script had to be written to fill in dates and times in the buoy data sets that did not exist and mark them to reflect that there was no data. After data availability analysis on all buoys, the M5 buoy (51.690425, -6.704336) located 56 km South of Hook Head was chosen. The location of the buoy is shown in Figure 6. The plot in Figure 7 gives an indication of the data availability and density. As can be seen by the red areas, the SWH is mainly in the range 0 to 3 metres. This is further illustrated by the histogram in Figure 8. The data also shows good availability across the time period in which Sentinel-1 has been active. The only major areas of missing data are in early 2014 and between late 2015 and the middle of 2016. This buoy data is available from the [Marine Institute data centre](#) under a Creative Commons license.

3.1.3 Image-Buoy Temporal Matching

Once the information from the SAR images in the region of interest had been acquired, the SWHs from the buoys had to be temporally matched with this information. Since the buoy data was available at hourly intervals in the best cases, image information was selected if its timestamp was within 1 hour of a buoy measurement timestamp. A script was written in Python to perform this step and the output was a CSV file of image data in the first set of columns, and the temporally matched SWH from the buoy in the final column. A subset of the data is shown in Table 2. Out of 818 images, at total of 742 were matched within 1 hour of a buoy measurement. A histogram of the matched data is shown in Figure 9. As expected from Figure 8, this matched data also displays a lack of data at higher sea-states. When applying this data to models, under-sampling at higher sea-states must be considered. Model performance may improve if data is limited to some maximum SWH cutoff. There is obviously a trade-off here in that prediction of higher sea-states is removed. An investigation will be performed in this light.

The ‘Orbit-pass’ column refers to whether the satellite was ascending or descending during the acquisition of the image. This information was encoded using binary values so they could be input to neural network models. Ascending was assigned a ‘1’ and descending was assigned a ‘0’.

3.2 Methodology

3.2.1 MLP Neural Networks

Although this project compares results from a number of model types, most of the work revolves around the Multilayer Perceptron (MLP) architecture. This is a type of feed forward neural network ([Haykin 1994](#)). Feed forward neural networks operate by propagating information in the direction

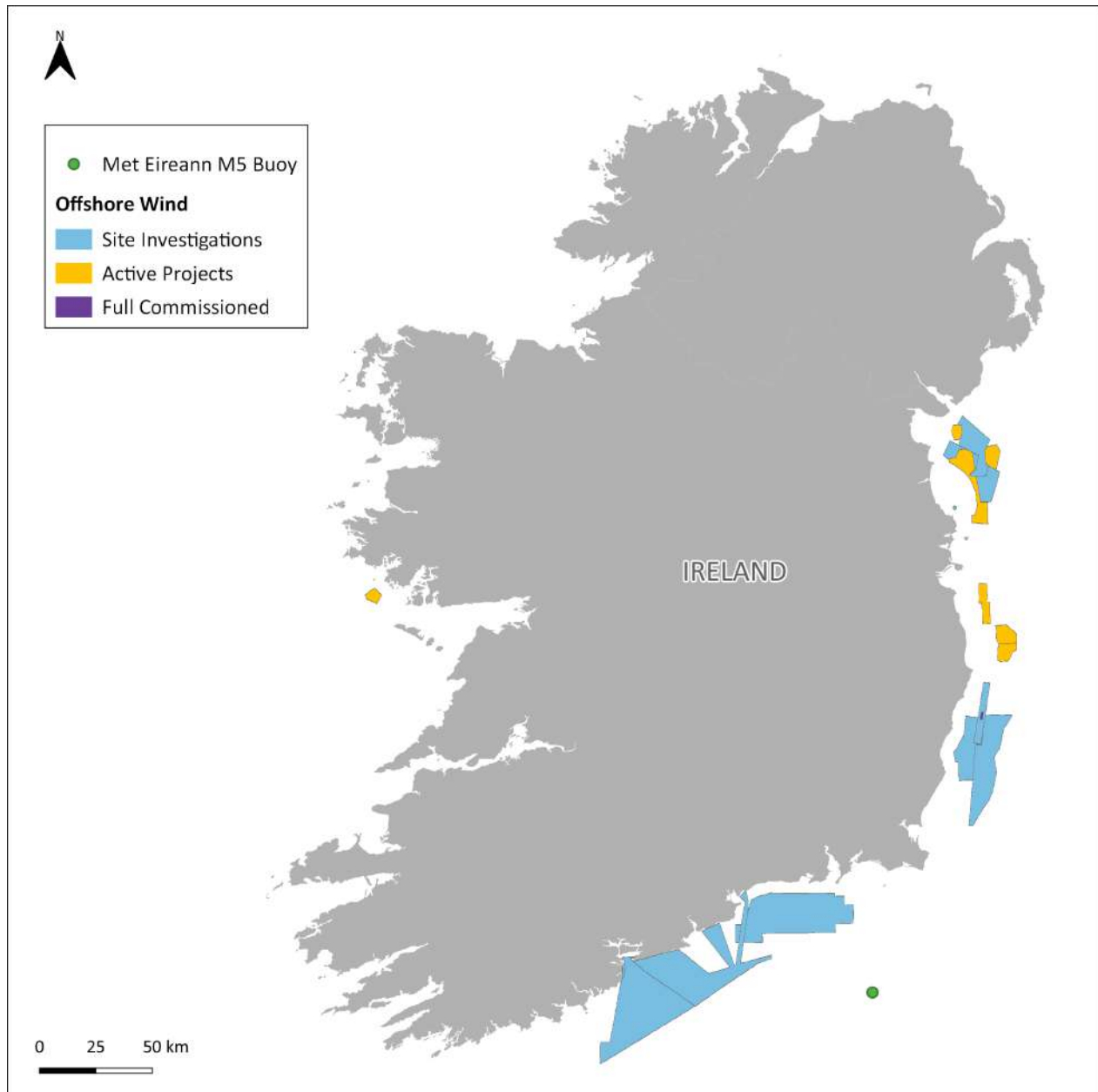


Figure 6: Map showing the location of the M5 Met buoy and its proximity to a number of planned offshore projects. *This map was created by my supervisor at Techworks Marine, Steve Coughlan.*

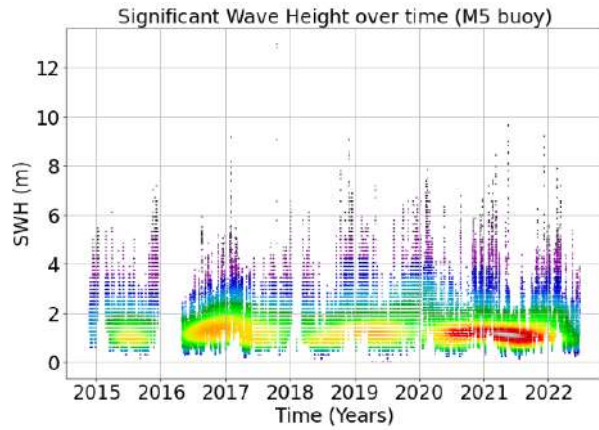


Figure 7: Significant wave height data for the M5 Met Eireann buoy since the launch of Sentinel-1A up to present day (at time of this work). Data density is shown with densest regions in red and lower densities in purple/black.

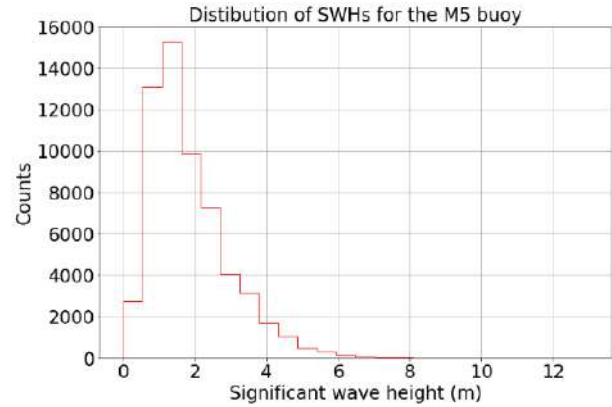


Figure 8: Histogram of the significant wave height distribution for the entire buoy data set for the same date range as in Figure 7.

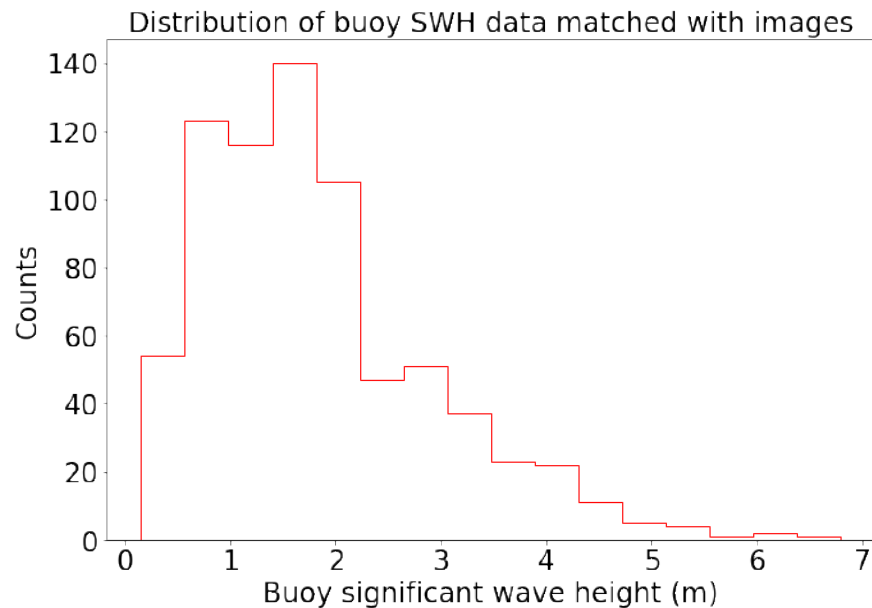


Figure 9: Histogram showing the distribution of SWH buoy data spatially and temporally matched within 1 hour of 742 satellite images.

from inputs to outputs through an architecture similar to that shown in Figure 10. The network comprises an input layer which consists of the input features derived from the SAR images, a number of hidden layers which are composed of nodes (shown in red) and an output layer. The nodes have weights and biases associated with them which are updated during training of the network to minimise loss. At each node, the dot product of the input to the node with the weight associated with that node is performed and the result is passed to a function called an activation function. Each hidden layer in the network has an associated activation function. The dot product works as follows:

$$\text{res}_i = W_i^T X + b_i \quad (3.3)$$

where the node i is the node index, res_i is the result to be passed to the activation function, W_i is the weight vector associated with the connection of the nodes from the previous layer to the nodes of the current layer, X is the input vector and b_i is the node bias value. The output of the node is then this result passed to the activation function as follows.

$$\text{out}_i = \text{act}(\text{res}_i) \quad (3.4)$$

where $\text{act}()$ is the activation function of the layer/node. The operation of a single node of a hidden layer as described here is outlined in Figure 11.

Since this is a regression problem, the activation function for the input layer and all hidden layers was chosen to be the Rectified Linear Unit (ReLU) function. This a piecewise linear function and is defined as

$$\text{ReLU}(x) = \max(0, x) \quad (3.5)$$

and is a common function to use in regression tasks. It has the advantage of being able to avoid the vanishing gradient problem and improve the speed of learning (Ide & Kurita 2017). Generally, nonlinear activation functions are desirable as opposed to linear because they can discover more complex relationships between a set of inputs and an output. However this function avoids some of the issues associated with traditional nonlinear activation functions, such as the sigmoid and hyperbolic tangent functions, due to its linear properties above 0. However it maintains the nonlinear properties below 0 to ensure complex patterns can still be recognised (Goodfellow et al. 2016).

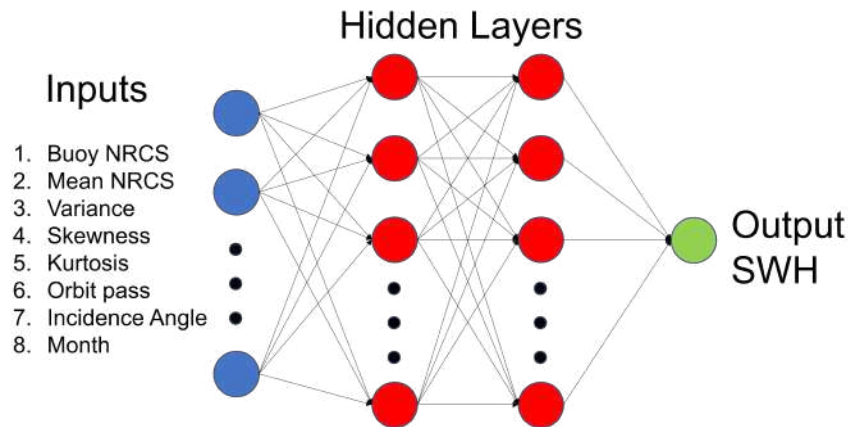


Figure 10: Schematic of a general Multilayer Perceptron Artificial Neural Network Architecture. Shown are the entire list of input features available in this work.

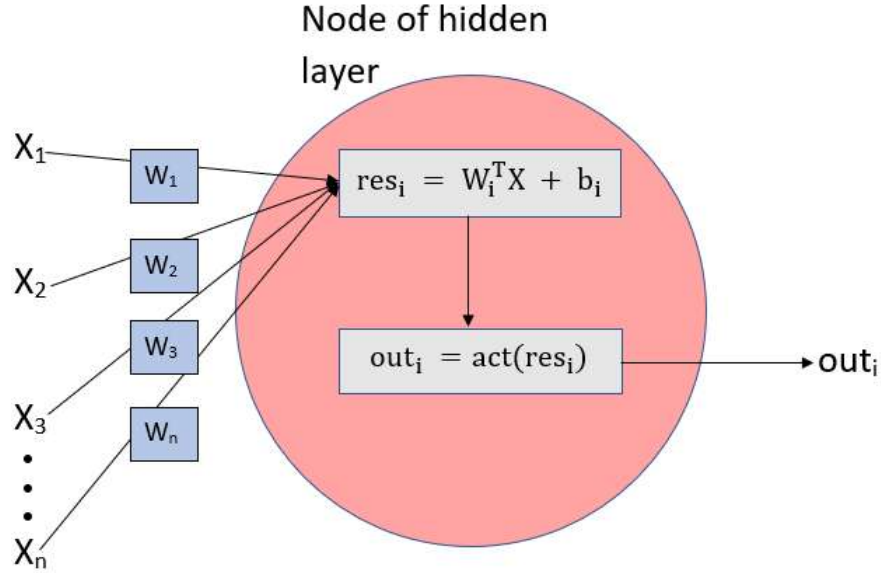


Figure 11: Operation of a single node in a hidden layer of a MLP architecture. This shows the series of steps applied to the input X to the node and how this generates an output.

Once information is propagated through the network, and an output is calculated, it is compared with the target SWH provided by the buoy. The difference between the predicted value and the target is used to update the weights of the network for the next iteration. The loss metric used in this case is the mean squared error (MSE). This error metric is default for many regression tasks and is common throughout the literature on this topic eg. [Wu et al. \(2021\)](#), [Quach et al. \(2021\)](#). The MSE is defined as follows:

$$\text{MSE} = \frac{1}{N} \sum (\text{Pred}_i - \text{Actual}_i)^2. \quad (3.6)$$

This model was implemented using the *keras* library in Python. This library allows for easy construction of an MLP network. The *Adam* optimizer ([Kingma & Ba 2014](#)) was used also. This optimizer was deemed a good choice because it extends the standard stochastic gradient descent algorithm to include adaptive learning rates per parameter and generally requires very little tuning. This means that focus could be put on the architecture of the network instead of fine tuning a learning algorithm. It also combines the advantages of other common stochastic gradient descent based algorithms, namely *RMSProp* and *AdaGrad* and has been shown to outperform them ([Kingma & Ba 2014](#)).

3.2.2 Data Filtering

Before the SAR image data could be applied to a neural network for training, a number of steps had to be taken to filter the data so as to give the network the best chance of predicting SWH within an acceptable margin of buoy measurements. As already described, a total of 742 images were temporally matched with 1 hour of a buoy measurement. Further to this, the following considerations were taken into account regarding the data.

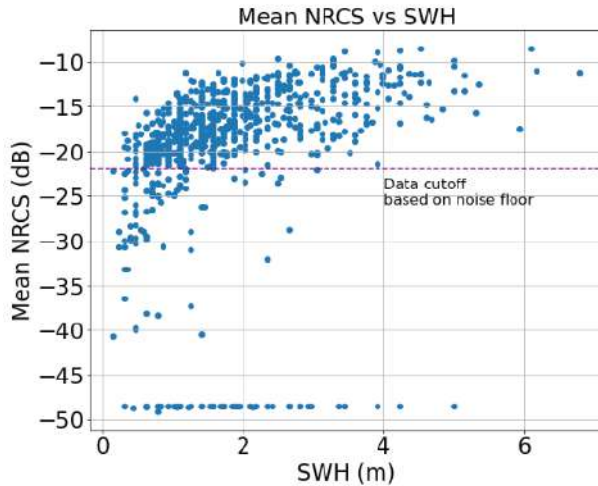


Figure 12: Mean NRCS distribution indicating the noise floor of the SAR instrument resulting in a filtering of data below -22 dB.

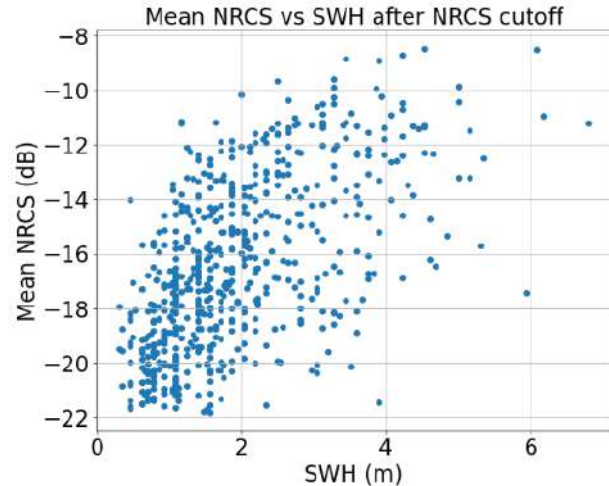


Figure 13: Distribution of the mean NRCS after the noise floor cutoff shown in Figure 12.

1. The noise floor of the CSAR instrument on Sentinel-1 is -22 dB.
2. There is a lack of data for high sea-states.

In the first case, images with mean NRCS below -22 dB were filtered out as it is not possible to establish the significance of data below this level. From the plot in Figure 12 it is clear that there are a number of data points corresponding to images that exhibit a mean NRCS below this level. Figure 13 shows that the remaining NRCS data seems to exhibit more of a relationship with SWH although there is a large scatter.

Regarding the second point, after the filtering carried out as described above, the remaining data distribution is shown in Figure 14. The filtering has reduced the dataset to 603. It is clear from this plot that there is very little data for SWHs > 4m. For this reason, filtering of data above this SWH is also performed. This leaves a remaining 568 data points. This is indicated by the purple line also shown in Figure 14.

3.2.3 Input parameters

The parameters listed in 3.1.1 are those extracted directly from the Sentinel-1 dataset available on GEE. However, some feature encoding was performed to best represent two of the features present in that list. The month of the year is included as a parameter to account for any correlation between the time of year and the sea-state. This was extracted from the date and time parameter. Since this is a cyclical parameter, simply including the number of the month is not sufficient. If this were the case, the model would think that January, represented by a '1', is furthest from December, represented by a '12', when in fact they are right next to each other in terms of time. Therefore it was necessary to perform an encoding that captured this periodicity of dates. A sinusoidal function was the obvious solution. The issue with encoding using a single sin or cos function is that not all

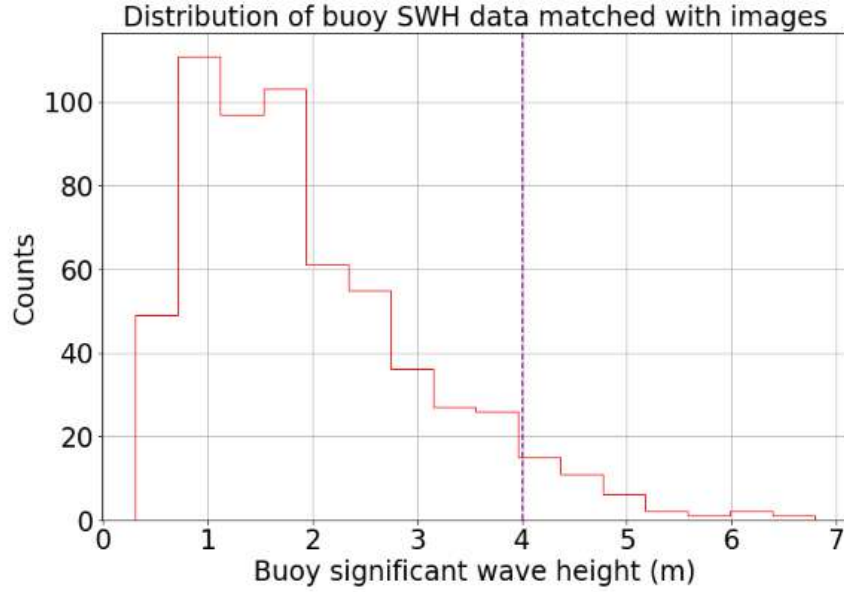


Figure 14: The distribution of SWH after removing data for mean NRCS < -22 dB. The purple line indicates the cutoff of SWH > 4m.

of the times are distinguishable due to the symmetry of these functions. This is solved by encoding the month of the year with both functions. The months are transformed as follows:

$$M_{\sin} = \sin\left(\frac{M}{12} * 2\pi\right) \quad (3.7)$$

$$M_{\cos} = \cos\left(\frac{M}{12} * 2\pi\right) \quad (3.8)$$

A similar encoding was implemented for the incidence angle. This left a total of 10 features that could be input to the model if the cos and sin terms of both month and angle are counted as single features. The number of features in the final list was reduced to 9. This is due to the very strong correlation between the NRCS at the buoy location and the mean NRCS in the region around the buoy. Originally, both were to be included, but the correlation score implies redundancy, meaning that no new information is provided to the model by including both. It also adds unnecessary complexity to the model in terms of its size and computation time. The correlation of the two variables is shown in Figure 16. The total list of input features is thus:

1. $M_{\sin}$
2. $M_{\cos}$
3. mean NRCS
4. NRCS variance
5. NRCS skewness

	Sin Month	Cos Month	Orbit Pass	Buoy NRCS	NRCS Kurtosis	Mean NRCS	NRCS Skewness	NRCS Variance	Incidence Angle
Sin Month	1.00	1.00	-0.01	-0.03	-0.01	-0.04	-0.06	0.04	0.01
Cos Month	1.00	1.00	-0.01	-0.03	-0.01	-0.04	-0.06	0.04	0.01
Orbit Pass	-0.01	-0.01	1.00	0.02	0.04	0.04	0.02	0.25	-1.00
Buoy NRCS	-0.03	-0.03	0.02	1.00	-0.13	0.89	0.15	-0.03	-0.02
NRCS Kurtosis	-0.01	-0.01	0.04	-0.13	1.00	-0.19	0.11	0.11	-0.04
Mean NRCS	-0.04	-0.04	0.04	0.89	-0.19	1.00	0.17	-0.07	-0.04
NRCS Skewness	-0.06	-0.06	0.02	0.15	0.11	0.17	1.00	-0.05	-0.02
NRCS Variance	0.04	0.04	0.25	-0.03	0.11	-0.07	-0.05	1.00	-0.25
Incidence Angle	0.01	0.01	-1.00	-0.02	-0.04	-0.04	-0.02	-0.25	1.00

Figure 15: Correlation matrix of input features. The strong correlation of buoy NRCS and mean NRCS is the only evidence of redundancy in the features.

6. NRCS kurtosis

7. Orbit pass

8. $\theta_{i\sin}$

9. $\theta_{i\cos}$

To demonstrate that, at least linearly, there is very little correlation between the rest of the features, a correlation matrix is provided in Figure 15.

3.2.4 Optimal Architecture, Hyperparameters and Training

The selection of the best model architecture is generally done by a process of trial and error as there is no set rule for this selection. The technique used to simultaneously select the best model architecture and to train the model was a grid search in combination with k-fold cross validation. This was implemented with the *GridSearchCV()* method from the *sklearn* library.

In the case of this project, the dataset was split into 80% training and 20% test data. The test data is held out as ‘unseen’ by the model to evaluate its’ ability to make predictions on data that it has not learned from in the training process. The 80% training data was used in the k-fold validation process. K-fold cross validation is a neural network training technique whereby the training set is split into a number of ‘folds’ of equal size and treating one of the folds as a validation set while training the model on the other $k - 1$ folds (James et al. 2013). The process is summarised as follows and an illustration is provided in Figure 17 for 5 folds:

1. Randomly split data into k groups.
2. For each group:
 - Hold out the group as a validation set.
 - Train the model on each of the remaining groups.
 - Evaluate the model on the held out group.
3. Use the results from the process above to describe the skill of the model.

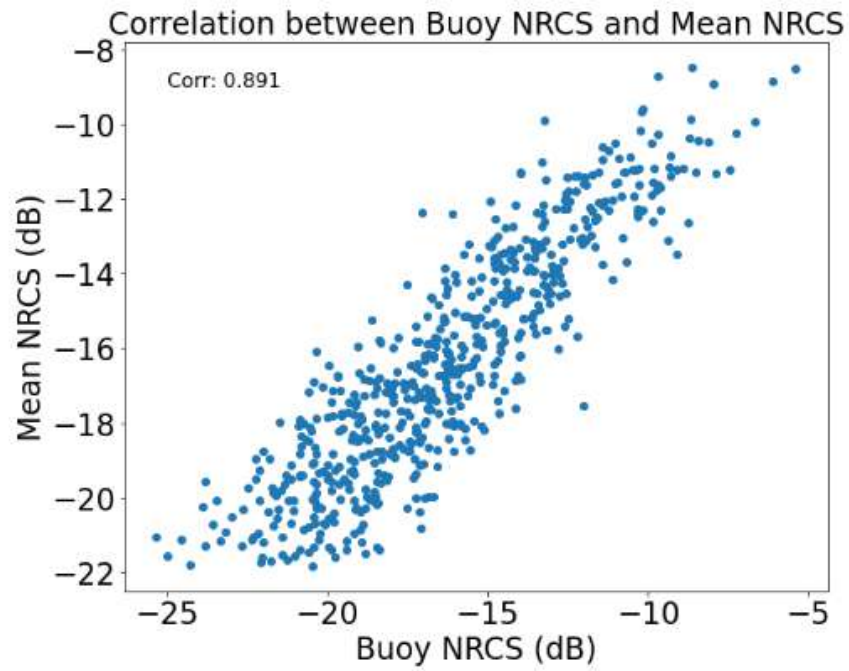


Figure 16: Correlation between mean NRCS and buoy NRCS.

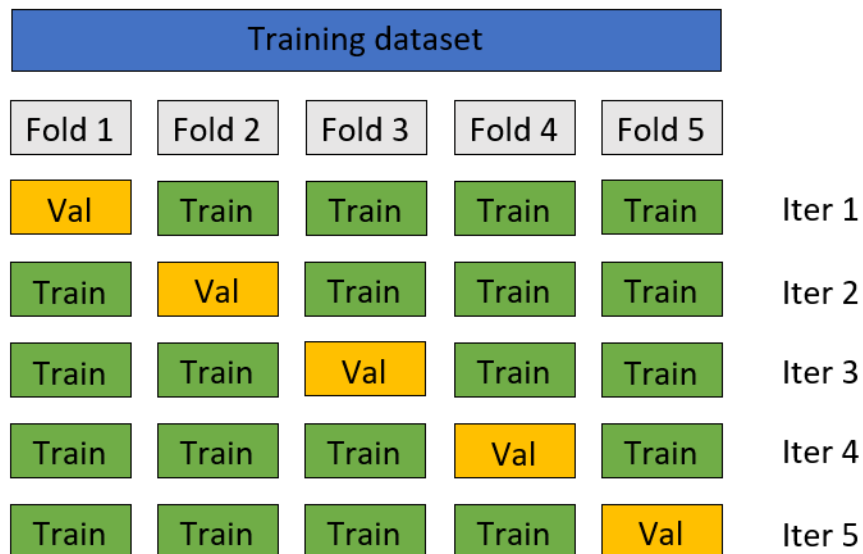


Figure 17: Diagrammatic view of K-fold cross validation. In this case the training set is split into 5 folds.

According to [Kuhn & Johnson \(2013\)](#), there is no formal rule for choosing k but it is usually selected to be 5 or 10. The bias (difference between estimated and real performance) decreases as k is increased but so does the computational requirements. Values of 5 or 10 are regularly used because it has been shown that they do not result in very high bias or variance ([James et al. 2013](#)). In this case, a value of $k = 5$ was chosen.

Neural network hyperparameters include batch size, epochs and learning rate. As previously discussed, the *Adam* optimizer was used and its parameters including learning rate were left largely unchanged. Batch size is the number of data samples that the model sees before its weights are updated. The number of epochs is the number of times the entire set of training data is worked through. The batch size was selected to be 16 and the number of epochs was set to 1000. Batch size selected based on a grid search of batch sizes of 8, 16 and 32. Similar performance was observed across all batch sizes so an intermediate value was selected. The number of epochs was set to 1000. Similar performance was observed for number of epochs ranging from 200 to 1000 and conducting multiple grid searches generally resulted in a different optimal number of epochs each time.

The grid search involved selecting the optimal number of hidden layers in the model and also the number of nodes in each hidden layer. Architectures with 1, 2 and 3 hidden layers were explored and their performance compared using the same set of training and test data. The grid is set up so that for each grid parameter combination, the same cross validation training procedure is performed. At the same time, the scores of each parameter set is tracked and the best performance is reported. In this way, training, testing and optimal architecture searching is done simultaneously.

4 Results and Discussion

This section presents a discussion on the results broken down in terms of the limits put on the data used in the training process. The results of the first set of data used whereby the SWH is limited to below 4m is discussed in [4.1](#). A further sub-setting of the data is investigated for data below 3m in [4.2](#). Lastly for the perceptron architecture, a NRCS variance cutoff is introduced to investigate its effect. This is discussed in [4.3](#).

4.1 Data limited to SWH < 4m

The dataset in this case was submitted to a 5-fold cross validation and grid search procedure. Throughout the grid searches of all architectures, the training-testing split was kept constant so as to best compare the results of these architectures. In other words, each of the architectures was trained and cross validated using the same 80% of the dataset, and the models were tested using the same remaining 20% of the data.

Firstly, a model with a single hidden layer was used. The grid search space consisted of 9, 10, 12, 15 and 18 nodes. The grid search determined that 12 nodes achieved the best performance of all single hidden layer architectures. The results of the cross validated training and testing for this architecture are shown in [Figures 18 and 19](#) along with their correlation, bias, RMSE and scatter index values as well as a line of best fit. The trend in both training and testing results demonstrates that the model has some ability to learn and predict SWH. The metrics of bias and

RMSE are somewhat comparable to those of previous works given in Table 1 however the rather weak correlation and high scatter is the downfall here. This is evident in the high scatter index as opposed to the best of the literature being around 20%. The model appears to overestimate small values of SWH and underestimate large values. This could be due to a combination of data availability and the features used as input to the model. It may also be that the model requires more hidden nodes to extract more information from the inputs. For that reason, a further node is added and the grid search procedure is repeated.

In the case of two hidden nodes, the first hidden layer was left with 12 nodes and a grid search space of 3, 5, 6, 9 and 12 nodes was applied. The optimal architecture was deemed to consist of a second hidden layer with 9 nodes. The training and testing results are presented in Figures 20 and 21. The training results show an improvement in all metrics while the testing does not exhibit improvement and results are largely similar. Despite the improvement in training, the similar testing results to the 1 hidden node case indicate that adding a second node does not improve the generalizability of the model. Meaning that its performance on unseen data has not been changed by adding another layer.

Finally, a further layer was added to the architecture. Although adding a second layer did not improve performance, it is worth investigating if there is a level of complexity that must be reached before an improvement is observed. A third hidden layer was added and nodes numbers of 3, 4, 6 and 9 were search using the grid. The previous layers were left unchanged so that the architecture now consisted of an input layer and hidden layers of 12, 9 and 3 nodes. The results as in the other cases are presented in Figures 22 and 23. In this case, the affect is even more minimal than that of changing from 1 hidden layer to 2. Testing results improved slightly on the 2 hidden layer case but have not shown any significant change on the testing results from the single hidden layer case. These results indicate that with the current data, features and model type, no improvement will come from continuing to increase the complexity of the architecture. A MLP model with a single hidden layer will perform as well if not better that a model with three hidden layers.

4.2 Data limited to $SWH < 3m$

The lack of data for high sea state conditions prompted the limiting of the data to SWH below 4m in the previous section. A further investigation was undertaken by limiting the data to SWH below 3m. This was done because the number of data points between 3 and 4 metres is still relatively low and this poorly sampled data may be hindering the performance of the model overall. A similar process was used as in the 4m case and so is described in less detail here. The six plots shown from Figures 24 to 29 follow the same training and testing grid search procedure as before. These results demonstrate that although there may be some differences in bias, RMSE and scatter index, the model architecture has very little bearing on the predictions. Limiting the data to 3m allowed the dataset to approach a more Gaussian distribution and yet no significant improvement was observed. The correlations are weaker than for the 4m results above but the bias, RMSE and scatter index are reduced. This may simply be as a result of their being less data. Since the model tends to perform poorly for high SWH, the limiting to 3m has reduced many of the error metrics but has not improved the correlation. The same overestimation at low SWH and underestimation at high SWH seems to imply that the model is not able to make strong enough connections between the input features and the output SWH from the buoy measurements.

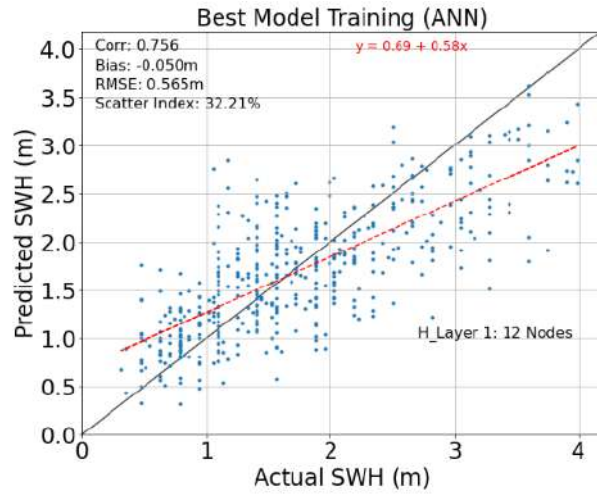


Figure 18: Results of the training of the best model obtained from the grid search using 1 hidden layer on data limited to $SWH < 4m$. The model consists of a hidden layer of 12 nodes.

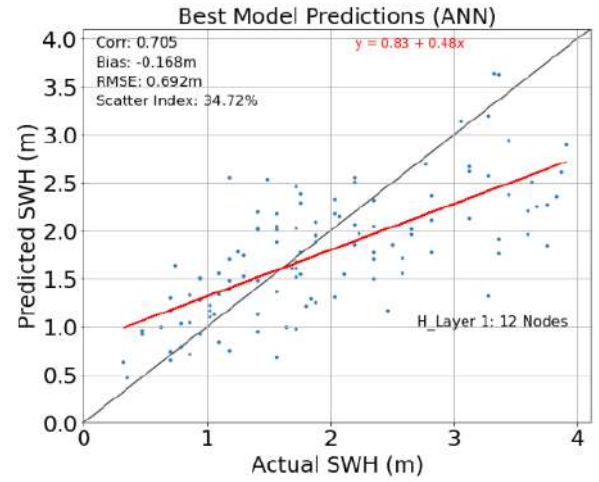


Figure 19: Predictions of the same model from Figure 18 on unseen testing data.

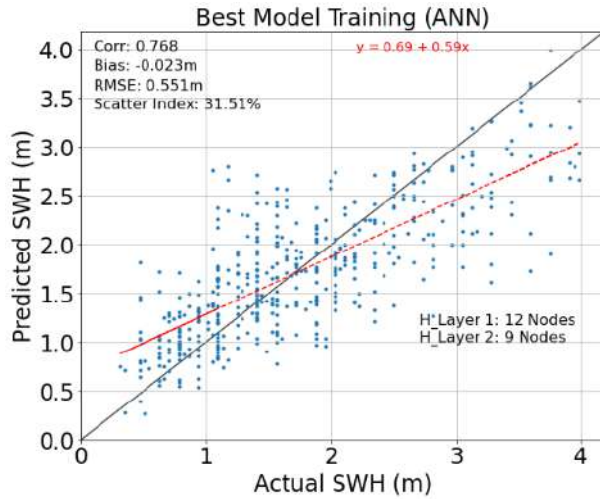


Figure 20: Results of the training of the best model obtained from the grid search using 2 hidden layers on data limited to $SWH < 4m$. The 2 hidden layers contain 12 and 9 nodes respectively.

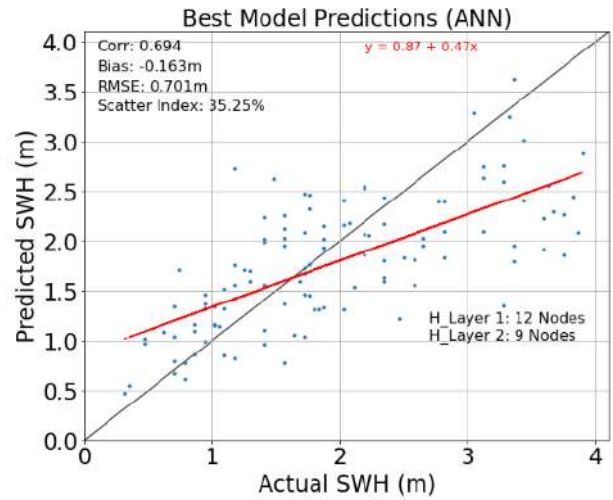


Figure 21: Predictions of the same model from Figure 20 on unseen testing data.

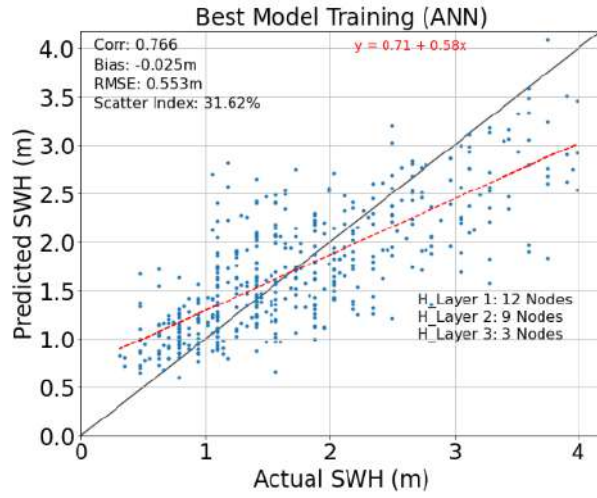


Figure 22: Results of the training of the best model obtained from the grid search using 3 hidden layers on data limited to $SWH < 4m$. The 3 hidden layers contain 12, 9 and 3 nodes respectively.

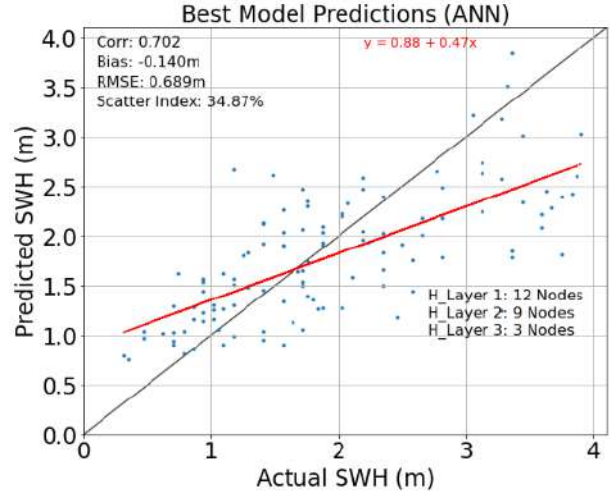


Figure 23: Predictions of the same model from Figure 20 on unseen testing data.

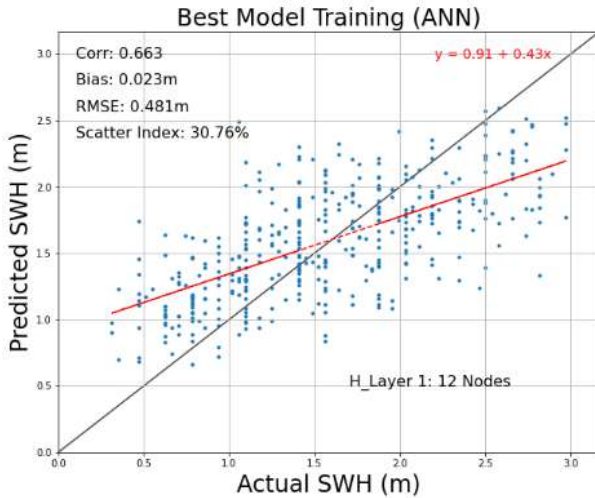


Figure 24: Results of the training of the best model obtained from the grid search using 1 hidden layer on data limited to $SWH < 3m$. The model consists of a hidden layer of 12 nodes.

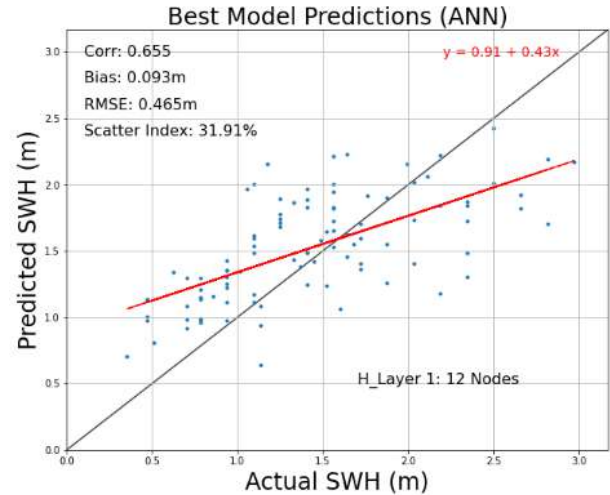


Figure 25: Predictions of the same model from Figure 24 on unseen testing data.

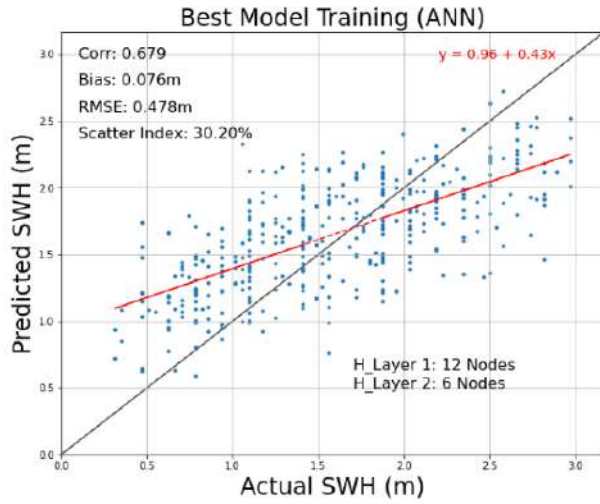


Figure 26: Results of the training of the best model obtained from the grid search using 2 hidden layer on data limited to $SWH < 3m$. The 2 hidden layers contain 12 and 6 nodes respectively.

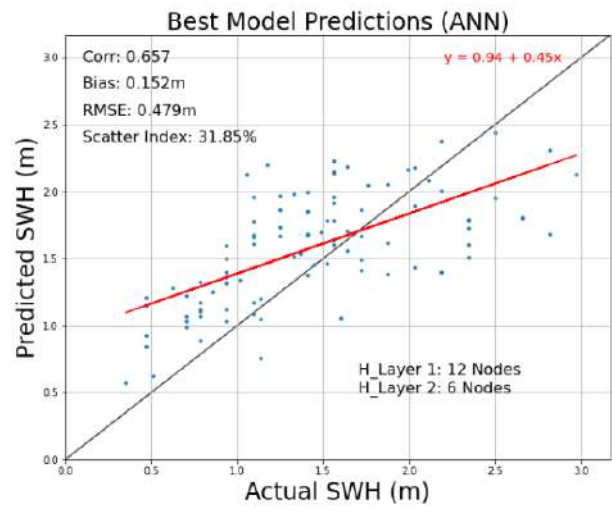


Figure 27: Predictions of the same model from Figure 26 on unseen testing data.

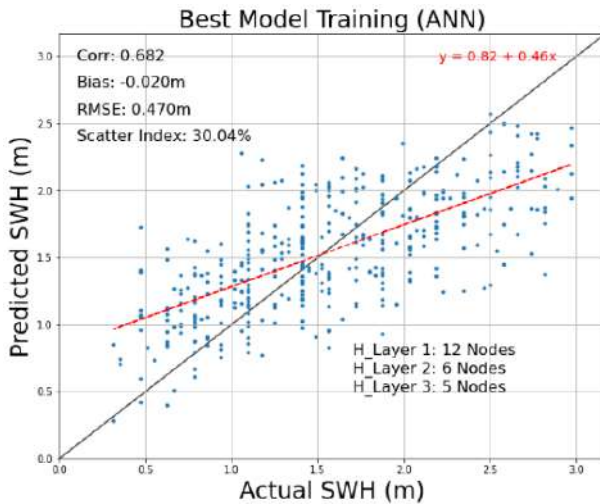


Figure 28: Results of the training of the best model obtained from the grid search using 3 hidden layer on data limited to $SWH < 3m$. The 3 hidden layers contain 12, 6 and 5 nodes respectively.

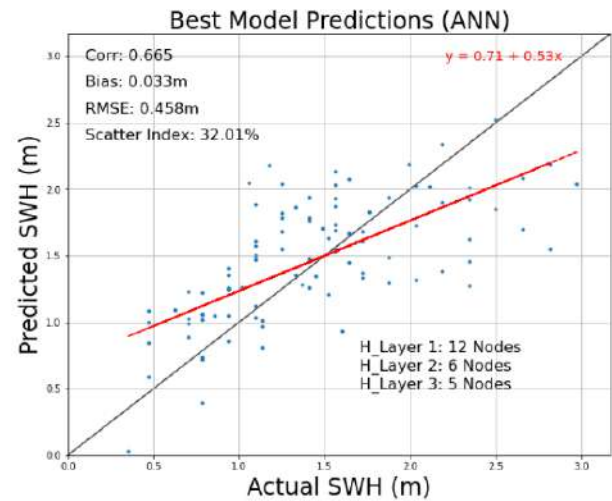


Figure 29: Predictions of the same model from Figure 28 on unseen testing data.

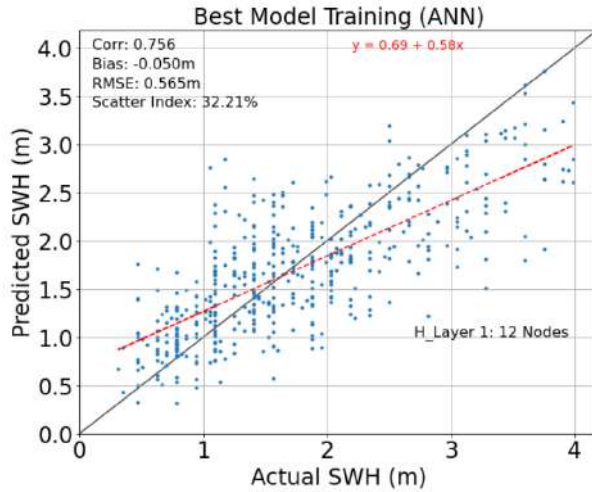


Figure 30: Results of the training of the best model obtained from the grid search using 1 hidden layer of 12 nodes on data limited to SWH < 4m and NRCS variance below 3.5 dB.

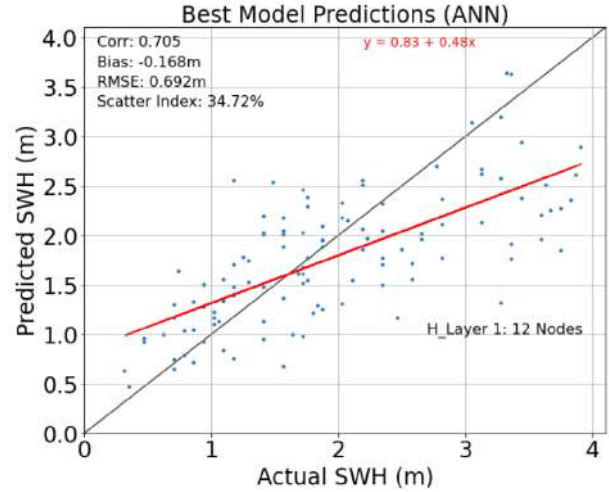


Figure 31: Predictions of the same model from Figure 30 on unseen testing data.

4.3 Introducing a Variance Cutoff

As a final effort to achieve improved performance from the MLP type model, a further limitation was introduced to the data. This is loosely based on the homogeneity tests used by many studies such as [Xue et al. \(2020\)](#) and [Wu et al. \(2021\)](#). These tests use the variance in the image power spectral density in calculating a homogeneity parameter used to determine the usefulness of an image to this kind of study. Although in this work it was not possible to apply these spectral techniques, it is worth investigating if a simpler approach has any effect. The ocean wave field shift-invariant on scales of several kilometres but some images may be affected by the phenomena discussed in 2.2.3. Larger NRCS variance was used as an indicator for images that may be disproportionately impacted by such effects. A cutoff was used for images that exhibited NRCS variance above a value of 3.5 dB. Since the best results out of the previous cases were achieved for the cutoff SWH of 4m, this variance cutoff was applied to this data. The same process of architecture testing was once again repeated as previously described. The best training and testing results from the grid searches were observed for a MLP architecture with a single hidden layer of 12 nodes. This is shown in Figures 30 and 31. No improvement was observed yet again when the number of hidden layers was increased. Very similar results are observed across all of the results presented so far. Reducing the dataset to only include data that was better sampled and reducing to only include what was deemed to be more homogeneous images had no effect. This indicates that either a different model should be applied to this data or that the input features do not have enough information about the sea-state in them to allow any model to achieve good performance.

4.4 Other Models

The previous results prompted an investigation into whether applying a different model would yield better predictions of SWH. Two model types were employed to test this. These were a ran-

dom forest regressor and extreme gradient boosting regressor. The random forest was implemented simply using the *RandomForestRegressor()* from *sklearn* while the *XGBRegressor()* from the *xgboost* library was used for the extreme gradient boosting. This was a small section of this work as a whole an considerably less time was spend on this task than the previous. For that reason, a detailed discussion of the theory of these model types is not presented. They were introduced for comparison mainly to validate a hypothesis that the suboptimal performance was due to the features and dataset rather than the specific model used.

4.4.1 Random Forest Ensemble

A random forest consists of an ensemble of decision tree models and implements bootstrapping. Bootstrapping or ‘bagging’ is a sampling technique whereby random samples may contain a data point more than once. It is also called sampling with replacement (Efron & Tibshirani 1986). Each tree in the ensemble is created from a bootstrapped sample. Variance is reduced over using a single decision tree by taking the average of predictions from all trees as the prediction value, thus improving performance. Bagging is not optimized to reduce variance since trees by arise from a similar structure (Kuhn & Johnson 2013). Random forests tackle this by selecting a reduced number of the total input features at each split point to consider when making predictions. This results in reduced tree correlation i.e the trees are more unrelated to each other (Kuhn & Johnson 2013).

A grid search with cross validation was implemented in this case as there are a number of hyperparameters of random forest models that require tuning. A forest of 200 decision trees, each drawing 200 samples from the training set, and each with a maximum depth of 35 nodes was determined to perform best based on mean squared error scoring. The number of features used to determine the best splits was deemed to be 8. The dataset used was that limited to SWH below 4m as in the first case presented above. The results of training and testing are presented in Figures 32 and 33.

The training results show that the model learned on the data rather well compared with the MLP model. This may be as a result of overfitting by the decision trees within the ensemble. The testing results indicate that this model performed worse than the MLP model in nearly all cases where data was limited to SWH below 4m with lower correlation and higher values across the three other metrics.

4.4.2 Extreme Gradient Boosted Model

Extreme gradient boosting or ‘XGBoost’ is a popular open source gradient boosting implementation that was developed by Chen & Guestrin (2016). It, like the random forest model, consists of an ensemble of decision trees. However the difference here is that trees are generated additively as training happens. The addition of new trees comprises a gradient descent process whereby the previous trees’ residuals are used to train the next iteration of the ensemble.

A grid search of hyperparameters determined that an ensemble of 1000 trees with a maximum depth of 20 nodes was to be used. 90% of the training data was used for each base sample for each tree and 50% of the input features were used in each tree. The training results showed that the model fit perfectly to the training data indicating overfitting despite the use of cross validation. The results of

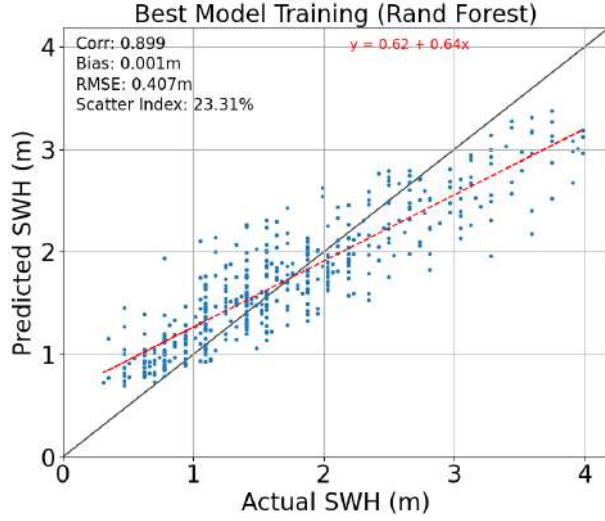


Figure 32: Results of the training of the best model obtained from the grid search of random forest models on data limited to SWH below 4m.

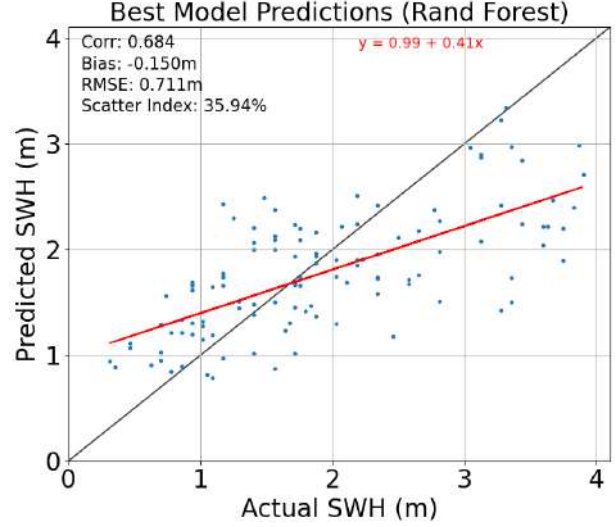


Figure 33: Predictions of the same model from Figure 32 on unseen testing data.

the testing of the model are presented in Figure 34. Once again, similar results are observed and the phenomena overestimation and underestimation at low and high SWH respectively have occurred.

5 Conclusions

This work has demonstrated that although correlations between SWH data that was measured by an in-situ instrument and that predicted from SAR satellite images are observed, the models developed here are not feasible for current use as a reliable forecasting tool for sea-state. Investigations in this project showed that a MLP network performed very similarly across a number of data subsets. These subsets were engineered in an attempt to provide the designed models with a set of data that was both well sampled and contained reliable information that could lead to predictions of the sea-state. The similar performance across all subsets as well as across model architectures consisting of up to 3 hidden layers led to an initial sense that the dataset and thus model input features may be lacking in the information necessary to allow for strong correlation with in-situ measurements.

In an effort to construct a test of this, a random forest model and an extreme gradient boosted model were implemented. The performance of the random forest model was very similar to that of the MLP model as seen by the comparison between Figures 19 and 33 although the MLP model still won out by a small margin. The gradient boosted model performed worst of all. As stated previously, there were gaps in the knowledge of these later two models given the time allotted to the final section of this work although some conclusions can be drawn here. Ultimately, the fact that no model outperformed the other by a significant margin despite measures such as dataset engineering, cross validation, hyperparameter tuning and architecture testing being implemented to best avoid many of the pitfalls in machine learning, leads to a conclusion that the performance is limited by the information provided to the models rather than the models themselves.

Despite the fact that the performance of the models was not to the level that was perhaps desired

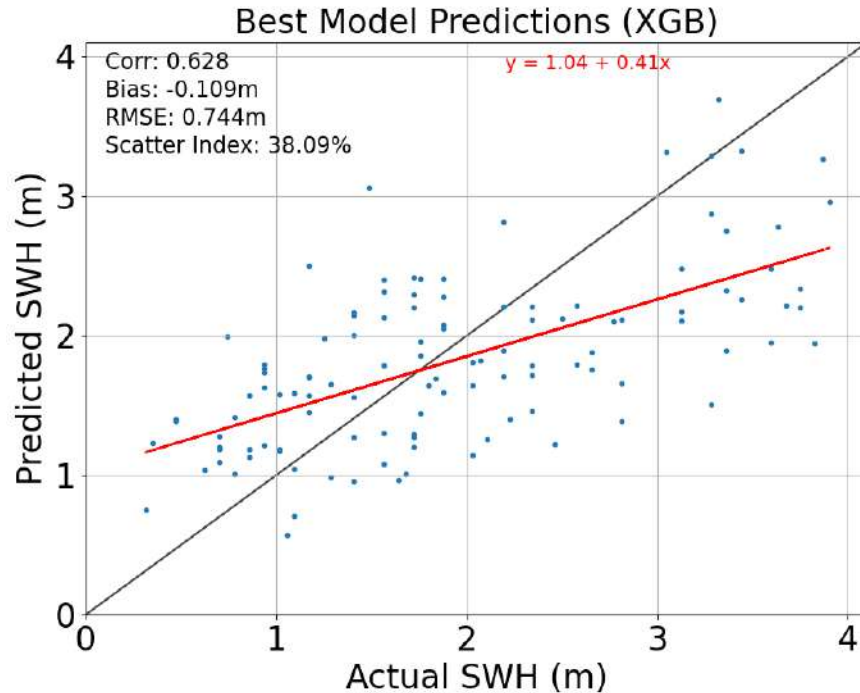


Figure 34: Predictions of the XGBoost model on unseen test data.

at the outset of this project, a great deal of satisfaction can be drawn from this work. A large amount of knowledge was gained in a number of aspects due to the work involved in this project. An understanding of the current state of the art of machine learning techniques used in sea-state prediction as well as the history and theory behind SAR has been gained. Secondly, the process of designing machine learning models from a relatively beginner knowledge base has been very rewarding and will no doubt be beneficial going forward. Finally, working on such a project in the setting of a workplace has been an extremely valuable experience.

6 Future Work

Although this work demonstrates the potential of artificial neural networks in predicting significant wave height in the waters around Ireland, there are a number of ways in which this project could be extended and the results improved upon. This work could be furthered generally by a more in depth knowledge of machine learning techniques. Areas such as hyperparameter tuning and learning algorithms could be explored given more time and experience. Another avenue that could be explored is the introduction of more data. The dataset provided on dataset provided on Google Earth Engine is limited to data from February 2015 onwards while Sentinel-1A was launched in 2014. There are a number of other buoys operated by Met Eireann at similar distances from the shore around Ireland. The study area could be extended to allow temporal and spatial matching of image and buoy data in other locations.

As discussed, some other studies made use of spectral techniques allowing for extraction of further features eg. CWAVE features, azimuth cutoff wavelength etc. Although complicating the architecture of the model, these features may introduce vital information to predicting SWH not present

in the NRCS data that comprised the majority of the features used in this work. This would require more time, a knowledge of image spectral processing techniques and access to the images themselves.

During the methodology planning phase of this project it was anticipated that the results from the machine learning models would be compared with SWH data generated by a wave model or hindcast. This would have been a good addition to go along with the results presented and would have likely strengthened the conclusion that these results demonstrated that the models here are not at the point of being feasible as a forecasting tool. Unfortunately, due to the amount of attention that was required in other parts of the project, this step had to be left out.

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A Google Earth Engine Image Processing Script

```

1
2 // region of interest
3 var roi = ee.Geometry.Point([-6.704336, 51.690425]);
4
5 // apply smoothing
6 var smoothing = function(image) {
7   var smoothing_radius = 200;
8   var smoothed = image.focal_mean(smoothing_radius, 'circle', 'meters');
9   // copy properties because some are lost after smoothing
10  //return smoothed;
11  return smoothed.copyProperties(image).copyProperties(image, ['system:
    time_start']);
12 };
13
14 // create a reducer combination for all desired statistics
15 var reducer = ee.Reducer.first().combine({
16   reducer2: ee.Reducer.mean(),
17   sharedInputs: true}).combine({
18   reducer2: ee.Reducer.variance(),
19   sharedInputs: true}).combine({
20   reducer2: ee.Reducer.skew(),
21   sharedInputs: true}).combine({
22   reducer2: ee.Reducer.kurtosis(),
23   sharedInputs: true
24 });
25
26 // Get the image date, orbit pass and polarization band and apply the
    reducer
27 // to the image within the region of interest
28 var multiStats = function(region){
29   var getStats = function(image){
30     // image date and time
31     var date = ee.Date(image.get('system:time_start')).format('YYYY-MM-dd-
        HH-mm-ss');
32     // orbit pass
33     var pass = image.get('orbitProperties_pass');
34     // band
35     var band = image.bandNames().get(0);
36     // calculate pixel statistics and make dictionary
37     var reduceDict = ee.Dictionary(image.reduceRegion({
38       reducer: reducer,
39       geometry: region,
40     }));
41     // make dictionary for date, pass and band
42     var dataDict = ee.Dictionary(
43       {'Date-and-time': date,
44        'Orbit-pass': pass,
45        'Band':band
46     });
47     //combine dictionaries
48     var Dict = dataDict.combine(reduceDict);
49     // create an image property with this data

```

```

50     var output = image.set({'Stats': Dict});
51     return output;
52 };
53 return getStats;
54 };
55
56
57 // get the mean of the incidence angle
58 var angleMean = function(region){
59     var getMean = function(image){
60
61         // single element dictionary with angle
62         var angleDict = ee.Dictionary(
63             image.reduceRegion({
64                 reducer: ee.Reducer.mean(),
65                 geometry: region,
66                 scale:50
67             }));
68
69         var date = ee.Date(image.get('system:time_start')).format('YYYY-MM-dd-
70             HH-mm-ss');
71         var dateDict = ee.Dictionary(
72             {'Date-and-time':date})
73
74         var Dict = dateDict.combine(angleDict);
75
76         // create an image property with this data
77         var output = image.set({'Angle':Dict});
78         return output;
79     };
80 };
81
82 // circle of specified radius around buoy
83 var circle = roi.buffer(200);
84
85 // filter the image collection by metadata properties and apply smoothing
86 var s1Images = ee.ImageCollection('COPERNICUS/S1_GRD')
87     .filterDate('2014-10-03', '2022-16-08')
88     .filter(ee.Filter.bounds(circle))
89     .filter(ee.Filter.listContains('transmitterReceiverPolarisation', 'VV'))
90     .filter(ee.Filter.listContains('transmitterReceiverPolarisation', 'VH'))
91     .filter(ee.Filter.eq('instrumentMode', 'IW'))
92     .map(smoothing);
93
94 // select VV band and calculate statistics
95 var vv = s1Images.select(['VV'])
96     .map(multiStats(circle));
97
98 // select VH band and calculate statistics
99 var vh = s1Images.select(['VH'])
100     .map(multiStats(circle));
101
102 // select angle band

```



```
103 var angle = s1Images.select(['angle'])
104     .map(angleMean(circle));
105
106 // select the new image properties that contain the required information
107 var vv_arr = vv.aggregate_array('Stats');
108 var vh_arr = vh.aggregate_array('Stats');
109 var angle_arr = angle.aggregate_array('Angle');
110
111 // exporting to a table requires a feature collection, so make one here
112 var generateFeatures = function(list){
113     var featureCollection = ee.FeatureCollection(
114         list.map(function(element){
115             return ee.Feature(null, element);
116         })
117     );
118     return featureCollection;
119 }
120
121 //export each set of data to my drive
122
123 Export.table.toDrive({
124     collection: generateFeatures(vv_arr),
125     description: 'VV-Statistics',
126     fileFormat: 'CSV'
127 })
128
129 Export.table.toDrive({
130     collection: generateFeatures(vh_arr),
131     description: 'VH-Statistics',
132     fileFormat: 'CSV'
133 })
134
135 Export.table.toDrive({
136     collection: generateFeatures(angle_arr),
137     description: 'Angle',
138     fileFormat: 'CSV'
139 })
```

UNIVERSITY COLLEGE DUBLIN

SCHOOL OF PHYSICS

MSC. SPACE SCIENCE & TECHNOLOGY

Construction of a Catalogue of Variable AGB Stars with Reliable Distance Estimates using *Gaia* DR3

Author

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August 19, 2022



Declaration of Authorship

I hereby certify that the submitted work is my own work, was completed while registered as a candidate for the degree stated on the Title Page, and I have not obtained a degree elsewhere on the basis of the research presented in this submitted work.

Abstract

AGB stars are the late stage evolution of low to intermediate mass stars. They demonstrate mass loss and long period variability as they transition toward the planetary nebula phase of their stellar lifetimes. This report details the ongoing campaign to develop a catalogue of Galactic AGB stars with reliable distance estimates using data from the latest *Gaia* data release; Data Release 3. The project is split into three stages each concentrated on a different subgroup of AGBs: high mass OH/IR sources, Carbon-rich AGBs, and low mass Oxygen AGBs. As of 2022 the first two portions are approaching completion with the low mass Oxygen sample currently being analysed. The report focuses on the analysis of the variability properties of the sample of OH/IR stars. Using virtual observatory tools, *Gaia* epoch photometry data was extracted for the OH/IR sources which was used to fit light curves to the sources. From a starting point of 1,505 stars, variability analysis was possible for a subset of 1,071. For the stars with light curve data, the model fits were sorted by quality and grouped into subclasses: Mira variables, Non-Mira variables, and Dubious sources. For the Mira subgroup, the variable properties including period, amplitude, and mean magnitude were assessed. Following this the Miras with reliable *Gaia* parallaxes were identified and used to estimate the stellar distances, find the absolute magnitudes of the stars, and plot the period-luminosity relation. This analysis was undertaken with the aim of creating an auxiliary table for the new *Gaia* catalogue containing variability information for the sources.

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List of Acronyms

2MASS - 2 Micron All Sky Survey
AF - Astrometric Field
AGB - Asymptotic Giant Branch
AGIS - Astrometric Global Iterative System
BAM - Basic Angle Monitor
CAB - Centro de Astrobiología
CCD - Charge Coupled Device
CSE - Circumstellar Envelope
DR2/DR3 - Data Release 2 (or Data Release 3)
ESA - European Space Agency
FPA - Focal Plane Array
ICRS - International Celestial Reference System
IR - Infra-red
IRAS - Infrared Astronomical Satellite
IVOA - International Virtual Observatory Alliance
LPV - Long Period Variable
RP/BP - Red Photometer/Blue Photometer
RVS - Radial Velocity Spectrometer
SCRA - Steady Circumstellar Radiative Acceleration
SED - Spectral Energy Distribution
SM - Sky Mapper
SRV - Semi-Regular Variable
SSiC - Sintered Silicon Carbide
VO - Virtual Observatory
VOSA - Virtual Observatory SED Analyser
WISE - Wide-field Infrared Survey Explorer

Acknowledgements

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I am extremely grateful to my team of supervisors in the European Space Agency: Fran Jiménez-Esteban, Belén López Martí, and Dieter Engels. The team were, and continue to be, vital guides in navigating these astronomical catalogues, and I look forward to the remaining work on the rest of the project.

I would also like to thank a number of faculty members in UCD. Firstly, Professor Lorraine Hanlon for all the guidance throughout the year, including helping me develop the application that would eventually land me this position at ESA. Secondly, Dr. Robert Jeffery who taught me and the class how to write better code and preemptively stopped us being put on some sort of Python watch-list. Also, thanks to Professor Sheila McBreen for the check-ins over the summer.

Finally I am beyond grateful to my parents, my sister, Lara, and my girlfriend, Megan, for keeping an eye on me throughout throughout this MSc. and repeatedly asking questions about stars that I didn't know to the point where I eventually had to go and find out the answers for myself.

1 Introduction

Stellar evolution is one of the most studied and diverse fields of astrophysics. The life cycles of the stars in our Universe have been extensively modelled, combining facets from nearly all branches of physics. Astrophysicists use fluid and statistical mechanics to derive the conditions needed for vast interstellar clouds of hydrogen to collapse and form protostars. Since the 1950's, nuclear physicists have built detailed nucleosynthesis models which explain how this hydrogen ignites in intense fusion reactions, getting converted to heavier elements and expelled into space (Burbidge et al. 1957). Quantum mechanics and Pauli's exclusion principle explains how the internal structures of stars are maintained by electron degeneracy while at the other end of the spectrum of physics, general relativity describes how the most massive stars eventually collapse to form neutron stars or in the most extreme cases, black holes (Ostlie & Carroll 2007).

As such, stellar evolution is relatively well understood but the validation of these models requires evidence which is where astronomy becomes necessary. Astronomical observations of stars not only help validate these models based on observed behaviour but new, unexpected behaviour can lead to the development of increasingly accurate theories. The interplay between astrophysics and astronomy is bi-directional with astronomy both proving theoretical predictions while simultaneously exploring new phenomena. The advent of multi-wavelength stellar catalogues means that now collections of stars from all evolutionary stages and their respective stellar properties can be studied over all wavelengths, from gamma to radio.

The Asymptotic Giant Branch (AGBs) is one stage of stellar evolution. AGB stars are low to intermediate mass post main sequence stars in the late stages of their journey towards becoming planetary nebulae. They are marked by their intense stellar winds leading to huge mass loss, dense circumstellar envelopes enriched with heavy elements, and strong maser emission visible in the radio band. AGB stars also demonstrate variability, with the regular pulsating Mira variables making up a subset of the branch. The evolution of AGB stars has been extensively modelled but previous astronomical studies of the stars, though successful, have depended primarily on observations of AGBs in the Magellanic Clouds. This is due to their known distances but means the results are constrained to the narrow range of masses and metallicities found in those regions (some examples include Cioni & Habing (2003), Ventura et al. (2016), Suh (2020)). Studies of galactic AGB stars are desirable but have not been fully exploited due to a lack of reliable distance estimates which are needed to properly define their spectral properties.

The project detailed in this report is part of an ESA Express Procurement to build a complete and comprehensive catalogue of AGB stars in the galaxy. The goal is to provide information on a galactic sample of AGB stars using accurate astrometry from ESA's *Gaia* mission which can then be made available for anyone wishing to study these stars (Prusti et al. 2016). The project will tackle two fundamental facets of AGB analysis: analysis of their spectral energy densities and analysis of their variability properties. The catalogue contains positional information, information derived from spectral analysis of the sample including bolometric fluxes and absolute magnitudes where possible. Variability information will be available for a select sample of the stars using photometry data made available in the latest *Gaia* release, Data Release 3 (DR3). This includes periods, amplitudes, mean apparent magnitudes and sub-classification based on their variability.

As of 2022, two stages of the project have been completed. *The Gaia Catalogue of OH/IR Stars* was constructed in 2020 and contains over a thousand high mass oxygen rich AGB candidates. Shortly after in 2021, *The Gaia Catalogue of Carbon-Rich AGB Stars* was constructed with around 3,000 carbon rich AGBs. All the sources in both catalogues had *Gaia* counterparts. These have not been officially published as of 2022 as one key portion of the project is not yet finished. The remaining task is the creation of a catalogue of low mass oxygen rich AGBs which is currently underway by the team at CAB in ESAC.

This report focuses on the variability analysis carried out on the *The Gaia Catalogue of OH/IR Stars* using the newly released light curve data from *Gaia* DR3. DR3 contains light curves for over 10 million stars taken over the course of 34 months. With the typical period of an AGB star being between 200 and 800 days, this new photometry data allows for the study of the variability properties of these AGB stars. It will also allow for the study of the AGB period-luminosity relation when paired with *Gaia*'s unparalleled astrometric

precision. The overall project has been ongoing for over two years now and is a collaborative effort between a number of astronomers based in ESAC and the University of Hamburg. A breakdown of who performed the specific work detailed in this report is provided in Appendix B.

The project also makes use of the international astronomical data collaboration; the Virtual Observatory (VO). The project requires the combination and cross-referencing large data-sets from of a number of surveys and catalogues, from infrared to optical and thus is a perfect application of the VO. The VO makes working with multiple catalogues of varying sizes and data types a straightforward task thanks to complementary VO analysis tools such as VOSA¹ and Aladin² and provides the ideal foundation for this task.

This report is structured as follows. Section 2 is split into three subsections, each focusing on a different aspect of the project in the form of a literature review.

The first subsection examines the Asymptotic Giant Branch including the physical properties of the stars, astronomical observables and historical background. This section includes a breakdown of the various types of AGB star and includes OH/IR stars and the branching of AGBs into Oxygen-rich and Carbon-rich types. Finally, this section will examine some of the recent work in the study of these stars as well as work done to build catalogues in the same manner as this project.

Following this is an introduction to *Gaia*, the ESA space observatory which makes up the backbone of this project. This introduction will briefly summarise the development of *Gaia* and its primary goals including the need for precise astrometry, the theory of stellar parallax, the technical capabilities of the observatory and previous data releases made by the *Gaia* team. However, the report will also focus on the latest data release: *Gaia* DR3, its contents, and the advances in astronomy made by the release. An important aspect of this project is how *Gaia* classifies variable stars and what analyses can be performed on the classified sources.

The final portion of Section 2 focuses on the VO and its role in modern astronomy. A key component of this project revolves around the combination of various IR and visible catalogues to which the international virtual observatory plays a key role.

Section 3 delves into the methodologies used during the project. Given that this is a highly data based project, this section includes descriptions of the software used throughout, a breakdown of the data collection, extraction, and cleaning processes and a summary of the previous work on the project up to this point. A concise account of the light curve fitting algorithm, error analysis, and complementary analysis being performed by other members of the team is also included at this point.

Section 4 details the the results of the analysis from the visual analysis of the light curves to summarising the variable properties of the sample, examination of the sources with good distance estimates, and establishment of a period-luminosity relation.

Again, given the relatively early stage at which this report was written with respect to the project, Section. 5 contains a summary of the work that is yet to be performed including a rundown of the science that can be accomplished with the results of the work up to this point. Finally, Section 6 consists of discussions and conclusions based on the results from section 4 and a look at some shortcomings and successes of the analysis.

¹<http://svo2.cab.inta-csic.es/theory/vosa/>

²<https://aladin.u-strasbg.fr/>

2 Theory

2.1 Asymptotic Giant Branch Stars

The Asymptotic Giant Branch phase is the late stage of evolution for low to intermediate mass stars of masses $\sim 0.5M_{\odot} - 8M_{\odot}$. Stars in the AGB branch are red supergiants, larger than the stars on the first red giant branch with high luminosities of $2,000 < L/L_{\odot} < 55,000$ but lower effective temperatures (Höfner & Olofsson 2018). According to the standard Morgan-Keenan stellar classification scale AGB stars are K and M type stars but are also described by newer spectral classes such as C-type for AGBs with Carbon in their atmospheres or even S-type. The AGB phase is one of the last stages of evolution of these stars. After this point they transition to post-AGB stars into planetary nebulae before finally ending their stellar lifetime as white dwarfs having shed their nebulae (see Section 2.1.3).

As these stars consume the helium in their cores following their departure from the main sequence and the first red giant branch, they are left with a small inert core made up of carbon and oxygen and dominated by electron degeneracy pressure. Compared to the 0.1-1.2 billion years that stars spend on the main sequence

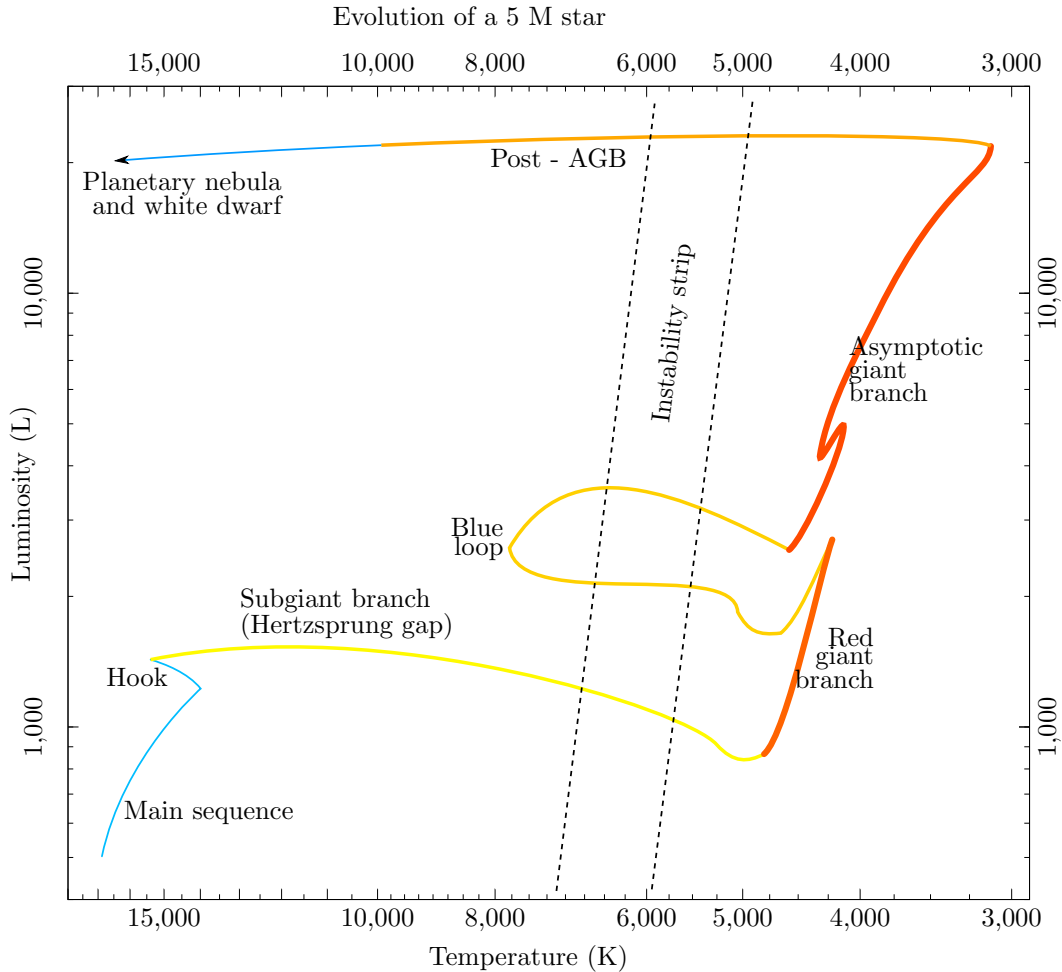


Figure 1: Evolution of an intermediate mass ($5M_{\odot}$) star on the Hertzsprung-Russell diagram. The evolution begins with the departure from the main sequence in the bottom left corner and ends with the transition from a planetary nebula into a white dwarf. The cool but luminous AGBs are in the top right corner of the diagram (Viking 2017).

they spend a comparatively short 1-15 million years on the AGB.

Chemically, AGB stars can be split into two categories: Oxygen-rich AGBs and Carbon-rich. The distinction is made with the carbon to oxygen ratio (C/O) found in the stellar atmosphere or the dust in the circumstellar envelope (CSE). All AGB stars begin with a C/O ≈ 0.5 which develops into a *Carbon Star* if C/O becomes greater than 1. O-rich stars will display spectra with features at $10\mu m$ and $18\mu m$ due to silicate dust and oxides such as titanium-oxide (TiO), water (H₂O), or hydroxyl (OH). C-rich AGBs on the other hand will display spectra with carbon based compounds such as diatomic carbon (C₂), carbon-monoxide (CO), ethyne (C₂H₂), and other more complex compounds such as graphite. In AGB stars with low mass loss and thin CSEs these features appear as emission lines in the stellar spectra whereas for stars with higher mass loss they appear as absorption features in the spectra due to the thick layer of circumstellar dust (Suh 2021).

Current modelling of AGB evolution and nucleosynthesis predict that the development of an O-rich or C-rich chemistry depends on the initial mass of the star as it enters the AGB phase.

Low Mass AGBs ($M < 1.5M_{\odot}$) are O-rich stars. The presence of carbon in an AGB star's atmosphere is the result of convection currents dredging carbon ash from the helium burning shell up to the surface. Low mass stars violently ignite their helium cores in a helium flash and do not sustain enough convection over their lifetimes to pull carbon to the surface. They exhibit O-rich signatures before ending their lives as a white dwarf (Karakas & Lattanzio 2014).

Intermediate Mass AGBs ($M < 1.5M_{\odot}$) become C-rich stars via a process known as the *Third Dredge-up*. The *first* dredge-up happens earlier in the star's lifetime, while it occupies the RGB. Red giants have convection in their outer layers which extend from the surface of the star down to the hydrogen shell. This allows for products of hydrogen burning to be transported to the surface (Priyalnik 2004). The *second* dredge-up occurs early in the AGB phase when the star begins to expand. The expansion of the outer envelope leads to a decrease in effective temperature. This causes the convective envelope to deepen and reach through the hydrogen burning shell into the helium shell. This second dredge up increases the amount of helium and nitrogen (from the CNO cycle) in the outer layers of the star (Ostlie & Carroll 2007). With the third dredge up, if enough carbon material is brought from the 3α burning regions, C/O becomes greater than 1 and the star becomes a carbon star. The increase in the C/O ration, brings with it an increase in *s*-process heavy elements which get dredged up with the carbon by the convective currents.

Higher Mass AGBs ($M < 1.5M_{\odot}$) revert to O-rich chemistry. This is because the highest mass AGB stars experience a phenomena known as *Hot Bottom Burning*. This occurs when the bottom of the convective envelope has a high enough pressure and temperature ($T > 30$ MK) to initiate nuclear burning (Ventura et al. 2015). For massive AGB stars this burning takes the form of proton capture nucleosynthesis which converts interior carbon to nitrogen. This C-N conversion keeps the C/O ratio < 1 and promotes an O-rich chemistry. Hot bottom burning has been verified through the concentration of Li observed in O-rich AGB stars in the Magellanic clouds. Li is produced when the ⁷Be generated in the nuclear reaction $^3\text{He}(\alpha, \gamma)^7\text{Be}$ is brought to the surface via convection from the base of the envelope and decays via electron capture to ⁷Li. This process, known as the Cameron-Fowler mechanism, necessitates the high temperatures indicative of HBB and thus is a good indicator of the process. A further indicator of HBB is the presence of heavy *s*-process elements such as Rb and Zr in the spectra (Garcia-Hernandez et al. 2013).

High-mass O-rich AGBs are progenitors for OH/IR objects. OH/IRs are the result of huge mass loss rates which fully obscure the central stars optical flux (Justtanont et al. 2013). They are known by their strong luminosity in the IR as well as strong OH maser emission. The first completed portion of this project, the high-mass O-rich *Gaia* catalogue is made up of OH/IR objects.

Outside of their chemistry AGB stars have three primary physical characteristics, as listed by Priyalnik (2004):

- A strong stellar wind which results in extensive mass loss (Section 2.1.2). The wind is the result of a dominating radiation pressure in the stellar envelope.
- A thermally unstable two shell structure consisting of a helium and hydrogen shell both undergoing nucleosynthesis (see Fig. 3). The unstable configuration of the shells leads to thermal pulsation causing

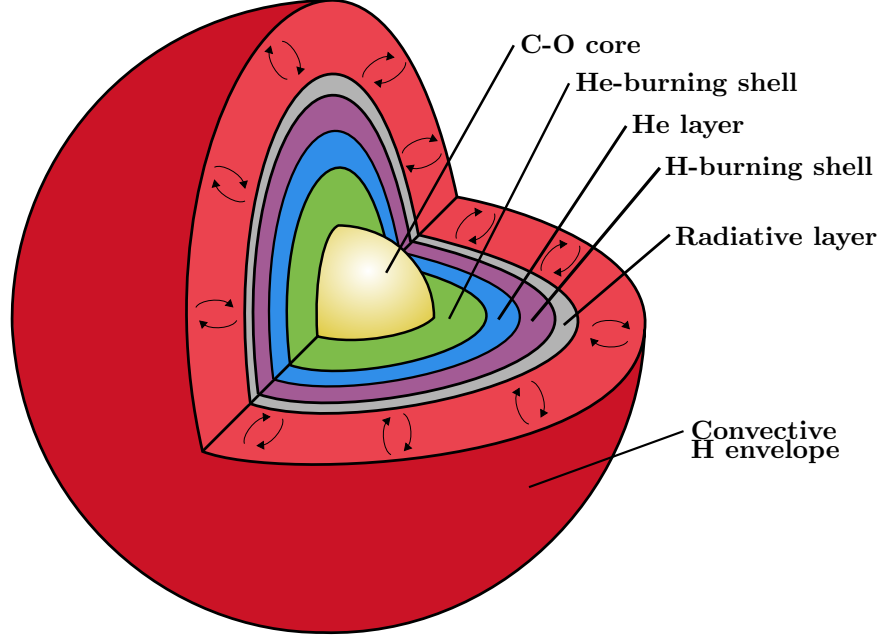


Figure 2: Cross-section of an intermediate mass AGB star. Note the many layers of hydrogen and helium in various states of nuclear burning which leads to thermal pulsation. The central C-O core is inert and supported by electron degeneracy pressure (Persson 2014).

variability in the stars luminosity.

- The overall luminosity of the star is proportional to the mass of the carbon and oxygen core rather than the overall mass of the star.

During the later portion of their time on the AGB, stars will under go a series of thermal pulsations. This portion of the stars lifetime is referred to as the thermally pulsing asymptotic giant branch (TP-AGB). These long-term pulsations are a direct result of the shell structure of AGB stars. As the hydrogen shell burns it creates helium which is dumped inward onto the helium shell, increasing its mass. Once the helium shell has enough mass, its lower levels become degenerate, increasing the pressure and temperature leading to a helium flash akin to those seen in the RGB phase with a peak luminosity of $\sim 10^8 L_{\odot}$ (Priyalnik 2004). This sudden increase in burning forces the hydrogen burning shell outward, cooling it and stemming nucleosynthesis. The lack of energy causes the layers to shrink back in on themselves before hydrogen burning restarts and the process repeats.

Thermal pulses dramatically modify a stars properties including luminosity, radius, and temperature. Unfortunately these changes occur over vast timescales ($\sim 10^4 - 10^5$ years) and thus can not be directly observed.

Paczynski (1971) first noted the correlation between core-mass and luminosity, modelling the relationship as

$$\frac{L}{L_{\odot}} = 59250 \left(\frac{M_{core}}{M_{\odot}} - 0.5 \right) \quad (1)$$

where L and M represent the luminosity and mass of the AGB star. This linear relationship was derived from numerical computations of stellar models and has since both been applied to observations of AGB stars, and validated with later reviews (Herwig et al. 1998). More recent investigations have shown that this relation breaks down for AGBs with $M > 3M_{\odot}$, a result of hot bottom burning massively increasing the luminosity (Ventura et al. 2022).

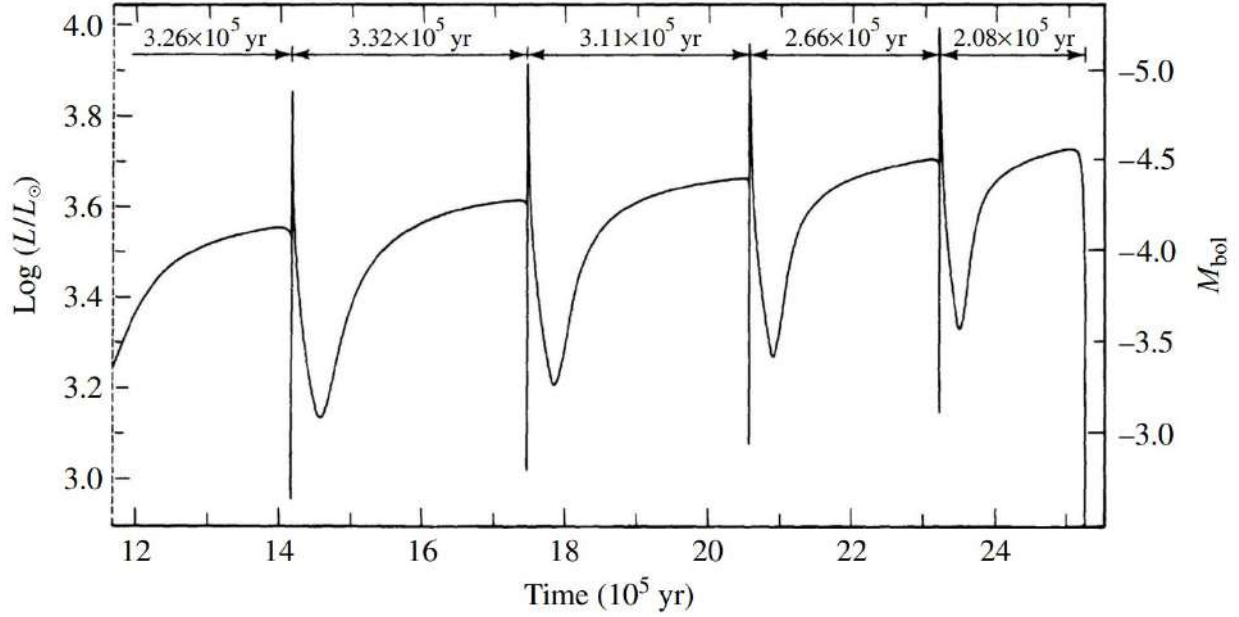


Figure 3: The luminosity of a $0.6M_{\odot}$ star as it occupies the TP-AGB. The surface luminosity while varying with the pulses increases slightly after each flash leading to an upward trend (Ostlie & Carroll 2007).

2.1.1 Identifying AGB Stars

AGB stars have a few unique identifiers that allow them to be detected and characterised. They were traditionally identified through their strong maser emission originating from their CSE. Maser stands for the Microwave Amplification of the Stimulated Emission of Radiation and is a form of spectral line emission. Maser emission is a non-thermal process and happens when a population inversion is induced in a molecule by an external energy source, in this case the radiated energy from the AGB star. A population inversion simply means that there are more electrons present in the higher energy levels of a molecule than the lower levels. If a photon interacts with a molecule with electron in the upper energy levels, stimulated emission occurs causing the molecule to emit two coherent photons in response (Gray 1999).

To achieve this state of population inversion the molecules need to be “pumped” which typically takes the form of an external IR radiation source or the chemical formation of molecules with electrons in higher energy levels. The former is the favoured process for AGB stars which emit cool red starlight, ideal for a pumping mechanism.

Maser emission is a powerful tool for AGB astronomy. They not only permit the detection of this stellar type, but can also provide constraints on the parameters needed to understand the AGB stellar wind and associated mass loss phenomena (Höfner & Olofsson 2018). Observations of AGB maser sources let astronomers spatially resolve the stellar atmosphere to distances comparable with the stellar radius, as well as measure bulk gas motion two quantities vital for the study of AGB mass loss.

Interestingly it appears that the mass loss rate affects the type of masers detected. Low mass loss stars show higher emission from H_2O and SiO (silicon monoxide) masers whereas stars with higher mass loss rates show strong OH emission (Engels 2005).

An observation of an AGB maser will often result in a twin peaked emission line structure. This is due to both the expansion velocity of the CSE and the radial velocity of the star incurring a Doppler shift in the observed wavelength. The emission from the approaching side of the CSE gets blue-shifted to higher wavelengths and the receding side gets red-shifted. The two peaks are separated by 30 ± 10 km/s. One can even determine the radial velocity of the star by finding the midway point between these two peaks and

calculating its Doppler shift (Etoka et al. 2015).

It should be noted that the distances over which maser emission takes place around a star are huge. SiO and H₂O masers are found in the dust formation zone typically ~ 5 AU from the star. They require higher excitation temperatures of 500K and 1,000K, respectively. OH maser emission happens at radii between $\sim 500 - 10,000$ AU requiring only 300K.

AGBs can be characterised by their colour specifically through the use of colour in diagrams such as the IRAS two-colour diagram or via plots of their Wesenheit function (van der Veen & Habing 1988, Lebzelter et al. 2018). The IRAS two-colour diagram has been highly successful for identifying AGB stars. Such a diagram consists of a plot of the flux difference in the 12 and 25 μm bands against that of the 25 and 60 μm bands. The flux difference between two bands is described by this expression:

$$[12] - [25] = 2.5 \log \left(\frac{F_{12\mu\text{m}}}{F_{25\mu\text{m}}} \right) \quad (2)$$

where $F_{i\mu\text{m}}$ denotes the source flux in a particular band i . The equation for [25]-[60] is the same but with the larger wavelengths.

By plotting [12]-[25] vs. [12]-[25] for a sample of AGB stars it was found that the position of the stars on the plot adhere to a curved trend. van der Veen & Habing (1988) fitted a line of best fit to a sample of AGB stars and found that AGBs follow a trend that can be described by

$$[25] - [60] = -2.15 + 0.35 \exp(1.5([12] - [25])) \quad (3)$$

van der Veen & Habing split the IRAS diagram into a set of areas along this line, each containing a subset of AGB stars with different properties. Therefore AGB stars can be identified and characterised based on where they fall on this diagram. For example, region *II* on the diagram contains variable stars with O-rich CSEs, while region *VII* contains variable C-rich AGBs. These regions are marked in Fig. 4b.

AGB stars can also be identified spectroscopically via their Spectra Energy Distributions (SEDs). An SED is the plot of an objects energy vs. wavelength. Using SEDs to identify astronomical objects is a technique that is used in most branches of astronomy and has become more prevalent in recent years thanks to the increase in observatories which allow for the construction of spectra from radio through to gamma. A classic example of such analysis is the characterisation of stellar populations in galaxies based on their emitted SED (Walcher et al. 2011). They can also be used to identify galaxy types, with active galaxies such as 3C 273 showing much flatter SEDs than optically strong spiral galaxies due to strong emission in the X-ray and gamma bands (Brown et al. 2019). More recently, studies such as that performed by Robitaille et al. (2007) have used SED to characterize the evolutionary stage of stars. By fitting spectral data to a large parameter grid of pre-computed theoretical SED models, they described the physical properties of 30 young stellar objects in the Taurus-Auriga star-forming region. Clearly SED analysis is a powerful tool for investigating stellar evolution.

AGB stars have a distinctive single peaked SED (Fig. 4a). The SED peaks in the IR at $\sim 1 \mu\text{m}$ due to the thermal re-emission of stellar radiation from the circumstellar dust around the star. CSE dust absorbs optical wavelengths leading to low spectral flux for wavelengths shorter than IR.

2.1.2 Mass Loss & the CSE

One of the defining features of AGB stars is the high mass loss rates, with indications of substantial mass loss first observed on the α Herculis binary system in the 1950's (Deutsch 1956). This mass loss makes them the largest contributors of dust to the interstellar medium (Herwig 2005). AGB stars are thought to be the main producers of carbon, oxygen, and nitrogen and are estimated to generate approximately half of

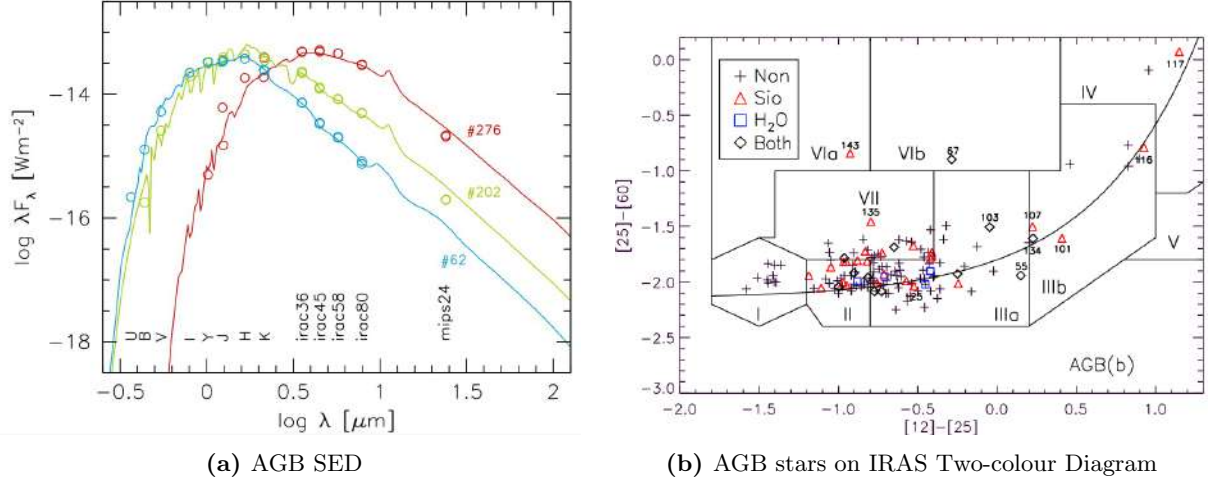


Figure 4: SED and IRAS Colour Diagram for AGB objects. Fig.(a) shows the single peaked SED characteristic of an AGB star fitted to three sample stars: one O-rich and two C-rich AGBs (Gullieuszik et al. 2012). Fig.(b) shows a sample of AGB stars plotted on the IRAS Two-colour diagram (Yoon et al. 2014)

the remaining elements heavier than iron making them essential to the chemical enrichment of the universe (Kamath & Van Winckel 2022).

AGBs demonstrate typical mass loss rates beginning at approximately $10^{-8} M_{\odot}/\text{yr}$ and increase up to $\sim 10^{-5} M_{\odot}/\text{yr}$ as the star transitions from a Mira variable to an OH/IR star (McDonald & Zijlstra 2016). Still, this is not a hard upper limit as observations on oxygen rich AGBs have yielded even greater mass loss rates. For example analysis of the O-rich star *IRAS 16342-3814* performed by Dijkstra et al. (2003) found that its mass loss rate may be as high as $\sim 10^{-3} M_{\odot}/\text{yr}$. At this rates it is estimated that AGB stars lose up to 85% of their mass due to this mass loss. In fact, unlike other stages of stellar evolution, the time scales of which are dependant on nuclear burning in the stellar envelope, the timescale of the AGB phase is influenced almost solely by the surface mass loss rates (Kemper et al. 2003).

Mass loss from AGB stars appears as a continuous stellar wind; a mix of gas and dust, elements that have been created in the star by nucleosynthesis and blasted out into space. These winds have velocities of 3 – 30 km/s (Goldman et al. 2016). In literature this mass loss is mathematically defined as the total mass loss rate in all directions and is normally determined by finding a local mass loss rate and integrating this over a spherical shell. Changes in mass-loss rate over time are relatively small but AGBs can undergo brief periods of intense mass loss. This was typically observed for C-rich AGBs and are potentially connected to the thermal helium pulses.

When the stars reach their highest mass loss rate, the central star becomes heavily obscured in the optical by the CSE, with the dust in the CSE effectively absorbing visible wavelengths.

Early attempts to quantise the mass loss in relation to physical properties of the star include Reimers’ law:

$$\frac{dM}{dt} = \eta \frac{LR}{M} \quad (4)$$

where L, R, and M denote the stellar luminosity, radius, and mass of the star, respectively, and η is a fitting parameter on the order of 10^{-13} (Reimers 1975). This empirical relation was derived from a set of observations made of population I red giants.

The success of this relationship has been proven many times since its discovery but in recent years has come under some scrutiny as further developments are made in the field. The two primary problems with the law are summarised by Schröder & Cuntz (2005). The main issue with the law is the fitting parameter η which

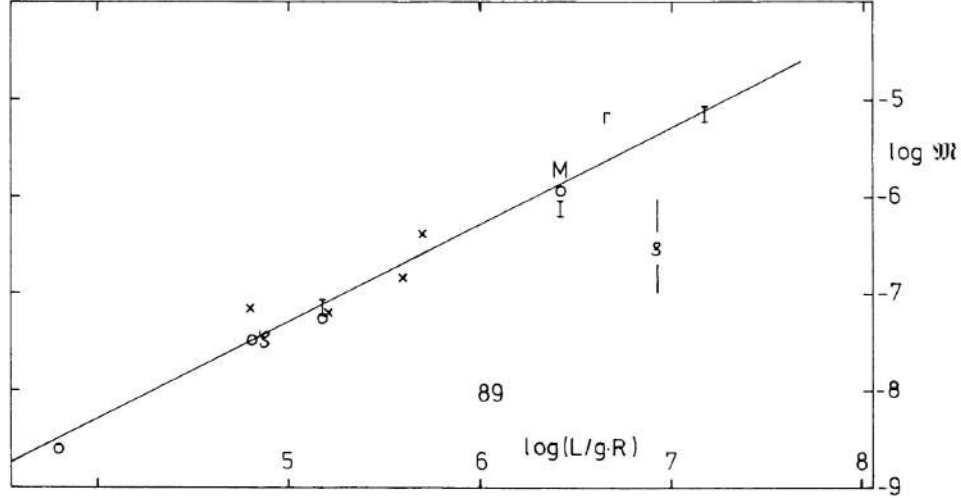


Figure 5: Reimers' original observations of AGB Stars which provided the basis for Eqn. 4 (Reimers 1975)

is not universal and needs to be scaled according to the type of giant stars being observed.

Secondly the law is based on dimensional analysis and lacks any real physical meaning. Other astrophysicists, such as Holzer & MacGregor (1985) were quick to note that for cooler stars the luminosity of the star is not actually expected to play a part in the mass loss process. Reimers' law appears to allocate too much importance to the luminosity. However, Reimers' law did provide the basis for many improved descriptions of mass-loss rates. Schröder & Cuntz (2005) derived a modified version of Reimers' law which considers the mechanical energy flux needed to generate non-radiative mass loss and its dependence on effective temperature (T_{eff}):

$$\frac{dM}{dt} = \eta \frac{LM}{R} \left(\frac{T_{eff}}{4000[K]} \right)^{3.5} \left(1 + \frac{g_{\odot}}{4300 \cdot g} \right) \quad (5)$$

where the variables have the same meanings as Eqn.4 with the addition of the stellar and solar surface gravities, g_* and $g_{\odot}$.

Estimations of the density profiles of AGB CSEs can be derived from the mass loss rates and flow velocities of the gas and dust in the CSE (Höfner & Olofsson 2018). Assuming a constant expansion velocity, the quantities can be linked with the continuity equation:

$$\dot{M}_i = 4\pi r^2 v_{\infty, i} \rho_i(r) \quad (6)$$

where r is the distance from the star, ρ is the density, v_{∞} is the terminal velocity of the matter, and i denotes the matter (gas or dust) being measured. These are typically kept separate because gas and dust expand at different velocities. All that is left is to determine is the expansion velocity of the CSE components. This is achieved through the examination of Doppler shifted radio emission lines from specific CSE masers dubbed *mass loss indicators*. It should be mentioned that this process can work in reverse with density and velocity profiles being used to estimate mass loss. The density profile of the gas/dust is modelled separately and follow an inverse square relation with the radius of the CSE. However, accurate density profiles are difficult to find and an inverse square law does not fully describe the behaviour of the CSE due to photodissociation or other chemical processes affecting the density close to the stars surface (Höfner & Olofsson 2018).

Mass loss indicators must have a set of key properties that allow them to be used as a standard indicator for all AGBs (Schöier 2007).

1. They must be observable and constant regardless of the CSE chemistry which will vary dependent on the C/O ratio.
2. They must have a strong emission so that observations can be made of faint/distant sources.
3. They must have a relatively straightforward decay/energy level profile with distinct easily separable wavelengths. That being said, almost counter-intuitively, the indicator should have a wide range of transitions which will allow for analysis of many different energy level transfers. The various possible emissions of the indicator should be well understood and characterised over a wide range of wavelengths

Historically the best indicator of AGB mass loss has been circumstellar carbon monoxide (CO) rotational line emission. CO is a good indicator for a few reasons. It is the most abundant element in CSEs after H₂ and is present regardless of internal CSE chemistry. It can be easily excited through collisions and radiative interactions which, when combined with its abundance means it is a strong emitter. Finally the CO transition lines are plentiful, well understood and the rotational lines probe relatively high energies meaning a wide range of energies can be analysed (Schöier 2007).

The 18cm OH Maser emission can also be used as an identifier of AGB stars as well as a mass loss indicator as seen recently in the work performed by Engels & Bunzel (2015). Oxygen-rich AGBs show strong maser emission from OH and H₂O and molecules which is excellent for the identification of stars with thick CSEs, obscured cores, and high mass loss. OH maser examinations are the basis for the sample of AGB stars in this study (Section. 3.4)

There is still much debate as to the precise mechanics of how the enormous mass loss rates of AGB and post-AGB stars are both triggered and sustained but much progress has been made in recent years thanks to the advent of high angular resolution instrumentation and the development of multi-band surveys.

It was understood early on that the process originates from the outward radiating pressure from the central star dominating over the gravitational potential (Höfner, S. 2008). The radiative acceleration of the dust and gas needs to be greater than the inward acceleration due to gravity

$$\frac{a_{rad}}{a_{grav}} = \frac{\langle \kappa \rangle L_*}{4\pi c G M_*} \quad (7)$$

where L_* and M_* denote stellar luminosity and mass, a_{rad} is the radiative acceleration, a_{grav} is the acceleration due to gravity, G is the gravitational constant, and c is the speed of light. The radiative acceleration is the result of scattering and collisions by photons and thus is critically dependent on the scattering cross-section which in turn is dependent on the opacity of the CSE. This implies there is a critical CSE opacity which allows radiative acceleration to dominate. By setting $a_{rad}/a_{grav} = 1$, an expression for the critical opacity can be derived

$$\langle \kappa \rangle_{crit} = \frac{4\pi c G M_*}{L_*} \quad (8)$$

It was found that basing the model solely on the molecular gas found in stellar atmospheres that it alone is not opaque enough at near-IR wavelengths to trigger an outflow. Dust grains such as the amorphous carbon and silicates can provide the collective opacity needed to trigger this accelerated outflow of matter. These grains condense at temperatures of approximately 1,500K and so need to be transported to a condensation radius, R_c in order to build up and exceed $\langle \kappa \rangle_{crit}$. An expression for R_c is provided by Lamers et al. (1999)

$$\frac{R_c}{R_*} \approx 0.5 \left(\frac{T_c}{T_*} \right)^{-\frac{4+p}{2}} \quad (9)$$

where T_c is the condensation radius, R_* and T_* are the stellar radius and effective temperature, and p is a constant which varies according to the condensed material and depends on the wavelength absorption by

the substance. By setting $T_c = 1,500$ K and $p = 1$, which is a value based on Mg and Fe bearing silicates, (Bladh & Höfner 2012) gives $R_c \approx 2 - 3R_*$.

The question now becomes how the material reaches these distances from the stellar surface. The answer lies in the radial pulsations of AGB stars (see Section 2.1.4). Modelling of the kinetic energy transferred by stellar pulsation shock waves found that they can easily levitate gas to R_c . Originally it was thought that perhaps the mass loss was fully powered by variable pulsation but this resulted in gross outflow, far exceeding mass loss estimates (Bowen 1988).

All this results in the Steady Circumstellar Radiative Acceleration (SCRA) model, the current thinking behind a dust-driven, radiative acceleration based, mass loss mechanism:

- Stellar pulsation levitates gas up to the condensation radius at which point it condenses into dust. This dust is made up of amorphous carbon and silicates with traces of heavier elements.
- Thanks to continuous pulsation from the star, the matter at this condensation radius eventually reaches a critical opacity (κ) which triggers an outflow by radiative acceleration.
- The accelerated dust collides with gas molecules imparting some of its momentum. The gas is expelled out along with the condensed material via these scattering interactions.

The result of these steps are a steady spherically symmetric outflow and an enrichment of the stars interstellar neighbourhood with the CSE material (Bladh & Höfner 2012).

2.1.3 Post AGB: Planetary Nebulae & White Dwarfs

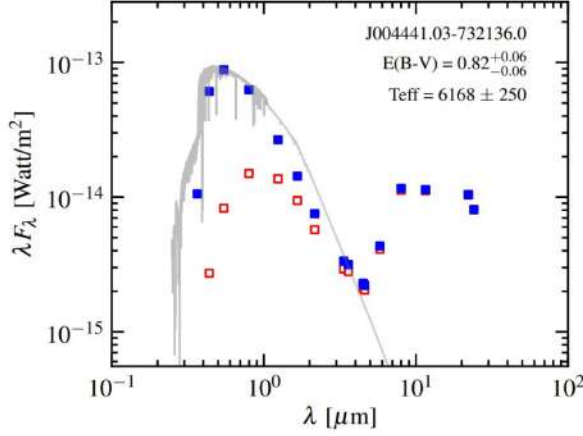
The Post-Asymptotic Giant Branch stars and proto-Planetary nebulae are what remain after the AGB phase. These objects, while well studied, are rapidly variable and very short lived ($\sim 10,000$ years) making them difficult to study and not well understood (Kamath & Van Winckel 2022). The transition from AGB to post-AGB occurs when the majority of the stellar envelope has been removed by the mass loss processes. What remains is the C/O core. At this point the stellar envelope will begin to shrink increasing the star's effective temperature from $\sim 3,000$ K to $\sim 30,000$ K, with the stellar luminosity staying relatively constant. As shown in Fig. 1 this causes the star to traverse the top of the Hertzsprung Russell diagram on its way to the planetary nebula phase.

As with their AGB precursors, post-AGB stars can be identified by their SED and their colour index. Post-AGB objects display double peaked SED with two local maxima (see Fig. 6a). The two peaks are distinct from one another with one arising in the optical and the other in the IR. This double-peaked distribution occurs as the matter in the CSE begins to subside and the interior central object becomes visible. This increase in visibility forms a peak at optical/near-infrared wavelengths. The other peak remains at far infrared wavelengths and is due to IR excitation of the cold circumstellar dust shell (Fujii et al. 2002). On the IRAS two-colour diagram post-AGB objects fall along the line defined by van der Veen & Habing (1988) in regions IIIb, IV and V (see Fig. 6b).

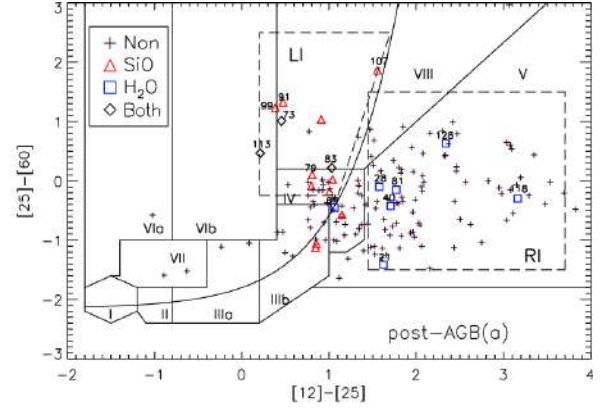
An auxiliary goal of this project will be to identify post-AGB stars from the sample of O-rich candidates which will be accomplished via analysis of the SEDs. In August, a project very similar to that described in this report by Oudmaijer et al. (2022) published a census of post-AGB stars using *Gaia* DR3 data. They established stellar properties for 154 post-AGB stars from an initial sample of 350, demonstrating the applicability of **Gaia** data for stellar studies.

2.1.4 Long Period Variables

One property of AGB stars that is important to the project is long period variability. This variability is a different process to thermal pulsation. Variable AGB stars are generally categorised into three classes: *Mira* variables, *Semi-regular* variables, and *Irregular* variables.



(a) Post-AGB SED



(b) Post-AGB stars on IRAS Two-colour Diagram

Figure 6: SED and IRAS Colour Diagram for Post AGB sources. Fig.(a) shows the two-peaked SED characteristic of a post-AGB object (Kamath & Van Winckel 2022). Fig.(b) shows a sample of Post-AGB stars plotted on the IRAS Two-colour diagram (Yoon et al. 2014).

Mira variables, as the names of the other classes would suggest, are the most regular pulsators of the three. They are named after the red giant Mira, the variability of which was first observed by the amateur astronomer David Fabricius in 1596. Fabricius originally mistook the star for a supernova event when it dimmed entirely from view a month after he first surveyed it (Ostlie & Carroll 2007). It wasn't for another 13 years, in 1609, that the star would be rediscovered by Fabricius, having been spotted in its bright phase. Shortly after this, in 1631, another astronomer Johann Fokkens Holwarda managed to determine Mira's period to be ≈ 11 months. It was around this time when the star was renamed from its original title *o Ceti* to *Mira* meaning wonderful (Hoffleit 1997).

Miras are mostly oxygen-rich AGBs (though some C-rich Miras do exist, Feast et al. (2012)) with a spectral type K or M. They have pulsation periods from $\sim 80 - 1,000$ days and exhibit variations in visual amplitude

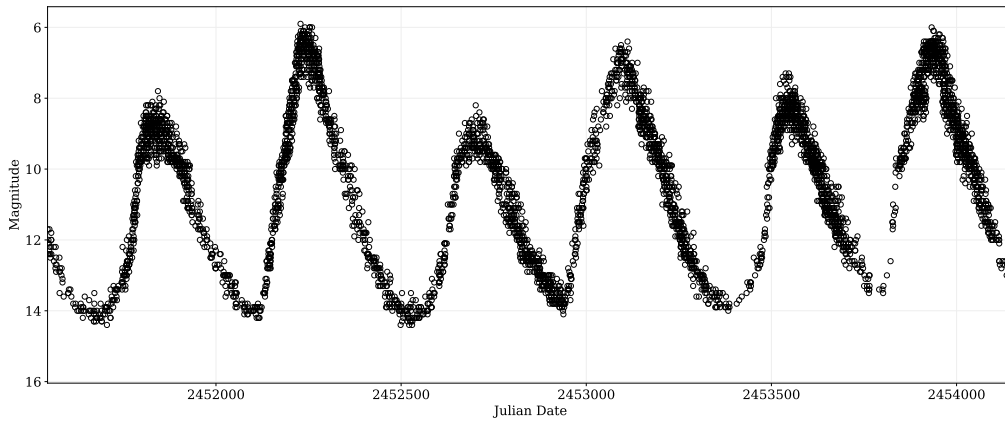


Figure 7: AAVSO Light Curve of the Red Giant star R Cygni constructed from observations performed between 2000 and 2007. R Cygni is a classic example of a Mira variable with a period of 429 days and a variation in magnitude of ≈ 6.5 mag from a maximum mean magnitude of 7.5 mag to a minimum of 13.9 mag (Kafka 2021).

larger than 2.5 magnitudes. They also pulsate in the IR band but with smaller magnitudes, ~ 1 magnitude (Jiménez-Esteban et al. 2021). They pulsate radially with an isotropically oscillating fundamental mode often complemented by overtones. They are the coolest and brightest AGB stars with effective temperatures $\approx 5500\text{K}$. Miras can be easily studied due to their large amplitudes, long periods, and intrinsic luminosity. As such they make up a large portion of the catalogued variable stars in the galaxy.

Mira light curves have a characteristic asymmetric shape, established by Campbell (1955). Campbell defined the ratio between rise-time of the curve and the period as an *asymmetry factor*, f , which, as seen in Section 3.6 is the key to modelling the light curves. Miras show f -values between 0.5 and ≈ 0.1 , with the lower values only observed in variables with comparatively long periods (Vardya 1988).

It is widely accepted that Mira oscillations are powered by the κ -mechanism, which is also the favoured explanation for the pulsation of younger variable stars such as cepheids but with a slight difference (Willson & Marengo 2012). For cepheid variables the pulsation is driven by the ionization of HII in the outer layers of the (still hydrogen burning) core in a region called the HII partial ionization zone. HeIII has a higher opacity causing it to effectively absorb energy and is pushed outward initiating the pulsation. In the Mira scenario, the oscillations are caused by the changing opacity (κ) of *hydrogen* around the helium shell in the hydrogen partial ionization zone. As the star contracts the temperature rises in this zone and begins to ionise the hydrogen and create free electrons. The byproducts of the ionization cause the opacity to increase. As energy radiates outward from the lower layers it gets blocked by this opaque layer and pushes it outward, causing the outer layers of the star to expand. This is what causes the increase in luminosity.

As the layers expand they cool and the electrons recombine with the ionized hydrogen which decreases the opacity. The momentum transfer to the hydrogen is reduced and gravity once again dominates causing the layers to fall back into hydrostatic equilibrium. With this, the radius of the star shrinks and the luminosity decreases. It should be noted that while it is well understood that the fundamental pulsation of Mira variables are driven by this mechanism, less is understood about how the pulsations interact with their thick atmospheres (Ostlie & Carroll 2007).

Much physical information can be gleaned from the period of a Mira variable. The period can provide information on the stars' distance and can even be tentatively used as predictors of a stars age. Feast (2008) found that for O-rich Miras their observed periods grew with decreasing age and increasing initial mass. Miras with ages ~ 12 Gyr had short periods ~ 150 days while those aged ~ 3 Gy or younger showed longer periods of ~ 450 days or more.

In regards to distance measurements, Miras demonstrate the same period-luminosity (PL) relationship seen in Cepheids and RR-Lyraes. For Miras, the PL-relation has been studied extensively in the IR (Glass et al. 1987, Feast et al. 1989, Knapp et al. 2003). Whitelock et al. (2008) found it to be

$$M_K = (-3.51 \pm 0.20) (\log P - 2.38) - (7.15 \pm 0.06) \quad (10)$$

based on observations of 31 LMC Miras in the infrared K-band where M_K is the absolute K-band magnitude and P is the period.

There have been many efforts to quantify the relationship between stellar properties and period, perhaps most notably by Ostlie & Cox (1986) during their attempt to constrain the AGB branch on the H-R diagram. Using theoretical models of LPV pulsation they tested approximately 100 stars. The team varied the stellar parameters, testing scenarios with stellar masses equal to 0.2, 1, 1.4, and $2M_\odot$, temperatures between 2,700K and 3,400K, and luminosities between 1500 and $7,000L_\odot$.

By fitting a least squares fit to the models, Ostlie & Cox derived the following relation between a Mira variable's mass and radius and the period of the fundamental and first mode of pulsation, respectively.

$$\log P_0 = -1.92 - 0.73 \log M + 1.86R \quad (11)$$

$$\log P_1 = -1.6 - 0.51 \log M + 1.59R \quad (12)$$

These two equations were initially heralded as a way to determine the pulsation mode of Mira variables, given the star’s stellar parameters but there are issues with this technique. The models depend on accurate measurements of the stellar masses and radii, both of which can be difficult to measure directly.

Therefore astronomers turned to other methods to determine pulsation mode, one of which is analysis of the Mira light curves. Miras oscillating in the fundamental mode will generate asymmetric, ”Sawtooth” light curves while those oscillating in their overtones will form a more symmetric sinusoidal light curve with a smaller amplitude (Willson & Marengo 2012).

It is also expected that the periods of Mira variables are themselves not constant and will change over time as the star evolves through the TP-AGB. AGB evolutionary theory predicts that the thermal pulses experienced by the stars throughout their time on the AGB significantly affect their mass and luminosity. Templeton et al. (2005) derived an expression for the expected rate of change in the period of a Mira’s fundamental mode oscillation in terms of the changes in its mass M and luminosity L .

$$\frac{d \log P}{dt} = -0.32 \frac{d \log M}{dt} + 0.55 \frac{d \log L}{dt} \quad (13)$$

Eqn. 13 is derived directly from Eqn. 11 and a relationship established in an earlier study by Iben (1984). This all indicates that the period of a Mira variable is the key to unlocking the properties of the star. Studying the variability of these stars will allow for a deeper understanding of their evolution and stellar properties.

In addition to the Mira variables, some AGB stars show irregular periodicity or *semi-regular* variability (SRV). These stars have variable amplitudes < 2.5 mag in the optical band and do not show the same regular pulsation as Miras. These stars are not well understood. It is thought that SRVs are stars between the higher mass Mira variables and lower mass stars such as G or K red giants originating from Sun-like stars. Their irregular periodicity is likely the result of non-radial, multi-mode pulsation. They are difficult to model accurately because of their irregular periods and so in the context of this project they are not relevant to the variability analysis other than being flagged as SRVs for potential future study.

2.1.5 Previous AGB Catalogues & Studies

This project will not be the first attempt to build a catalogue of AGB candidate stars. The 21st century has seen an uptick in the number of catalogued AGBs especially LPVs and Mira variables. A number of large sky surveys have collected data on LPVs with new variables being discovered regularly. For example, the Large sky Area Multi-Object fibre Spectroscopic Telescope’s (*LAMOST*) fourth data release in 2017 contained a catalogue of 191 new Mira candidates with information on their spectral emission lines, Balmer emission lines, and temperatures (Yao et al. 2017). The All-Sky Automated Survey for SuperNovae (*ASAS-SN*) classified 10,239 variable stars as out of a sample of 412,000 of as Miras in its *ASAS-SN catalogue of Variable Stars* (Jayasinghe et al. 2019). From the Southern Hemisphere, the *Catalina Surveys Southern periodic variable star catalogue* contains around 110,000 variable sources with $-20^\circ < \delta < -75^\circ$, built from photometric observations performed between 2005 and 2013, 1,286 of which have been labeled as LPVs (Drake et al. 2017).

One major recent example is the *First Gaia Catalogue of Long Period Variables* by Mowlavi et al. (2018). This much larger catalogue was built from *Gaia* DR2, released in 2018 and contains over 151,000 stars that have been identified LPVs by *Gaia*’s variable star classifier (Section. 3.1). The catalogue was groundbreaking in that it was the first time that the galactic plane was properly represented in an AGB study. It was built solely from *Gaia* observations. However, the catalogue suffers from several key issues that hinder it being a key source for reliable identification of AGB stars.

Firstly when compared with the scale of *Gaia* DR2, the catalogue is relatively incomplete. This is due to a focus on eliminating contamination in the sample due to other variable sources. This priority route was chosen as it is well documented that the periods of LPVs can extend upwards of 1,000 days which is much larger than DR2’s observational baseline of 668 days (Gaia Collaboration et al. 2018). Despite a rigorous anti-contamination campaign the catalogue is still marred by contaminant variables, particularly young stellar objects (YSOs) in the solar vicinity. It is estimated that around 7% of sources in the *Gaia* Catalogue of Long Period Variables are contaminant sources (Holl et al. 2018).

The catalogue is unable to distinguish between O-rich AGB stars and C-rich AGB stars as the variable classifier used (discussed further in Section 3.1) operates using photometric input. On top of this, the magnitudes of these LPVs were derived by applying bolometric correction factors that have been calculated for M-type stars with G-band amplitudes less than 3 mag. Furthermore these bolometric corrections do not take extinction effects into consideration. This final point is of particular relevance given how AGB stars already have reduced flux in the optical band.

Since 2021, there have been two very significant additions to this collection of catalogues. The first is the work completed by Suh (2021) and their *catalogue of AGB stars in our Galaxy*. This catalogue contains 11,209 O-rich stars and 7,172 C-rich stars. The project has been ongoing since 2009 and is being constantly updated making it a invaluable resource for the study of AGBs (Suh & Kwon 2009). The completeness of the catalogue led it to form the basis of the *Gaia Catalogue of C-rich Stars*.

The second is the 2022 update to the *OGLE Collection of Variable Stars* which holds measurements of 65,981 Mira type variables (Iwanek et al. 2022). This catalogue is discussed further in Section 5.3 as it will be used to assess the quality of the variability analysis.

The primary advantage the *Gaia Catalogue of Galactic AGB Stars* will have over these aforementioned samples is the purity of the AGB sample, thanks to a rigorous selection process based on SEDs, in tandem with variability analysis. The catalogue will allow for the study of well characterised AGB stars over the entire spectrum of masses, AGB types, and evolutionary stages.

2.2 Astrometry

With the rapid advancements in astronomy and astrophysics taking place in the 20th century came the realisation that increasingly precise measurements of position and distance would be necessary (Perryman 2012). Here enters the field of astrometry, the branch of astronomy that deals with the accurate measurements of the positions and motions of celestial objects, from stars up to clusters of galaxies (Kovalevsky & Seidelmann 2004).

A full understanding of an object’s astrometry requires at least five parameters: right ascension (α) and declination (δ), or an equivalent coordinate system, distance from Earth, and the two proper motion components (in the direction of α and δ , respectively) (Lindgren et al. 2016).

Astrometry is a field that is driven by advancements in technology. The earliest astronomers used basic lens telescopes and, later, made use of photographic plates to log the positions of the stars. The arrival of CCD cameras pushed astrometry into the modern age, with moderately-sized telescopes now providing swaths of data with millisecond positional accuracy and similar accuracy for the proper motions of the stars (Anderson et al. 2006).

Historically astrometric measurements were primarily ground based, made using observatories and telescopes on Earth. There had been some limited successes with this approach but, as discussed in Section 2.2.1, ground-based astrometry has serious obstacles that hamper positional accuracy.

When determining the distance to celestial objects there is no one golden technique that is applicable to all sources at all distances. Instead astronomers employ a chain of techniques that allow them to probe deeper and deeper into the universe (Fukugita et al. 1993). This interconnected series of techniques, known as the Cosmic Distance Ladder makes use of a range of astronomical approaches from the period-luminosity

relationship of variable stars to the standard candles like 1a supernovae used for extra-galactic distance estimations.

But, turning back to the galactic neighbourhood and nearby sources, the first rungs on this distance ladder are the fundamental distance measurements and stellar parallax.

2.2.1 Stellar Parallax

Stellar parallax is the oldest known method of estimating stellar distances, with Freidrich Bessel's parallax measurements of the binary system 61 Cygni in 1838 often cited as the first reliable measurements of distance to a star (Reid & Menten 2020). An object's parallax is its apparent angular displacement with respect to an observer due to the movement of the observer. In the context of astronomy, this apparent angular displacement manifests as the position of nearby stars seeming to shift amongst the background of more distant sources when observed from the Earth at different points during the year.

The reason for the apparent shift is demonstrated in Fig. 8. As the Earth orbits the Sun, observations are taken of a star every six months, utilising the maximum limits of the Earth-Sun orbit as the baseline for the measurement. The position of the target star will oscillate back and forth against the background of more distant stars.

Considering the scenario described in Fig. 8, an expression for d , the distance between the Sun and the target star in terms of the parallax angle p can be defined using simple geometry:

$$\tan p = \frac{1 \text{ [AU]}}{d} \quad (14)$$

Because parallax angles measured from Earth are minuscule, the small angle approximation $\tan p \approx p$ holds. This method of determining distances using stellar parallax gives rise to the fundamental astronomical unit of distance the *parsec*. A parsec is the distance to a source that has a parallax angle of $1''$ when measured from the Earth-Sun baseline. Applying these units to the expression for p gives a tidy formula for estimating stellar distances using parallax.

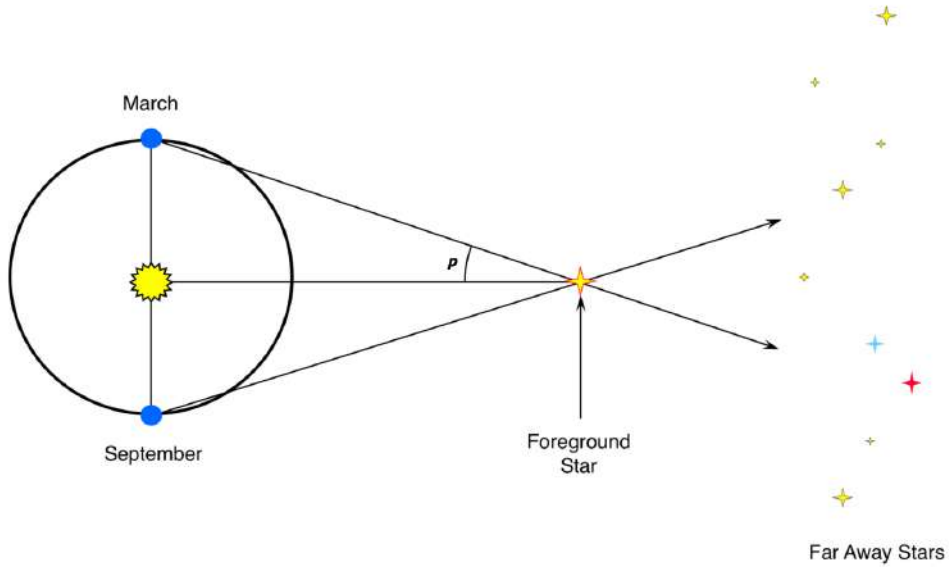


Figure 8: Visual diagram of the principle of stellar parallax. As Earth orbits the Sun, the apparent position of the observed star changes (Diodati 2020).

$$d \text{ [parsec]} = \frac{1}{p \text{ [arcsec]}} \quad (15)$$

Using stellar parallax to estimate distance benefits from needing no information about the intrinsic properties of the observed source unlike other methods such as the period-luminosity relation or supernovae.

However, as with any scientific measurement, stellar parallax has its own limitations especially for ground-based observations. Given that stellar parallax is inversely proportional to distance, very distant stars have very small parallaxes which can be difficult to accurately measure. The nearest stars to Earth such as Proxima Centauri are just over a parsec away meaning that to measure the distances to even nearby stars requires the ability to measure angular displacements down to sub-arcsecond precision (Benedict et al. 1993).

For example, even since the 1990’s with newer CCD equipment (rather than the photographic plates traditionally used) the minimum measurable parallax from the ground is $\sim 0.002''$ (Monet et al. 1992). It was also a difficult task to measure sources with a relative error of less than 10% which limited usable parallax measurements to stars up to ~ 12 pc. Even this small relative error can lead to absolute magnitude errors of 0.1 mag, two orders of magnitude worse than standard photometric techniques.

Throughout history many astronomers have compiled catalogues of the stars’ positions in the night sky. The Greek astronomer Hipparchus of Nicaea is considered to have compiled the first astrometric catalogue which was later absorbed into Ptolemy’s catalogue of 1,022 sources (Vanin 2022). Centuries later in 1598, Tycho Brahe published a manuscript containing the positions and magnitudes of 1,004 stars. Although groundbreaking, Tycho’s catalogue had poor accuracy. Astrometric discrepancies on the order of arcminutes are prevalent throughout the catalogue and around 15% of the stars have positional errors $> 10'$ (Verbunt & Van Gent 2010). As technology advanced the size and accuracy of the catalogues increased accordingly. A number of projects in the mid-to-late twentieth century managed to build astrometric catalogues using ground-based trigonometric parallax measurements. For example, the *General Catalogue of Trigonometric Stellar Parallaxes* by Jenkins (1963) contained 10,000 parallax measurements of 6,400 sources.

Earth-based telescopes are limited in that they have a fixed position on the Earth meaning they can only observe parts of the sky at a time. A telescope based in the Northern or Southern Hemisphere will only ever be able to observe their respective skies, meaning a single ground-based survey will always suffer from being half-completed. An all-sky survey can be constructed by patching together the surveys of many telescopes from around the world but this is a non-trivial undertaking as, unless the surveys were all performed at exactly the same time, the stars in each sector will have moved slightly.

Earth’s atmosphere poses another large challenge. Turbulence in the troposphere caused by the convective heat flow from the ground can significantly distort the images of astronomical objects. Even for telescopes constructed at high altitudes where the atmosphere is thinner, these fluctuations are still on the order of $1''$ (Perryman 2012). On top of the astronomical seeing restrictions, light pollution from populated regions further inhibits the study of faint sources for all but the most remote observatories.

Another major issue with ground based parallax measurements is that the background reference stars are also subject to apparent displacement due to parallax. The effect is dependent on the distance to the reference stars and thus is often minuscule for distant sources. However, this means that the measurement of a star’s parallax over a number of observations is actually a measurement of the star’s parallax relative to the averaged parallax of the background reference stars. A conversion from relative to absolute parallax can be achieved by modelling the background parallax but as stated by Van Leeuwen (1997) this modelling is a non-trivial undertaking.

All these issues guided astronomers toward the realisation that accurate astrometry lies in space-based observations.

2.3 Gaia

Launched in 2013, *Gaia* marks the current pinnacle of astrometry, a space based observatory with the ability to deliver absolute parallaxes and 3-dimensional astrometry down to microarcsecond accuracy (Prusti et al. 2016). However, aside from its unmatched accuracy, what sets *Gaia* apart from other astrometric surveys is the scale of the survey (Fig. 9). The mission aims to survey more than a billion stars in the sky, building an accurate map of all observable celestial bodies down to a magnitude of ≈ 20 mag in the optical. Although astrometry is *Gaia*'s primary aim it has a plethora of significant secondary science goals including investigating star formation, galactic structure, binary star systems, and, as seen in this project, stellar variability.

The spacecraft operates in the visible range with an effective bandwidth between 300 and 1050nm. Over the course of the nominal five year mission time, each source was measured approximately 70 times however this number has likely increased dramatically with a series of mission extensions, the latest of which ensuing *Gaia* will remain operational until at least 2025.

Gaia operates using the same principles as *Hipparcos*, an earlier ESA space-based astrometry mission operational between 1989 and 1993, that managed to produce a catalogue of $\sim 118,000$ stars using the principle of scanning astrometry (Van Leeuwen 1997). The basis of scanning astrometry is the conversion of temporal measurements into spatial measurements by tracking the times at which the image of a source is at a well-defined location. *Gaia* does this by tracking a source across an array of CCD pixels whereas *Hipparcos* used a series of slits (Lindgren & Bastian 2010).

The satellite orbits the second Sun-Earth Lagrange point in a Lissajous orbit, an orbit that was chosen for its uninterrupted observations, stable thermal environment, and low radiation. A circular sun shield (see Fig. 10a) further reduces light spillage from the Sun. At the L2 orbit the spacecraft reels and spins along an intricate path known as the *Gaia Scanning Law* which was chosen to maximise sky coverage. The spacecraft rotates at a rate of 1° per minute completing a full circle every 6 hours. The spin axis of the spacecraft is tilted at a constant angle of 45° with respect to the Sun. Thanks to the spinning of the craft and the slow

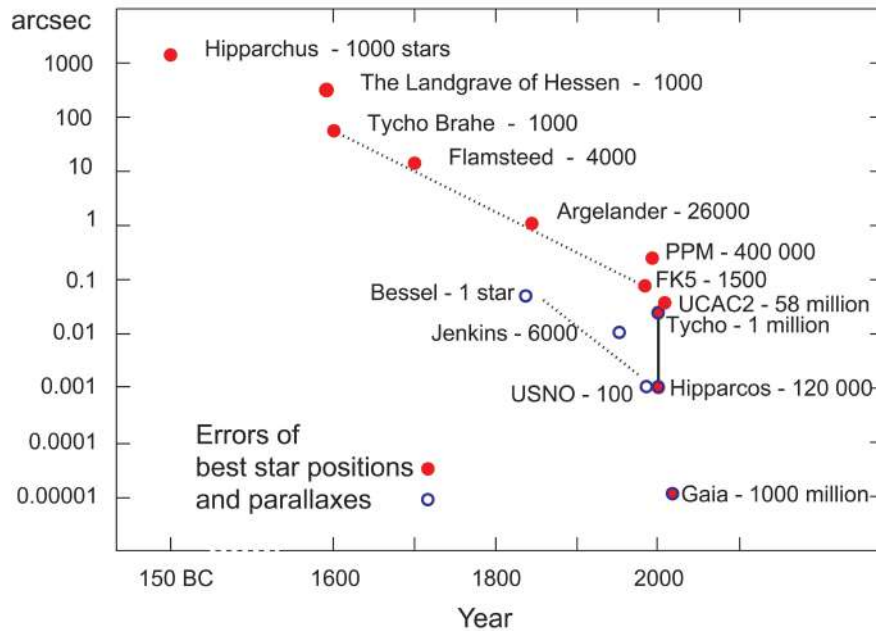
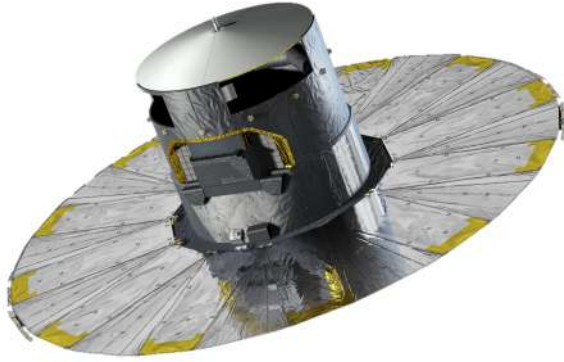
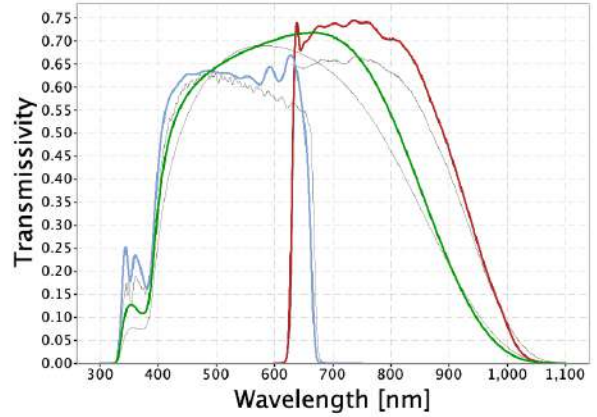


Figure 9: Historical advancements in astrometry charted as position and parallax accuracy over over time. *Gaia* is by far the most accurate and with the most measurements. The mission is more accurate than pre-twentieth century surveys by around 8 orders of magnitude (ESA 2003).



(a) *Gaia* Spacecraft



(b) *Gaia* pass bands.

Figure 10: *Gaia* Spacecraft and Scanning law. Fig.(a) shows the *Gaia* spacecraft with the cylindrical payload which houses the mirror array and CCD camera shown in Fig. 11. Fig.(b) shows the bands covered by *Gaia*'s three photometric instruments. The Green line is *Gaia*'s G-band, and red and blue denote the G_{RP} and G_{BP} , respectively (Montegriffo et al. 2020).

precession of the spin axis around the Sun, a full trace of the sky is achieved approximately every month. The spacecraft scans the same portions of the sky repeatedly and with different orientations and feeds it to the **A**strometric **G**lobal **I**terative **S**olution (AGIS) (Lindegren et al. 2016). AGIS is an iterative least squares algorithm which searches for an astrometric solution for input sources. Every time new information for a source is captured, AGIS re-runs the solution using the new data in combination with the previous solution, each iteration converging on a more accurate source. However, AGIS does not only solve for a source's position. Each cycle solves for the astrometric parameters of the source, the attitude of the spacecraft, a set of calibration parameters, and a set of global parameters. The calibration parameters are needed to translate the movement of light along CCDs to angular motion relative to the spacecraft axis. Finally the global parameters correct for deviations from General Relativity (Lindegren et al. 2012). They are solved in this order with the output of the previous parameter serving as input for the successive ones.

Gaia's payload consists of two telescopes, fixed at an angle of 106.5° to one another (the *basic angle*). The images collected by the two FOVs are focused onto a single focal plane. The fixed angle means that large angle measurements can be performed on sources with the same accuracy as small angle measurements. The parallax of a star can now be measured against a totally separate set of reference stars which removes the relative parallax uncertainties found in small field angular measurements. This effectively transforms relative parallax measurements into absolute parallax measurements (Jordan 2008, Turon et al. 2010).

The *Gaia* payload is built primarily out of Sintered Silicon Carbide (SSiC). There are multiple reasons for choosing this compound. It is lightweight yet very stiff, with a very high specific stiffness, $E/\rho = 1.35 \times 10^7$ Nm/kg, which is around five times higher than other regularly used spacecraft materials such as Aluminium or Titanium. Additionally, the material demonstrates exceptional thermal properties with a high thermal conductivity of $\lambda = 180$ W/m/K and low thermal expansion coefficient of $\alpha = 2 \times 10^{-6}$ (Bougoin 2017). All these properties combine to make a material that is incredibly thermally stable which won't warp or change its shape with temperature. A secondary benefit is that it can be polished to reflect light. As such nearly all of the payload, mirrors, toroidal support structure, and focal plane are all made from SSiC.

Images of the sky are focused onto *Gaia*'s focal plane array (FPA) by six sets of rectangular mirrors. The FPA is an array of 106 CCDs arranged into groups, each with a different astrometric or photometric speciality (see Fig.11). As the spacecraft spins, the images move across the FPA, encountering each of the different rows of CCDs. The read-out rate of the CCDs is synced with the rate at which objects travel across the

FPA.

The *Gaia* CCDs are quite large, each measuring 4.5×5.9 cm. A CCD has 4500×1966 pixels each of which have dimensions 10×30 μm . This pixel size was chosen as a compromise between pixel resolution and full well capacity (Gaia-UK 2014). They are p-type silicon wafers doped with n-type silicon channels. *Gaia*'s CCD pixels have novel technology. The pixels are equipped with an Anti-Blooming Drain (ABD), a feature unique to *Gaia* which prevents excess electrons from flowing into adjacent pixels if they become saturated by bright sources. The CCD silicon also has a doped layer called the supplementary buried channel which improves the charge transfer efficiency and protects against loss of charge from radiation damage. Noise and dark current are managed by keeping the CCDs passively cooled to 163K (Crowley et al. 2016).

The sources are first focused by the telescopes onto the two rows of Sky Mapper (SM) CCDs. The two telescopes focus their images onto the FPA with the stars from each telescope being recorded by a designated row of seven SM CCDs. A physical barrier ensures the light from telescope 1 strikes SM1 and the same for telescope 2 and SM1. These detector use object detection algorithms to scan in incident images for local flux maxima. Unlike the rest of the FPA, the SM CCDs are fully read out but with a reduced resolution of 2×2 pixels. The SM assigns windows around the regions of interest in the images that contain sources which are carried over to the other FPA CCDs.

Nice rows out of the entire FPA, 62 CCDs, are devoted to broadband astrometric measurements. These CCDs are called the Astrometric Field (AF) and collect images of the sources in the *Gaia*'s G-band. Using the AF *Gaia* can decipher 5 of the 6 astrometric phase space parameters. The AF CCDs are standard broadband silicon CCDs, approximately 16 μm thick and have a peak sensitivity centred at 650 nm (the G-Band) (Crowley et al. 2016). These AF-type CCDs are also used for the SM and WFS portions of the FPA. The first row filters out any false readings that could have been made by the SM. As the images move across the AF, the position and brightness of the stars are measured in the small windows designated by the SM CCDs. This method of assigning windows to the sources is a necessary optimisation tool that decreases the amount of data that is read from the CCDs and downlinked from the spacecraft. Rather than continuously reading out the entire 8.8 million pixels, the CCDs only output the windows, which each contain ~ 12 pixels

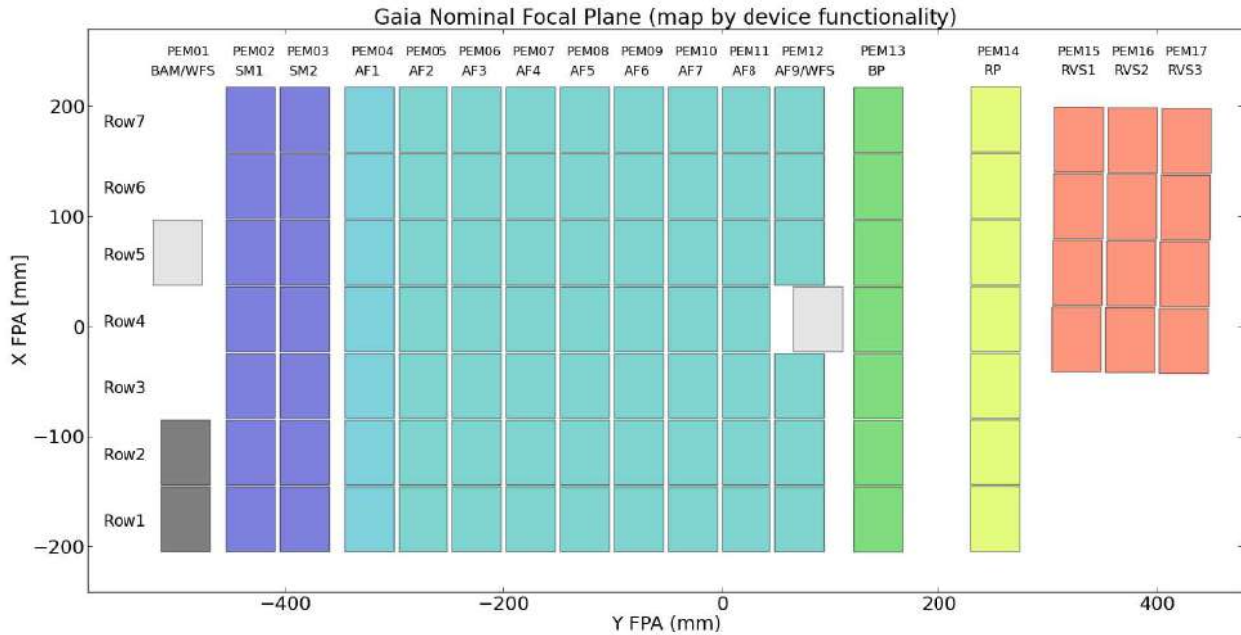


Figure 11: Diagram of *Gaia*'s FPA. Images are first cast onto the RHS of the FPA on one of the two SM CCD columns. As the spacecraft turns these images travel from left to right across the plane (Crowley et al. 2016).

depending on the source. The AF instrument is extremely sensitive being able to resolve sources in dense regions up to 1.05 million objects per degree (Prusti et al. 2016). The AF handles the modelling of source point spread functions for the sources which are used as the input for AGIS.

Gaia has a two spectrophotometers. The Blue Photometer (BP) studies the 330 – 680 nm wavelength range and the Red Photometer (RP) studies the 640-1050 nm range. These CCDs have an anti-reflective coating sensitive to shorter wavelengths for the BP and longer for the RP. This increases the instrument sensitivity to its respective wavelength range. Before reaching the photometers the images are cast through two fused-silica prisms. The RP and BP instruments will be able to infer stellar properties that are typically based on measurements of colour including temperature, chemistry, and in some cases directly label source types (see Sec. 3.1 for an example with C-rich AGBs). The spectrometers have a relatively low spectral resolution ($\lambda/\Delta\lambda$) between $\approx 20 - 100$.

Finally, the FPA has the radial velocity spectrometer which provides the final elusive astrometric parameter: the velocity of stars in the line of sight. The Radial Velocity Spectrometer (RVS) instrument consists of 12 CCDs that measure spectra for select bright sources over the near-infrared 847-874 nm wavelength range. This range spans the CaII triplet and was chosen to ensure all of the triplet lines were captured to ensure no radial velocity information is lost. Care was taken when choosing this narrow band as too wide a wavelength range would introduce additional cosmic background, and increase the change of overlapping spectra. The RVS can only collect spectra for bright objects ($G_{RVS} < 16$ mag) but for these select sources the instrument provides radial velocities via Doppler-shift spectral measurements, information on reddening, rotational velocities and stellar atmospheres, and basic stellar parameterisation (David et al. 2014, Recio-Blanco et al. 2016).

Four additional CCDs are devoted to image quality management. The Wave Front Sensor consists of two CCDs which monitor the optical quality of the two telescopes. They measure and correct any optical aberrations caused during launch or by gravity. The correction is calculated via centroid determination and wavefront reconstruction.

Complementing them is the basic angle monitor (BAM) made up of two CCDs, one nominal and one redundant. BAM ensures the “basic angle” of 106.5° between the two telescopes remains constant (Gielesen et al. 2013). BAM is able to measure tiny fluctuations in the basic angle with an accuracy $\approx 0.5\mu$ arcseconds. Even with the thermal stability of SiC, variations of up to 7μ as can affect the base angle due to tiny thermal fluctuations. BAM operates using interferometry. An artificial star is generated by a laser diode for each telescope. The laser beams are split in two by a set of beam splitters and are reflected by a series of mirrors onto the BAM CCDs. When the beams recombine they form an interference pattern on the monitor. If the base angle shifts, the path length of one or both of the beams shifts accordingly and the interference pattern on the CCDs changes. This allows *Gaia* to correct for the tiny variations in line-of-sight that happen due to thermal fluctuations (Mora et al. 2014).

The combination of all these FPA devices provide *Gaia* with the means, not just to take an accurate census of the sources in the nearby universe, but also to accurately identify their physical properties and answer a plethora of scientific questions about the nature of the galaxy’s inhabitants.

2.4 The Virtual Observatory

As stated by Borne (2013), “*Astronomy is primarily a forensic science*” given that it involves the collection and analysis of physical evidence to infer the properties and evolution of our universe. The data collected by astronomers is vast and varied. It can be photometric, astrometric, or spectroscopic. It can be raw or processed. It can take the form of images or time series or other data types. And thanks to the abundance of missions and observatories the rate at which data is being produced is growing exponentially.

At the turn of the 21st century it became more and more apparent that the sheer influx of astronomical data would begin to become an issue, with the increasing volume and complexity known to astronomers of the time. Szalay et al. (2000) cited the volume of data to be increasing twice every 1.5 years, at the same rate

predicted by Moore’s Law. This connection to Moore’s law, the idea that the number of transistors in dense integrated circuits doubles approximately every two years, is not a coincidence (Moore et al. 1965). The increase in technological complexity observed by Moore is being applied to astronomical detectors such as CCDs providing increased sensitivity, observational capabilities and allowing us to gaze further and further into the depths of Space. By 2005 the rate of astronomical data being collected daily was estimated to be on the order of ~ 1 TB (Djorgovski & Williams 2005), already large but tiny compared to the amount collected by today’s standards.

LOFAR, the Dutch radio telescope array generates, a massive 13 Tbits/s (van Haarlem et al. 2013). Looking forward to the future, upcoming survey telescopes such as the Large Synoptic Survey Telescope (renamed in 2019 as the Rubin Observatory) in Chile are estimated to have a data collection rate of 20 TB of data per night. Not all of this directly translates into tabular data as the data tends to be significantly reduced by the time it is released but it demonstrates the scale of the data being collected especially as these are only two of 600 observatories currently operating as well as ≈ 20 space telescopes.

Another matter that needed to be addressed was the combination of astronomical data. For example, an astronomer may wish to combine radio, X-ray, IR, and optical observations of a sample of sources, utilising data from different surveys, from different eras, all with different table formats.

Thus the Virtual Observatory was formed. The aim of the VO is to facilitate a standardised platform for efficiently accessing astronomical data. The International Virtual Observatory Alliance (IVOA) was formed in 2002 to develop and monitor the VO. It acts as a liaison between the 22 international components of the VO (see Fig. 12). Their work is focused on sharing expertise from the member VOs and developing common standards for the VO (Quinn et al. 2004). The vision of the IVOA is to create a digital, multi-wavelength sky that can be accessed by anybody, anywhere. It was envisioned to be an “*enabler of science with massive and complex data sets*” (Djorgovski & Williams 2005).

To further the aims of the VO, the IVOA have defined a set of protocols that make it easy to access and manipulate VO data. Examples include the Simple Cone Search, which searches a circular section of the sky for sources, the Simple Image Access, which finds all images within a region of the sky, and the Table Access Protocol (TAP), that allows users to retrieve tabular data with SQL-like query languages (Graham et al. 2012).

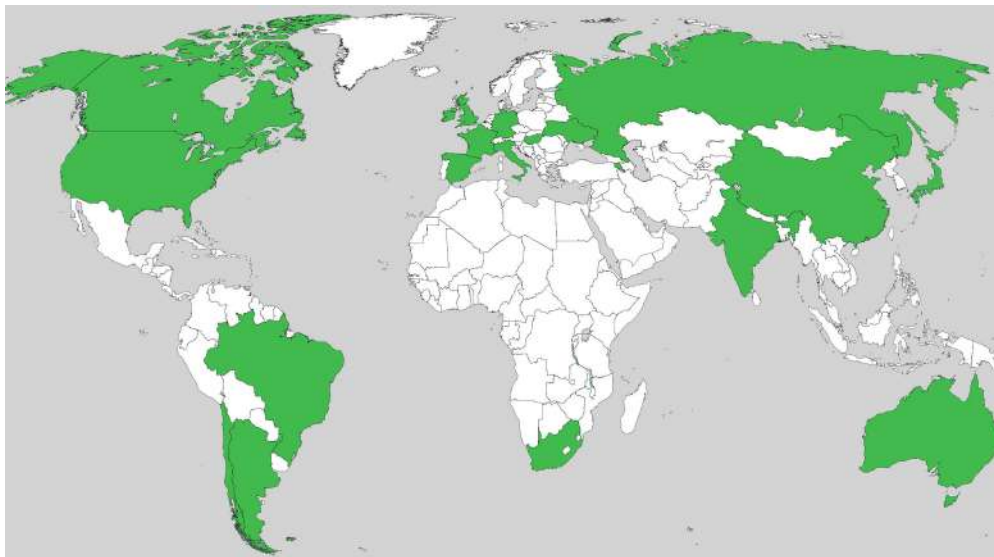


Figure 12: Countries that have contributed VO programs to the IVOA. Note the distribution over both hemispheres leading to full sky coverage that could not be achieved by a solely European VO program, for example. (Araya et al. 2015)

IVOA also introduce the *VOTable* standard, an XML data format that permits easy storage and exchange of tabular data between VO components (Ochsenbein et al. 2011). *VOTable* profits from the heritage of XML, a widely used markup language that is used throughout the internet. A *VOTable* consists of a combination of meta-data and table data. The metadata portion contains information on the parameters, units, any relevant descriptions of the the data fields. Essentially the metadata ensures the table can be transferred between systems while maintaining the key data information i.e. float values remain as floats etc. The corresponding table data is a classic array of rows and columns with each row representing a source or observation, while the columns represent the properties of the row entry.

On top of the ease of access to this wealth of data and the IVOA protocols, the VO is complemented by a set of community developed tools, which can interface directly with the VO and perform analysis on the accessed data (Araña et al. 2015). Some examples include *Aladin*, developed by the French VO which is a tool used to search images of the sky and that can create mosaics of regions of interest from sourced images. Another tool, used throughout this project, is *VOSA* which is a tool created by the Spanish VO for SED analysis.

Unfathomable amounts of data are now available to scientists, with a range of analysis tools and flexible formats. A secondary benefit of the VO is that astronomers no longer need to allocate time on observatories or travel to obtain data. If they wish to study a target sample, based on position or source type, they can query the VO and access multi-wavelength target data over many epochs and surveys. It is clear that the VO is a hugely powerful tool for the modern astronomer.

3 Observations & Methodology

3.1 *Gaia* Data Release 3

The primary development in the field of astronomy that has allowed for progression in this project is the release of *Gaia* Data Release 3 (Gaia Collaboration et al. 2022). Released on the 13th of June 2022, it is the second installment of *Gaia*’s third data release following the publication of Early Data Release 3 (EDR3) in December 2020 (Gaia Collaboration et al. 2021). Both releases contain data collected between the 25th July 2014 and the 28th May 2017.

The release is by far the most extensive data collection published by *Gaia* up to this point, with information on 1,811,709,771 sources. This includes parallax measurements for 1.46 billion sources, object classifications for 1.59 billion sources with astrophysical parameters such as temperature, metallicity and gravity, thanks to the BP/RP CCDs. The number of sources with radial velocity has increased by a factor of 5 compared to DR2 thanks to measurements performed by the RVS.

However, the data that is of key relevance to this project is the information *Gaia* has compiled on variable stars. DR3 improves on the DR2 variable star analysis in almost every way, having amassed data on 10,509,536 variable sources compared to the 550,737 sources present in DR2.

The photometric data for the variable sources was collected over the course of 34 months. As seen in Section 4, this places a hard upper limit on the quality of variable sources with $P > 1,000$ days. The sampling rate of the sources is set by *Gaia*’s scanning law: the spacecraft can only measure a star as it passes through the telescopes line of sight. Part way through the observational campaign, *Gaia* changed its scanning law to one that favours high latitude observations meaning that sources with $\beta \pm 45^\circ$ were scanned more often than those in the galactic plane (Prusti et al. 2016).

Gaia Data Release 3	
Source Type	Number of Sources
Total Sources	1,811,709,771
2-parameter Astrometry	343,964,953
5-parameter Astrometry	585,416,709
6-parameter Astrometry	882,328,109
Variable Sources	
Total Variable Sources	10,509,536
Classified with Machine Learning	9,976,881
Photometric Time series	11,754,327
Long Period Variables	1,720,588

Table 1: Condensed summary of *Gaia* DR3 data products with a focus on the data relevant to this project, in particular the data DR3 provides on variable sources.

Gaia separated 9,976,881 of the variable sources in DR3 in a set of 24 variable source types using a supervised machine learning model called **nTransits+5** to perform the classification. Data from literature on variable stars was used to train the model. The selection process and classifier input is detailed extensively in Holl et al. (2018) and Lebzelter et al. (2022). Several of the selection criteria for LPVs include:

- A period longer than 60 days;
- Number of observations in the G-band ≥ 12 ;
- Ratio of RP observations and G-Band observations > 0.8 ;
- 5%-95% interquartile G-band variability greater than 0.1 mag;

- $G_{BP} - G_{RP} > 0.5$ mag.

An added bonus of DR3’s classification of LPVs is that it provides a (tentative) identifier for Carbon star candidates: the `is_cstar` flag. Identification of C-stars is based on narrow-band photometry using the RP photometer. The RP photometer has relatively low resolution but is adequate for the task of differentiating between the two stars based on spectral features (Mowlavi et al. 2019).

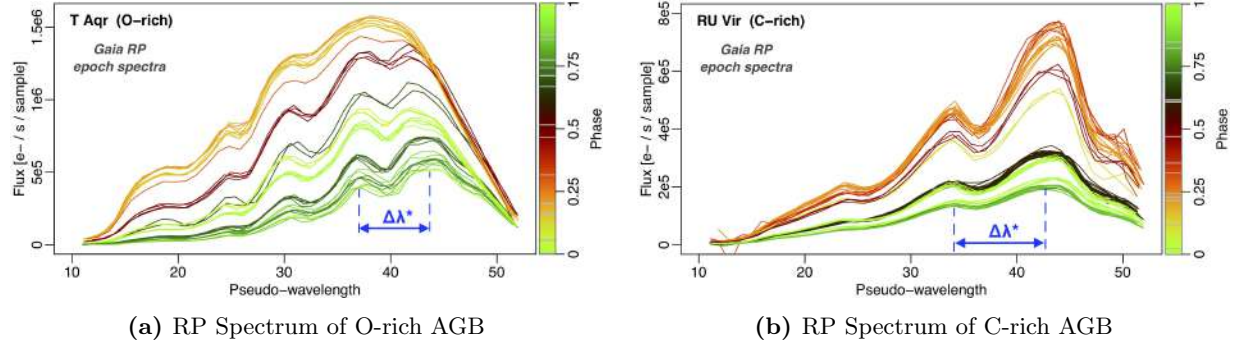


Figure 13: Comparison between *Gaia* RP spectra for an O-rich AGB star (*T Aqr*) and a C-rich AGB star (*RU Vir*). On the X-axis is the dimensionless *pseudo-wavelength*, an arbitrary binning of the measured spectrum. The median pseudo-wavelength difference between two peaks is what *Gaia* uses to identify Carbon stars. Note that $\Delta\lambda^*$ is larger for *RU Vir* due to the absorption features.

As discussed in Section 2.1, O-rich and C-rich stars have distinct spectral signatures. The spectra of O-rich stars display absorption features due to TiO molecular bands. C-rich stars have wide absorption features due to the presence of CH and C₂ molecules in the CSE. For O-rich, M-type stars the two largest peaks are from these TiO absorption features which is in line with spectral observations of AGB stars with O-rich chemistry.

Such analysis needs to consider the changes in temperature observed in variable stars. LPVs have temperatures that can vary by hundreds of Kelvin depending on their pulsation phase, which can affect the strength of the absorption features found in the spectra. This can be seen in Fig. 13a where, as the star gets hotter (the yellow spectra in the plot), certain absorption features around $\lambda \approx 40$ become smoothed out. *Gaia* combats this by collecting RP spectra for each epoch and then assessing the median pseudo-wavelength separation between the two highest peaks as the baseline for its classification.

If the median $\Delta\lambda_{RP} > 7$, the source is marked as a C-star candidate i.e. `is_cstar` = TRUE. This characterization via pseudo-wavelength separation was verified by Mowlavi et al. (2018) using a sample of 12 bright and well monitored LPVs.

This new data release is ready to be exploited and contains a wealth of photometric and astrometric information that can now be utilised to study variable AGB stars.

3.2 *Gaia* Archive

Gaia’s archive of data can be accessed in a number of ways: web interface, specialised Python modules, and Virtual Observatory protocols (Salgado et al. 2017, Taylor 2018). Aside from the rare exceptions where a source ID is known and can be manually extracted from the archive’s web interface, accessing tabular data for multiple sources requires the use of *Astronomical Data Query Language* (ADQL).

ADQL is a specialised variation of structured query language (SQL), the programming language used to communicate with and manage databases. ADQL was developed from SQL92 specifically designed to accommodate the retrieval of astronomical data-sets (Plante 2007). The syntax remains essentially similar to

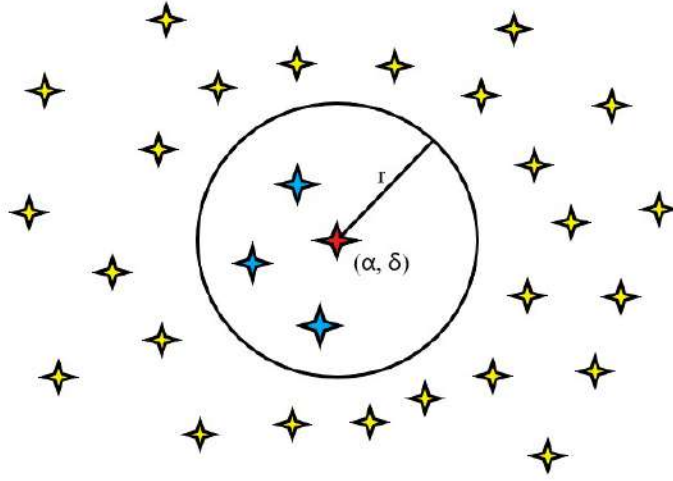


Figure 14: 2-D representation of a simple cone search on a source with an RA of α and a declination of δ . The search centres on the coordinates of the input source (highlighted in red) and searches for complementary sources from the matching catalogue within an area enclosed by the radius r (these chosen sources are highlighted in blue).

that of SQL with the addition of a number of astronomy related extensions which assist in the extraction of astronomical sources via their observable properties.

A typical example of an ADQL request used in the project would be the following:

```
SELECT ohir.*, gaia.source_id, gaia.ra, gaia.dec, gaia.phot_variable_flag
FROM gaiadr3.gaia_source as gaia
JOIN user_cdebarra.ohir_sample as ohir
ON 1=CONTAINS(
    POINT('ICRS', ohir.ra,ohir.dec),
    CIRCLE('ICRS',gaia.ra, gaia.dec, 2./3600.))
```

The above request performs a cross match between *Gaia* DR3 (`gaiadr3.gaia_source`) and a custom generated catalogue of OH/IR stars (`user_cdebarra.ohir_sample`). The cross match is carried out via a simple cone search query (see Fig. 14). A simple cone search is one of the most basic forms of astrometric based cross matching and was one of the first protocols used by the IVOA. The simple cone search queries an area centered on the sky position of an input source and searches a given radius around this point. The search then returns any sources located within this specified region from the given catalog. The sky position is provided as the RA and Dec of the source in ICRS coordinate system (Plante et al. 2008).

This request is only a basic example and can be easily edited to be more specific such as only searching for cross matches in the radius that have relative parallax errors less than 30% or with mean magnitudes brighter than 10 mag.

3.3 Additional Software

3.3.1 Python & IDL

Over the course of the 21st century Python has become an invaluable tool in the astronomer’s arsenal. This is for a variety of reasons. It is a free, general-purpose programming language, that is widely documented and

maintained. Python can be utilised in almost all aspects of astronomical data analysis thanks to continuous community input and its ever-growing array of specialised libraries (Price-Whelan et al. 2018). It also benefits from being highly flexible and relatively easy to use for casual programmers and scientists (Faes 2012).

Python was the primary programming language utilised throughout the project for the filtering and compilation of the data post-extraction from the *Gaia* archives. Python benefits from a set of libraries that allow for the editing and visualisation of tabular data. In the context of this project it contains a set of invaluable modules that can handle VOTable/XML data and extract them to a data-frame for such analysis. These include `astropy`³, `pandas`⁴, and `matplotlib`⁵.

On top of these libraries, a number of bespoke modules were developed specifically for this project specialising in handling the VOTables from the *Gaia* archive and extracting the light curve data directly into plots. The functions were designed so that they could take the *Gaia* bands (G , G_{RP} , G_{BP}) as arguments and return only the photometry in that band.

Alongside Python, the curve fitting of the photometry described in Section 3.6 was performed by a script written in Interactive Data Language (IDL) by F. Jiménez-Esteban. IDL is a programming language originally designed for data analysis purposes. Although not designed specifically with astrophysical applications in mind it gained favour with the astronomy community initially because it was easier to use than the programming languages of the time such as FORTRAN (Landsman 1993). IDL had been previously used by the team for a number of successful studies and thus it was opted to utilise it again for this project. Although an IDL script was used to fit the light curves the script was also translated to Python by the author in order to perform quick testing of additional parameters e.g. testing alternative periods.

The Python code written for this project can be accessed in a designated GitHub repository⁶.

3.3.2 TOPCAT

In addition to Python, TOPCAT was a vital tool employed throughout the project. TOPCAT⁷ (Tool for **OP**erations on **C**atalogues **And** **T**ables) is a free software capable of visualising and editing large tables. It was originally developed by Mark Taylor, an astrophysicist from the University of Bristol in 2003. The application is open source, written in Java, and is able to easily handle tables of data with thousands if not millions of rows, as encountered regularly in astronomy. It is most typically used with astronomical source catalogues but can in fact be adapted to handle any sort of tabular data from any field.

One of the many benefits of TOPCAT is the versatility of its I/O, which is capable of handling most, if not all, of the file types commonly encountered in astronomy; FITS, CSV, ascii, XML etc. It is also designed to be compatible with the VO. This is seen most clearly through its TAP capabilities. Catalogue counterparts can be found for any user generated table from catalogue services like VizieR using ADQL requests, all executed in the TOPCAT application. It is by far the quickest way to cross-reference source catalogues rather than downloading catalogues and filtering them with a Python script for example.

The software also has highly practical visualisation capabilities (demonstrated further in Section 4). This includes the ability to manually highlight subsets of data in scatterplots based on visual inspection, identifying outliers and the software will highlight them in the table, and filter catalogues based on input boolean and algebraic expressions. Perhaps the most useful feature is the ability to highlight a single point in a scatterplot (such as Fig. 19) at which the row in the catalogue will be highlighted enabling easy identification of outliers or sources of interest.

Much of the exploratory data analysis was done with TOPCAT and it was especially useful for the rapid

³<https://www.astropy.org/>

⁴<https://pandas.pydata.org/>

⁵<https://matplotlib.org/>

⁶<https://github.com/cuandeb/Thesis-Code>

⁷<http://www.star.bris.ac.uk/~mbt/topcat/>

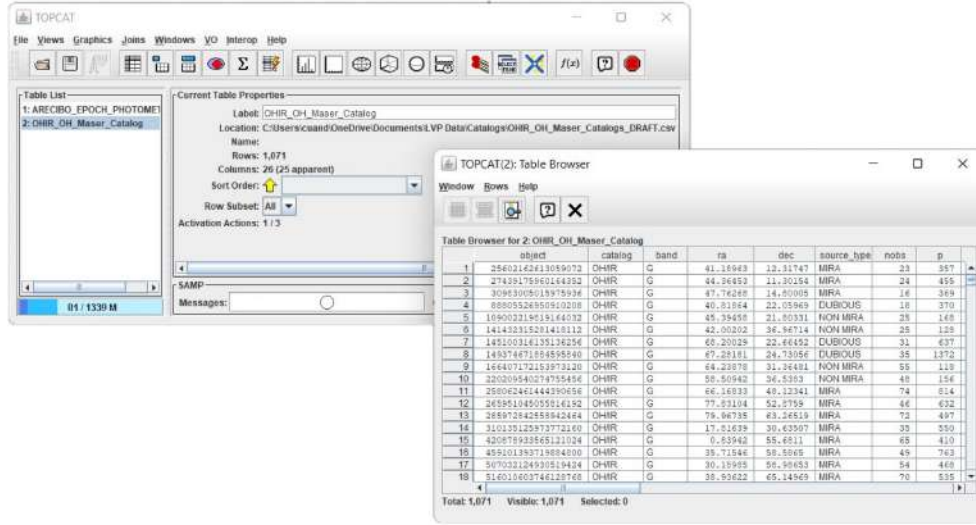


Figure 15: Examples of TOPCAT windows. The window on the LHS shows the basic TOPCAT console while the RHS shows an astronomical catalogue viewed through TOPCAT.

identification and characterisation of outlying LPVs in the analysis outlined in Section. 4.

3.4 Construction of the OH/IR Sample

The preliminary *Gaia Catalogue of Galactic OH/IR Stars* is used as the input for the variability analysis. This catalogue, first prepared by B. Lopéz-Martí, D. Engels, and F. Jiménez-Esteban in 2020 was the first stepping stone to what would eventually become the *Gaia Catalogue of OH/IR Stars*.

The initial sample of OH/IR sources used to construct the input catalogue was the *Database of Galactic OH Masers* which contains 13,655 radio observations of 2,341 sources compiled by Engels & Bunzel (2015). The catalogue is a compilation of almost all the stars in the galaxy that have been detected using the OH maser transitions of 1612, 1665, or 1667 MHz. The basis of this sample was two previous maser catalogues. The first is the *Catalogue of Stellar 1612 MHz Maser Sources* by te Lintel Hekkert et al. (1989), a sample of 442 OH maser sources compiled from literature between 1963 and 1986. The second is the *Catalogue of Observations for Stellar Masers* by Benson et al. (1990) which contains a further 713 sources.

The rest of the sources in the catalogue were taken from literature between 1984 and 2014. These additional sources were taken mostly from surveys done using single-dish radio telescopes but also some surveys from the Australian Telescope Compact Array and the Very Large Array radio telescopes.

In regards to limitations of the catalogue, the observations are restricted to galactic masers, however given the project aim of establishing a catalogue of galactic AGB stars, this is not an issue (Engels & Bunzel 2007). There are also limitations with regards to the detected lines. There are no sources before 1984 in the catalogue that have measurements using the main 1,665 MHz and 1,667 MHz transition lines, but again this does not affect the use of the catalogue for this project.

The catalogue is also prone to contamination by sources that also display maser emission. The primary contaminants for OH maser sources are post-AGBs and young stellar objects (YSOs) which have been found to have OH and H₂O molecules in the surrounding star formation regions (Braz et al. 1990).

The first stage of filtering this OH maser sample was to verify their astrometric accuracy and to find cross-matches with *Gaia* sources. Because the original catalogue has been compiled from a wide range of literature, the accuracy of the source positions also varies accordingly. Also, given that the OH maser database has

been constructed from radio observations, it is likely that when cross matched with *Gaia*, an optical mission with greater astrometric precision, the result will be contaminated with duplicate matches. In order to combat this, intermediate cross-matches were made with two IR/Mid-IR catalogues: *2MASS* and *AllWISE*.

The *AllWISE Source Catalogue* is a mid-infrared catalogue built from observations made by the **Wide-Field Infrared Survey Explorer** (WISE). WISE was a NASA medium class explorer mission that set to image the universe in the infrared. The spacecraft (see Fig. 16) had onboard, a 40cm telescope which mapped the sky using 4 infrared bands: 3.4, 4.6, 12, 22 microns (referred to W1, W2, W3, W4, respectively) (Wright et al. 2010).

Each band had designated focal plane detector arrays which were cooled by a two-stage hydrogen cryostat (Naes et al. 2008). WISE's first observation campaign lasted from 7th January to 6th August 2010 in which all four detectors were fully operational. Once the first layer of hydrogen in the cryostat was depleted the second campaign began with three of the bands operating at full sensitivity in what was known as the 3-Band Cryo Phase. Finally, once the cryostat was fully exhausted of hydrogen, WISE began the *NEOWISE Post-Cryo phase*, which saw the W1 and W2 detectors operating at full sensitivity. The *AllWISE* data release combines the data from all three campaigns into one release.

The final *AllWISE* Source catalogue contains positions, proper motions, and IR photometry for over 747 million sources and is considered $> 95\%$ complete for sources with W1 magnitudes brighter than 17.1 mag. For high SNR sources, *AllWISE*'s accuracy is ≈ 50 mas.

Complementing *AllWISE* is the **Two Micron All Sky Survey** (*2MASS*) (Skrutskie et al. 2006). *2MASS* is a near-infrared survey built from observations carried out by two telescopes, one in Arizona and one in Chile. The location of the two telescopes in the Northern and Southern hemispheres enabled coverage of the entire sky.

2MASS has 3 near-infrared photometric bands *J*, *H*, and *K* with wavelengths 1.25, 1.65, and 2.61 μm , respectively. These particular wavelengths were chosen as they allowed *2MASS* to fulfill its role as a near-infrared survey while utilising the atmospheric windows at which thermal background is minimised. The *K* band was found early on to be the ideal infrared band where the atmosphere is most transparent to

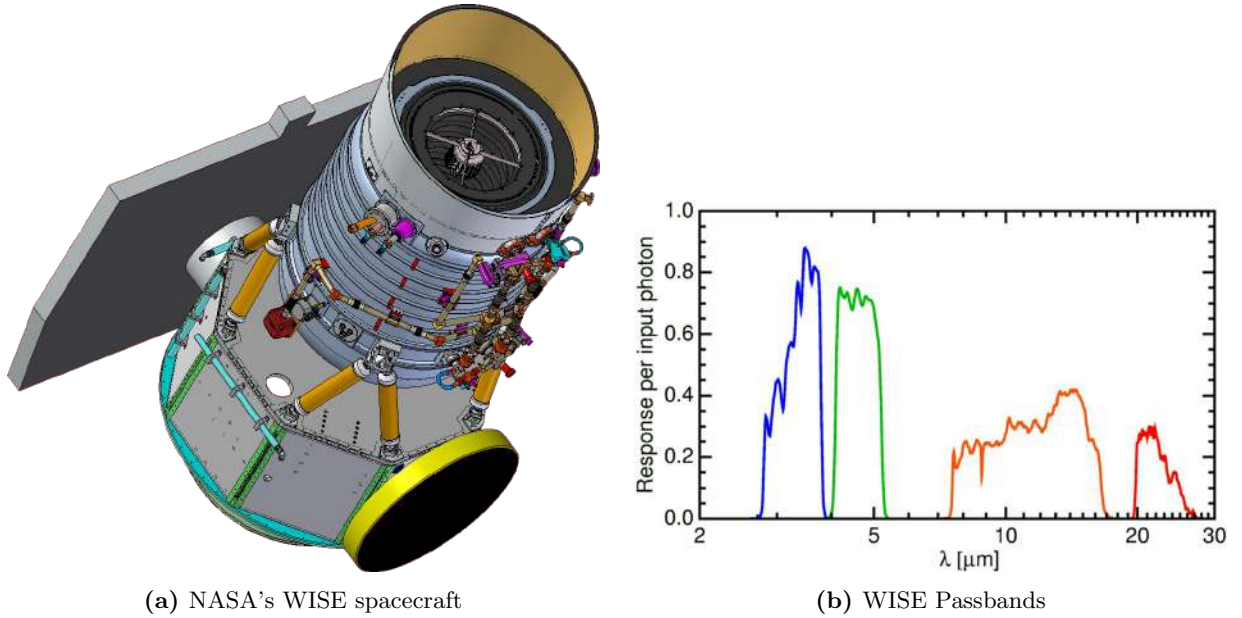


Figure 16: NASA's WISE spacecraft and Passbands. Fig.(a) shows the *WISE* space telescope with its 40cm aperture. Fig.(b) The WISE passbands with W1 through W4 shown left-to-right with W1 in blue and W4 in red (Wright et al. 2010)

IR radiation (Harwit 1999). The J , and H bands allow for delineation between stellar and extra-galactic sources as well as calculating extinction (Skrutskie et al. 2006).

The *2MASS* Point Source Catalogue contains over 470 millions sources, covering an estimated 99.998% of the total sky. This makes it an invaluable tool for cross-matching sources. *2MASS* also has higher spatial resolution than *AllWISE* meaning sources that can not be resolved in the latter may be resolvable in the former.

The basic procedure was as follows. An OH maser source was first cross-matched with the *AllWISE* catalogue with a relatively wide search radius of $15''$. If a counter-part existed it was then cross-matched against the *2MASS* catalogue with a smaller search radius of $3''$. For these counterparts a final cross match was performed with *Gaia*. Finally, the centre of the galaxy was removed. This region was an area with a radius of 2° centered around the galactic core ($l = 0^\circ$, $b=0^\circ$). This region was removed as the concentration of dust and sources meant it was impossible to ensure that the *2MASS/AllWISE/Gaia* counterparts were accurate.

Throughout the construction of the catalogue a number of issues were faced that diluted the amount of sources remaining in the final catalogue.

- The cross-match with the *AllWISE* catalogue did not return a match or returned the wrong counterpart. This depended primarily on the accuracy of the OH maser sample, which, as previously stated, suffered from inaccuracies in source position due to the range of literature it was built from.
- Sources had inaccurate counterparts when matched with *2MASS*. The majority of the time this was due to maser database/*AllWISE* sources being resolved into multiple sources when matched with *2MASS* and the correct source being misidentified due to positional uncertainties.
- The sources did not have any *Gaia* counterparts. This is because despite having a strong IR flux, the source is too dim in the optical to be detected by *Gaia*. This is to be expected as OH masers are expected to occur in sources with thick CSEs that hide the source in the optical.
- The source had multiple *Gaia* counterparts in the cone search radius. This again was the result of *Gaia* being able to resolve sources at a greater resolution than the previous two catalogues.

The final OH/IR table contains 1,505 sources with *Gaia* cross-matches. This catalogue will serve as the basis for the OH/IR photometry study that is discussed below in Section 4.

3.5 Extraction of *Gaia* Epoch Photometry

Gaia's epoch photometry data is not included in the main *Gaia* tables (e.g., `gaia_source`) and must instead be accessed via the archive's DataLink protocol. DataLink can be looked at as an IVOA web service which provides additional data products (mean spectra, Markov chain samples, epoch photometry etc.) for objects outside the main catalogues. The sources to be analysed are passed to the *Gaia* archive via an ADQL request based on the RA and Dec of the sources. For the sources with DataLink supplements, a compressed file can be downloaded with the data.

The format used for this project is the RAW table (seen below in Table 2). This format outputs a table where each row is a separate source containing the transit data: photometry in the G , G_{RP} , G_{BP} bands, transit time in Julian Date etc. Each column in the table contains a separate array with the time series data. For example, each entry in the `g_transit_time` column is an array of all the times when observations were made. To generate the light curves for each source these arrays need to be unpacked, from the table and plotted against one another.

There is no magnitude error included in the tables but the uncertainty on the flux is provided and can be converted to magnitude error via error propagation leading to a magnitude uncertainty of

source_id	n_transits	g_transit_time	g_transit_flux	...
4307923326892561408	22	(1723.98236107013,...	(231830.70138386745,...	...
4506519900232457472	15	(1723.9870306415412...)	(188780.59119516777,...	...
2047207019324667520	28	(1764.6947961481644, ...	(17286.843392325416, ...	...
1859282958664240000	45	(1765.6132025083677,...	(252321.9209190612,...	...
2019915770633603584	31	(1762.3649957376633,...	(4084291.453766633,...	...
...				

Table 2: Abridged example of RAW photometry data extracted from *Gaia* DR3 archive. Taking the **g_transit_time** column as an example; each row contains an array of values with a length equal to **n_transits**. To use the data these arrays need to be unpacked. In total there are 48 columns in the RAW photometry files with information on all portions of the CCD array.

$$\Delta m \simeq \frac{2.5}{\ln 10} \left(\frac{\Delta f}{f} \right) \quad (16)$$

The *RAW* table also contains several flags for the quality of the photometry points. For each row of CCDs (Two SM, 9 AF, and a BP and RP array) there is a Boolean value for a good or not-good photometric measurement. A bad photometry point is one that has been rejected by *Gaia*'s photometry processing software. An example of such would be a point with a negative flux value which is considered unphysical, despite being prevalent throughout the catalogue as a result of large uncertainties (Bailer-Jones et al. 2021). However, it was opted to perform the filtering based on a set of independent criteria rather than the *Gaia* status.

The data was accessed using a Python script that read the RAW photometry table, unpacked the time series from each row and saved the observation times, magnitude, and magnitude error as an ascii .DAT file labelled with the source ID. The structure of such a file is shown in table A.

t_{obs} (JD)	m (G-Band)	Δm
2456904.3634844637	10.144052819389278	0.0324049851669844
2457021.043132047	7.8168248378570695	0.012965337128150895
2457021.117132476	7.917117472138013	0.01540686189293751
2457040.866032465	8.105690301108083	0.010190689026033454
...	...	...

Table 3: Example table that is used as the input for the Python fitting algorithm

3.6 Construction of Light Curves from Epoch Photometry

With the epoch photometry data collected for the sources, the next stage of the analysis involved fitting a light curve to the epoch photometry points in order to analyse the variability properties of the stars. For AGB stars there are several possible curves that can be fitted to the data.

The first is a symmetric sinusoidal light curve. This is the most basic model and is described by the function

$$m(t) = \overline{m} + A \sin \left[2\pi \left(\frac{t}{P} + \phi \right) \right] \quad (17)$$

In this equation $\overline{m}$ represents the mean magnitude around which the star's magnitude oscillates, A is the peak to peak amplitude of the star, ϕ is the phase, P is the period of the pulsation and t represents time.

To fit a light curve to this function the various fitting parameters need to be established. Eqn. 17 can be broken down further into the following equation by applying the trigonometric relationship $\sin(A+B) = \sin A \cos B + \cos A \sin B$.

$$m_i = \overline{m} + A \cos(2\pi\phi) \sin\left(\frac{2\pi t_i}{P}\right) + A \sin(2\pi\phi) \cos\left(\frac{2\pi t_i}{P}\right) \quad (18)$$

In the above equation m_i represents the magnitude of the i th photometry point for a given source, t_i is the time at which m_i is observed. This can be further broken down into

$$m_i = a_1 + a_2 \sin\left(\frac{2\pi t_i}{P}\right) + a_3 \cos\left(\frac{2\pi t_i}{P}\right) \equiv m(t_i, a_1, a_2, a_3) \quad (19)$$

where the constants a_1, a_2, a_3 denote

$$a_1 = \overline{m}$$

$$\phi = \arctan\left(\frac{a_3}{a_2}\right)$$

In order to fit this function and its parameters, a series of trial plots were tested using a set range of periods and the ideal fit was found by minimizing the sum of the squared residuals of the fit where the sum of the squared residuals are defined as

$$\chi^2 = \sum_{i=1}^n \frac{(k_i - m(t_i, a_1, a_2, a_3))^2}{\sigma_i^2} \quad (20)$$

This method of least squared fit is used commonly throughout the sciences for the curve fitting due to its flexibility and the ease of calculating uncertainties on fits via the covariance matrix of the fitted parameters.

In order to aid with the light curve analysis and verification of well fitted parameters an auxiliary plot of the χ^2 value was graphed against the trial periods was generated alongside the fits. This plot (called a *Periodogram* from this point on) allows for two things. Firstly it facilitates an easy method of visually assessing the quality of a fit (See Section 4). Secondly the method also provides a way of determining the uncertainty on the light curves' periods. Around the minima of the periodograms the shape of the minimum can be approximated to a parabola. The uncertainty on the period is then taken to be the half-width of this parabola, extrapolated up to the maximum χ^2 value. The width is given by

$$\Delta P \approx \frac{P^2}{2T} \quad (21)$$

where T is the time baseline over which observations of the source were made. For fits where the period is unambiguous, the periodogram will have a sharp, thin minimum and the uncertainty will be small. Conversely an ambiguous period will have a wide shallow minimum surrounded by other possible minima with comparable χ^2

This method provides a neat and easily calculable way of determining the error on the period however as mentioned later in Section 4 it is not without its issues, especially for sources with large periods which may exceed the base time.

Previous studies performed on AGB stars highlighted an issue with the symmetric model from Eqn. 23. In work performed by Jiménez-Esteban et al. (2006), it was observed that some AGB type objects display large fluctuations in their mean magnitudes over large time intervals (on the order of 1,000s of days). This can

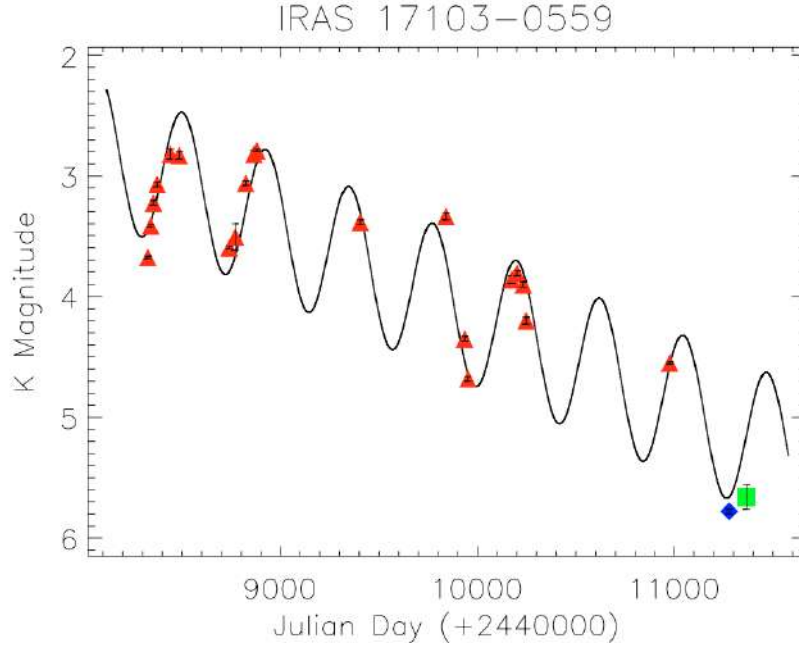


Figure 17: K-Band photometry of the source IRAS 17103-0559 from a survey performed from 1999 to 2005. As well as the regular pulsations of the star the slowly decreasing mean magnitude obvious from the global decrease in magnitude.

be seen below in Fig. 17 which shows K-band observations of the source IRAS 17103-0559. The changing magnitude results in a light curve that slopes downward.

This phenomenon has been widely predicted to occur amongst AGB stars and thus necessitates a different model. This changing mean magnitude is the second type of light curve that can be fitted to AGB stars. As a function this is written as

$$m(t) = \overline{m}(t) + A \sin \left[2\pi \left(\frac{t}{P} + \phi \right) \right] \quad (22)$$

where $\overline{m}(t)$ is a linear function defining the increase/decrease in mean magnitude over the observation period. This model has been used for many AGB studies to g...

For fitting the DR3 data a third option was utilised. It was decided that an asymmetric cosine function best described the pulsations of the stars. This asymmetric model is defined as follows

$$m(t) = \overline{m} + \frac{A}{2} \cos(2\pi\Omega(t)) \quad (23)$$

where the variables have the same meaning as in equation 17 with the addition of Ω which describes the asymmetric nature of the light curve.

$$\Omega(t) = \begin{cases} \frac{t-t_0}{2Pf} & \text{if } 0 \leq \frac{t-t_0}{P} < f \\ \frac{t-t_0-P}{2P(1-f)} + 1 & \text{if } f \leq \frac{t-t_0}{P} < 1 \end{cases} \quad (24a)$$

$$(24b)$$

where P is the period of the pulsation, t_0 is the time of the first epoch measurement and f is the asymmetry factor of the cosine function and has a value between 0 and 1. If $0 < f < 0.5$ the function will rise steeply before dropping off slowly or vice versa for the case where $f > 0.5$. In the case of a perfectly symmetric cosine function, $f = 0.5$, the ascent is equal to the descent.

When performing the fitting the periods between 100 and 2,500 days were tested in steps of 1 day. This range was chosen mainly due to the documented periods of long period variable stars which typically vary between 100 and $\sim 1,000$ days (Trabucchi et al. 2019, Mattei 1997). The asymmetry factor was also tested in steps of 0.01 in the range $f = 0.1 - 0.9$

In order to optimise the fitting a two-step process was implemented. First the photometry points were fitted with a symmetric, sinusoidal light curve function. The period of the symmetric fit was then taken and input into the asymmetric fit as an initial starting parameter. The asymmetric fitting model then fit a light curve in the range $P_{sym} \pm 100$ days. This greatly reduced the time taken to fit the light curves which was beneficial as for certain samples over 2,500 sources were being analysed.

Due to the enormous number of LPVs that *Gaia* DR3 has access to, a key component of the analysis is to determine a set of parameters and properties of the stars with which one can confidently classify the light curve fits as good or bad. Accurate classification will allow for the differentiation between actual LPVs and other spurious sources. This task is performed manually through visual inspection of the plots but due to the large number of sources this is not a straightforward undertaking.

4 Results

4.1 OH/IR Sample

The first sample of stars analysed for the project was the set of OH/IR stars. The sample was taken from *The Gaia Catalogue of Galactic OH/IR Stars* which was compiled from a catalogue constructed by [Engels & Bunzel \(2015\)](#). See Section 3.4 for a breakdown of how this catalogue was generated.

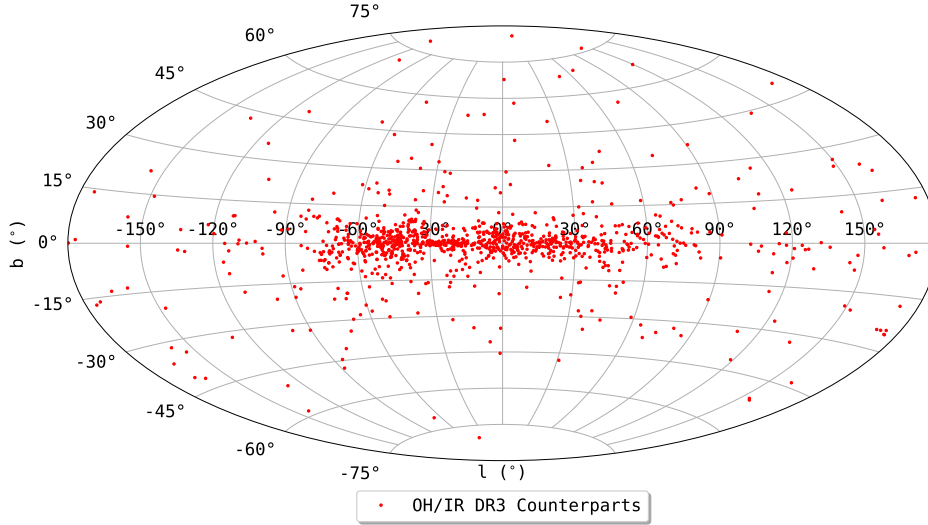


Figure 18: Sky Map of the OH/IR sources used in the first stages of analysis. Here the sources are presented in Galactic coordinates. There is no bias toward any particular region of the sky due to the OH/IR sample having been compiled from a range of surveys. There is a natural expected density along the galactic plane.

The input catalogue contained 1,505 OH/IR sources which originally had *Gaia* DR2 counterparts. As mentioned in Section 3.1 the primary difference between DR2 and DR3 being the addition of light curve data for more than 10.5 million objects. The *Gaia* DR3 counterparts were sourced by performing a cone search of DR3 based on the coordinates provided in the sample. Three searches were checked: 1, 2, and 3 arcseconds. Despite the wider radii returning a larger number of cross-matches, each search yielded the same number of variable sources. Out of the original 1,505 sources, *Gaia* DR3 provides variability data for 1,071 stars.

Fig. 19 shows the results of the light curve fitting code, with the period of best fit for each source plotted against the fitted amplitude (peak to peak, technically $2 \times A$). This plot shows the first hints of a relationship between period and amplitude of the light curves. There appears to be a general upward trend in the range of periods between 200 and 700 days and the amplitudes between ≈ 1.5 and 4 mag. Care must be taken not to infer too much from this over-density of points for two reasons. Firstly, these are only the amplitudes/periods that the algorithm returned and could be the result of an inaccurate light curve. Secondly, the photometry is G-band (i.e in the Optical range) which for the case of OH/IR stars is massively suppressed by the circumstellar extinction. This could lead to amplitudes which are smaller than expected for the stars.

The initial stage of the analysis was to visually inspect the light curves and mark each as a *GOOD* fit, *BAD*

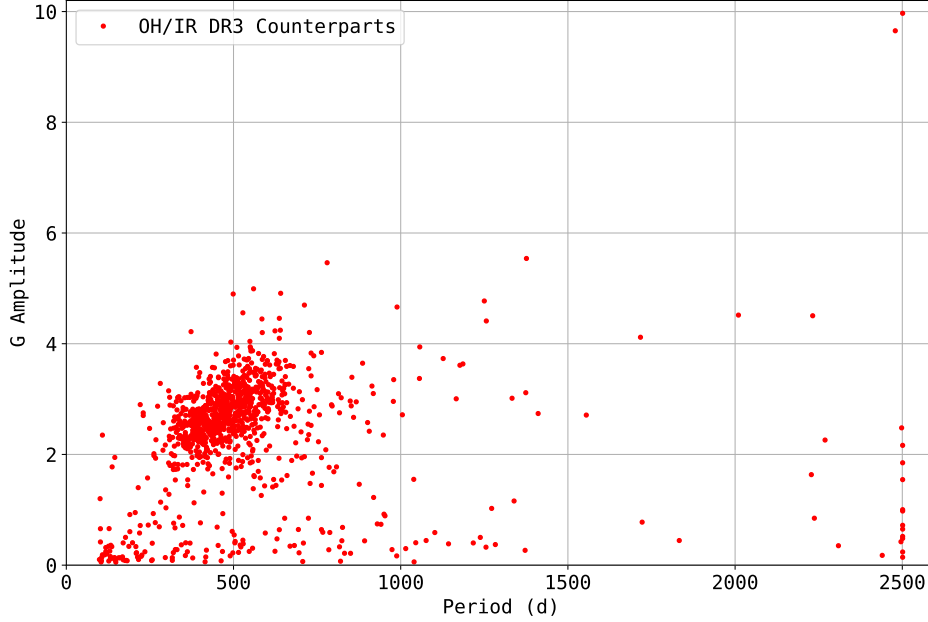


Figure 19: Plot of the period and amplitude of the fitted light curves for the 1,071 OH/IR sources. The first indication of a relationship between the quantities is the over density of points at $200 < P < 700$ days and $2 < A < 4$.

fit, or *DUBIOUS* fit. This portion of the analysis was time consuming and repetitive but was initially the only way of classifying the fits and beginning the search for properties of the curves that would allow for the classification of good or bad fits.

An example of a *GOOD* fit is provided in Fig. 20. Good fits are the easiest to classify as they tend to have a few obvious traits. A *GOOD* plot shows minimal scatter from the curve of best fit. The sources with *GOOD* fits tended to be well sampled and evenly distributed, with the number of observations typically $> 20 - 30$ photometry points. The periodograms of these fits only confirm their quality with the plots showing sharp distinctive periods, with $\chi^2 \sim 1$ (see Fig. 21).

An example of a *BAD* fit is provided in Fig. 22. The visual features that define a bad fit are clearly demonstrated in the figure. The photometry points are scattered and do not sit well on the light curve. The folded phase diagram and periodogram further substantiate this poor fit. The folded light curve appears as noise overlaid on a light curve, while the periodogram has no recognizable period.

Additionally in some cases, such as *Gaia* 4053283512051388800, the light curve that is fitted to the photometry point is the wrong shape, with a slowly rising side and a sharp drop-off, a shape that is not associated with AGBs (Vardya 1988).

There are outlying cases, where the fit appears very good but does not make physical sense or, at the very least, is too ambiguous to mark as a good fit. This is the case for source 5940907619822698752 (shown in Fig. 23) which visually demonstrates a great fit to the data, although an inspection of the periodogram reveals issues. There is no clear period and the χ^2 has reached a minimum of 2,500 and seems to be decreasing still with a large trailing edge. This renders the error analysis obsolete for the period. The uncertainty calculations for the periods assume that around a minimum the periodogram can be approximated to a parabola. This is clearly not the case and results in non-physical uncertainties. For example, 5940907619822698752 has a period of $P = 2,500 \pm 4,552$ days.

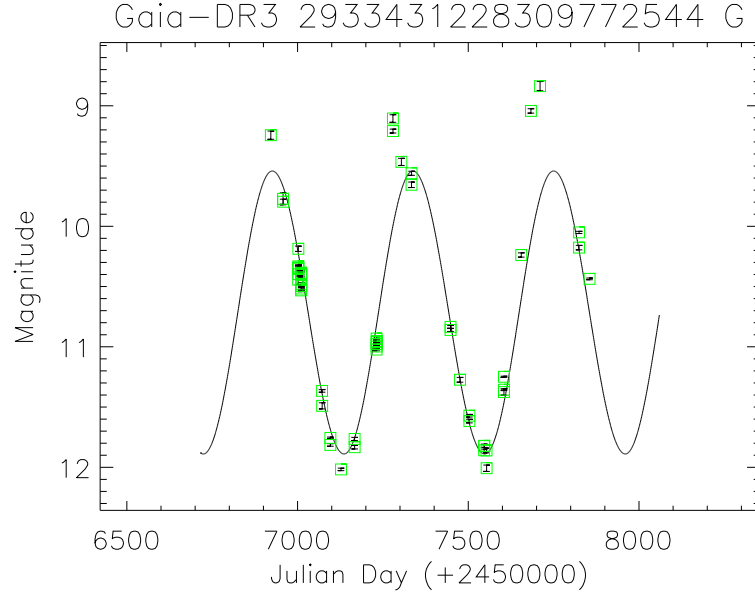


Figure 20: Example of an OH/IR light curve (*Gaia* source 2933431228309772544 with a good fit. Note the high number of data points and minimal deviation from the fitted light curve.

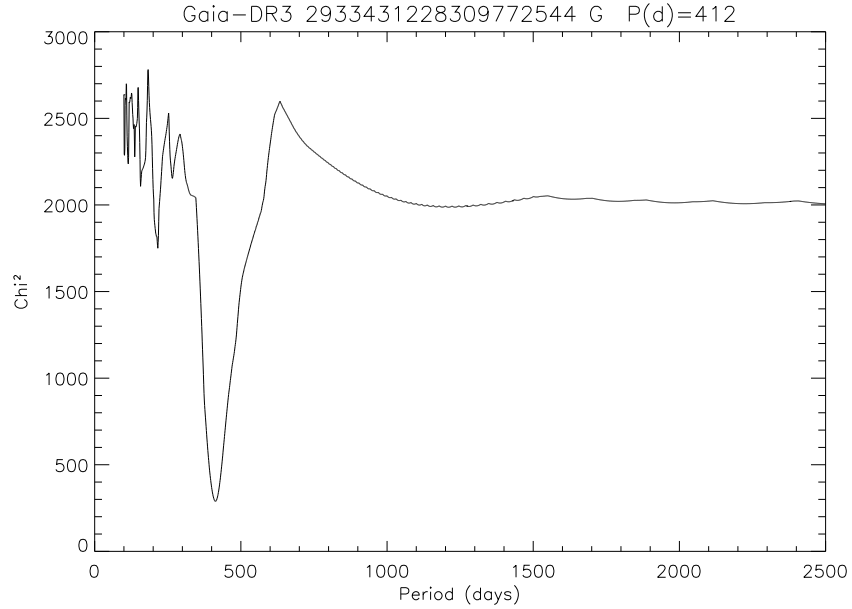


Figure 21: Periodogram for *Gaia* source 2933431228309772544. The optimal period for fit presents itself as a distinct minimum χ^2 value at $P = 412$ days. This period was used to fit the light curve shown in Fig. 20

Out of the three categories, *DUBIOUS* fits were the most subjective of the three and are hard to pinpoint quantitatively. They are characterised by plots showing a general adhesion to the light curve but with a lot of scatter, or curves that seem to be well fitted, but with low sample sizes or missing portions. TOPCAT was particularly useful for these sources as their placement on the period-amplitude diagram offered extra information. *DUBIOUS* sources could be located in the catalogue by their source ID and by highlighting their catalogue entry, their location on the plot would also be highlighted.

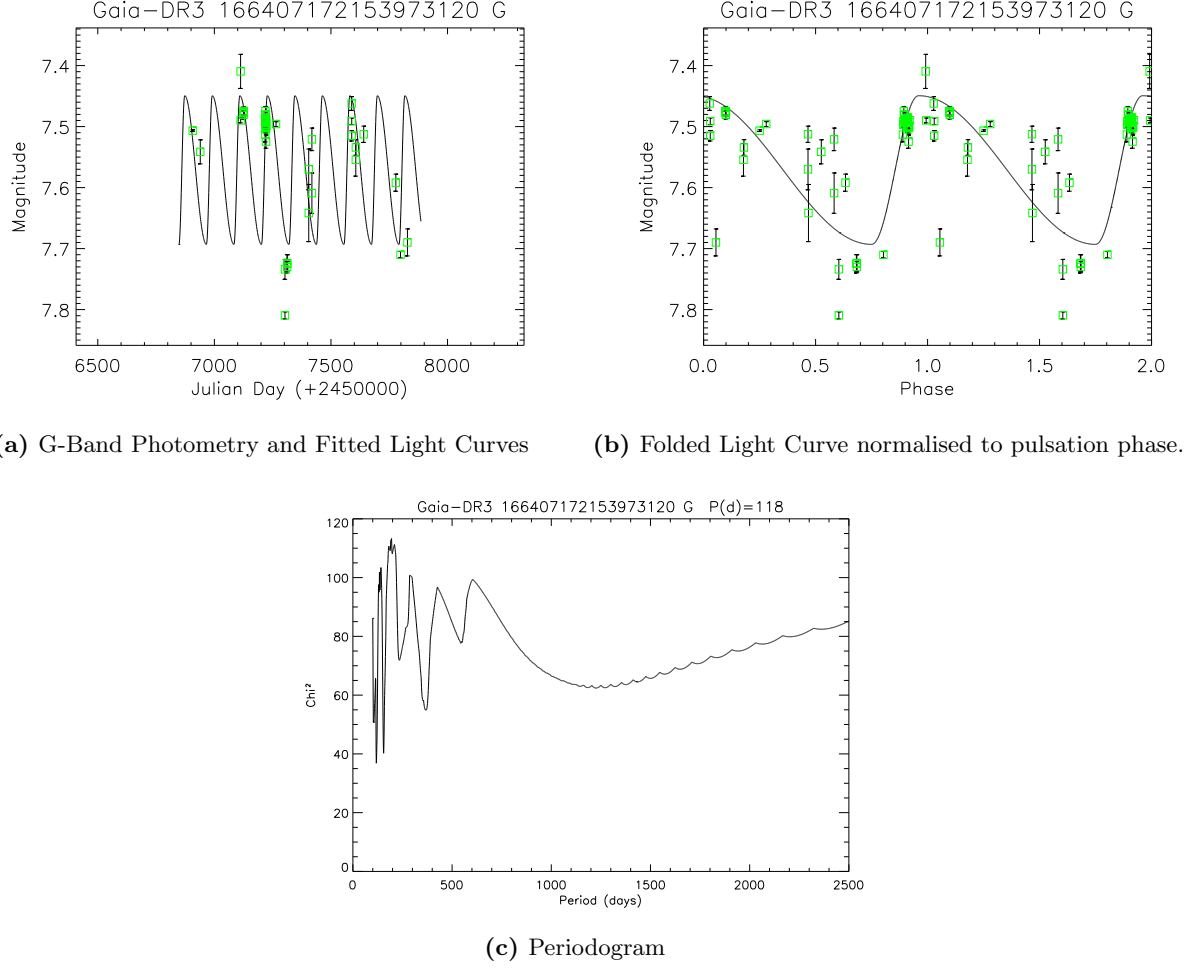
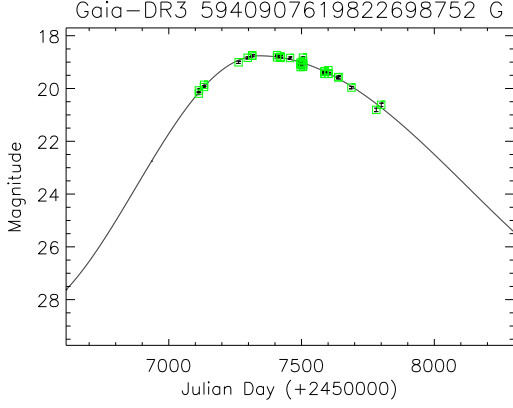


Figure 22: Light curve and associated plots for *Gaia*-166407172153973120. The fitted light curve showing a poor correlation between the photometry points and the curve of best fit. This is further corroborated by the folded phase diagram which shows the photometry points distributed sporadically around the light curve. Finally the periodogram shows no distinct period for the sources.

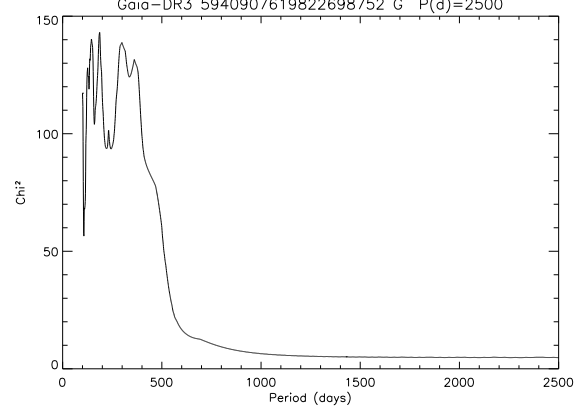
With the *GOOD* fits marked, the next step was to assess the quality of the period measured by the fits. Photometric observations made by *Gaia* were made over a period of 1,038 days (see Section 3.1). This means accurate periods containing information about the full cycle of stars were limited to those with periods below this limit. Therefore, any source with a period greater than the time interval of its photometric time series was flagged as having an incomplete period.

These sources were simply flagged rather than discarded for several reasons. As seen in Fig. 24 some sources that were initially labelled as *GOOD* were flagged as having bad periods. These sources all shared similar properties in that it appears that *Gaia*'s observation campaign accidentally observed the maximum and minimum of the peaks. For the case of an object with a non-variable mean magnitude this would provide enough data to sufficiently constrain the period and amplitude of the source but given the evidence to the contrary seen in Section 3.6, these can not be inferred from the data. The periodograms also show an apparent minimum, though lacking the clarity of the peaks seen in Fig. 21. Therefore, these few sources were marked as *PRELIM*, indicating that these may represent the real periods and amplitudes of an LPV but have incomplete photometry data to confirm this.

The second step in filtering the data was to examine the lower end of the spectrum i.e. amplitudes less than 1 mag and periods less than 1,000 days. The main characteristics of these sources are a complete lack of



(a) G-Band Photometry and Fitted Light Curve



(b) Periodogram

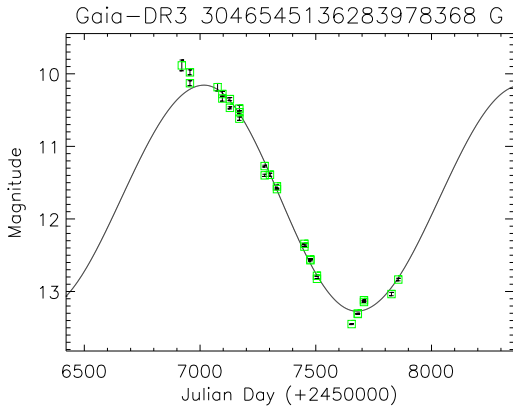
Figure 23: *Gaia* Source 5940907619822698752

periodicity, the photometry points appearing almost as noise. Any source conforming to these criteria were flagged as a semi-regular variable.

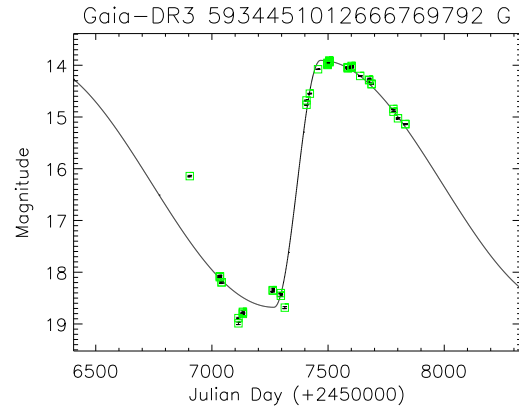
The issue with using this method of characterising these plots, the characteristics of the data are often vastly misrepresented by the fit. The points are denoted by their position on the period amplitude plot but given the fact that the fits themselves are dubious there is no guarantee that this period is representative of the actual variable. Due to this, it was opted to use only the amplitude as a limiting factor for the SRVs: if $A < 1$, the source is classed automatically as an SRV.

Figure 25 shows the *BAD/DUBIOUS* fits for the OH/IR sample with explanations for the poor quality of the fits. The sources with periods above approximately 800 days are all accounted for by this condition and indicate that this may be an effective first step at filtering out dubious light curves. The plot also demonstrates where these SRVs fall within the remaining dubious plots. The majority of the poor fits are explained by these two criteria.

What remains are a cluster of seemingly unrelated bad fits with amplitudes greater than 1 mag and periods less than 1,000 days. Many of these points were located in the cluster of good fits. A visual analysis of the remaining fits revealed two general types of poor fit.



(a) *Gaia* 3046545136283978368



(b) *Gaia* 5934451012666769792

Figure 24: Examples of two sources flagged as *PRELIM*. The fits appear to be of good quality but do not cover a full pulsation period.

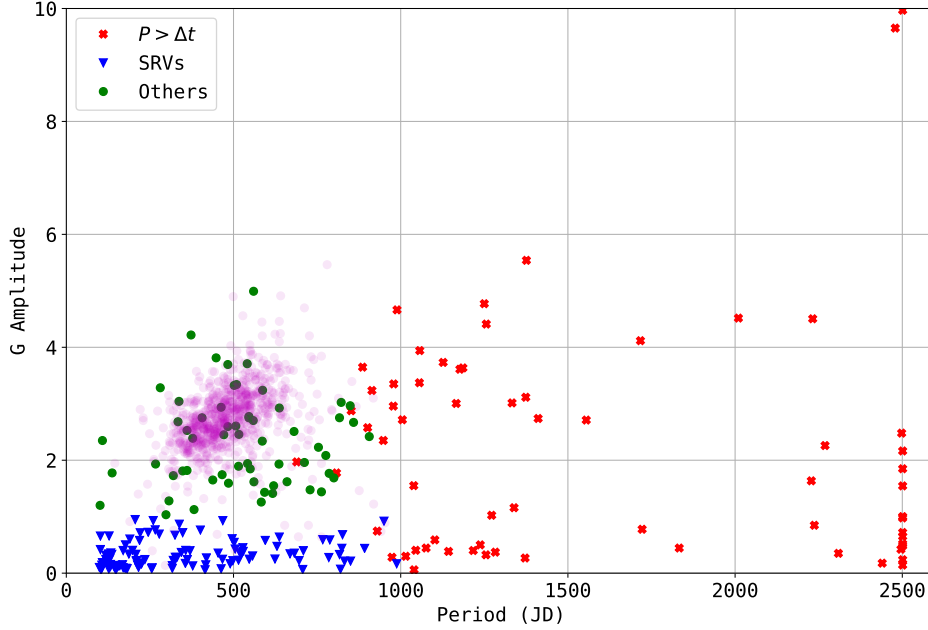


Figure 25: Fig. 19 with the sources categorised according to the quality of the light curves. The *GOOD* fits are included in magenta. A legend is provided for the remaining *BAD/DUBIOUS* fits. The sources marked *Others* are the Type 1 and Type 2 outliers.

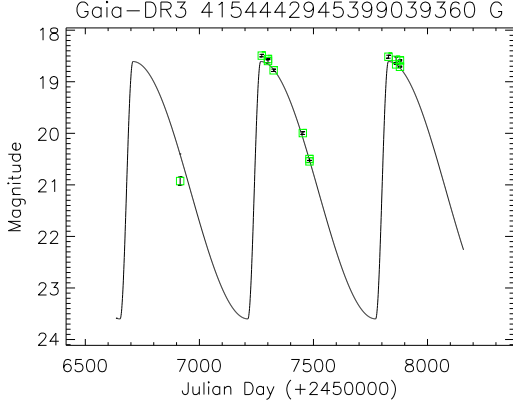
The first (“Type 1”, shown in Fig. 26) are defined by three properties:

- The photometry time series is poorly sampled. The number of observations in the series was less than 20 for all fits of this type.
- The series is missing large chunks of photometry between the peaks. Note in Fig. 26b the missing portion of the series.
- The sources are very dim with a mean magnitude $G > 20$ mag.

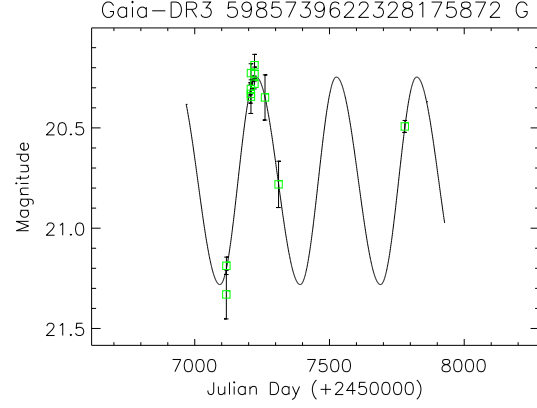
The second property arises due to the filtering of the original raw epoch photometry from the *Gaia* archive. As stated in Section 3.2, any point with a relative flux error greater than 5 was not included in the light curve fitting. Given that the flux error increases for dimmer sources and for the dimmest sources will have missing photometry points that have been removed due to large errors.

The sources with these three properties are clearly a consequence of the source lying on the upper limit of magnitudes detectable by *Gaia*. Even if the source has points with G-band magnitudes brighter than 20 mag, some or many of the dimmer points may lie below this limit and are either removed by the data processing steps or distort the fit with their contribution. To identify these faint sources a flag was implemented that would return true if the dimmest point in a source’s light curve had a G-band magnitude greater than 20 mag.

The second type of outlier (Type 2) are shown in Fig. 27. These sources have better coverage than the Type 1 outliers with between $\approx 20 - 100$ observations for each. There is good coverage over the entire curve with no gaps as seen in the Type 1 outliers. They are also brighter than the Type 1 examples with an average G magnitude ≈ 10 . Interestingly, these sources seem to fit the profile of a SRV quite well, demonstrating



(a) *Gaia* Source 4154442945399039360



(b) *Gaia* Source 5985739622328175872

Figure 26: Two examples of the Type 1 outlying fits demonstrating the low observation count and dim magnitudes that are typical of these fits

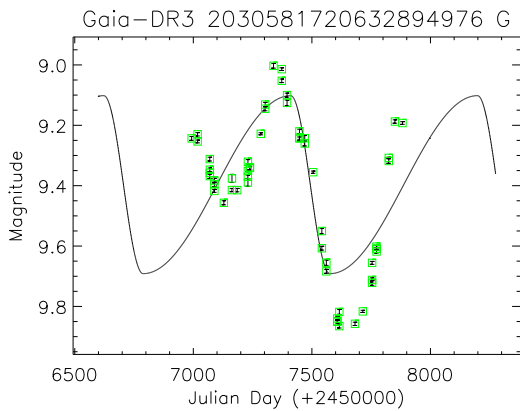
a general oscillation but are not well fit by a sinusoidal light curve. The one issue with this categorisation being that their amplitude is greater than the 1 mag amplitude typical of an SRV.

Ultimately, it was decided that visual inspection, though tedious, was the ideal method for assessing the quality of the curves and categorising the sources. There were too many outlying cases to confidently implement a filtering system for the different subgroups mentioned above. For example, there were many *GOOD* fits with amplitudes less than 1 mag, meaning that assuming any source with a small amplitude is an SRV or an irregular variable would remove many good quality fits from the sample.

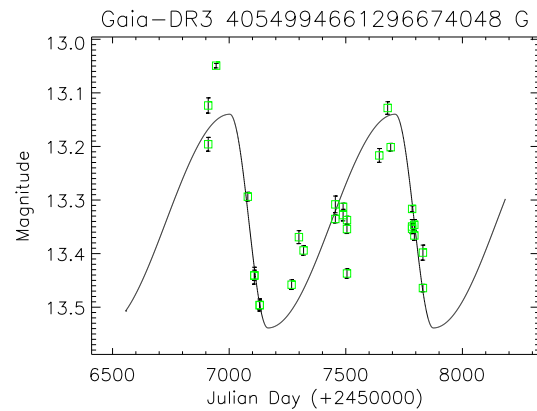
4.2 Final Classification of OH/IR Sample

It was decided that the sample should be classified into three subgroups:

- **Mira Variables:** These sources are denoted by a well sampled light curve and a clear minimum in the periodogram. The sources are clearly LPVs and were marked as *GOOD* in the previous visual



(a) *Gaia* Source 2030581720632894976



(b) *Gaia* Source 4054994661296674048

Figure 27: Two examples of the Type 2 outlying fits demonstrating the semi-periodicity, good coverage but generally poor fits to the data.

analyses.

- **Non-Miras:** These sources are likely semi-regular variables or irregular variables. They demonstrate a lack of periodicity and are poorly fitted by the least squares fit. The periodogram of these sources show either no discernible minima, or a range of minima (particularly at low periods) implying that the source is so scattered that it can only be fitted by arbitrarily small periods that eventually hit each point. The key difference between non-Miras and the dubious sources is the amplitude limit of $A < 1$.
- **Dubious:** These are the remaining *BAD/DUBIOUS* fits that do not fall under the banner of Non-Miras. They consist of sources with poorly sampled light curves, periodograms displaying clear secondary minima, sources with periods longer than the time series baseline (*PRELIM* sources), and dim sources with missing photometry points at the dimmest portions of the oscillation (Despite showing a possible good fit; see source 5985739622328175872, Fig. 26b).

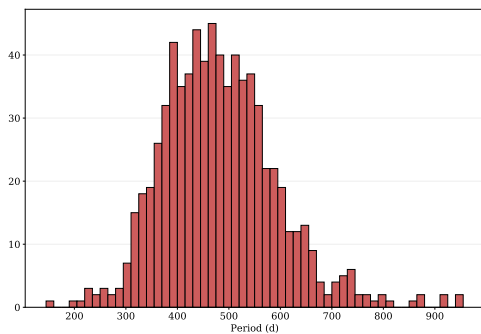
The final classification of the OH/IR sample according to these three types is tabulated below in Table 4.

OH/IR Sample	
Total Sources	1,071
Mira Variables	839
Non-Mira (Irregular Variables, SRVs)	110
Dubious Sources	122

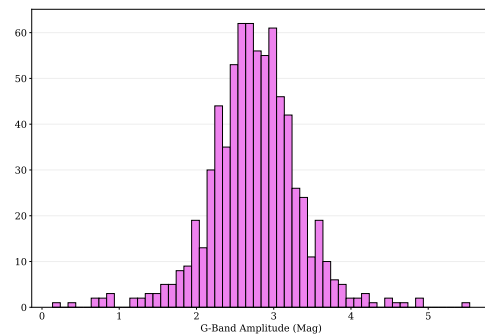
Table 4: OH/IR Final Classification

4.3 Variability Properties

Analysis of the Mira variability properties shows that the sample aligns quite well with the expected behaviour from the literature. The mean period of the sample is 484 days with the smallest period being 145 ± 11 days and the largest being 953 ± 471 days. As previously stated, Gaia’s 34-month observational baseline places an upper limit on the measurable period but it is likely that many of the Dubious sources are viable AGB candidates with periods greater than 1,000 days. The same is seen for the amplitudes. The mean amplitude of the sample is 2.73 mag with a minimum and maximum amplitude of 0.13 ± 0.001 mag and 5.46 ± 0.08 mag, respectively. Of these sources, 67 of them have an amplitude in the G-band less than 2 mag. It is unclear why so many sources have such low amplitudes given that Mira variables are known to have large visual amplitudes with $A > 2.5$.



(a) Histogram of Fitted Periods for the Sample



(b) Histogram of fitted amplitudes for the Sample

Figure 28: Variability properties of the Mira Sample

As seen in previous studies such as [Urago et al. \(2020\)](#), Mira periods and peak-to-peak amplitudes are correlated, with longer periods showing larger amplitudes. Figure 29 confirms that this sample of Miras is no exception. A linear function was fit to the sample and the two quantities were found to be related by

$$A_G = (0.94 \pm 0.19) \log P - (3.09 \pm 0.51) \quad (25)$$

where A_G is the peak-to-peak amplitude in the *Gaia* G-band.

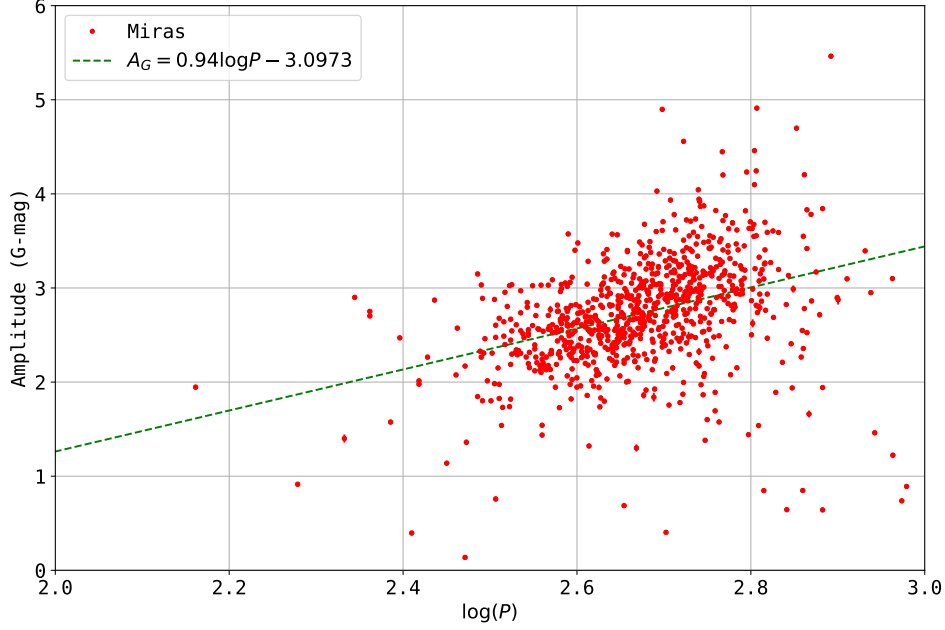


Figure 29: Plot of the fitted Mira periods compared with the Mira peak-to-peak amplitudes. A correlation between the two is apparent and a linear function has been fitted to the data. 1σ errorbars are included but are too small to see.

Gaia provides the period for a subset of its variable sources. Out of the total sample, *Gaia* has period information for 512 sources. The periods calculated from the fitting algorithm agree well with the *Gaia* counterparts as shown in Fig. 30. The only points that deviate drastically from the main line (i.e. $|P_{Gaia}/P_{Fitted}| \gg 1$) are sources with poor fits.

4.4 Parallax & Distance Measurements

With the sources classified according to source type, their distances can now be estimated using *Gaia* parallaxes. Not all the parallaxes listed in *Gaia* can be considered reliable and thus a set of criteria was established to filter sources that had reliable or unreliable parallaxes. If a source met any of the following criteria it was flagged as having a poor parallax measurement.

- The parallax was negative. Out of the original sample, 209 sources were found to have a negative parallax which are considered unphysical and are thus rejected. In fact, it is possible to have “physical” negative parallax values as they represent a true positive parallax calculated in the presence of large

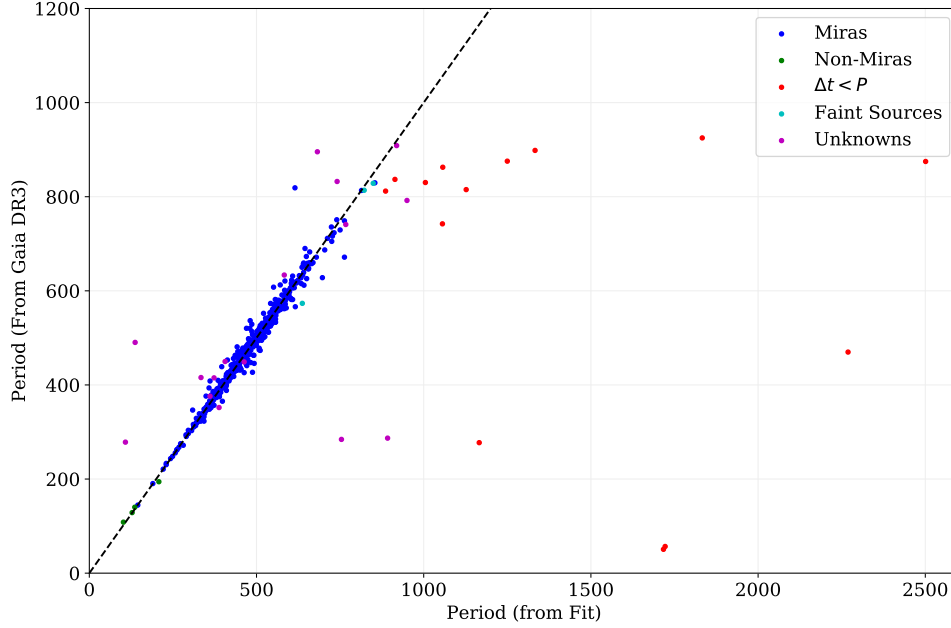


Figure 30: Plot of the fitted Mira periods compared with the periods as calculated by Gaia. The two show a good agreement with nearly all of the available periods aligning very well with the fitted counterparts. The few sources that deviate from the line are all *DUBIOUS* but are coloured to show their specific issue.

amounts of noise. They can not be used in the same manner as a traditional parallax measurement and are thus discarded.

- The relative parallax error is less than 30%. 365 sources had a relative parallax error greater than 30%.
- The astrometric excess noise is less than 1.5 mas. In the Mira sample, 367 sources had an astrometric noise greater than 1.5 mas. The astrometric excess noise is the difference between a sources measured position and the best fitting astrometric model. It is expressed as an angle in mas and contains information on the modelling errors that are not associated with observational noise (Lindegren et al. 2012).

If a source fulfils one or more of the above criteria it is flagged as having an unreliable parallax and discarded from the following analysis stages.

When working with large distances ($d > 100$ pc) the inverted parallax becomes a poor estimation of distance (Bailer-Jones et al. 2021). In the complete absence of uncertainties the inverted stellar parallax, i.e. Eqn. 15, can be taken as the true value of a star’s distance. Unfortunately, the *Gaia* parallaxes are measured quantities and thus have an associated measurement uncertainty. For sources brighter than ≈ 14 mag, the uncertainty $\sigma_{\bar{\omega}} \approx 0.04$ mas which increases to ≈ 0.7 mas for fainter 20 mag sources (Luri et al. 2018).

To combat this, Bailer-Jones et al. (2021) employ a probabilistic approach to calculate a more accurate parallax for a source. They assume the probability density function of a source over a distance r to be the product of two probabilities: a posterior likelihood and a distance based prior probability.

$$P_g^*(r|\bar{\omega}, \sigma_{\bar{\omega}}, p) = P(\bar{\omega}|r, \sigma_{\bar{\omega}})P(r|p) \quad (26)$$

In the above equation $\bar{\omega}$ is the source parallax, $\sigma_{\bar{\omega}}$ is the uncertainty on the parallax, and p is the HEALpixel number. Hierarchical Equal Area isoLatitude pixelation is the division of a spherical surface into set of pixels each with identical area. HEALpixels were developed with the goal of facilitating astrophysical analysis of huge astronomical datasets, spanning the whole celestial sphere (Gorski et al. 2005). Bailer-Jones et al. divide the sky into 12,288 pixels with equal areas of 3.36°^2 .

The likelihood of a source, $P(\bar{\omega}|r, \sigma_{\bar{\omega}})$, is assumed to follow a Gaussian probability distribution of the form

$$P(\bar{\omega}|r, \sigma_{\bar{\omega}}) = \frac{1}{\sqrt{2\pi}\sigma_{\bar{\omega}}} \exp \left[-\frac{1}{2\sigma_{\bar{\omega}}^2} \left(\bar{\omega} - \bar{\omega}_{zp} - \frac{1}{r} \right)^2 \right] \quad (27)$$

where $\bar{\omega}_{zp}$ is the parallax zeropoint which was taken to be -0.029 mas. This value was first determined by Lindegren et al. (2018) through parallax measurements of around 500,000 quasars which have such small parallaxes ($< 0.0025 \mu\text{s}$) that they can be taken as 0.

The distance prior is a three parameter generalised gamma distribution.

$$P(r|p) = \begin{cases} \frac{1}{\Gamma(\frac{\beta+1}{\alpha})} \frac{\alpha}{L^{\beta+1}} r^\beta e^{-(r/L)^\alpha} & \text{if } r \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad (28a)$$

$$(28b)$$

where $\alpha > 0$, $\beta > -1$, $L > 0$ and Γ is the gamma function: $\Gamma(n) = (n-1)!$. The values for α , β , and L are dependent on which HEALpixel is being observed. Therefore the dependency on p in Eqn. 28a really translates to a dependency on α , β , and L . The distance prior is different for each HEALPixel where the parameters for each pixel are derived from a simulated model of the galaxy.

A source's accurate parallax was taken to be the median of the probability distribution and the uncertainties on this parallax are the 16th and 84th percentile of the distribution. The results of these new distance estimates were compiled into a catalogue: *Geometric and Photogeometric distances to 1.47 billion stars in Gaia Early Data Release 3*. The sources from the OH/IR catalogue with reliable parallaxes were cross-matched with this catalogue to obtain distance measurements. Although the Geometric distance is cited by Bailer-Jones et al. as being the less accurate of the two measurements, it was decided to use it over the Photogeometric distance as the latter is less complete and prone to missing values which would further dilute the already heavily reduced sample.

By choosing sources with reliable parallax only, the sample gets significantly reduced and this galactic plane density seen in Fig. 18 is reduced considerably. The even distribution of these AGBs across the sky would imply that the sample is a collection of nearby stars. This is confirmed by Fig. 32, which shows the distance distribution of the good parallax sources. All the sources are within 4.5 kpc indicating that they are all relatively close by, in the solar neighbourhood.

Once a star has a reliable distance estimate, its absolute magnitude can be determined using the standard conversion from apparent to absolute magnitudes:

$$M = m - \log(d) + 5 \quad (29)$$

where m is the apparent magnitude of the source and d is the distance to the source in pc. The mean absolute magnitude of the samples was -0.48 with the brightest measured star having a magnitude of -2.9 mag and the dimmest with 6.8 mag. These absolute magnitude values are in line with the expected values from both the literature and the location of the AGB on the HR diagram (Robertson & Feast 1981, Kharchenko et al. 2002).

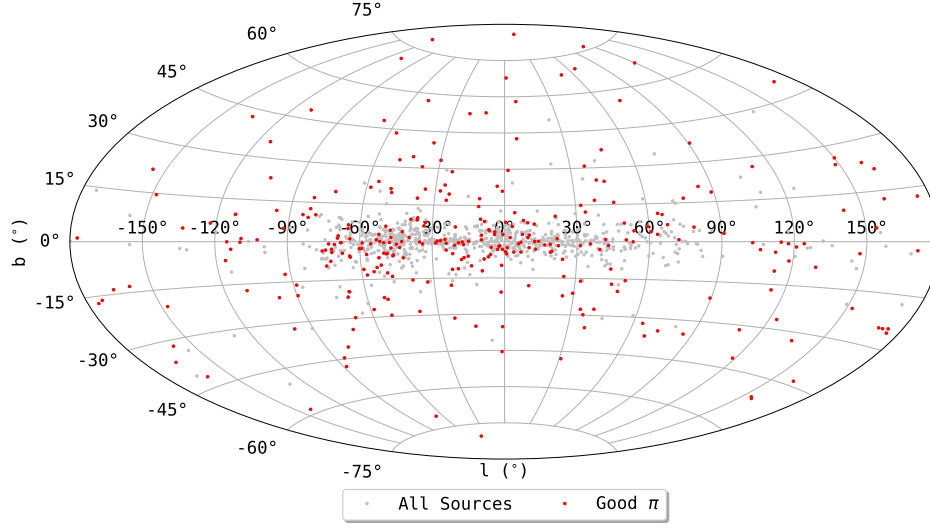


Figure 31: Sky-map showing the locations of the Mira variables with good parallax measurements. The central galactic density seen in Fig. 18

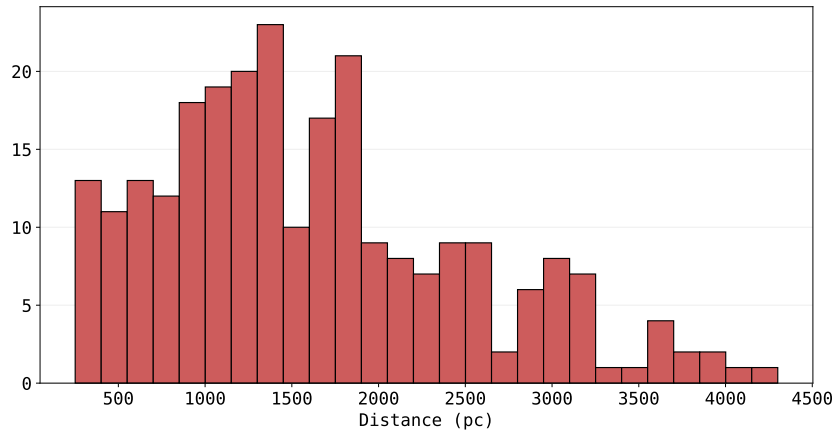


Figure 32: Distance distribution of the OH/IR Mira candidates with good parallax measurements. The sources are all within 4.5 kpc of the Solar System.

4.5 Period - Luminosity relation

As a final task to the preliminary data analysis of the OH/IR sample, an attempt was made to confirm the Mira period-luminosity relation. As detailed in Section 2.1.3 Mira variables have historically shown a very reliable period-luminosity relationship akin to that seen in other variable stars such as cepheids and RR-Lyraes. Plotting the absolute magnitude of a sample of Mira variables against the log of period of the

variables should, in theory, yield a linear relationship. By fitting a linear function to this relationship, one can theoretically extract a method of translating observed periods to absolute magnitudes and hence the distance to any arbitrary OH/IR star.

Fig. 33 shows the result of plotting this relationship for the sample of Miras with reliable distance estimates. The sources do not show a strong PL relationship with the sources actually displaying a downward trend, atypical of variable PL observations.

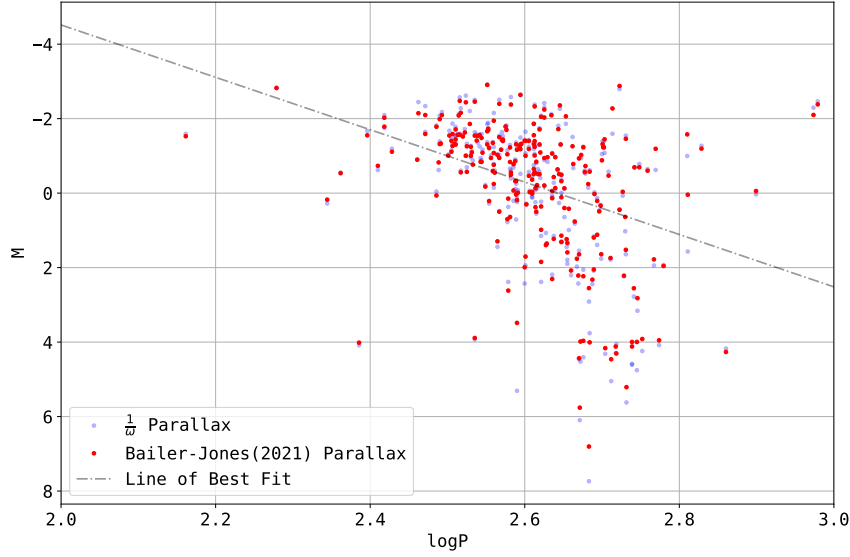


Figure 33: Result of plotting the period of the Mira variables with reliable parallax measurements against the absolute magnitude calculated by the [Bailer-Jones et al.](#) distances. The line of best fit yields a linear function $(-7.03 \pm 1.06) \log P + (18.58 \pm 2.7)$

It is unclear why these Miras do not follow the normal PL trend. One possibility is that they are heavily obscured by circumstellar extinction meaning the G-band magnitude (and subsequent variation) from *Gaia* is not actually representative of their true magnitude.

5 Additional Project Work

On top of the variability analysis detailed in Sections 3 & 4 there is still analysis and science to be performed on the three AGB catalogue samples.

5.1 Carbon Rich Sample

Variability analysis has begun on a sample of C-rich AGB candidates. The analysis is the same as that performed on the OH/IR O-rich sample, both in the filtering of the catalogue and the fitting of the light curves with an asymmetric cosine function.

The input sample of C-rich stars was also built in essentially the same manner as the OH/IR sample, with the only difference being the original catalogue from which the sample was derived. The C-rich sources were taken from the *Catalogue of AGB Stars in Our Galaxy* compiled by Suh (2021). This catalogue contains 7,172 C-rich AGB candidates which have been taken from the *IRAS* point source catalogue (Beichman et al. 1988) and the *AllWISE* catalogue.

At this stage of the project, light curves have been fitted to the stars and the visual inspection task is underway. Once the C-rich light curves have been classified, the variability properties of the final set of AGBs, the low-mass O-rich sample will be analysed. This sample will act as an excellent test of *Gaia*'s `is_cstar` flag.

It is expected that when the C-rich variability analysis has been completed, the three samples, O-rich and C-rich, will be combined into a full comprehensive catalogue of variable AGB stars ranging from low-mass O-rich sources, to intermediate mass C-rich stars, up to high-mass O-rich stars.

5.2 SED Analysis

Alongside the variability analysis, work has been carried out by other members of the team (by B. López-Martí in CAB, and D. Engels in the University of Hamburg) with regards to the SED analysis of the AGB samples. As discussed in Section 2.1.3, AGB stars can be differentiated from other sources such as Post-AGB stars by the shape of their SED. For the high-mass OH/IR samples and the C-rich sample this portion of the project has been completed prior to the commencement of the variability analysis covered in this report. SED analysis for the low mass O-rich sample is currently underway.

To construct the SED of an AGB star from the sample, additional photometry was required as *Gaia* only operates in narrow optical band whereas much of the key spectral information of AGBs comes from IR photometry. This task was performed with the **V**irtual **O**bservatory **S**ED **A**nalysers (*VOSA*). *VOSA* is highly equipped for this analysis as it is able to

- Read input tables of source identifiers, astrometry, photometry, and extinction.
- Provide additional photometry from other catalogues which can be accessed by the VO. *VOSA* is able to then construct a full SED based on the amalgamation of photometry data over different bands.
- Derive the physical properties of the source based on theoretical models that can be fitted to the SEDs. These properties include the effective temperatures, and luminosities. The software can also plot the sources on the H-R diagram based on these properties.

Fig. 34 shows an example SED for the AGB star *IRAS 17338+0051* (designated source *4375374585638670976* by *Gaia*). The SED shows the expected single-peak profile with a maximum $\approx 1 \mu\text{m}$ emblematic of an AGB star. The aim of this portion of the project is to filter out Post-AGBs and other spurious sources from the sample through visual analysis of the SED shape. For the remaining AGB stars, the bolometric flux can be

computed and, for the sources with reliable parallaxes, absolute luminosities can be calculated from F_{Bol} and the distance estimates from *Gaia* using the equation for luminosity $L = 4\pi d^2 F_{Bol}$

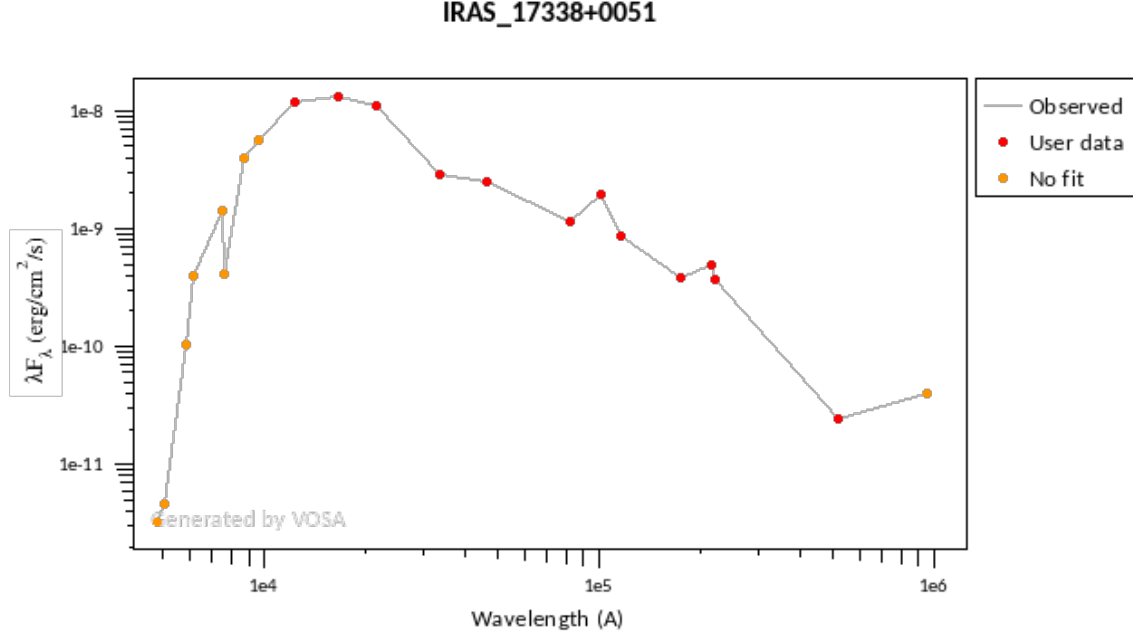


Figure 34: Example of single-peaked SED obtained during the *VOSA* analysis of the OH/IR sample by López-Martí & Engels

5.3 Further Assessment of Light Curve Quality

The release of *Gaia* DR3 has put pressure on other astronomical survey to provide catalogues of comparable scientific breadth. 2022 saw the release of the *OGLE Collection of Variable Stars* which, on top of observations of most known types of variable star, also contains measurements of 65,981 Mira type variables (Iwanek et al. 2022). Measurements were taken predominantly in the Johnson *I*-band. The sample covers the whole sky, with Miras observed in the Magellenic Clouds, Galactic Bulge and the Galactic disk. The completeness estimated to be $\approx 96\%$. The quality of the light curves is high for a few reasons. The photometry time series from *OGLE* span two decades meaning coverage is guaranteed for even the longest period sources. Secondly, the light curves were inspected multiple times by experienced astronomers with a focus on completeness of the sample while removing contaminant sources. Iwanek et al. cite the quality of the sample as being so exceptional that it can likely be used as training data for machine learning classifiers in order to automate the identification of valid light curves and classification of variable stars in large surveys.

OGLE light curves can be accessed online⁸ alongside a tabulated summary of the source properties which can be accessed using TOPCAT or any TAP enabled platform. Upon completion of the preliminary analysis of the C-rich sample, the quality of both groups will be assessed through comparison with this high quality *OGLE* survey. This comprises of period and amplitude comparisons, assessment of *OGLE* counterparts for Dubious sources as well as possible identification of any selection parameters that could aid in the quantisation of reliable fits.

⁸https://ogledb.astrouw.edu.pl/~ogle/OCVS/lpv_query.php

6 Discussions & Conclusions

This project was developed with the aim of creating a catalogue of Galactic AGB stars utilising data from *Gaia* DR3. The goal was, and continues to be, to create a global AGB catalogue spanning every chemistry, age, and variability type.

At this stage of the project, variability analysis has been performed on a sample of OH/IR stars. Using *Gaia*'s archive and VO tools variable photometry data was extracted for 1,071 sources out of a total 1,505. Using visual verification, the sources were evaluated, with the results determining whether they could be classed as Mira variables or not. Out of the 1,071 variable sources, 839 are Miras, 110 are non-Mira SRVs, and 122 are dubious fits that either have periods greater than the time series baseline or have dim photometry points.

One of the primary issues with the *Gaia* data revealed at this stage of the analysis was that it can't be used to study LPVs with periods greater than 1,000 days. Out of the sample, 65 sources had periods that were too long to be properly assessed with *Gaia* photometry.

The Mira subgroup was then inspected and those with dependable *Gaia* parallaxes were assembled based on a set of filtering criteria. For these sources, distance estimates from statistically determined geometric parallax values were extracted from Bailer-Jones et al.'s catalogue *Geometric and Photogeometric distances to 1.47 billion stars in Gaia Early Data Release 3*. These distance values revealed that the sources with reliable parallax values are confined to within 4.5 kpc of *Gaia*. They are distributed evenly across the celestial sphere from *Gaia*'s perspective but their parallaxes reveal that they are all nearby sources.

The distances were then used to derive the absolute magnitudes of the Mira sample. The absolute visual magnitudes lay in the $-2.8 - 6$ mag range which is in line with previous visual observations of Miras. The mean magnitude is low, but this is likely a result of the choice of AGB being examined. OH/IR are the most visually obscured AGB stars.

The optically weak source choice is also likely the reason for the odd shape of the PL relation plot. There is not even a semblance of the traditional PL relation for variable stars i.e. larger periods resulting in brighter stars. A probable cause is the absorption of the optical light by the thick OH/IR CSEs. A future task may be to use the *Gaia* distance estimations and the periods measured in this project but in conjunction with cross-matches from an IR survey such as the IRAS Catalogue of Variable sources (Spear et al. 1993) to see whether the lack of correlation is also present in the IR.

The severe dilution of the catalogue by the selection of sources with reliable parallaxes is somewhat frustrating especially given *Gaia*'s famed astrometric accuracy. A potential solution to this would be use a more relaxed set of criteria. A starting point would be addressing the minimum astrometric excess noise limit. This choice is supported by the *Gaia* literature which states that the quality of the astrometric noise can be quantified via the *significance* of the excess noise, D (`astrometric_excess_noise_sig` in the *Gaia* main table). If $D \geq 2$ it can be assumed that the source is astrometrically accurate even if the excess noise parameter is large. Therefore, this may be a preferable filter for parallax reliability. However, at this point in the project the previous analyses on AGB stars using *Gaia* has been performed using this fixed set of parallax criteria. Changing the selection process at this stage in the project would lead to inconsistency amongst the sources in the catalogue.

Despite these final PL-relationship results and the shortcomings mentioned above, a preliminary catalogue containing the sources and their associated variability information has been constructed from the variable counterparts from the OH/IR sample. This will later be combined with the results of the analysis of the C-rich sample and the low mass-O rich sample. This final concatenated table will be joined with the completed *Gaia Catalogue of Galactic AGB Stars*. The structure of the catalogue with the associated columns is detailed in Appendix A.

In regards to any changes that should be made to the project methodology, it may be prudent to approach the curve fitting with a slightly altered technique. One idea would be to use the method of Phase Dispersion minimization developed by Stellingwerf (1978). This method has been utilised successfully in previous AGB

studies as it well suited to the fitting of non-sinusoidal time series that are represented by sparse data points (Nakashima et al. 2000). The method uses a similar principle to the least-squares fit method but with a few key differences. The data is folded according to a test period to create a phase plot. The folded data is binned and the variance of the amplitude in each bin is computed. Ideally, if the folded curve has resulted in a neat periodic light curve, the variance will be minimal. Therefore the period that results in the smallest total variance from all the bins is the ideal fit. The method is versatile as the bin sizes can be varied or even overlapped to provide full phase coverage.

A totally different approach would be to abandon the fixed model described in Eqn. 23 and fit the light curves using a Fourier model. This approach has been successfully employed by many studies for light curve fitting and period reconstruction and even classification of variable types (Mantegazza & Poretti 1992, Ngeow et al. 2003, Percy & Golaszewska 2020). Because light curves are periodic they can be expressed as a sum of sine and cosine functions (Deb & Singh 2009).

$$m(t) = A_0 + \sum_{i=1}^N a_i \cos(i\omega(t - t_0)) + \sum_{i=1}^N b_i \sin(i\omega(t - t_0)) \quad (30)$$

where a_i and b_i are the amplitude of the $(i - 1)^{th}$ harmonics, $\omega = 2\pi/P$ where P is the period, N is the order of the fit, and t_0 is the epoch of maximum brightness. A further option would be to use principal component analysis which is able to replicate the results of Fourier analysis with less parameters and has been shown to be more effective than the Fourier alternative (Kanbur & Mariani 2004). However at this stage of the project, implementation of these methods would require a full overhaul of the fitting code and would once again lead to inconsistency in the catalogue when compared with the earlier fits.

It should not be forgotten that *Gaia* provides photometry in the G_{RP} band which appears to offer an untapped wealth of information about AGB stars. Given that the sample of stars analysed are cool, red, supergiants it is likely that they are emitting strongly at redder wavelengths. By studying the variability and epoch photometry of the AGB stars in the G_{RP} band, there is a possibility of more significant observations. This would become even more useful for dimmer sources, many of which had poor fits due to missing photometry points that were removed due to large errors in the data extraction process. Recently *Gaia*'s G_{RP} photometry has been used as an analysis tool for studying AGB stars as seen with Lebzelter et al. (2018) and Marigo et al. (2022). However, these works focused on using the G_{RP} flux as a tool to characterise the colours of AGBs and hence classify them. It would be interesting therefore to examine the quality of the G_{RP} epoch photometry and ascertain whether it can be utilised for future variability studies of AGB stars. Analysis would need to consider the reduced resolution of the RP instrument but this should not significantly affect the time series analysis.

An exciting prospect is that this project can be developed and improved with further *Gaia* data releases. Data Release 4, tentatively set for release in 2024, will contain photometric data recorded over 66 months (Gaia Collaboration 2022). This would open the doors to the possibility of studying all but the longest period LPVs. This is particularly enticing given the results of this project so far, which seem to imply solid observations of these LPVs, let down only by a lack of photometry data.

Debatably, the main outcome of this project, aside from catalogues and light curves, is a demonstration of the many tools that are now available to the modern astronomer and how they work efficiently together. The project combines astrophysics, programming, data science, virtual observatory tools, spectral analysis, and observations from 4 current and historic survey missions. The VO and its associated tools have proved invaluable for a project which is based primarily around the extraction and manipulation of tabular data but it also illustrated how readily available and accessible the data is to any astronomer with a computer. Similarly, the use of programming languages like Python and IDL have permitted the automated and almost instantaneous execution of data extraction, filtration, and visualisation.

Gaia DR3 offers a veritable wealth of astronomical data and this project utilises only a fraction of its potential. With Data Releases 4 and 5 on the horizon, the future of observational astronomy is bright and *Gaia* is the light shining the way.

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Appendices

A Catalogue Contents

Table 5 provides a summary of the contents of the preliminary catalogue of *Gaia* Variable AGB stars. This is only a draft version of the final catalogue and as such may not represent the final prepared table. A potential addition is a column with the names of the original cross-matches from the base catalogues i.e [Engels & Bunzel \(2015\)](#) or [Suh \(2021\)](#). At this point it is undecided whether this table’s columns will be concatenated with the final *Gaia Catalogue of Galactic AGB Stars* or whether it will be provided as an auxiliary table.

Table 5: Contents of the preliminary *Gaia* Catalogue of Variable AGB Stars

Gaia Catalogue of Variable AGB Stars			
Name in Catalogue	Type	Units	Description
<i>object</i>	string	-	Object source identifier in DR3 (<code>source_id</code> in the <code>gaia_source</code> table)
<i>catalogue</i>	string	-	Sample from which the object was taken e.g. OH Database, Suh21
<i>band</i>	string	-	Gaia photometric band in which the observations were performed
<i>ra</i>	float	deg	Right ascension of object (Gaia 2016 epoch)
<i>dec</i>	float	deg	Declination of object (Gaia 2016 epoch)
<i>source_type</i>	string	-	Object classification according to this study (MIRA/NON-MIRA/DUBIOUS)
<i>nobs</i>	int	-	Number of epoch photometry observations
<i>p</i>	int	d	Period of the fitted light curve
<i>err_p</i>	int	d	Uncertainty of the fitted period
<i>amp</i>	float	mag	Amplitude of the fitted light curve
<i>err_amp</i>	float	mag	Amplitude uncertainty of the fitted light curve
<i>m</i>	float	mag	Mean magnitude of the fitted light curve
<i>err_m</i>	float	mag	Uncertainty of the mean magnitude
<i>f</i>	float	-	Asymmetry factor f
<i>err_f</i>	float	-	Uncertainty on the asymmetry factor
<i>t_0</i>	int	JD	Julian date of first epoch photometry observation
<i>freq</i>	float	d ⁻¹	Object variable frequency as measured by Gaia
<i>err_freq</i>	float	d ⁻¹	Uncertainty on Gaia frequency
<i>gaia_p</i>	float	d	Period calculated by inverting the Gaia frequency
<i>gaia_p_error</i>	float	d	Uncertainty on Gaia period
<i>parallax</i>	float	arcsec	Parallax measured by Gaia
<i>relative_parallax_error</i>	float	-	Relative parallax error σ_π/π
<i>parallax_flag</i>	string	-	Flag denoting reliable Gaia parallaxes based on the criteria
<i>distance</i>	float	pc	Distance estimate obtained by inverting the Gaia parallax
<i>abs_mag</i>	float	mag	Absolute magnitude of the object
<i>r_med_geometric</i>	float	pc	Geometric distance from the Bailer-Jones catalogue
<i>r_geo_16</i>	float	pc	16th percentile of the Bailer-Jones geometric distance
<i>r_geo_84</i>	float	pc	84th percentile of the Bailer-Jones geometric distance
<i>abs_mag_geo</i>	float	mag	Absolute magnitude calculated with the Bailer-Jones geometric distance

B Tasks and Associated Personnel

As this is an ongoing and collaborative project, tasks were divided, shared, and performed by a few different astronomers based in Madrid, Hamburg, and Dublin. Table. 6 summarises the work described in this report and the personnel responsible for each task.

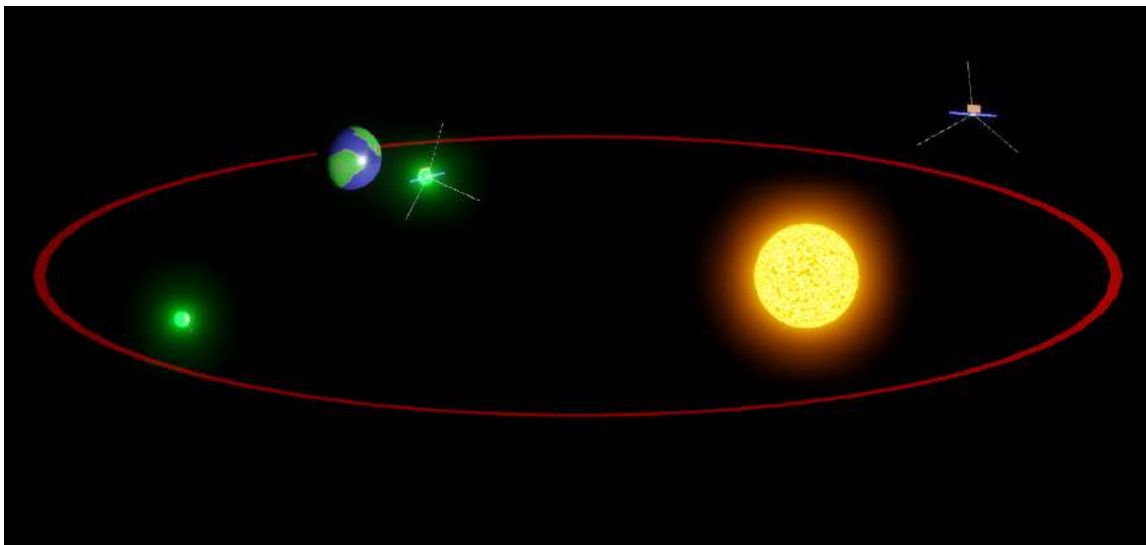
Table 6: Breakdown of the work detailed in this report with the associated personnel

Task	Personnel
Construction of sample OH/IR and Carbon star catalogues	D. Engels B. López-Martí
Extraction & Filtering of Epoch Photometry	C. de Barra
Fitting light curves to Epoch Photometry	F. Jiménez-Esteban
Visual Inspection of Light Curves	C. de Barra
Variability/Parallax Analysis & Catalogue Preparation	C. de Barra
SED Analysis	B. López-Martí

Evaluating Deep Space CubeSat Communication and Developing Conceptual Designs for the SURROUND Mission

Jack Cullen

August 19, 2022



Signed Statement of Original Authorship

‘I hereby certify that the submitted work is my own work, was completed while registered as a candidate for the degree stated on the Title Page, and I have not obtained a degree elsewhere on the basis of the research presented in this submitted work.’

Signed: Jack Cullen

Date: 19th August 2022

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Abstract

The SURROUND Project will provide a hazardous space weather forecasting service by tracking the radio bursts emitted by CME shocks and energetic particles as they travel through the inner solar system. These early warnings allow operators to prepare for the danger posed by these events to important, every-day technologies including global navigation systems and power grids; technologies, that if interrupted could cause severe disruption. To provide a forecasting service SURROUND will use a constellation of nanosatellites operating at a range of positions, L1, L4, L5 and in an ahead and behind orbit relative to Earth. These positions result in a significant variation in Earth-spacecraft distance resulting in the need to implement communication across deep space distances. In this report the options available in the pursuit of the required performance are examined. This including an exploration of the various high gain antenna designs suitable for CubeSat platforms, the power that can be provided to a CubeSat's antenna feed and the impact that ground station selection has on estimated downlink data rates. The findings of this report point to a trade-off between using expensive, high demand ground stations to achieve rapid downlink rates or lower cost, low demand ground stations that require lengthier transmission windows to obtain all data collected by SURROUND spacecraft. The findings of this study can then be used in conjunction with ground station schedules and cost to determine the best compromise. An additional component of this project has been the development of various visualisations, static and animated, to support the publicising and explanation of the SURROUND mission including the constellation and how this constellation works to forecast space weather events.

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List of Acronyms

AU	A stronomical U nit
GPS	G lobal P ositioning S ystem
MarCO	M ars C ube O ne
EDL	E nter D escent L anding
STEREO	S olar T errestrial R elations O bservatory
CME	C oronal M ass E jection
TOA	T ime O f A rrival
TDOA	T ime D ifference O f A rrival
JWST	J ames W ebb S pace T elescope
BER	B it E rror R ate
DSN	D eep S pace N etwork
CAD	C omputer A ided D esign
LEO	L ow E arth O rbital
HGA	H igh G ain A ntenna
EIRP	E ffective I sotropic R adiated P ower
FSPL	F ree S pace P ath L oss
BR	B it R ate
STL	S tandard T essellation L anguage

1 Introduction

Extreme space weather events caused by solar activity pose a significant threat to important technologies that are used day to day by the public and industry including global navigation and power grids. To provide forecasting of these events the SURROUND project will deploy a constellation of CubeSat spacecraft that will detect type II and III radio bursts produced by coronal mass ejection shocks and electron beams respectively. Three spacecraft in the SURROUND constellation will be placed at Lagrange points 1, 4 and 5, one spacecraft will be placed in an orbit ahead of Earth and another will be placed behind, an out-of-ecliptic spacecraft is also under consideration. Using a constellation of spacecraft to detect radio burst enables the application of trilateration methods to localise the origin of the radio burst and therefore the location of the space weather event. This enables an estimation of when these events will arrive at Earth.

To enable this constellation of satellite to function as an operational forecasting service the data collected by these spacecraft must be transmitted to Earth. Due to the limitations of a CubeSat configuration and the 1 AU and larger distances between some of the spacecraft and Earth, enabling an effective downlink is a significant challenge for the project. This report examines the feasibility of and possible approaches available to SURROUND to achieve capable and reliable data transfers. The various CubeSat compatible antenna configurations are examined, considering their radio communication capability but also their ease of deployment and their technology readiness level. To examine the performance of a proven CubeSat communication system implemented for SURROUND, the specifications of the MarCO interplanetary CubeSat have been applied in an estimation of the link budget across the orbits and resultant distances that will be experienced by SURROUND spacecraft. This link margin estimation also involves the testing of various ground stations to develop a picture of what data rates are achievable with what configuration, facilitating a judgement on if a high demand, high operational cost ground station is required to enable the SURROUND mission.

A second focus of this report is the development of material to aid in the visualisation and explanation of the SURROUND mission. This involved the development an early concept design for a SURROUND spacecraft. The produced design was then developed from a static representation to a deployment animation. In addition to spacecraft visualisation the overall mission concept was demonstrated in an animated representation of the mission's operating principle of radio burst detection by a constellation of spacecraft.

2 SURROUND Project Overview

2.1 Space Weather

The purpose of the SURROUND constellation is to provide a forecasting service for space weather as space weather events, due to solar activity, can negatively impact important technologies (Maloney et al. 2021). These space weather events can compromise the operation of navigation systems, power grids, spacecraft instruments and radio communications. In addition to the hazard to technology, these events also pose a threat to the health of astronauts and the occupants of aircraft operating in the polar regions. Of these hazards the current focus is the impact of large geomagnetic storms, initiated by CMEs,

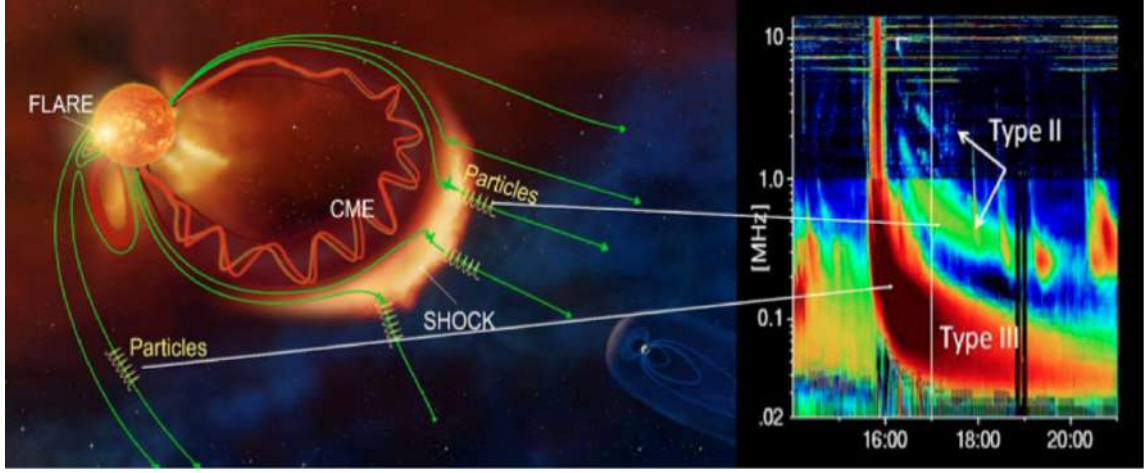


Figure 1: Left: Illustration of eruptive solar events and associated features. Right: Dynamic spectra of type II and type III radio burst detected by STEREO’s radio spectrometer payload (Maloney et al. 2021).

on terrestrial power grids. The types of relevant events are solar flares, coronal mass ejections and solar energetic particle events (Maloney et al. 2021), Figure 1. Forecasting these events is a valuable service so that the disruption caused can be anticipated thus allowing appropriate measures to be taken to mitigate or avoid damage.

There are currently a number of operational space weather forecasting services that can be used by the industries that would be impacted by these extreme space weather events. The most matured forecasting is that of solar flares. This is done by observing and then analysing the magnetic field of the solar surface using a variety of mathematical methods. Solar energetic particle events are also forecasted by using datasets of observations of the solar surface and previous flare activity. At present, forecasts of coronal mass ejections using probability do not exist. Forecasting of CMEs is instead carried out via coronagraph imagery. From images of the coronagraph, kinetic and geometric parameters are derived and then used as inputs in models for heliospheric propagation such as the Enlil model that is used operationally by both UK and US national space weather forecasters (Maloney et al. 2021). However, the field of space weather forecasting is considered to be in the very early stages of development with significant innovation required to improve solar event forecasting.

As stated, CMEs are tracked via observations of the heliosphere. Mission that provide these observations include the STEREO mission which provides observations from 2.5-30 solar radii and up to 1AU respectively (Maloney et al. 2021). Using these observation the motion of a CME can be tracked, potentially enabling the development of 3D trajectories. However, to estimate an arrival time at Earth a more common one-dimensional approach is used. In addition to the Enlil model, ballistic propagation and models using solar wind drag can be used. Despite the wide selection of available techniques, the arrival times generated by these approaches can have uncertainties of up to 20 hours. An additional means of detecting and tracking coronal mass ejections is by monitoring the radio emissions associated with their shocks. The mechanism by which these emissions are generated by shocks driven by CMEs involves the acceleration of particles to relativistic speeds resulting in plasma emission bursts, classified as type II radio bursts (Maloney et al. 2021). Dynamic spectra of these events are characterised by bands of

emissions drifting gradually towards lower frequencies over a range of timescales, from tens of minutes up to a number of hours. To a distance of approximately 1 AU these radio bursts can drift from 15 MHz to frequencies less than 100 kHz. Therefore these radio bursts are useful for tracking CME shocks into large heliospheric distances. However, this technique is limited due to uncertainty in the direction of the CME propagation and is dependant on the choice of radial density model which influences the calculated motion of the CMEs (Maloney et al. 2021).

Solar flare and CME events can lead to solar particle emissions where energetic particles are accelerated and propagate on open magnetic fields into the heliosphere (Maloney et al. 2021). These high energy particles have energies potentially in the GeV energy range, and so, pose a significant threat to technological and biological objects and are therefore identified as a high priority for detection and forecasting (Maloney et al. 2021). Unfortunately, the forecasting of these emission events is one of the most challenging aspects of this field. A potential avenue towards accomplishing this goal is developing an understanding of the relationship between X-ray flare characteristics and particle emission arrival times and using these flare properties to train models for arrival time estimation.

Type III radio bursts are produced by propagating energetic electrons and have a large frequency range from hundreds of MHz to around 1 kHz and so these, and by extension, their source phenomena, can be tracked from the low corona to 1 AU. Type III radio emission tracking is the only way to monitor the propagation of electron beams through the entire heliosphere and so can be used to estimate the motion and arrival time of solar particle emissions (Maloney et al. 2021). These methods of CME and SEP events using radio burst tracking are limited due to their singular reliance on dynamic spectra and use unconstrained and simplistic models to determine the locations of the radio bursts and therefore their associated space weather event. A potential more powerful method of leveraging radio emission detection for forecasting would be by the detection and locating of these emissions in a 3D space via trilateration (Maloney et al. 2021).

2.2 Radio Spectrometer Payload

Radio spectrometers have flown as payloads of spacecraft since the 1970s. Despite some evolution, these instruments are comprised of the same basic elements: antennas, receivers, and associated electrical devices such as low noise amplifiers. Current instruments cover a frequency range from a few kHz up to tens of MHz at time resolutions from seconds to tens of seconds. A recent development in radio payloads is the software defined radio which replaces components that were analogue in earlier designs with digital components that software can control and configure (Maloney et al. 2021). This change enables a single radio system to be configured via software alone to transmit and receive using either FM or AM modulation schemes. Therefore, it is possible that software defined radios will enable a reduction in development cost and reduce the complexity of systems as the tailoring of these radios to the needs of a specific mission will no longer require the design and manufacture of a unique system but instead a relatively simple software configuration of an off-the-shelf hardware solution (Duncan 2015). The technology readiness level of this technology has reached the point where space qualified solutions are becoming commercially available (Maloney et al. 2021).

To date, the selected payload of each SURROUND spacecraft will be a radio spectrometer with two potential approaches under consideration. The first is a version of Solar Orbiter’s Radio and Plasma Waves, RPW instrument simplified and miniaturised for SURROUND spacecraft’s 6U CubeSat configuration. This approach is attractive as an instrument derived from the RPW would carry forward its benefits of being a low mass, small footprint, low power system (Maloney et al. 2021). Additionally, a derived technology will reduce the development time and cost. The second approach would use a commercially available software defined radio which would avoid the costs of hardware development, trading this difficulty for the easier process of developing software. Existing designs of these software defined radios also have a small footprint and do not require high power. Despite these benefits, this approach would require the procurement of a suitable existing and available SDR which may be challenging given that this is a new technology.

2.3 Localisation of Solar Radio Bursts Via Trilateration

As discussed in 2.1 a potential powerful method for space weather forecasting is trilateration. Typically, this method determines the target’s position by using the times-of-arrival (TOAs) of signals that emanate from the target and have a known waveform and propagation speed and arrive and are detected by multiple detectors at different positions. The detectors must have known positions and the detectors’ clocks must be synchronised. After the signal is detected at each of the locations the location of the source position can be calculated. There are two approaches that can be used for this calculation. The time-of-arrival, TOA, method and the time-difference-of-arrival, TDOA, method, Figure 2.

The TOA approach is the simplest of the two approaches. It uses the time of signal emission and time of signal arrival to calculate the distance between a detector and the source. This is described by the following equation where t_e is the time of emission, t_r is the time of arrival and c is the speed of the signal, for radio emissions, the speed of light. The resultant distance can be represented by a circle where the calculated distance is its radius and the detector is at the centre, this is because a signal picked up by a single detector could originate from any direction. To obtain improved spatial information additional detectors are needed resulting in more circles, the points where these circles intersect are potential source locations. Following this, in two dimensions a minimum of three detectors is required as for one circle the source could be anywhere along the circumference, for two circles there are two intersecting points and for three circles there is one. To resolve the source position in three dimensions the circles are spheres and so another sphere is required. This TOA method is used in the Global Positioning System whereby a terrestrial receiver receives signals from GPS satellites that carry the time that the signal was sent, using this information and the time on the receiver’s clock the time difference and therefore distance to each of the GPS satellites is calculated and so the position of the receiver is found. Unlike signals from GPS satellites, space weather radio emissions do not carry information on their time of emission and so the TOA method cannot be used for this purpose.

$$d = c(t_e - t_r) \tag{1}$$

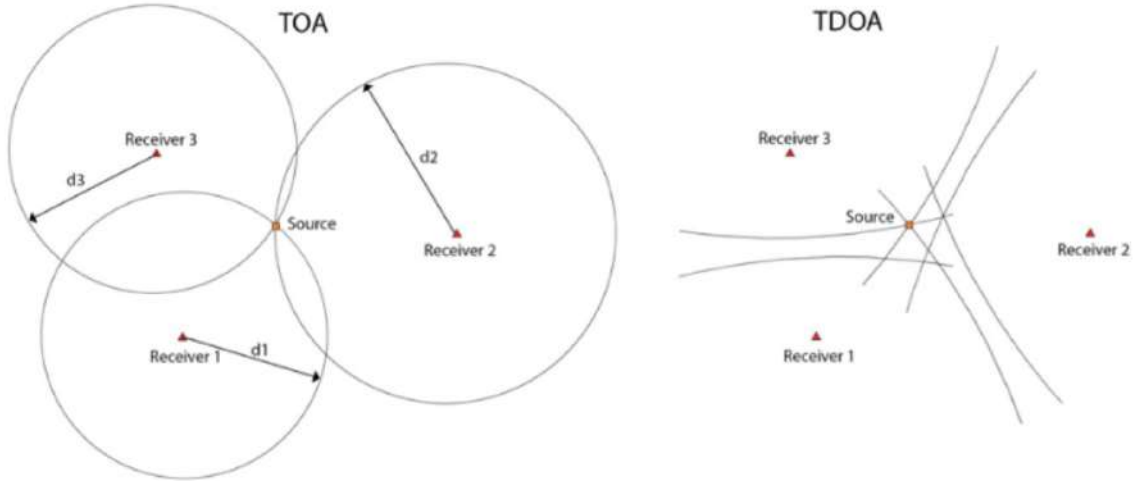


Figure 2: Left: Time-of-arrival, TOA, trilateration method, circle radii is defined by the time taken for the signal to reach the receiver and the speed of the signal. Right: Time-difference-of-arrival, TDOA method, the hyperbola are generated from the difference in the time at which the receiver pairs detect a signal (Maloney et al. 2021).

The alternative to the TOA method is the TDOA method which produces a similar result however uses a different process. By using the difference in the time that the signal is detected by a pair of receivers a hyperbola is generated between them. Although this method has the advantage of not requiring the emission time it is limited as the hyperbola contain two branches meaning that the source location cannot be narrowed down as readily as in the TOA method. An additional advantage of this method is its robustness, as it is still viable for low signal to noise ratio detections and does not require complex antenna. However, for this method to be successful the detectors must have a large angle of separation and so the configuration of the SURROUND constellation is of significance. Another method that is under consideration is that of Goniopolarimetry which uses analytical methods on wave parameters to gain directional information. This method has the advantage of requiring fewer satellites, however, it is highly dependent on antenna sensitivity and calibration.

2.4 Constellation Configuration

The SURROUND project concept suggests a constellation of six CubeSats, Figure 3, with the goal of maximising scientific return and to improve the accuracy of trilateration methods by reducing error. A spacecraft will be located at L1, L4 and L5 Lagrange points in the Sun-Earth system. These spacecraft are 'locked' in this configuration relative to Earth and so orbit virtually in sync, maintaining their relative positions. In addition to these spacecraft, another two spacecraft will be placed in an 'Ahead' and 'Behind' configuration where the 'Ahead' spacecraft will travel faster around the Sun compared to Earth due to an orbital radius of less than 1 AU while the 'Behind' spacecraft will travel slower than Earth due to a larger orbit radius. The final of the proposed spacecraft will be placed in an orbit out of the ecliptic plane in an inclined orbit to enable full three dimensional localisation. Although several missions have operated at L1 which is 0.01AU from Earth, the most recent and well known being JWST, L4 and L5 missions would

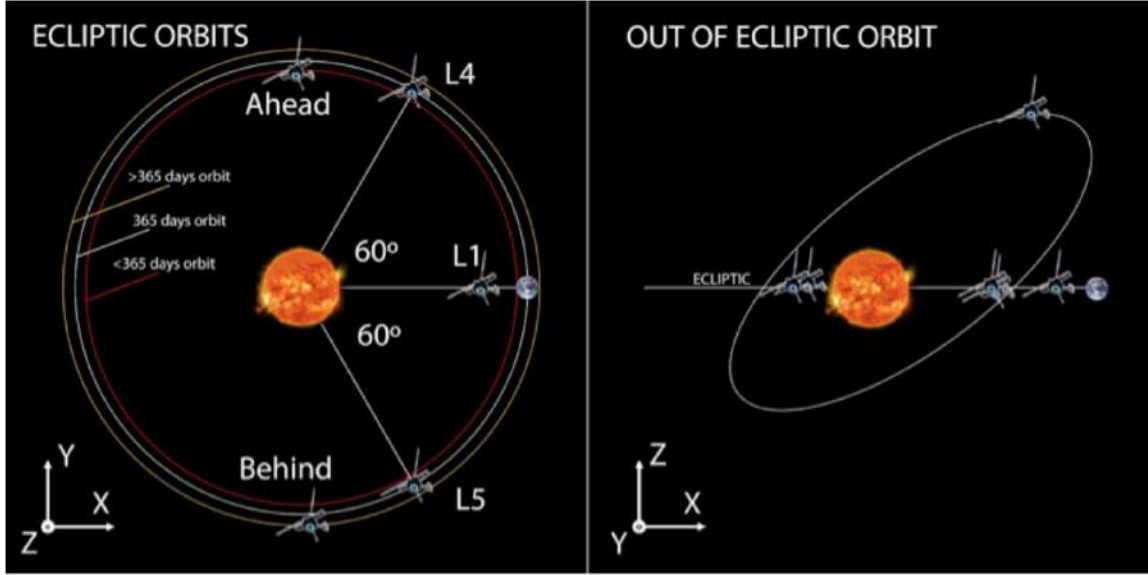


Figure 3: Proposed SURROUND Constellation (Maloney et al. 2021).

require a large delta V with the inclined orbit being the most demanding. To achieve these orbits a range of options are under consideration including electric propulsion, solar sails or a hybrid system (Maloney et al. 2021). In addition to the delta V requirement to obtain these orbits it is also necessary to accomplish these transfers within a reasonable transfer time.

3 Previous Mission Review

The CubeSat form factor has been chosen for the SURROUND mission. These spacecraft are small in size and have standardised forms which reduces the cost of developing and manufacturing these spacecraft. However, the majority of CubeSat missions have been constrained to operate in the Low Earth Orbit regime due to their limited capabilities for power generation, communication and propulsion (Kulu 2022). Only one CubeSat mission has been deployed on a deep space mission, the interplanetary MarCO spacecraft.

The Mars Cube One mission was a piggyback mission that flew to Mars with the InSight mission and provided the real-time transmission of telemetry data during the entry, descent and landing phase of the InSight lander's mission, Figure 4. The MarCO mission is of special relevance to SURROUND as the pair of spacecraft that carried out this mission were of the 6U CubeSat configuration, the configuration which SURROUND will use. While there are other 6U Cubesat Designs, the MarCO spacecraft is unique in operating in a deep space mission profile and is capable of high data rate transmissions at a distance of approximately 1 AU.

To do this it was equipped with a High Gain, X-band, Reflectarray antenna, which, when supplied with 5 Watts of power, provided a downlink data rate of 8 kb/s to NASA's Deep Space Network. MarCO's solar arrays provided the spacecraft with 17 Watts of power when situated in Mars Orbit. These same solar arrays would provide 35 Watts at Low Earth Orbit, which is the same distance from the Sun as the L4 and L5 Lagrange points. Therefore, it could be assumed that around double the power could be provided to

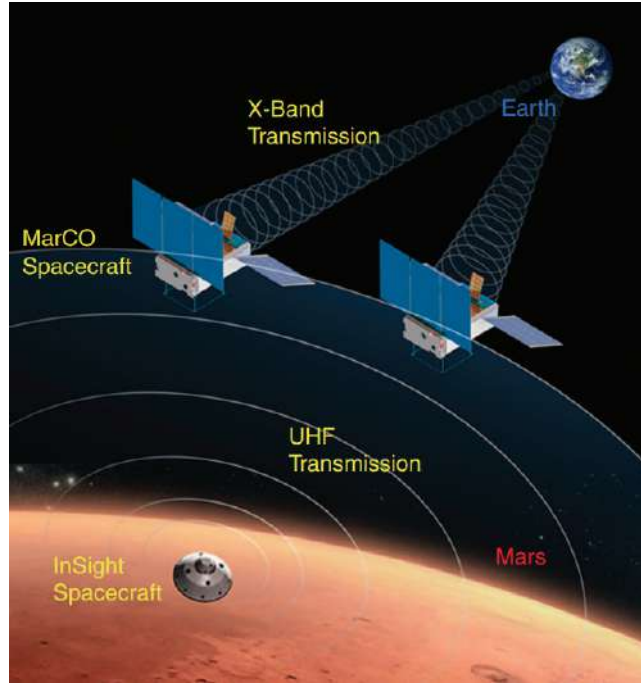


Figure 4: The two MarCO spacecraft provided a relay for near-real-time telemetry during the EDL phase of the InSight mission (Hodges et al. 2017).

the communication system. MarCO was also equipped with a SDR demonstrating that his technology is applicable and capable in deep space mission profiles (Duncan 2015) (Kobayashi 2017).

4 Communication Antenna Designs

A significant factor that makes deep space application of the CubeSat platform is the small size of the spacecraft. This small size constrains the space available for power systems and constrains the size of communication antennas. CubeSats deployed in Low Earth Orbit missions utilise ultra-high frequency dipole antennas or S-band patch arrays which, although having low gains they are sufficient for communication with large ground stations given that the distance between the ground station and spacecraft may only be a few thousand kilometres (Chahat et al. 2019). Meanwhile, a deep space mission will operate over distance of at least several million kilometres. As the power systems of a CubeSat are constrained, the development of higher gain antennas is vital to compensate for this limitation.

There are several key requirements for deep space high gain antenna, HGA, designs. These antennas must operate using either the X or Ka bands as these frequencies are those allocated to deep space communications (Chahat et al. 2019). Additionally, mechanical deployment should be minimised as mechanisms are a potential source of failures, if necessary these mechanisms should be very simple. For high volume science returns high gain antenna's must have narrow beamwidths. HGA designs under active development include reflectarrays, mesh reflectors and inflatables.

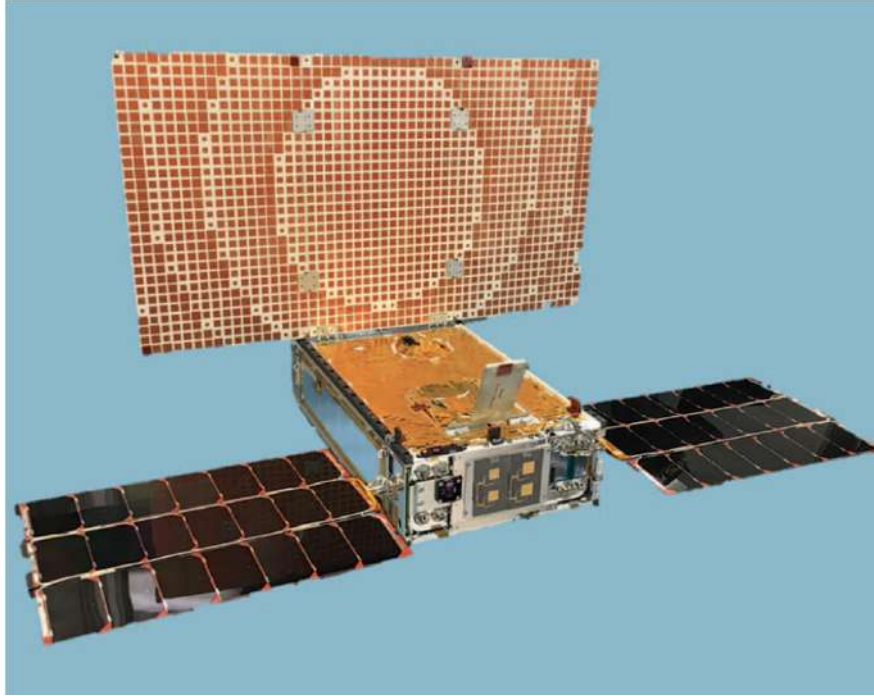


Figure 5: MarCO spacecraft deployed reflectarray (Hodges et al. 2017).

4.1 Reflectarray

The concept of the reflectarray was first introduced in 1996, (Chahat et al. 2019). This concept consisted of flat reflecting surfaces that deploy as panels using simple mechanical hinges. The first implementation of this design was used for LEO communication and had an efficiency of 26%. This early design was limited by a low-efficiency feed and large gaps and hinges resulting in a reduction in gain. Despite this, the antenna was successfully deployed in orbit and the produced signal was measured from Earth. The LEO design was then developed to an X-band communication system for the 6U MarCO CubeSat. This design improved significantly from the LEO system by achieving a gain of 28.8 dBi translating to an efficiency of 42% (Chahat et al. 2019). This was accomplished by reducing the gap size between the reflectarray panels and improving the efficiency of the feed. This system allowed MarCO to downlink 8 kb/s of near-real-time data from the InSight Mars lander during its entry, descent and landing. This type of high gain antenna was selected for the MarCO spacecraft as it is particularly suitable to meet the challenges of CubeSat platforms as it is deployed from a large physical aperture from a very limited volume due to the folding design. Additionally, it was capable of providing the high quality downlink when supplied with a low power of less than 5 Watts. To summarise, the key advantages of selecting a similar reflectarray configuration to MarCO include its low cost, volume, simplicity of deployment and high technology readiness level. The drawbacks of this design consist of a narrow bandwidth, requiring spacecraft pointing, a limited maximum aperture, low efficiency and thermal sensitivity. This reflectarray design is also limited to transmission only (Chahat et al. 2019).

A Ka-band HGA reflectarray has also been developed for the purpose of remote sensing applications yet has the potential to be adapted for communication, Figure 6. The design of this system utilises Cassegrainian optics which enables this array to be stored in a highly constrained volume, compatible with the 6U CubeSat form factor (Chahat et al.

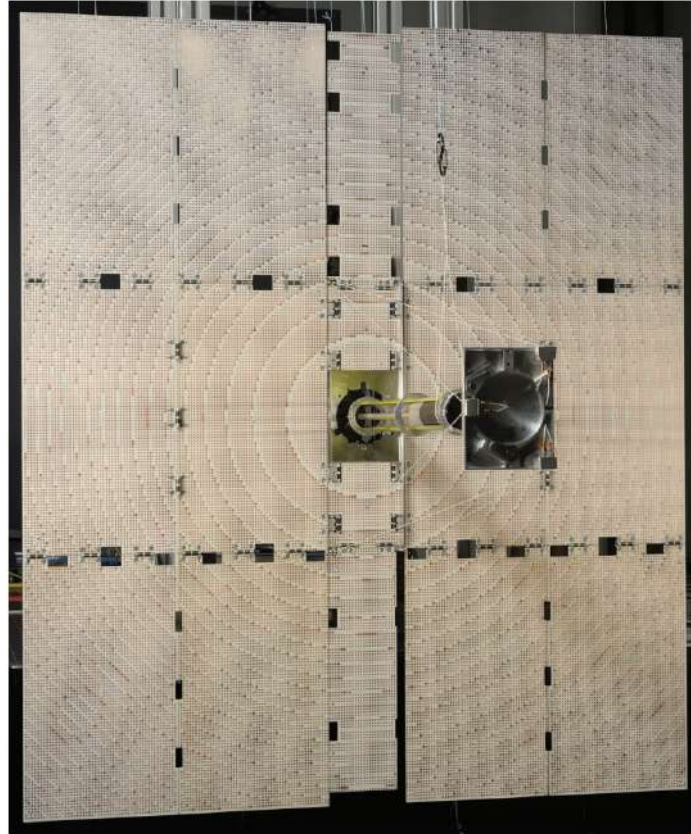


Figure 6: Ka-band 1 metre reflectarray uses Cassegrainian optics to enable compact storage (Chahat et al. 2019).

2019). This design possesses excellent performance; at a frequency of 35.75 GHz it has a gain of 47.4 dBi. Despite the improvement in performance over the MarCO reflectarray this array is more complex and has a greater cost.

4.2 Mesh Reflector

Mesh reflectors involve the deployment of a stowed mesh into a dish configuration. Several mesh reflector designs have been developed for a range of frequency bands. These include a 0.5m Ka-band design compatible with the 6U CubeSat configuration (Chahat et al. 2019). Undeployed, this design is contained in a volume of 1.5 U and provides a gain of 42.4 dBi and an efficiency of 56%. This design was proven operationally in LEO in 2018 (Chahat et al. 2019), 7. Mesh reflectors tend to have excellent efficiency and infinite bandwidth but require larger storage volume, have a higher mass, and greater risk of failure due to their relatively complex deployment mechanism. Similarly to reflectarrays the aperture of mesh reflectors is limited.

4.3 Inflatable Antenna

Inflatable antennas have been developed and deployed for use in S and X band communication, these deployable antennas resemble foil balloons, Figure 8. Due to the inflatable deployment mechanism this antenna type requires minimal storage space and deploys from this space in a simple and low risk way. This antenna type also has an infinite

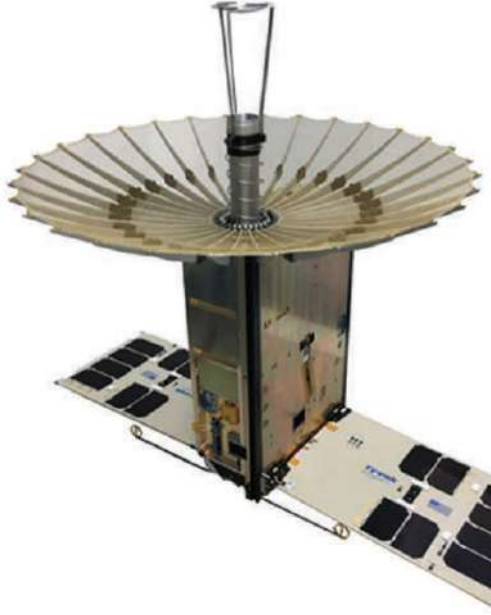


Figure 7: Deployed 0.5 metre diameter mesh reflector antenna mounted to NASA’s LEO RainCube spacecraft (Chahat et al. 2019).

bandwidth and can have a large aperture. Despite these advantages this antenna type is not without drawbacks, the inflatable antenna is a new technology and so has a low technology readiness level and has poor surface accuracy at high frequencies. Also, the side lobe level of the far field radiation pattern of this antenna type are high, reducing the efficiency of the signal (Chahat et al. 2019).

4.4 Patch Array

A patch array is mounted to the main body of the CubeSat and so the maximum aperture of the array is constrained by the dimensions of the satellite, thus limiting its performance (Chahat et al. 2019). However, this configuration does result in efficient storage volume and low mass and its lack of a deployment mechanism limits the risk of failure. This simple and established design results in a low cost and high TRL. However, the gain of this antenna is limited by feed losses.

4.5 Metasurface

Typically, the biggest challenges to increasing an antenna’s aperture, however, the metasurface technology could provide large apertures without the need for a deploying feed, only involving simple hinge deployment of the surface. However, as this technology is currently under development, its readiness level is low. This antenna type also has a low efficiency, a limited maximum aperture and a narrow bandwidth.

4.6 Membrane Antenna

These antennas can achieve a large aperture while requiring low storage volumes, valuable for 6U CubeSats. The membranes are either designed as patch arrays or reflectarrays where instead of no deployment or hinge-based deployment, the membrane is stretched

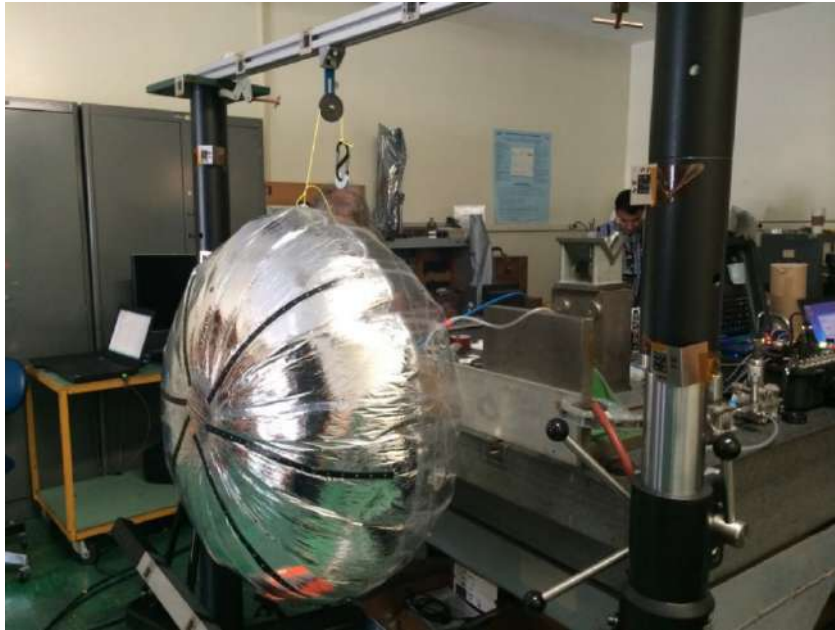


Figure 8: Inflatable antenna under development (Babuscia et al. 2016).

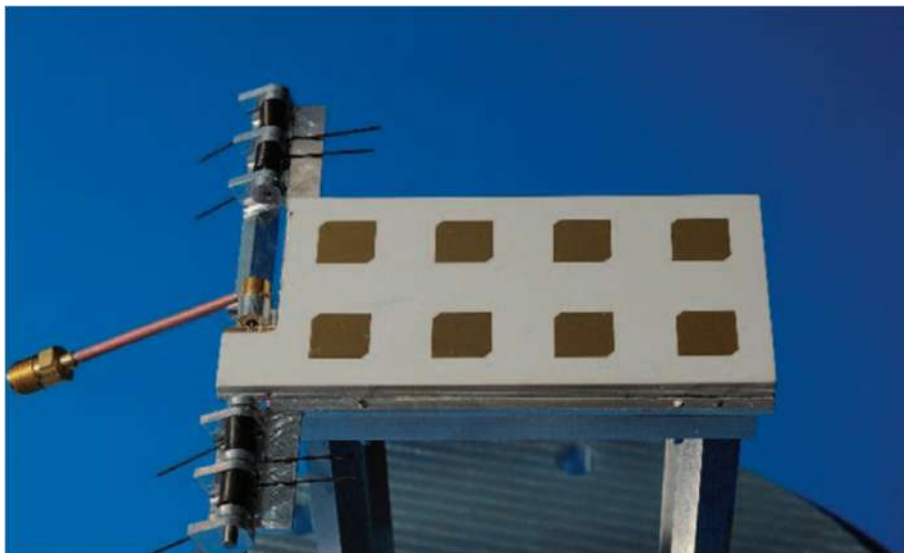


Figure 9: MarCO HGA feed to the reflectarray is a patch array design (Hodges et al. 2017).



Figure 10: Ka-band metasurface antenna (Chahat et al. 2019).

out into a large aperture (Chahat et al. 2019). Although a large aperture is desirable the deployment process is complex and is therefore a potential source of failure. Additionally, this technology has a low readiness level and has limitations in surface accuracy and efficiency.

4.7 Slot Array

This technology has several positive qualities including very good efficiency and simple deployment (also using hinges); however, these antennas operate using a narrow band at a single frequency and have yet to be implemented for spacecraft of the CubeSat form factor, having been developed for 100-kg Smallsats, Figure 12. Both this technology and membrane antennas are in their infancy, providing new directions for research (Chahat et al. 2019).

5 Communication Performance Theory & Methodology

A vital component of determining the feasibility of a space mission is the ability of a spacecraft to transmit its collected data. The calculation of a maximum data-rate is dependent on several factors, these are used in a set of equations, which together, make up the link budget (TutorialsWeb 2021). The downlink link budget can be considered in three sections: the spacecraft transmitter characteristics, the transmission losses and the characteristics of the receiver. To correctly develop the link budget method I tested the equations using inputs from the specifications of the Mars Cube One spacecraft communication system (Kobayashi 2017), Table 13. I fed these specifications into the link budget code developed in python and verified the performance of the code by obtaining outputs matching those of the MarCO literature, Figure 13.

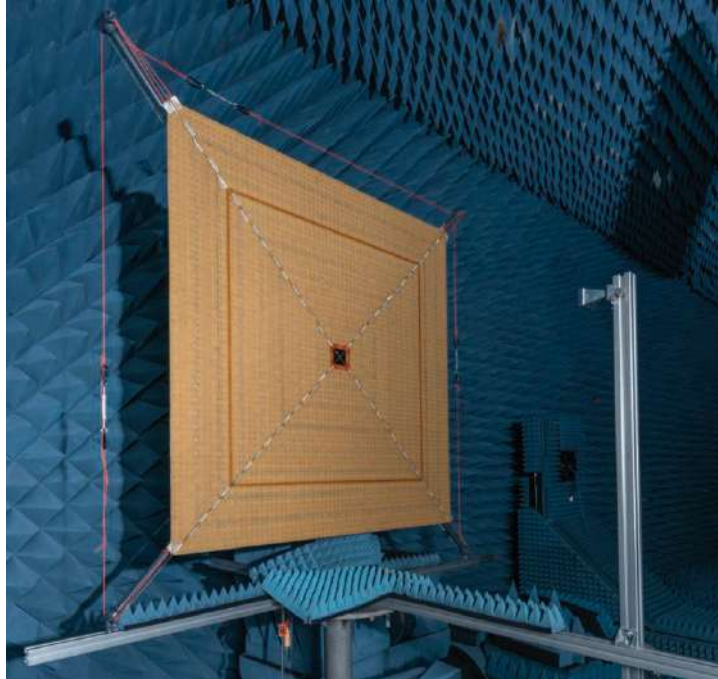


Figure 11: Membrane reflectarray undergoing development (Chahat et al. 2019).

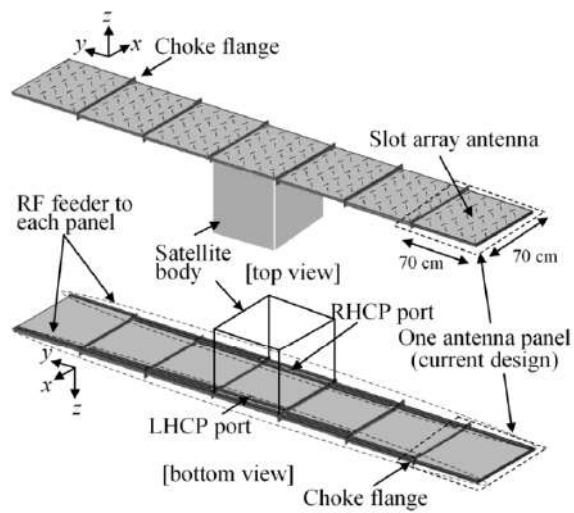


Figure 12: Slot array antenna design for 100 kg small satellite mission (Akbar et al. 2015).

Downlink Parameter		Units	Value
MarCO Tx.	Transmitter Power	W	4.6
	Passive/Pointing Losses	dB	-4.2
	Transmit Antenna Gain (HGA)	dBi	28.8
	EIRP	dBm	61.2
Path	Atmospheric Attenuation	dB	-0.15
	Free Space Path Loss	dB	-275.0
Receiving Station	DSS-63, 70m Antenna Gain	dBi	74.3
	Antenna/Passive Losses	dB	-0.3
	Total Received Power (Pt)	dBm	-139.8
	System Noise Temp	K	25.5
	Noise Spectral Density (No)	dBm/Hz	-184.5
	Received Pt/No	dB-Hz	44.7
Data Channel	Data Supp. (mod index = 77°)	dB	-0.23
	Data Rate (8000 bps)	dB-bps	39.0
	System Losses	dB	-0.8
	Th. Eb/No (Turbo-1/6, FER=1e-4)	dB	-0.1
	Req'd Pt/No	dB-Hz	40.0
Downlink Margin		dB	4.7

Figure 13: Table of MarCO Communication System's Downlink Parameters (Kobayashi 2017).

A significant challenge was at the 'tail end' of the link budget calculation where the received power was to be developed into an estimate for the downlink data rate. I have explored various methods to accomplish this. Initially, I attempted to use Shannon's Theorem to obtain a prediction of the maximum data rate theoretically achievable by a system, however, this involved the application of the system's bandwidth which could not be found in literature. As an alternative I decided to use link margin calculations which defines the link quality for this configuration clearly demonstrating at which segments in the orbit the quality of the downlink is adequate to achieve a selected data rate. My initial link budget code was implemented using linear values; however, it became apparent that using log space decibel values for link budgeting matched convention and so was more compatible with values available in literature (TutorialsWeb 2021) (NASA 2022) (Centre 2022).

Once I had successfully obtained the same output values produced in the MarCO literature 13 I was tasked with estimating if a similarly successful communication system could be achieved using different ground stations to those used for the MarCO mission which were the large 70m Deep Space Network antennas operated by NASA, these antennas are expensive to operate and are under high demand and so I also investigated the results achieved from using 35 m and 13 m ESA operated Estrack antennas.

1. EIRP

The Effective Isotropic Radiated Power is a measure of the 'output' of a radio communication system and is calculated from the sum of the power supplied to the transmitter (dBm) and the gain of the antenna (dBi), both in decibel units (dB).

$$EIRP = P_{Tx} + G_{Tx} - P_L \quad (2)$$

2. FSPL

The Free Space Path Loss is the greatest loss typically experienced by radio communication systems. This is the loss experienced by an electromagnetic wave as it propagates through a vacuum. The magnitude of this loss is dependent on the frequency of the transmission, f , and the distance which the wave travels, r .

$$FSPL = \left(\frac{4\pi r}{\lambda} \right)^2 = \left(\frac{4\pi f r}{c} \right)^2 \quad (3)$$

3. Received Power

The received power is the signal power that reaches and is 'captured' by the receiving antenna. There are three terms in the calculation of received power, these are the Effective Isotropic Radiated Power of the transmitter, the free space path loss experienced by the system and the gain of the receiving antenna.

$$P_{Rx} = EIRP - FSPL + G_{Rx} \quad (4)$$

4. Carrier/Noise Ratio

The carrier to noise ratio describes the ratio between the received power of a system and the noise power of a system. The terms of this calculation are the EIRP, FSPL, the gain to noise ratio, G/T of the receiver and the Boltzmann constant, k .

$$C/N_o = EIRP - FSPL + G/T_{Rx} - k \quad (5)$$

$$G/T_{Rx} = G_{Rx} - T_{sys} \quad (6)$$

5. E_b/N_o Available

The available energy per bit to noise power ratio is the carrier power to noise power ratio normalised for the input data rate, BR , of the system. In the same way as the other quantities, the bit rate in bits per second is converted to decibels.

$$\text{Available } E_b/N_o = C/N_o - BR \quad (7)$$

6. Available Link Margin

The available link margin of a system can be calculated from the difference between the available E_b/N_o and the E_b/N_o required for a specified Bit Error Rate, BER. For these calculations the required E_b/N_o is 0.1 dB as used by the MarCO communications system downlink which used a 1/6 Turbo code (Kobayashi 2017). The available link margin represents the additional attenuation that can be experienced by the system before the system would cease to function. A link margin of zero would indicate a barely working system where any additional attenuation would render the system inoperable.

$$\text{Link Margin} = \text{Available } E_b/N_o - \text{Required } E_b/N_o \quad (8)$$

6 Communication System Performance Estimation: Results and Discussion

6.1 Variation Due to Orbit

As mentioned in section 5 the distance between spacecraft and ground station is a factor that contributes to the data rate capable of the spacecraft. The SURROUND mission will involve spacecraft situated at several different positions, the purpose of which is to provide trilateration functionality. There will be a spacecraft situated at L1, 0.01 AU from Earth. There will also be individual spacecraft situated at both L4 and L5 Lagrange points which can be assumed to have a constant distance from Earth. Additionally, there will be an 'Ahead' and a 'Behind' spacecraft which will operate in a heliocentric orbit. The advantage of this configuration is that these spacecraft can provide views of the far side of the sun thus providing a more advanced warning of a space weather event that will appear as the Sun rotates on its axis. Spacecraft in the ahead and behind configurations will vary in distance to Earth and so the free space path loss will also vary.

I began the link budget calculation using the single distance of 1 AU which is both the distance which MarCO operated across and the distance to L4 and L5 from Earth. Once I had set up this scenario, I contained this code as a function where one of the input variables is the distance between transmitter and receiver. Initially, to model the ahead and behind downlink rates I generated orbital coordinates for Earth and the ahead and behind satellites by using circular orbits of radii 1AU, 0.9AU and 1.1AU respectively. I chose this configuration as by the laws of orbital mechanics the period is proportional to the radius of the orbit. At each time interval the distance between the positions of the satellites and Earth was calculated with this value then fed into the link budget code thus determining the value of free space path loss experienced by the signal. To generate predicted downlink rates the link budget python code was combined with an orbit position code whereby the downlink could be calculated across a given orbit.

The plotted circular orbital configuration plotted for three quarters of a year is seen in Figure 14. Although the spacecraft start along the Sun-Earth line these spacecraft gradually diverge as the ahead spacecraft at the smaller radius orbit has a faster orbital velocity compared to the Earth and the behind spacecraft at the greater radius orbit has a slower orbital velocity. Figure 15 displays the Earth-spacecraft distances across a longer time scale. The maxima and minima of the ahead and behind Earth-spacecraft distances occur at different times during the mission lifetime and the phase between the two spacecraft changes throughout the mission as the distance between the spacecraft and Earth also changes at different rates.

It was suggested that I should model the potential SURROUND ahead and behind orbits as the same orbital configuration as the STEREO mission's STEREO-A and STEREO-B spacecraft. To model these orbits, I made use of NASA's SPICE kernel system in which, amongst other information, spacecraft positions are archived (Semenov & Baalke 2022), code provided by Alberto Canizares facilitated the retrieval of this information. Using this system I collected the historical and projected positions of STEREO-A and B in the HEE coordinate system for both the lifetime of the STEREO Mission since its launch in October 2006 and into 2023.

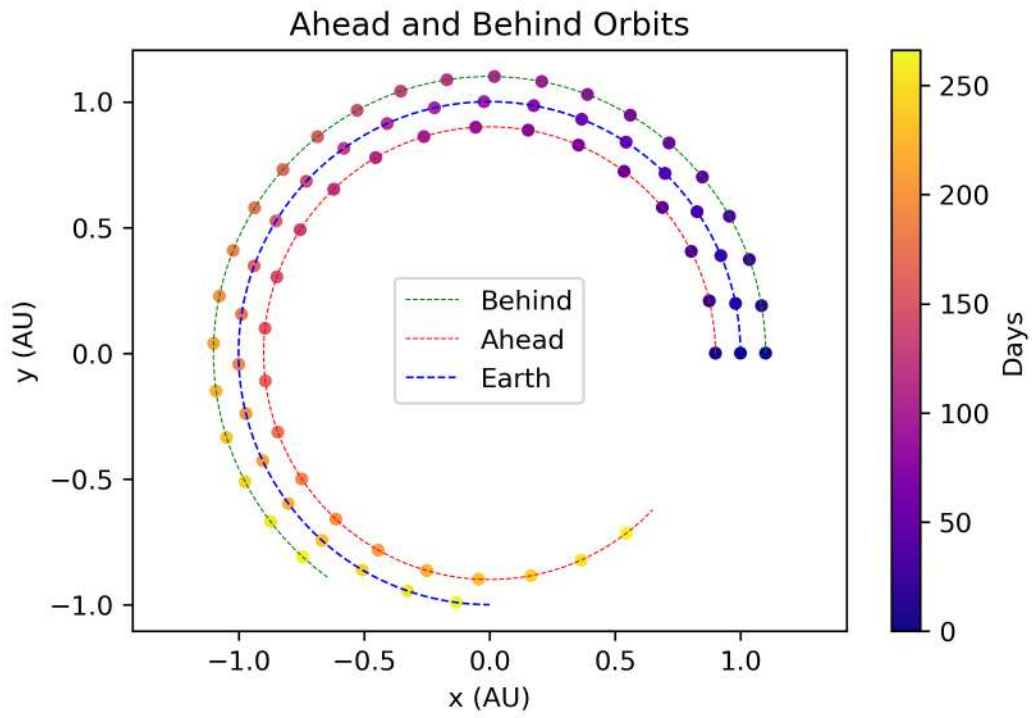


Figure 14: First implementation of ahead and behind spacecraft as circular orbits.

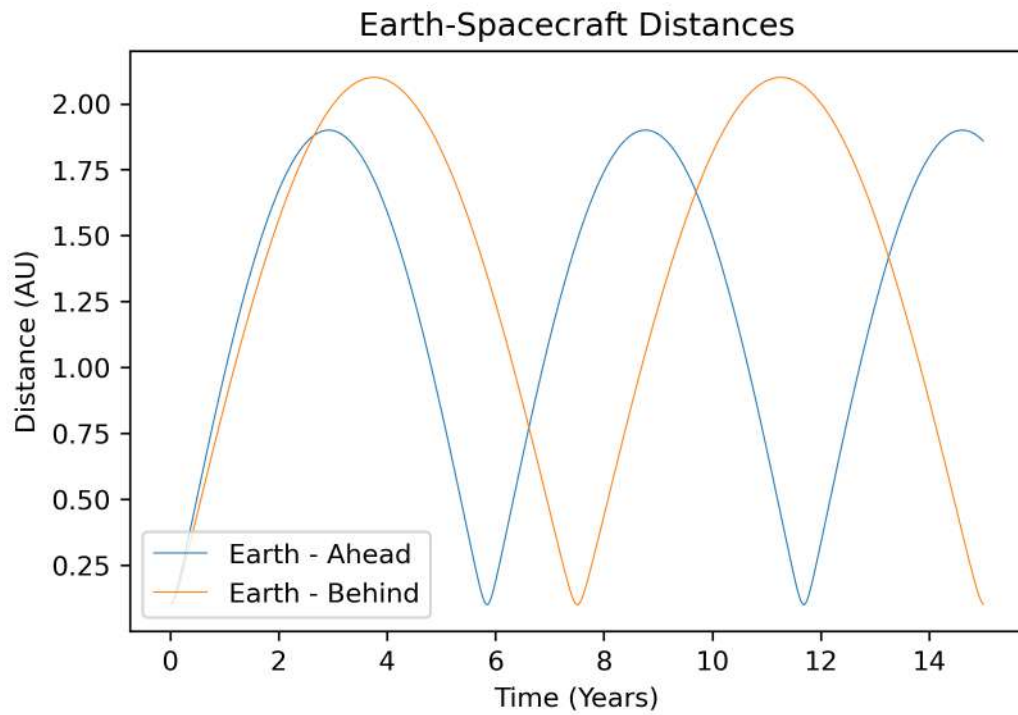


Figure 15: Circular orbit Earth-spacecraft distance progression.

Spacecraft Location	Earth - Spacecraft Distance (AU)
L1	0.01
L4 & L5	1.00
Ahead	0.01 - 1.966
Behind	0.01 - 2.085

Table 1: Earth-Spacecraft distances for the various SURROUND spacecraft positions.

STEREO Spacecraft Coordinates: 27 / 10 / 2006 - 7 / 12 / 2023

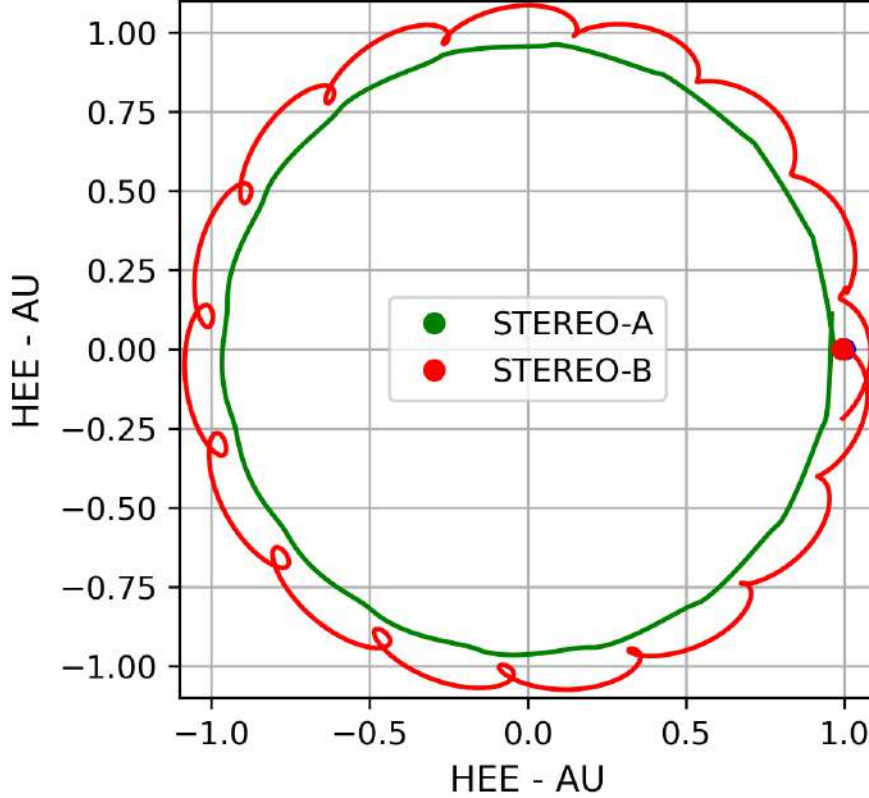


Figure 16: STEREO A and B positions in HEE coordinate system from launch date.

For each entry in these data sets I calculated the Earth-spacecraft distance for the ahead and behind craft using trigonometry, these distances are plotted in Figure 17. As expected, the STEREO spacecraft began their mission close to/at Earth and travelled further away from Earth over time at similar rates. A key difference between the distances of this configuration and that of the simple circular orbits is that the period of the change in distance for both spacecraft is $\tilde{17}$ years whereas in the simple orbit the period for the ahead spacecraft is $\tilde{6}$ years and for the behind spacecraft, $\tilde{8}$ years. In both cases, however, this period is large, meaning that for a significant time these spacecraft will be over 1 AU from Earth.

The impact that these distances have on the Link Margin is demonstrated in Figures 18 and 19 for a spacecraft using the same power and configuration of the MarCO spacecraft using a 70m DSN antenna to downlink at a data rate of 10kb/s. It can be seen that as this distance increases the downlink margin drops meaning that the middle portion of the mission will be most challenging for the ahead and behind spacecraft. For the other

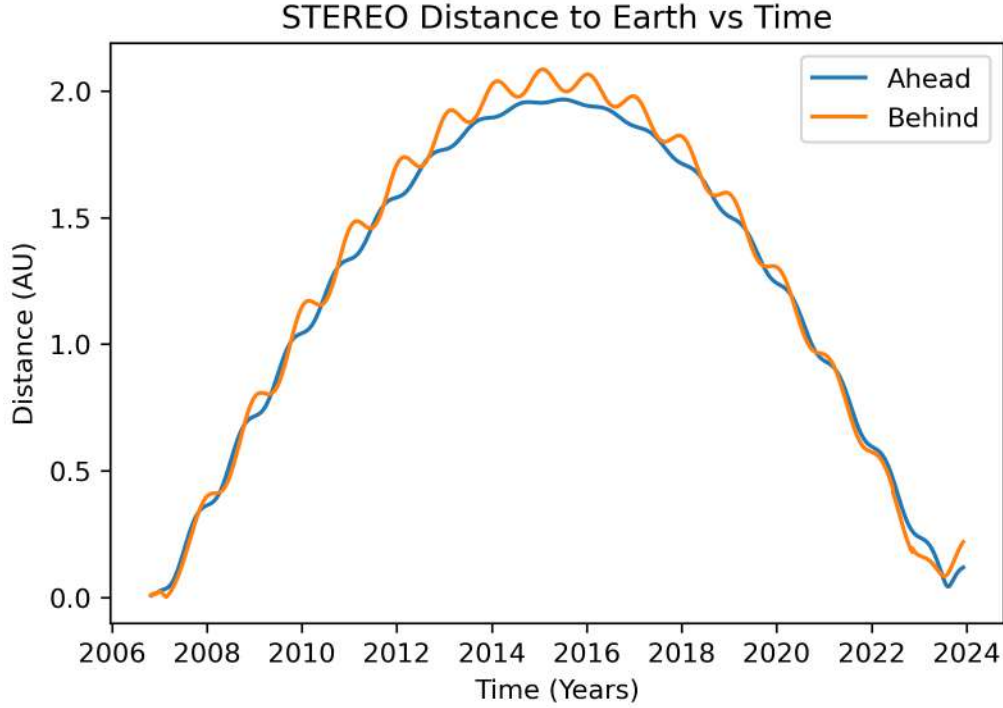


Figure 17: STEREO A and B Variation in Distance to Earth.

spacecraft at L4 and L5, their fixed distance of 1 AU results in a reduced margin however this is not as severe as the reduction at the maximum distances experienced by the ahead and behind spacecraft. The spacecraft positioned at L1 will be the easiest spacecraft to achieve data from in large volumes thank to it relatively small distance.

6.2 Spacecraft Power

Figure 20 shows the relationship between the power supplied to the spacecraft's transmitter and the downlink margin. An increase in power provided to the spacecraft antenna increases the downlink margin. The MarCO spacecraft when operating in Martian orbit was capable of providing 4.6 Watts to its antenna, power generated from MarCO's pair of 30x30 cm solar arrays. Although MarCO operated at 1 AU from Earth and so experienced the same free space path loss as a spacecraft operating at L4 or L5, the SURROUND spacecraft will also be 1 AU from the sun whereas MarCO operated at a greater distance; this is also the case for the ahead and behind spacecraft as although their distance to Earth will reach ~ 2 AU they are in a heliocentric orbit where their distance to the Sun is approximately 1 AU. Therefore, if the 6U SURROUND CubeSats are equipped with the same power system as the 6U MarCO spacecraft, they will be able to generate more power, potentially increasing the antenna feed power. In the literature describing MarCO's systems (JPL 2018) it is stated that MarCO, operating near Earth, would generate approximately 35 Watts over double that it is capable of generating near Mars, 17 Watts. By scaling the antenna feed power near Mars orbit with the increase in power generation a theoretical feed power of approximately 9.5 Watts could be reached. As can be seen in Figure 20 this increase in power has the potential to improve the link margin by a worthwhile amount. The case featured in this figure is a spacecraft operating at either L4 or L5 using the MarCO communication and power systems, attempting to

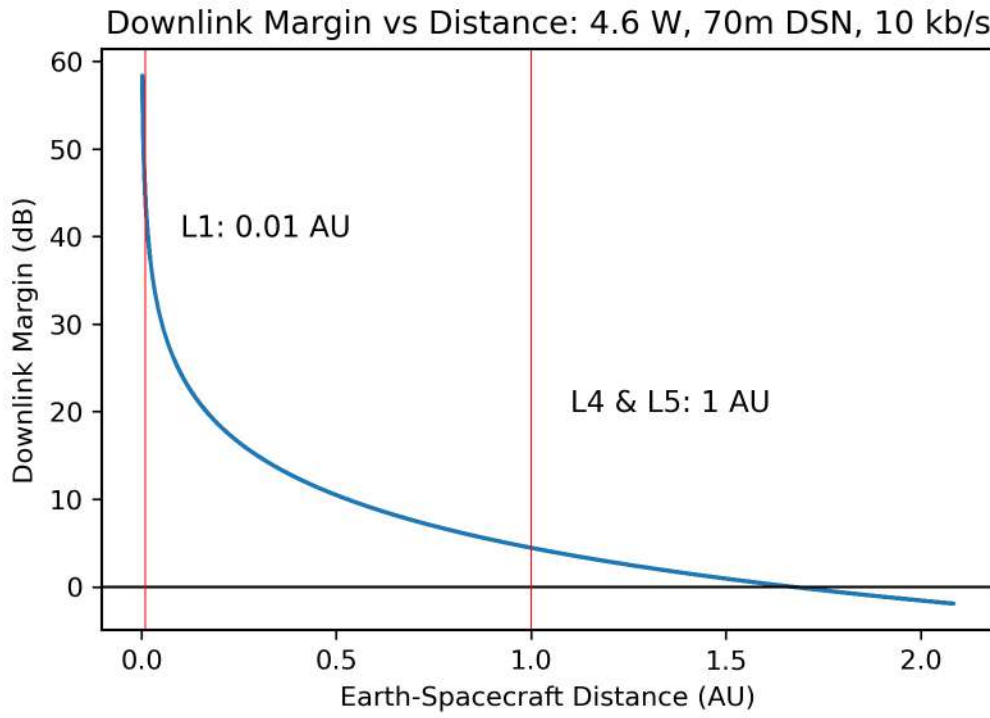


Figure 18: Relationship between link margin and transmitter-receiver distance for selected spacecraft configuration and link regime.

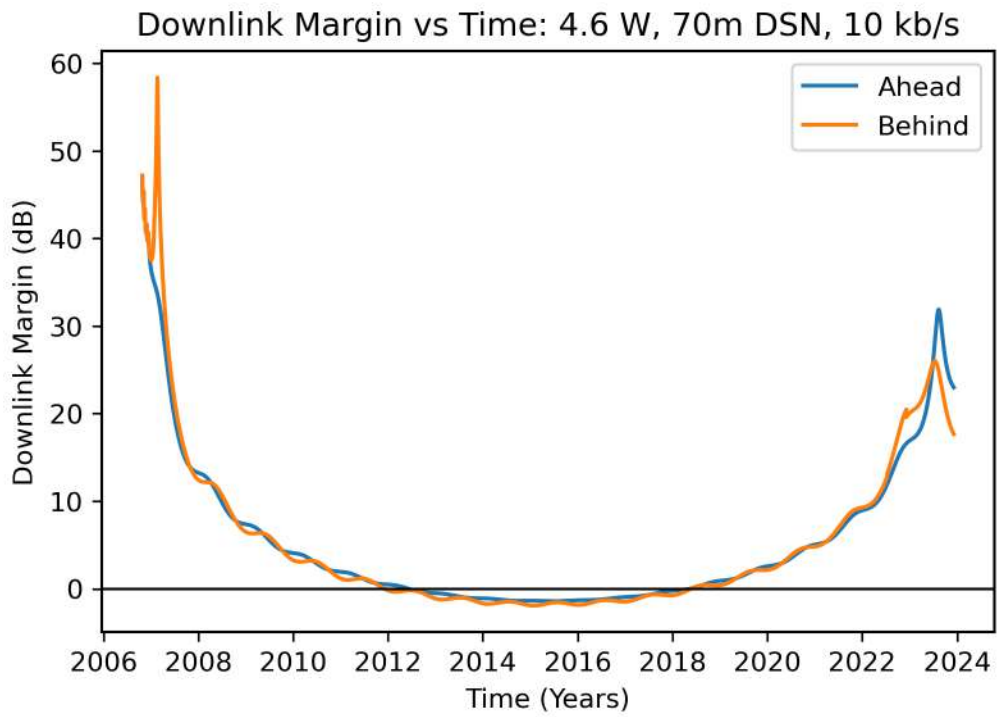


Figure 19: Prospective link margin across STEREO orbit for selected spacecraft configuration and link regime.

Transmitter Power vs Downlink Margin: 13m ESTRACK, 100bit/s, 1AU

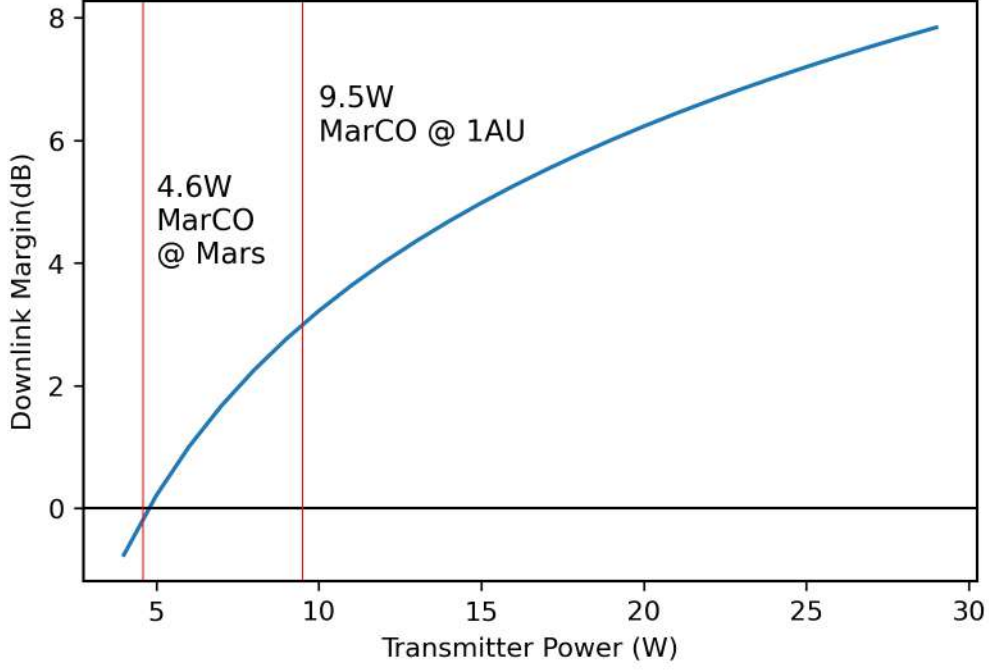


Figure 20: Relationship of the downlink margin to the power provided to the spacecraft’s transmitter.

downlink at a rate of 100 bit/s to the smallest ground station examined, a 13 m Estrack antenna. This increase in power takes the link quality from inadequate to a margin of 3dB.

6.3 Ground Segment

An important component of spacecraft communication is the ground station made available for a mission to downlink its data to. Ground stations are constrained by their availability as these facilities are under high demand and their running costs. The performance of the ground station is a significant component of the link budget calculation, in particular the antenna’s gain and the system operating temperature from which the gain-to-noise temperature ratio, G/T can be derived. Figure 21 shows the relationship between receiver G/T and the resultant downlink margin for a spacecraft equipped with MarCO’s antenna at 1AU, fed with 9.5 Watts with an input data rate of 100 bit/s. It can be seen that the selection of ground station is a very important factor in providing sufficient link margin with clear benefits to using a more capable system.

Figure 22 estimates the performance of the MarCO configuration across a range of distances that will be experienced by SURROUND spacecraft including those placed in the Stereo A and B heliocentric orbits. The downlink margin has been calculated for this configuration across the distances to Earth experienced by the spacecraft. This has been done for the selection of ground stations described above with their key characteristics provided in table 2. From this table it can be seen that the larger antennas have a higher gain to noise temperature ratio, beneficial to the link quality. This is reflected in the results presented in Figures 21 and 22 where the larger antennas produce significantly

Dish Diameter (m)	G/T (dB/K)	Agency
70	60.2	NASA
35	50.8	ESA
13	35.6	ESA

Table 2: Table of Earth Ground Station Characteristics (NASA 2022) (Centre 2022)

larger link margins.

For the spacecraft positioned at L1, 0.01AU from Earth, the choice of ground station is not as critical a factor in comparison to spacecraft at greater distances, however, the L4, L5, ahead, and behind satellites are located at 1AU and so there are greater demands on the link budget due to the relationship of path loss to distance. Therefore, for these spacecraft ground stations that have greater gain and low system noise temperatures are more important to enable high link quality. These systems are part of deep space networks which are operated by a number of space agencies including ESA and NASA and are distributed globally to enable wide, inclusive coverage. The largest antennas operating in this role are the 70 metre NASA DSN antennas and while these can be made available to ESA, the demand and operating costs of these antenna are high. ESA’s Deep Space Network, Estrack, is comprised of a series of 34 and 13 metre dishes. The demand on this network is also very high, therefore, it is necessary to formulate a downlink regime to minimise the time required to downlink the required data.

6.4 Communication Comparison

Figure 23 brings together the previous discussed factors to demonstrate the various combinations that could be selected for SURROUND spacecraft communication. Column-wise the matrix varies which of the examined ground stations is used while each row applies a different supplied power to the antenna feed. Each plot displays the margin-distance curve for a selection of bit rates so that the potential of each set-up can be visualised.

The lower left plot displays the resultant link margin when applying the lowest featured power of 4.6 Watts and using the smallest deep space antenna, the 13 m Estrack. It can therefore be seen that using this limited configuration would enable a spacecraft operating at L1 to comfortably downlink data at a rate of at least 10 kbit/s. However, this situation is much bleaker for those spacecraft positioned 1 AU from Earth where communication using this limited set-up is not possible, with negative estimated link margin values. This can be improved upon by increasing the power supplied to the spacecraft’s antenna to the ‘MarCO at 1 AU from the Sun’ value of 9.5 Watts, enabling 100 bit/s communication from 1 AU distances, however, this power increase and a further increase to 15 Watts are insufficient for enabling either 1000 bit/s rates for 1 AU craft or any communication from the farthest distances experienced along the STEREO orbits. Therefore, it can be concluded that to achieve higher data rates than 100 bit/s from L4 and L5 spacecraft an antenna with a greater G/T ratio should be used.

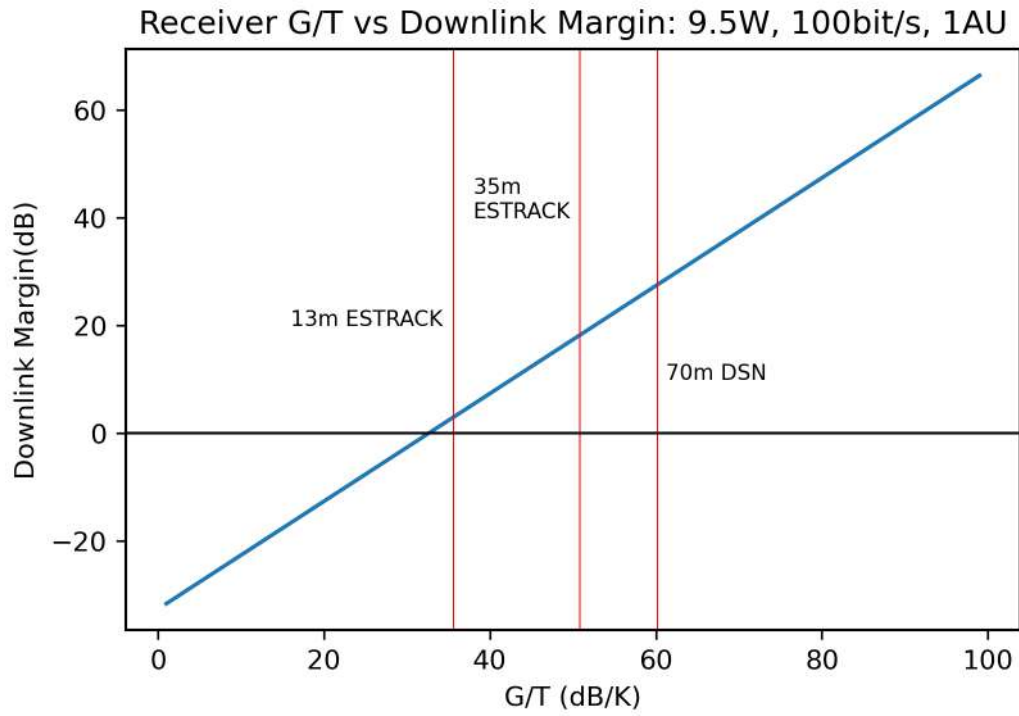


Figure 21: Link Margin variation with Ground Station G/T at a transmitter-receiver distance of 1 AU.

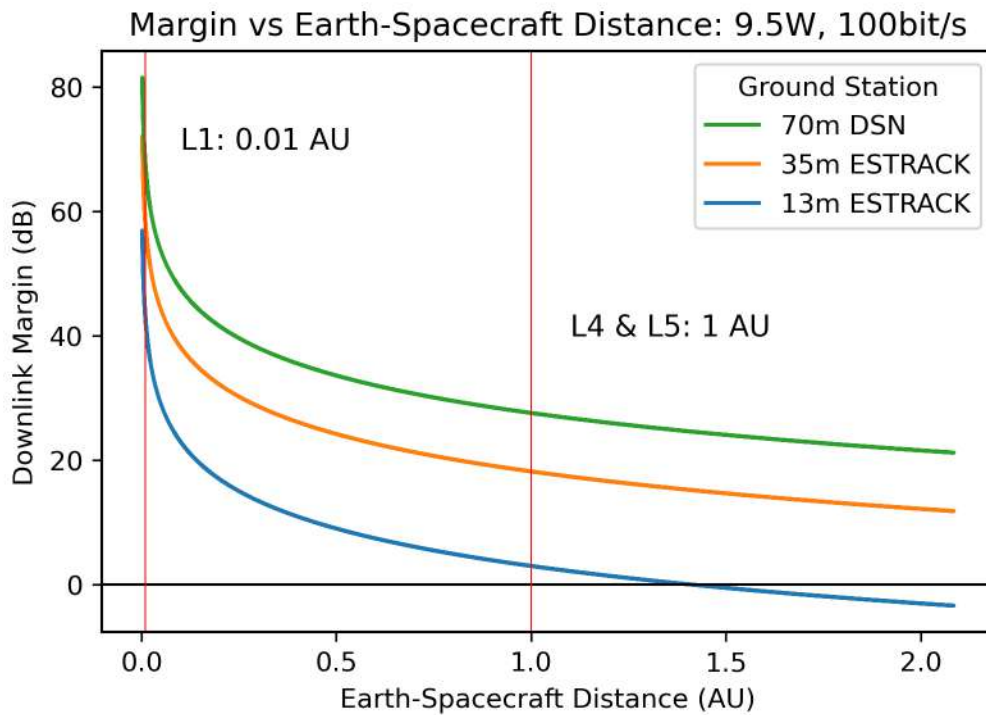


Figure 22: Link Margin variation with Earth-Spacecraft Distance for selection of Ground Station G/Ts.

The central column of this matrix displays the estimated link margins when using 35 m Estrack antennas. When using a low power of 4.6 Watts, a data rate of 100 bit/s is possible across all Earth-spacecraft distances to be experienced by SURROUND mission spacecraft, increasing the spacecraft's antenna feed power will increase the margin of this signal further. At this low power it is also possible to achieve a data rate of 1000 bit/s out to distances past 1 AU although not out to the greatest distances. However, with an increase in power to 9.5 Watts the link margin for a 1000 bit/s signal is lifted into positive values across all distances, although at the extremities the margin is relatively small, potentially risking interruption from additional sources of attenuation. A data rate of 10 kbit/s is not reliably achievable at 1 AU, even when using a feed power of 15 Watts, as this configuration's link margin would only be slightly positive, a barely working signal.

The rightmost column features plots of margin-distance curves for configurations using the most capable of the antennas, NASA's 70m DSN. For all of the power levels tested in this estimation the resultant link margins are an improvement over previous configurations. At the maximum distance of approximately 2 AU, downlink at rates of 100 bit/s and 1000 bit/s are readily accomplished, however, achieving and maintaining a 10 kbit/s downlink is challenging at the two larger powers at this distance as the resultant margin is relatively close to zero and not possible for the lowest power which results in a negative link margin. At 1 AU the link margin for the three data rates is positive, therefore it can be concluded that if 10 kbit/s is required to downlink SURROUND data this is likely to only be possible if a 70m DSN antenna, or equivalent, is available.

The radio spectrometer payload of the SURROUND project is assumed to sample 128 channels at 8 bits per channel every 10 seconds. This results in a data collection rate of 102.40 bits per second and a resultant daily data volume of 8,847,360 bits. The transfer windows for the examined data rates are shown in table 3. Therefore, a clear trade-off can be seen. Using an ground station-power combination that only offers 100 bit/s results in a lengthy required transfer window and although these antenna will have a lower operational cost and less competitive scheduling this long transfer window may offset these benefits. Meanwhile, using a ground station-power combination that offers 10,000 bit/s data rates enables a much more compact transfer window and although the required ground station will be costly and under high demand, this low transfer window time could possibly be accommodated. The 13 metre Estrack antennas would only be capable of applying the large transfer window to spacecraft at distances of 1 AU while the 35 metre Estrack antennas should be capable of enabling 1000 bit/s communication across all SURROUND distances, however, these antennas will not be able to achieve a higher 10,000 bit/s downlink, only made possible using a 70 metre DSN. To resolve this trade-off, the scheduling of these various networks needs to be assessed and the costs per unit time of using these systems. Only then can a final judgement be made on what deep space capable antenna can be used for the SURROUND mission.

7 Spacecraft Preliminary Design

At the phase 0 stage of SURROUND it is important that a clear vision of the project is conveyed to ensure confidence in the mission. A component of this is formulating profes-

Communication Matrix

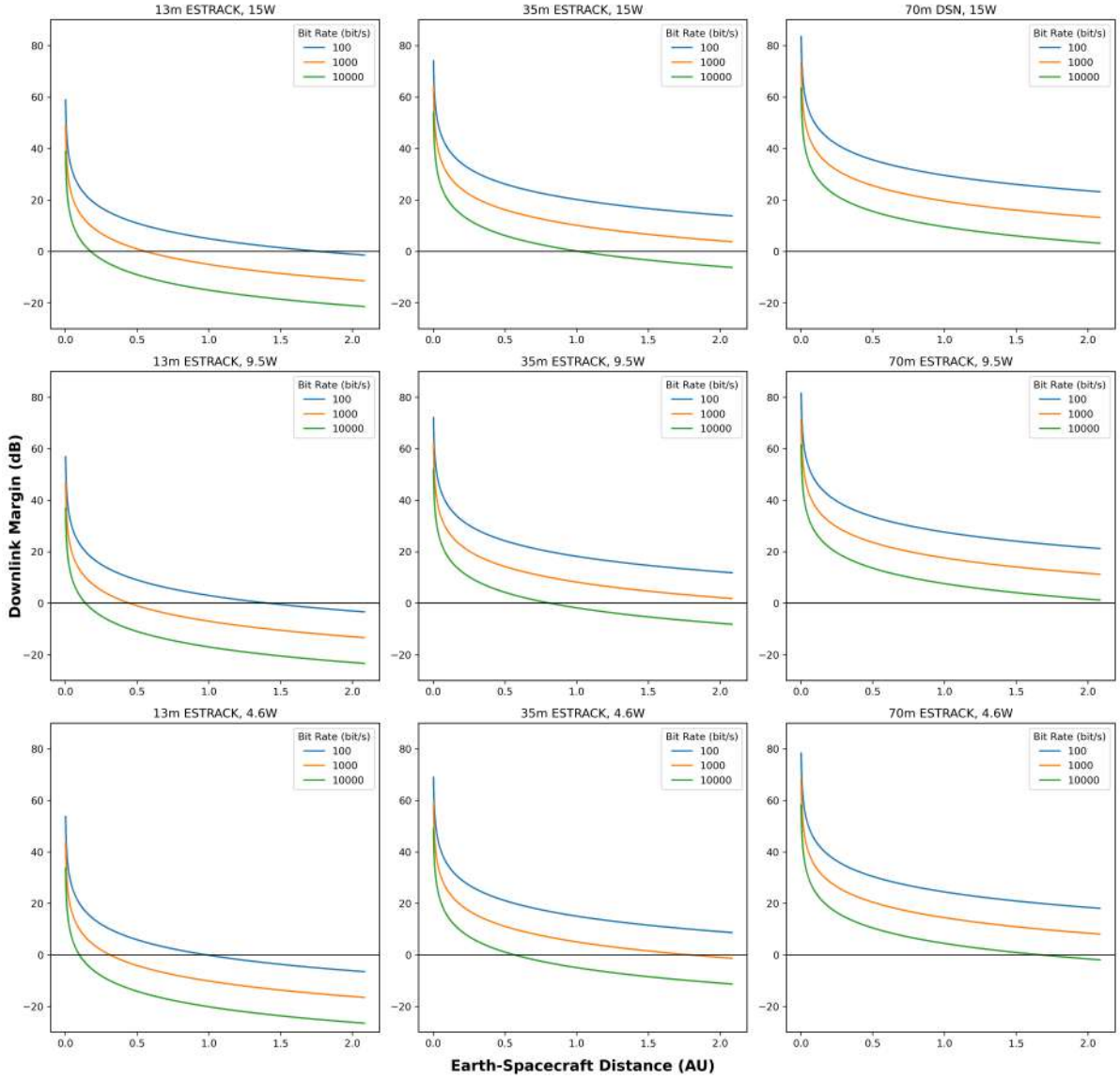


Figure 23: Matrix of estimated link margins for SURROUND spacecraft across the Earth-spacecraft distances involved in the SURROUND constellation for 13m, 35m and 70m ground stations and antenna feed powers of 4.6W, 9.5W and 15W.

Downlink Rate (bit/s)	Transfer Window (mins)
100	1474.56
1000	147.46
10000	14.75

Table 3: Table of transfer windows required to downlink daily collected radio spectrometer data at different downlink rates.

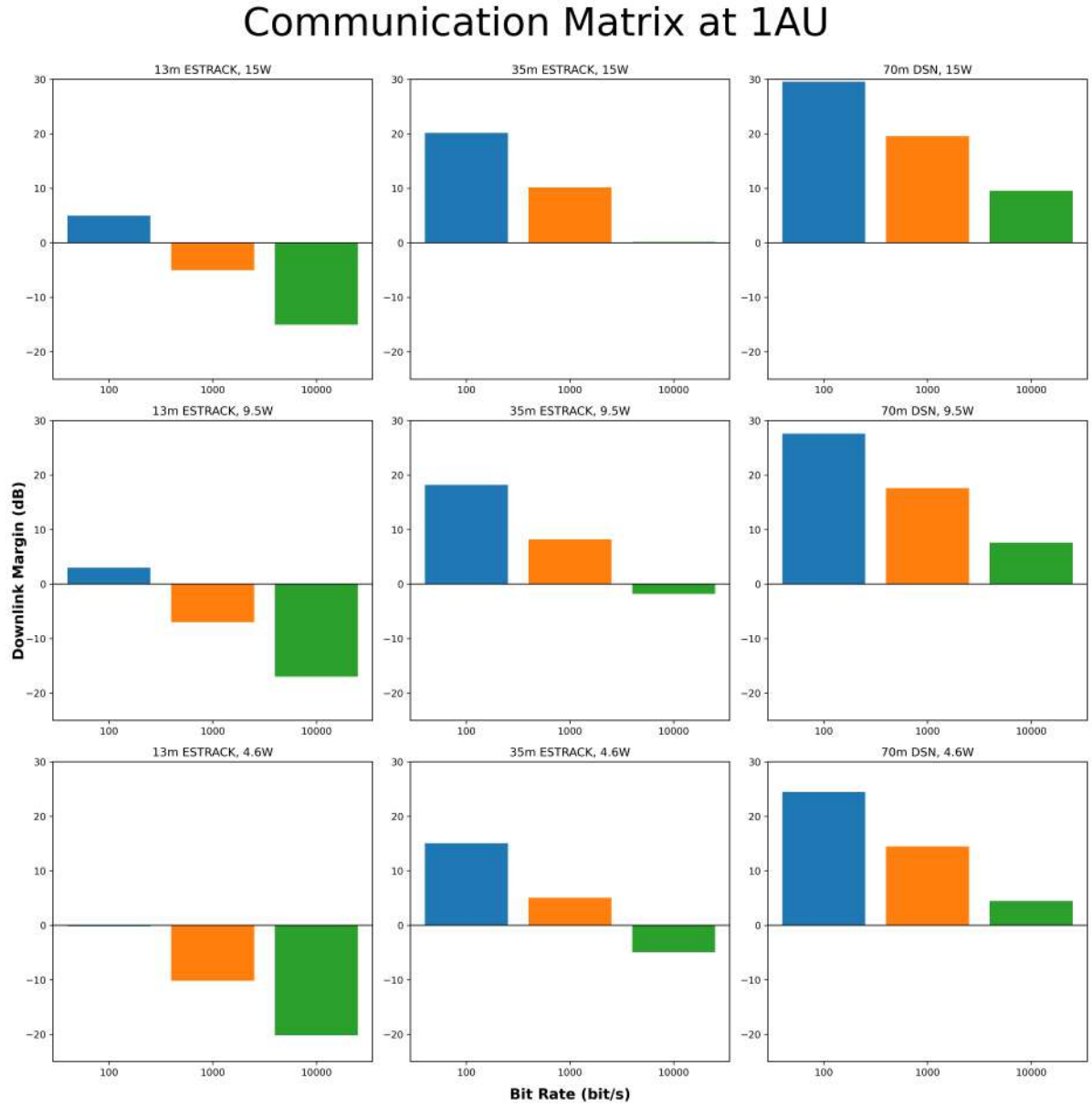


Figure 24: Matrix of estimated link margins for SURROUND spacecraft at 1 AU for 13m, 35m and 70m ground stations and antenna feed powers of 4.6W, 9.5W and 15W.

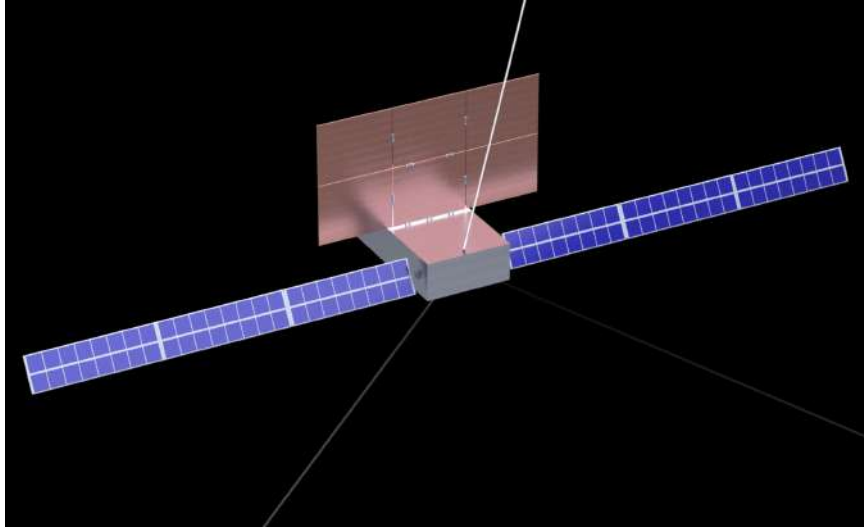


Figure 25: SURROUND spacecraft preliminary design.

sional graphics to accompany scientific and operational components of the project. To accomplish this, I produced a preliminary yet reasoned spacecraft design that is somewhere between an artist's impression and an engineering design.

To produce a reasonable, realistic design the generated representation includes the key, externally visible components of a spacecraft. These include the power, communication, propulsion and payload sub-systems. In addition to the inclusion of these component the placement of these systems must be considered so that they are appropriate to the mission context. These components must also be included in such a way that allows for the deployment of the spacecraft from a stowed configuration which involves the changing of component geometry, for example, solar panel unfolding and pointing. With regards to this element of the design, the relative placement of the large reflectarray and solar panels was an important factor to ensure that these components do not collide during deployment or operation.

As the SURROUND mission will use the 6U CubeSat form factor, I used the successful interplanetary 6U MarCO spacecraft as inspiration for this design (Hodges et al. 2017) (Kobayashi 2017) (JPL 2018). Initially, power generation system consisting of solar panels were to be mounted in the style of the MarCO CubeSat where arrays mounted on either side consist of two outer panels joined and folded out from the sides of a central panel that is connected to the main spacecraft body. However, this configuration when implemented for SURROUND would not be compatible with the placement of the deployed reflectarray.

The reason for this is that at an L1 position the solar panels and reflectarray do not face the same direction as would be the case in a spacecraft located at Mars where both the Earth and the Sun are visible from the same orientation. Therefore, for this L1 spacecraft concept, the reflectarray is pointed in the opposite direction to the solar panels which also determines the location of the antenna feed. A reflectarray consists of set of panels that fold up and out from the spacecraft into an array (Hodges et al. 2017), and as the name suggests, this array reflects the transmission signal which is provided by the antenna feed. For this system to function the reflectarray's feed must be located on the side of the spacecraft facing the receiver so that the signal is directed at the reflectarray

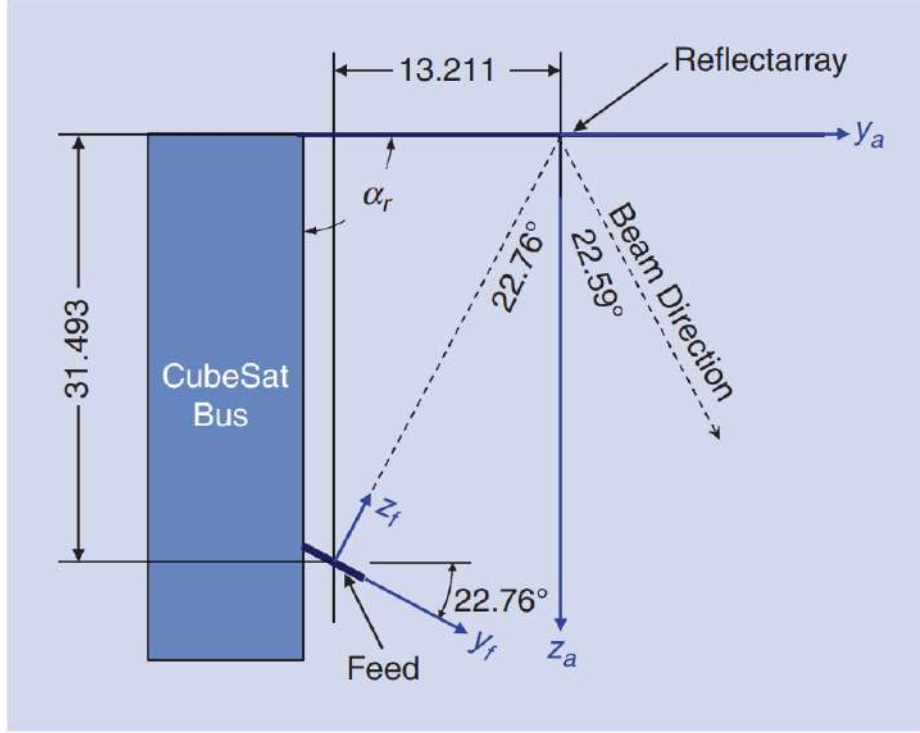


Figure 26: MarCO reflectarray antenna design (Hodges et al. 2017).

and then reflected back towards Earth, as seen in Figure 26. This necessary placement precludes the positioning of the solar arrays or the payload radio spectrometer dipole antennas on the same side of the spacecraft as the antenna feed, the Earth facing side of the spacecraft. The solar panels are also placed at the Sun facing side of the spacecraft to avoid shadows caused by the spacecraft body or other components.

Although the antenna feed is on the opposite side of the spacecraft to the solar panels the reflectarray is located towards the centre of the craft so that the reflectarray design can function but also to allow space for the solar panels to rotate. To ensure no collision could occur between the solar arrays and reflectarray the square array configuration found on MarCO was rearranged so that the root panel supported a second panel which in turn supported an outer panel creating a strip of solar cells. This configuration reduces the radius swept by the rotation of the solar panels when pointing, thus avoiding contact with the reflectarray.

On the underside of the model a representation of an electric thruster is found. However, the propulsion system of SURROUND is yet to be determined with the two methods under investigation consisting of electric propulsion and solar sails.

7.1 CAD Modelling

The design concept was developed into a CAD model using Dassault Systèmes Solidworks package. The first stage of producing the deliverables of this work package was to produce a static model that featured all necessary elements to represent an early concept for a SURROUND spacecraft. A key element of this process was to accommodate the

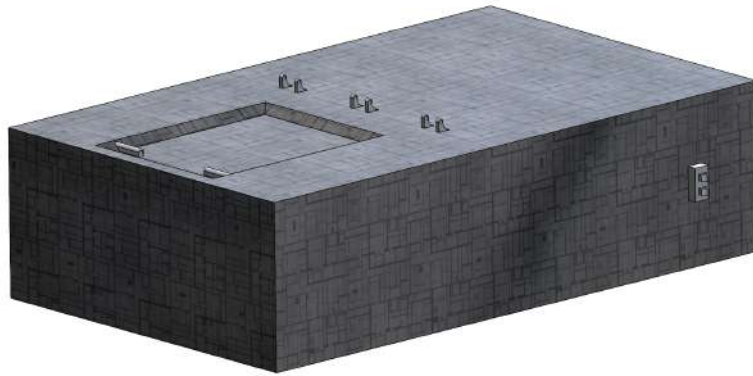


Figure 27: SURROUND spacecraft concept body.

adjustment of the spacecraft configuration from stowed to deployed both for the purpose of achieving realism and to prepare for the next work package which was to produce an animation of the spacecraft deployment.

The first step towards producing the model was to define the central spacecraft body, that of a 6U CubeSat. Although the overall form factor of this component is a simple rectangular box there are numerous additional features to accommodate the other components of the spacecraft. The most prolific of these features are the various connection points. On the top surface of the spacecraft body near the centre there are three extruding supports, Figure 27, that provide anchorage and pivot axis to the deployable reflectarray. Additionally, on this surface there is a recess within which the reflectarray feed is stowed and deployed from. On either side of the spacecraft there is a mounting point by which the root of the solar panel assembly is connected to the spacecraft. This is done via a pivoting T-shaped hinge that permits the assembly to rotate from a parallel alignment along the spacecraft sides to a perpendicular position. The root panel of the solar array connects via socket to the T-shaped hinge this enables the array to rotate about this axis to adjust the pointing of the solar arrays towards the Sun. The solar arrays on either side of the spacecraft are each composed of three panels: the root panel, a central panel and an outer panel. These panels are connected together via hinges, two across each gap. The construction of these hinges involves identical brackets on the edge of each panel and a joining part which is connected through both brackets, this connecting part defines the gap distance between panels in the deployed position and importantly results in an offset between the panels in the folder configuration, preventing the surfaces of the panels from touching when folded. These arrays are stowed by in a 'z' or 'concertina' fold and deploy in this manner.

The most complex structure in the design is the reflectarray and several different CAD techniques were used to create this system. Due to the decrease in available surface area at the top of the spacecraft body, the SURROUND reflectarray must be folded to a greater degree compared to the inspiring MarCO design. Therefore, instead of a design where the array is split into left, centre and right and where the left and right panels fold to the front or back of the central panel, the SURROUND reflectarray is split into a 3x2 array. This complicates the stowed configuration and by extension the deployment process. The top set of panels and bottom set of panels folded to the central panel in

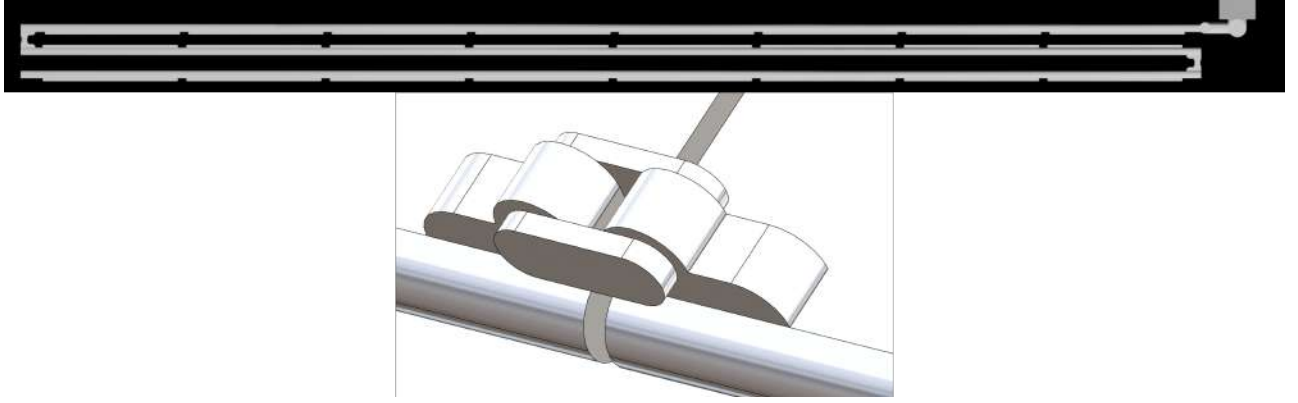


Figure 28: SURROUND spacecraft concept folded solar array (top), inter-panel hinge (bottom).

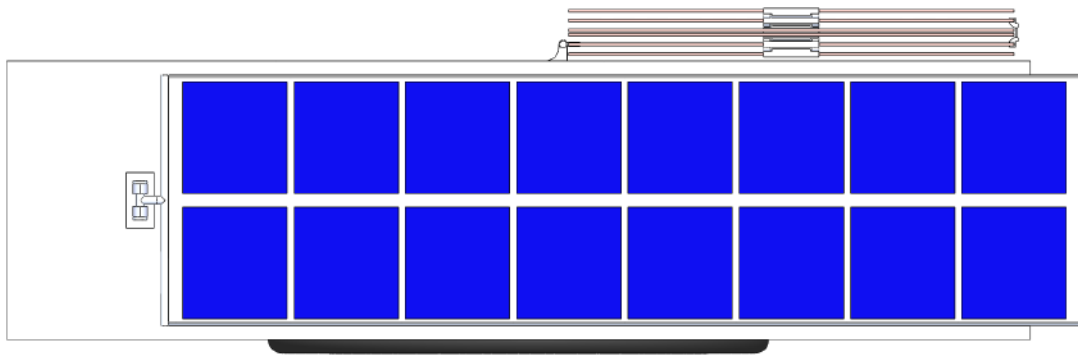


Figure 29: SURROUND Stowed Configuration Side View

each set forming a 1x2 profile, for each set this was done in the same way as the MarCO reflectarray where one of the side panels would fold to either the back or to the front of the central panel and the opposite panel would fold to the unoccupied side of the central piece. To fold this 1x2 pattern to a 1x1 pattern the top panels must be folded down onto the folded lower panels. Simply rotating the upper panels downward would not be sufficient as due to the increase 'thickness' of the folded upper and lower sets these would clash and so an intermediate part is required to accommodate the pivoting of the upper set outwards and downwards to avoid colliding with the lower set. This now entirely folded reflectarray can then be pivoted downward to lie parallel to the spacecraft body.

A key visual feature of a reflectarray and a vital functional feature are the pattern of elements, the design of which is complex and determines the performance of the system, therefore this complex task was outside of the scope of my internship in this phase 0 project. None the less, this pattern is a distinguishing visual feature of such a device and so it has been included in this preliminary design. Initially I used the 'autotrace' function to obtain a Solidworks sketch from a schematic of MarCO's reflectarray, 30, which I then extruded out to create a physical part from this sketch.

However, it was judged that the reflectarray phase wrap pattern for use in the SURROUND representation should be entirely original and so a different approach was needed.

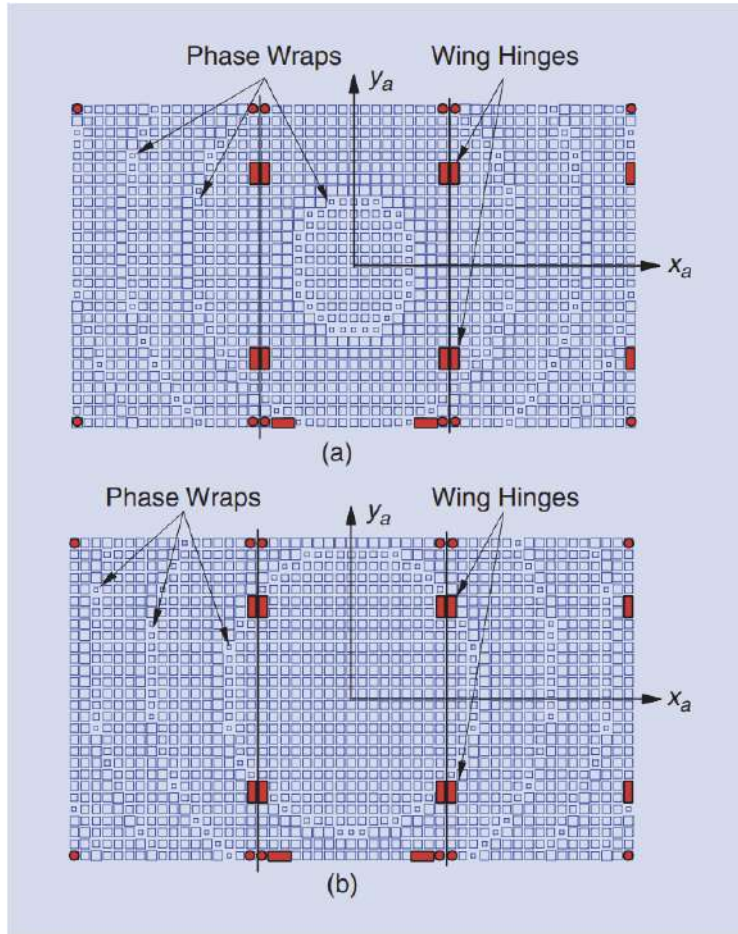


Figure 30: MarCO reflectarray designs (Hodges et al. 2017).

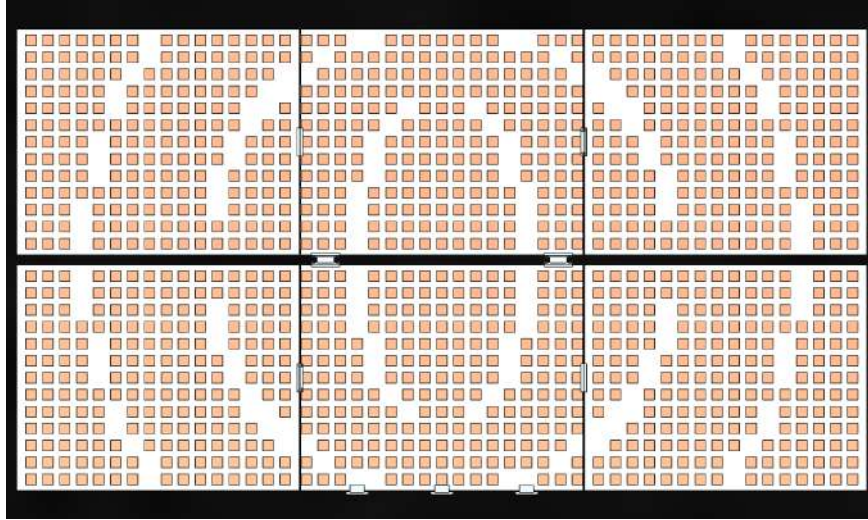


Figure 31: SURROUND Concept Reflectarray.

A reflectarray consists of an array of square elements arranged in a pattern of concentric circles/ovals. To implement a recognisable pattern I first defined the outer dimensions of the reflectarray by sketching a large rectangle. I then sketched out a series of concentric ovals of increasing radius and then used the trim tool to define the boundaries between the oval sketches. I then defined a single square offset from the centre of the reflectarray and extruded this square by an almost negligible distance to form a reflectarray element (If kept at the centre, this would have resulted in a reflectarray element in the panel gap). By selecting this single element and one of the oval boundaries the boundary can be populated by evenly spaced copies of this single element using a fill pattern. Once this had been done for the entire pattern I then sliced this pattern into the 3x2 configuration of the reflectarray panels and applied and fixed each segment to the corresponding panel 31.

7.2 Animation, Visualisation and Rendering

7.2.1 Spacecraft Visualisation

The first type of visualisation that I created of the design were static renders. Initially, I attempted to use the Solidworks Add-in 'PhotoView 360' to generate these photo-realistic images, however, I found that I could not obtain the desired reflectivity for the solar panels and so I discovered a new programme, also by Dassault Systèmes, 'Solidworks Visualise'. This rendering software produced high-quality, reflective, renders with the capability to preview the renders on lower quality before committing the computer's resources to a high quality render. In addition to static rendering there was also the capability to make the visualisations more dynamic by implementing a turntable and rendering a video which is done by generating static renders for each frame of the video and compiling these into an mp4 file format. These outputs are significantly more intensive for computers to work on and so took longer to generate. The final, and most complex of the render types that I used was the 'interactive image' type which produced a 360 degree, rotatable output that could be viewed via a web browser.

To accomplish the stated goal of producing animated representations all moving parts

must be affixed via angular mates whereby the angle between the part and a reference is defined. These mates, once created, can be loaded into a 'mate controller' whereby the angle of each mate can be specified within a particular configuration. This can be leveraged to create a sequence of configurations that start with the stowed spacecraft where all structures are in the fully folded position. The next configuration involves the adjusting of mates to change the alignment of the root solar panel to the spacecraft body. With each progressive configuration the orientation and/or position of discrete sections of the spacecraft are adjusted until a series of configurations are set up so that the final step is of the fully deployed spacecraft. These configurations are set at particular times past the start of the animation as nodes known as 'keys' along the timeline, at each key the spacecraft will match the selected configuration while in the space between the keys the positions of the parts that change between the configurations are interpolated linearly. A challenging aspect of this process was ensuring that the swivelling parts would rotate in the correct direction to get to their position set in the next configuration rather than phasing through a part by rotating the opposite direction. While this could be addressed by defining a reference location for the angular mates, it is a known issue in the Solidworks program that the angular mate type is generally unstable. I experienced this instability on numerous occasions which forced me to manually adjust the affected mates to the correct form and to restart the computer's rendering process.

While almost all of the animation mates used involved changing the orientation of components the deployment of the dipole antennas required a different approach as these structures extend rather than swivel, changing the geometry of the component. The unique approach I used to animate this stage of deployment is an advanced mate type, a path mate. This approach involves two parts, one is an extruded part to act as the root of the dipole antenna, the second is virtual guide curve. The centres of these two components are then aligned along the same axis in the spacecraft assembly and the path mate is added. In the assembly, the antenna root is edited to have a swept boss which extrudes the part outwards along the path that is taken from the guide curve. This approach enables the length of the swept boss to be controlled by the path mate by defining the percentage distance along the path that the boss will follow. Therefore, in the animation until the antennas are to be deployed the distance is set to be within the spacecraft body and then a key is set where the antenna is fully deployed and the positions between this key and the last are interpolated.

Deployment Stages

1. Fully folded spacecraft: solar panels, reflectarray, feed and dipole antennas retracted, Figure 32.
2. Folded solar panels swing perpendicular to the side of the spacecraft body, Figure 33.
3. Folded solar panels rotate 45 degrees to vertical, Figure 34.
4. Central solar panel folds out 180 degrees from root panel, Figure 35.
5. Outer solar panel folds out 180 degrees from central panel, Figure 36.
6. Folded reflectarray assembly rotates 90 degrees to upright position, Figure 37.

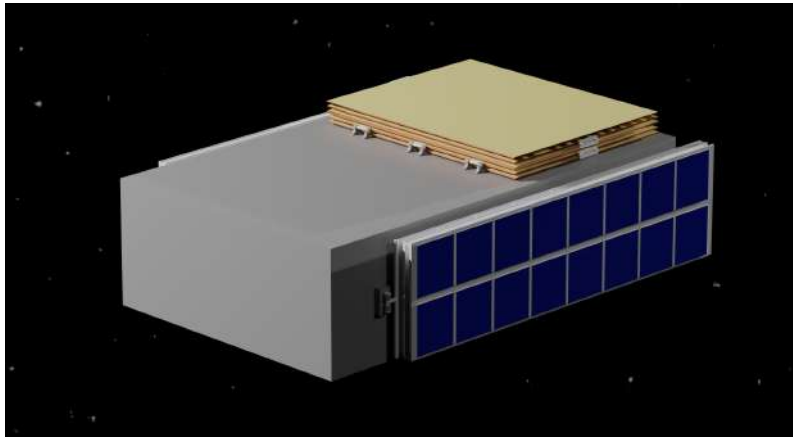


Figure 32: SURROUND Concept - Deployment Phase 0: Stowed

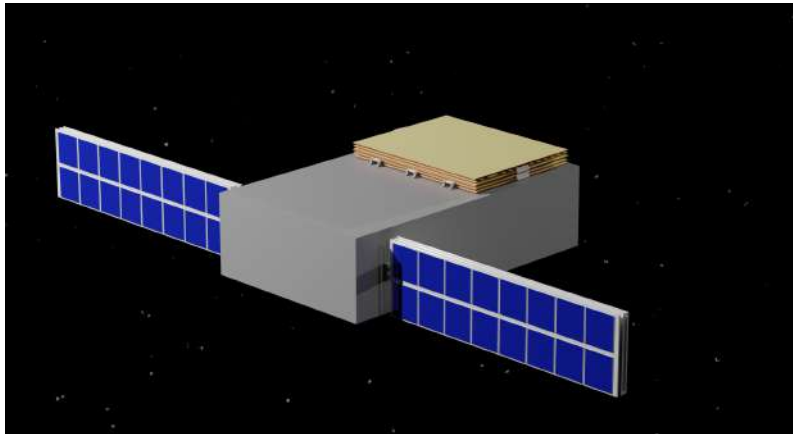


Figure 33: SURROUND Concept - Deployment Phase 1: Solar arrays swing outward.

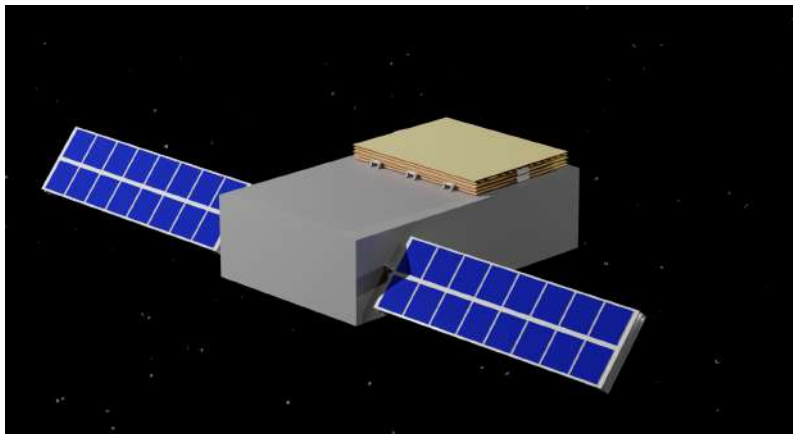


Figure 34: SURROUND Concept Deployment Phase 2: Solar array pointing.

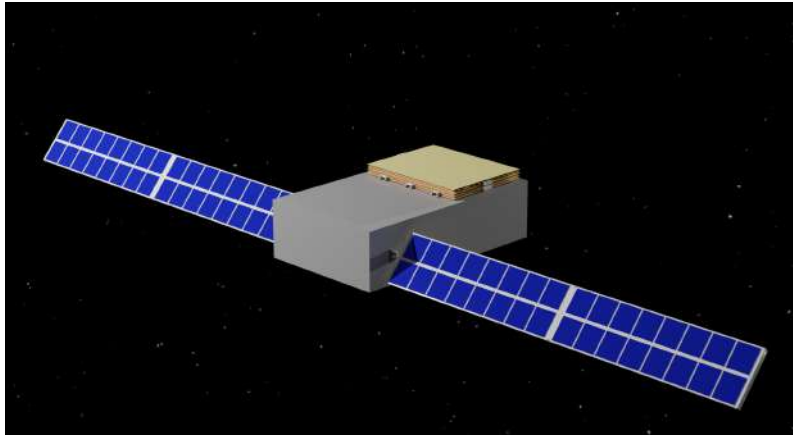


Figure 35: SURROUND Concept Deployment Phase 3: Solar array's central panel unfolds.

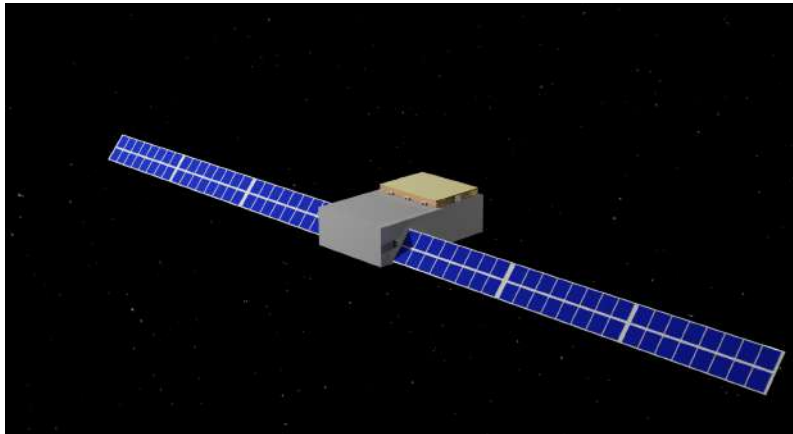


Figure 36: SURROUND Concept Deployment Phase 4: Solar array's outer panel unfolds.

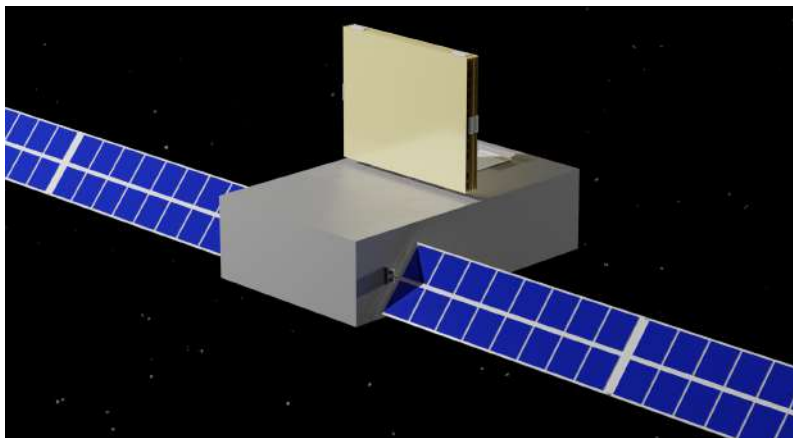


Figure 37: SURROUND Concept - Deployment Phase 5: Folded reflectarray swings upwards.

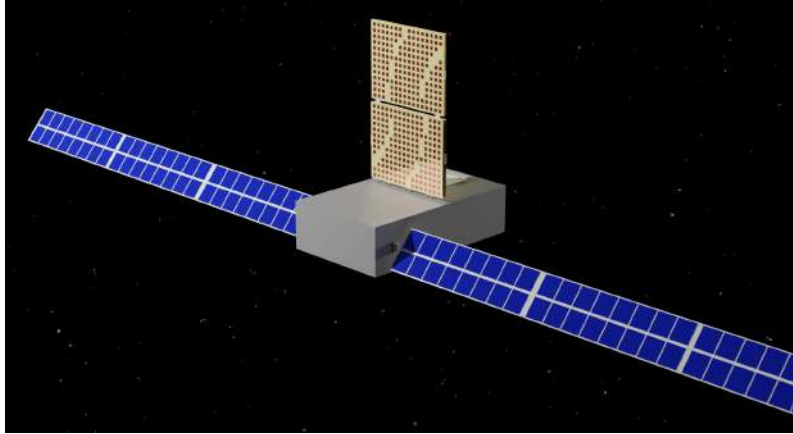


Figure 38: SURROUND Concept - Deployment Phase 6: Upper reflectarray panels swing upwards from lower panels.

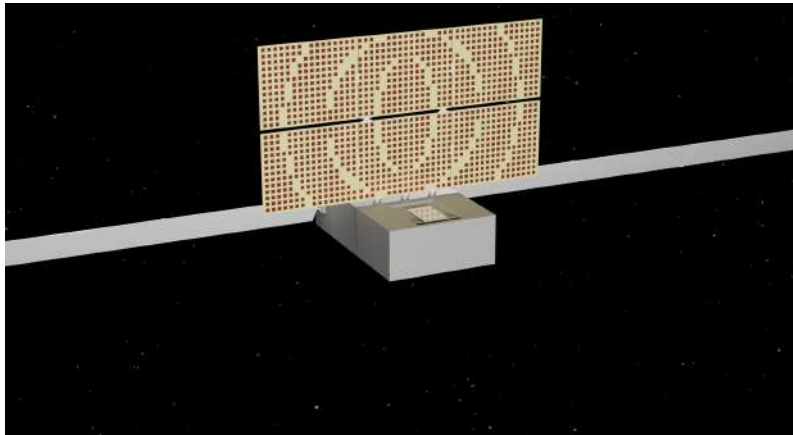


Figure 39: SURROUND Concept - Deployment Phase 7: Outer reflectarray panels swing outwards from central panels.

7. Upper reflectarray panels fold upwards from behind the lower centre panel, Figure 38.
8. Left and right panels fold out from both the upper and lower panel, Figure 39.
9. Reflectarray feed panel folds upwards from recess in spacecraft body, Figure 40.
10. Dipole antennas deploy outwards, Figure 41.

The Solidworks CAD software allows the user to animate assemblies via the creation of a 'Motion Study'. To produce the spacecraft deployment animation, I loaded the mate controller settings into the motion study and proceeded to set the lighting conditions for the animation and creating a camera whereby the viewing angle could be set and changed in a precise manner to generate spin and zoom movements.

To produce photo-realistic renders of the animations I initially attempted to use the Solidworks Visualize programme, when I first did this, I had not implemented an animation for the extension of the dipole antennas. I produced renders of the deployment animation using visualise both from a set viewpoint and using a turntable rotation whereby as the spacecraft deployed the camera view rotated around 360 degrees. Once I was able

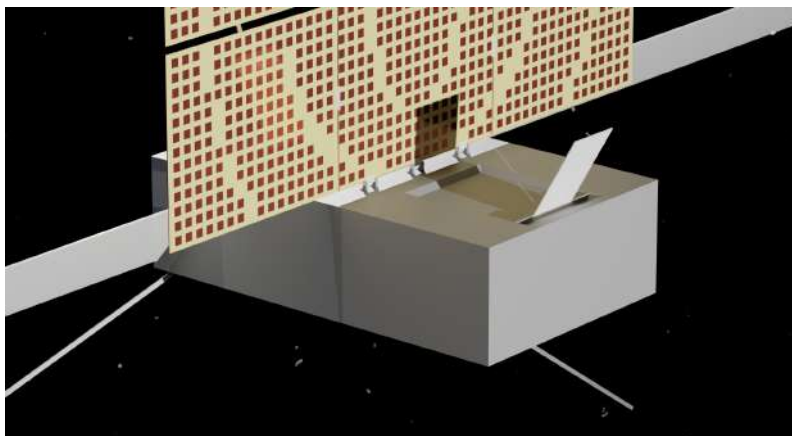


Figure 40: SURROUND Concept - Deployment Phase 8: Reflectarray antenna feed deploys outwards from recess in spacecraft body.

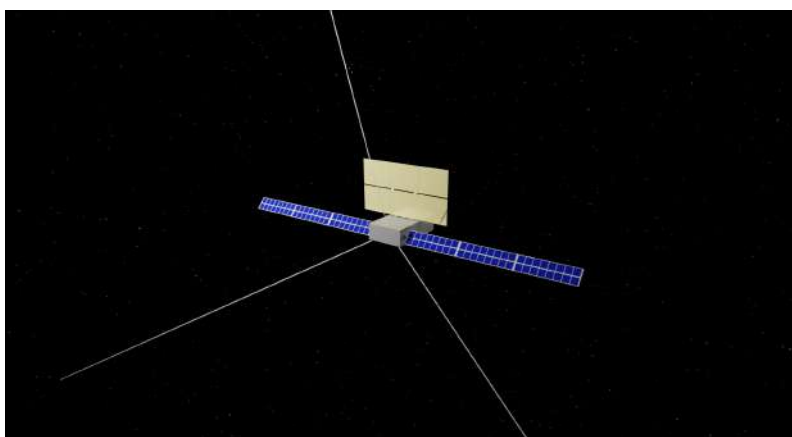


Figure 41: SURROUND Concept - Deployment Phase 9: Radio spectrometer's dipole antennas extend outward.

to animate the extension of the dipole antennas, I attempted to again use Visualize to generate animation renders, however, it then became apparent that this programme did not support the change in component geometry during an animation, only the movement of parts. Therefore, I switched to using the Photoview 360 add-in to the main Solidworks programme.

7.2.2 Mission Context Visualisation

To publicise and make clear the operating principle of the mission I created animations of the satellite formation on the scale of the Sun-Earth system and a representation of radio burst detection by the different spacecraft. This can then be used to aid in the explanation of how the mission will function including the method of determining the location of the radio emission.

I produced this animation using Blender software. I began with importing a STL file of the deployed preliminary design and applied materials to the plain object, a global material of reflective metal and a blue reflective material for the solar panels and a copper material for the reflectarray panels and feed.

I then defined the Sun and Earth objects by creating two spheres, the sun object was kept at the centre and the Earth object was moved along an axis. I proceeded to define the materials for these two objects to resemble their respective bodies. For the sun I applied a noise texture combination of orange and black with a high emissivity value to add glow. For the Earth object I applied another noise texture using a three-colour combination of blue green and brown. To animate the orbit of the Earth around the Sun I added an invisible cube at the Sun and parented the position of the Earth to this object. This enabled the Earth orbit to be implemented by rotating the cube 0-360 degrees.

To position the spacecraft, I imported into the Blender project in an L1 orbit I position the spacecraft close to the Earth object on the Earth-Sun line and then parented the spacecraft to Earth so that it would follow in the orbit. Next, I duplicated the spacecraft and placed one at L4 and another at L5, due to the duplication these objects were already parented to the Earth object and so would maintain their positions relative to Earth throughout the object's orbit. Additionally, I implemented a rotation of the Earth and Sun to simulate Earth's spin and the changes in the Sun's surface.

To visualise the detection of the CME radio bursts I set up a torus object which expands outwards in all directions from a position set near the sun. I also created three objects that I intended to use as 'notifiers' whereby they would indicate the passing of the 'wave' via a change in appearance. To implement this function, I created a sphere and matched and parented its centre with that of the L1 spacecraft. The notifiers should each 'activate' when the radio wave passes over the spacecraft and then remain on. Blender provides a method of scripting the appearance of objects via geometry using a system of nodes 42, the key node I used to accomplish this task is a geometry proximity node that can output the distance between an object and the nearest mesh face of another object. In the initial implementation of this approach the notifier would be activated whilst within the geometry of the torus object, but this was too brief a moment. To keep the 'notifier' activated after the signal passes, I used an inequality math node which activated the appearance change if the distance from the torus origin to the torus face

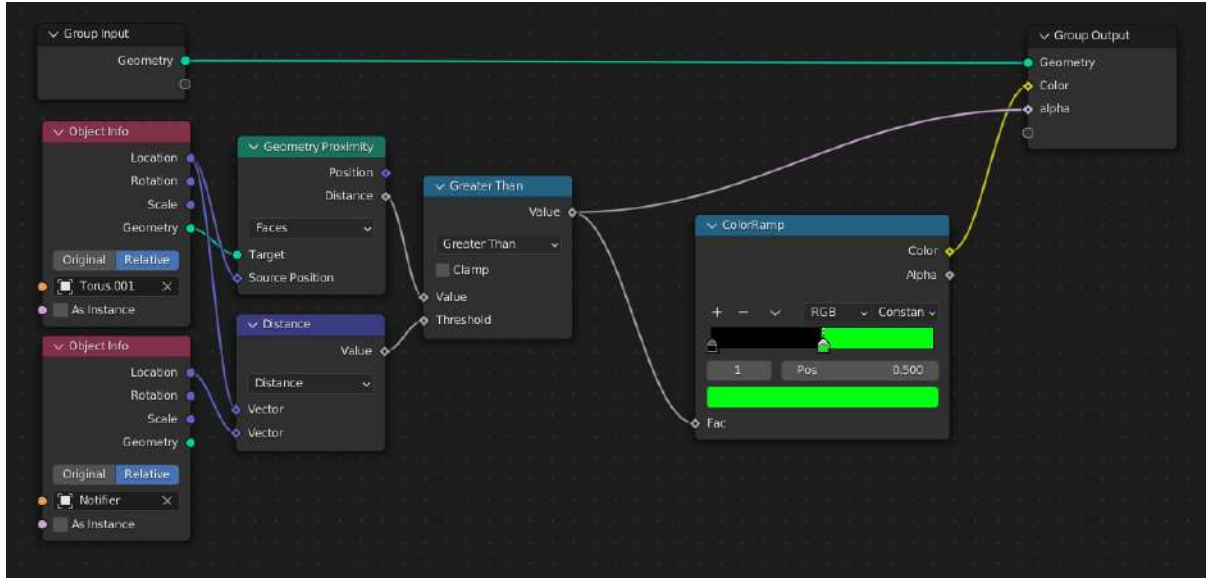


Figure 42: Geometry nodes for the animation of the 'notifier' that indicates the detection of the radio burst by the spacecraft's radio spectrometer

exceeds the distance between the notifier and the torus origin which is the condition reached once the torus ring has passed over the notifier.

8 Conclusion

This project has focused on assessing the performance of CubeSat communication systems to evaluate if a 6U CubeSat can provide a sufficient downlink to operate in a deep space mission regime by transmitting collected data to Earth. Of the currently operational antenna designs available to CubeSats, the reflectarray antenna is the most reliable option as it has been proven via an interplanetary mission and this reliability is accompanied by good performance, although, when competing technologies mature, this assessment could be revisited.

By using the hardware specifications of the MarCO CubeSat, a known system could be applied and assessed for the needs of the SURROUND project. This analysis evaluated the factors contributing to the downlink margin and estimated the performance of this configuration across the proposed orbits of the SURROUND mission. It was also important to assess the necessity and impact of using large diameter deep space antennas to obtain data transmitted by a SURROUND spacecraft, as these large antennas are costly to operate and are under high demand. From this analysis, it can be concluded to communicate with all spacecraft in the constellation throughout their orbits, a 35m Es-track antenna would be required, however, this antenna would still require several hours to obtain a day's worth of data, therefore a trade-off arises whereby the largest deep space antennas, NASA's 70m DSNs could facilitate this downlink volume within 15 minutes yet at greater cost per unit time of operation. If a descope is necessary a lower demand and cost thirteen metre antenna could provide a low-rate downlink to spacecraft at L4 and L5, however, this would prevent the use of an ahead-behind configuration. Communication with the spacecraft at L1 is the easiest to accomplish, with high link margins reached for this spacecraft using all examined configurations.

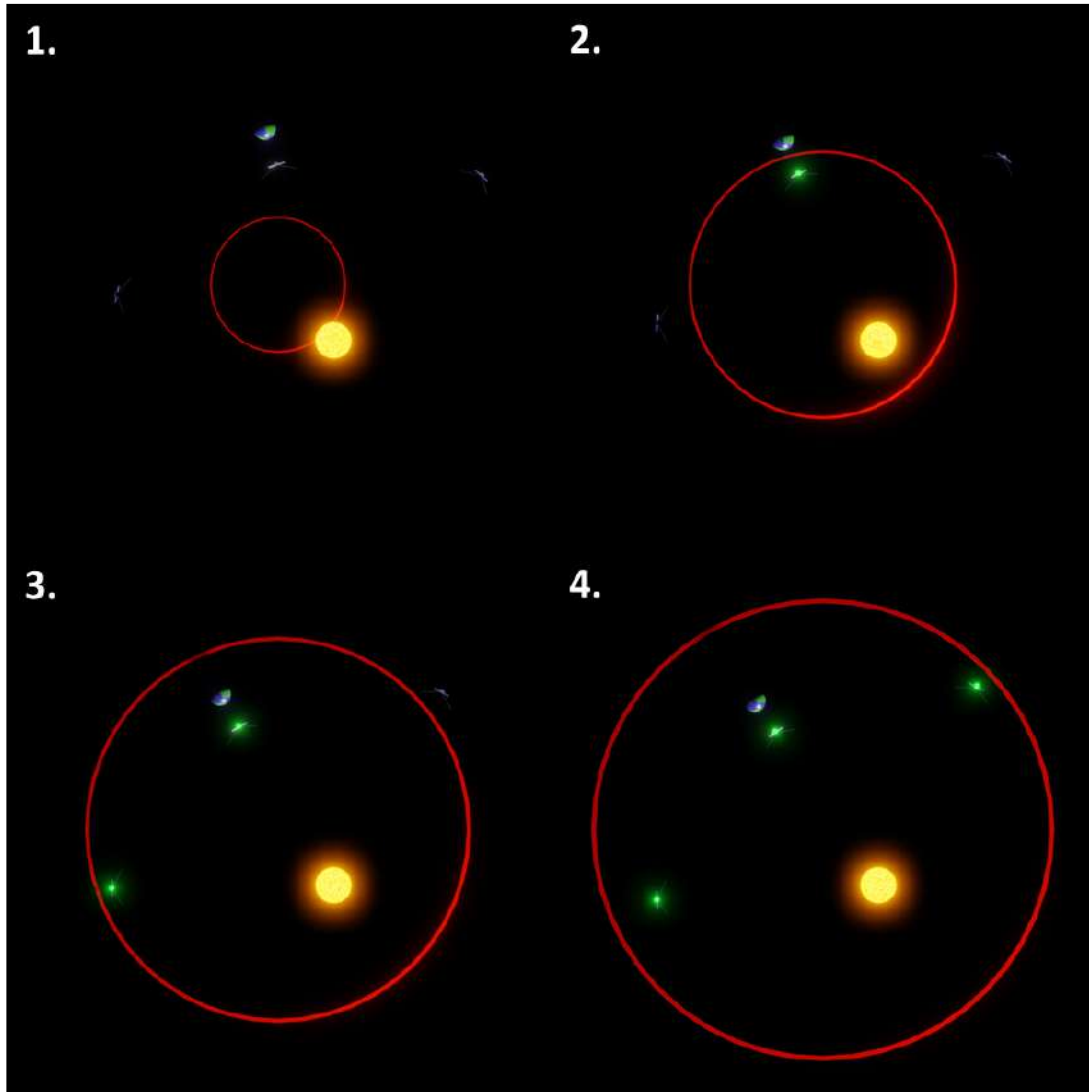


Figure 43: The four phases of the SURROUND animation: 1. A radio burst is emitted from a space weather event and propagates outwards. 2. The spacecraft closest to the radio burst detects the signal. 3. The second closest spacecraft now also detects the radio burst. 4. All three satellites featured in this animation detect the radio burst and so, by using the differing times of arrival of the signal the TDOA method can be applied to localise the origin of the radio burst.

The next steps in defining the communication solution for the SURROUND project would be to examine the schedules and monetary cost of using the various ground stations so that the trade-off can be resolved. The link margins when using other CubeSat-compatible HGAs should be estimated with the goal of enabling the use of a lower cost ground station while achieving the downlink achievable via large ground stations as outlined in this report. Additional factors influencing communications were outside the scope of this report, in future work the impact of spacecraft pointing and resources required to maintain pointing could be factored in.

Success in developing and implementing a CubeSat configuration capable of >1 AU communication could herald in a new age of solar system exploration whereby areas of the solar system that once were exclusive to large space agencies with budgets capable of fielding large, complex spacecraft are now open to lower budget organisations, now able to reach out into the solar system to collect and transmit data back to Earth.

Additionally, in this project several visualisations for the SURROUND mission were created. These included an early spacecraft concept design and the animation of this design deploying from a stowed configuration. A visualisation of the SURROUND constellation has also been produced to demonstrate the geometry of the constellation configuration and the detection of a radio burst by the different spacecraft at different times which will lead on to an explanation of the TOA and TDOA methods. The next step to be taken would be to incorporate a visualisation of the TOA and TDOA methods into the constellation animation and then developing a comprehensive video for the SURROUND project whereby the deployment process defined in Solidworks would be integrated into the blender environment and subsequently the animation of the SURROUND constellation. Ultimately this video would aid in publicising the mission by explaining the SURROUND mission in an approachable and dynamic way.

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9 Appendix

Link Margin Python Function

```
226 #####
227
228 def margin(BR, Gain, Dist, Power):
229
230     import math
231
232     #Desired Bit Rate, bit/s
233     BR = BR
234
235     #####Transmitter####
236     P_Tx = Power
237     G_Tx = 28.8
238     P_Tx_dBW = 10*math.log10(P_Tx)
239     PL = 4.2
240
241     EIRP = P_Tx_dBW + G_Tx - PL
242
243     #####Path Losses####
244     #Frequency
245     f = 8.425e9
246     #Speed of light
247     c = 3e8
248
249     R = Dist
250
251     FSPL = 10*math.log10(((4*math.pi*f)/c)**2)
252
253     #####Receiver####
254     #GT
255     GT_rx = Gain
256     #Boltzmann
257     k = -228.6
258
259     CNo = EIRP - FSPL + GT_rx - k
260
261     EbNo_av = CNo - 10*math.log10(BR) - 0.8 - 0.23
262
263     #####Margin###
264
265     #Eb/No required for BER 1/10^7
266     EbNo_req = 0.1
267
268     Margin = EbNo_av - EbNo_req
269
270     return Margin
271
```